Supplemental Materials for Local Structural-After-Measurement (LSAM) and Traditional Approaches for Nonlinear Effects among Latent Variables: A Simulation Study

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Table S1 includes all the information for the calculation of all performance measures. Tables S2 contains the computation time for the different approaches for simulation 1 and simulation 2. Tables S3-S72 include the numerical results for large part of those metrics. Specifically, for the trimmed mean-based metrics, we did not include them in the tables as interpretation remains the same, but these can be retrieved on the GitHub repository ('results_study_1.rds' and 'results_study_2.rds').

Figures S1-S9 are the complete plots for simulation 1. For Simulation 2, we split the plots by distribution. Hence, we show part of the measures under normality, right-skewness, and uniform conditions respectively (Figures S10-S33).

Table S1

Complete performance measures for evaluating parameter estimates in the simulation studies

Performance Measure	Empirical Calculation	MCSE
Mean-Based Metrics		
Mean Estimate	$\bar{B} = \frac{1}{K}\sum_{k=1}^{K} B_k$	—
Bias	$\bar{B} - \beta$	$\sqrt{S_B^2/K}$
Relative Bias	$(\frac{\bar{B}}{\beta}) - 1$	$\sqrt{S_B^2/(K\beta^2)}$
Percent Relative Bias	$(\frac{\bar{B}}{\beta}) - 1 \times 100\%$	—
MSE	$\frac{1}{K}\sum_{k=1}^{K}(B_k - \beta)^2$	$\sqrt{\frac{1}{K}[S_B^4(k_B - 1) + 4S_B^2 g_B(\bar{B} - \beta) + 4S_B^2(\bar{B} - \beta)^2]}$
RMSE	$\sqrt{\frac{1}{K}\sum_{k=1}^{K}(B_k - \beta)^2}$	$\sqrt{\frac{K-1}{K}\sum_{j=1}^{K}(\text{RMSE}_{(j)} - \text{RMSE})^2}$
Relative MSE	$\frac{(\bar{B}-\beta)^2+S_B^2}{\beta^2}$	$\sqrt{\frac{1}{K\beta^4}[S_B^4(k_B - 1) + 4S_B^2 g_B(\bar{B} - \beta) + 4S_B^2(\bar{B} - \beta)^2]}$
Relative RMSE	$\sqrt{\frac{(\bar{B}-\beta)^2+S_B^2}{\beta^2}}$	$\sqrt{\frac{K-1}{K}\sum_{j=1}^{K}(\text{rRMSE}_{(j)} - \text{rRMSE})^2}$
Mean SE	$\overline{SE} = \frac{1}{K}\sum_{k=1}^{K}\widehat{SE}_k$	—
SE Ratio	$\overline{SE}/S_B$	—
Rejection Rate	$r_\alpha = \frac{1}{K}\sum_{k=1}^{K} I(P_k < \alpha)$	$\sqrt{r_\alpha(1 - r_\alpha)/K}$
Coverage	$c_\beta = \frac{1}{K}\sum_{k=1}^{K} I(A_k \leq \beta \leq B_k)$	$\sqrt{c_\beta(1 - c_\beta)/K}$
Average Width	$\bar{W} = \bar{B} - \bar{A}$	$\sqrt{S_W^2/K}$
Median-Based Metrics		
Median Estimate	$Mdn(B)$	—
Median Bias	$Mdn(B) - \beta$	—
Relative Median Bias	$\frac{Mdn(B)-\beta}{\beta}$	—
Percent Relative Median Bias	$\frac{Mdn(B)-\beta}{\beta} \times 100\%$	—

Performance Measure	Empirical Calculation	MCSE
MAD	$Mdn(\lVert B_k - Mdn(B) \rVert)$	—
Median RMSE	$\sqrt{\text{Bias}^2_{Mdn} + \text{MAD}^2}$	—
Trimmed Mean-Based Metrics		
Trimmed Mean Estimate	$\bar{B}_{\text{trim}}$	—
Trimmed Bias	$\bar{B}_{\text{trim}} - \beta$	—
Trimmed RMSE	$\sqrt{(\bar{B}_{\text{trim}} - \beta)^2 + \text{MAD}^2}$	—
Trimmed Mean SE	$\overline{SE}_{\text{trim}}$	—
RSB (Robust SE Bias)	$\overline{SE}_{\text{trim}}/\text{MAD} - 1$	—

Note. B = parameter estimator; β = true parameter value; K = number of replications; S_B^2 = sample variance; k_B = standardized kurtosis; g_B = standardized skewness; $I(\cdot)$ = indicator function; P_k = p-value from replication k; α = significance level; A_k, B_k = CI endpoints; S_W^2 = variance of CI widths; $\bar{B}_{\text{trim}}$ = 20% trimmed mean of estimates; $\overline{SE}_{\text{trim}}$ = 20% trimmed mean of standard errors; MAD = median absolute deviation with scaling factor 1.4826; RSB = robust relative standard error bias; MCSE = Monte Carlo Standard Error. Dashes indicate MCSEs not computed for these measures.

Table S2

Computation time by approach (in seconds)

					Study 1								Study 2					
					LMS		QML		UPI		LSAM		LSAM		QML		UPI	
Condition	Model	Distribution	N	Reliability	Mean	Mdn	Mean	Mdn	Mean	Mdn	Mean	Mdn	Mean	Mdn	Mean	Mdn	Mean	Mdn
1	Full	normal	400	0.4	11.00	10.14	0.95	0.83	0.23	0.18	0.81	0.80	3.20	3.20	7.09	5.71	6.19	5.53
2	Full	normal	1000	0.4	14.81	14.82	1.39	1.37	0.18	0.18	1.88	1.87	7.76	7.77	8.46	8.33	5.38	5.33
3	Full	normal	400	0.6	9.50	9.49	0.86	0.84	0.16	0.16	0.83	0.82	3.41	3.45	5.41	5.36	5.40	5.29
4	Full	normal	1000	0.6	14.05	13.98	1.49	1.47	0.16	0.16	1.89	1.88	7.84	7.84	8.89	8.82	5.29	5.23
5	Full	normal	400	0.8	13.54	13.57	1.14	1.14	0.20	0.20	0.82	0.81	3.32	3.30	6.66	6.62	6.11	6.11
6	Full	normal	1000	0.8	22.38	22.27	2.07	2.07	0.20	0.20	1.89	1.88	7.78	7.77	10.69	10.62	5.99	5.94
7	Full	right-skewed	400	0.4	12.55	10.38	1.13	0.94	0.27	0.18	0.82	0.81	3.12	3.11	7.72	6.49	6.04	5.35
8	Full	right-skewed	1000	0.4	14.64	14.27	1.67	1.65	0.19	0.18	1.89	1.88	7.46	7.47	10.87	10.71	5.34	5.25
9	Full	right-skewed	400	0.6	10.72	10.41	0.97	0.94	0.18	0.16	0.82	0.81	3.13	3.12	6.82	6.68	5.42	5.37
10	Full	right-skewed	1000	0.6	15.62	15.39	1.71	1.69	0.17	0.17	1.88	1.88	7.45	7.45	12.02	11.89	5.29	5.29
11	Full	right-skewed	400	0.8	15.39	15.39	1.29	1.28	0.21	0.21	0.81	0.81	3.12	3.12	8.47	8.33	6.36	6.35
12	Full	right-skewed	1000	0.8	25.01	24.92	2.33	2.31	0.21	0.21	1.87	1.87	7.41	7.42	15.42	15.26	6.27	6.27
13	Full	uniform	400	0.4	15.70	11.15	1.69	0.96	0.87	0.23	0.81	0.80	3.11	3.08	11.37	8.69	10.48	7.64
14	Full	uniform	1000	0.4	16.46	14.87	1.50	1.35	0.27	0.19	1.91	1.90	7.63	7.62	11.01	8.77	6.54	5.50
15	Full	uniform	400	0.6	9.47	9.48	0.82	0.81	0.17	0.16	0.84	0.83	3.37	3.36	5.40	5.36	5.70	5.53
16	Full	uniform	1000	0.6	13.69	13.68	1.53	1.52	0.16	0.16	1.92	1.91	7.89	7.88	8.97	8.90	5.35	5.33
17	Full	uniform	400	0.8	12.20	12.14	1.26	1.25	0.20	0.19	0.83	0.82	3.37	3.35	6.91	6.85	6.11	6.07
18	Full	uniform	1000	0.8	20.11	20.18	2.35	2.34	0.20	0.19	1.91	1.90	7.84	7.84	12.03	12.00	6.04	6.00
19	Linear	normal	400	0.4	10.80	9.95	0.87	0.74	0.27	0.18	0.83	0.82	3.27	3.25	7.21	5.60	6.63	5.62

Table S2

Computation time by approach (in seconds) (continued)

					Study 1								Study 2					
					LMS		QML		UPI		LSAM		LSAM		QML		UPI	
Condition	Model	Distribution	N	Reliability	Mean	Mdn	Mean	Mdn	Mean	Mdn	Mean	Mdn	Mean	Mdn	Mean	Mdn	Mean	Mdn
20	Linear	normal	1000	0.4	13.39	13.30	1.18	1.16	0.18	0.18	1.92	1.91	8.00	8.03	7.63	7.56	5.51	5.51
21	Linear	normal	400	0.6	8.68	8.66	0.68	0.66	0.16	0.16	0.85	0.83	3.41	3.41	4.70	4.66	5.49	5.44
22	Linear	normal	1000	0.6	11.97	12.01	1.12	1.09	0.16	0.16	1.93	1.92	7.96	7.99	7.44	7.41	5.41	5.40
23	Linear	normal	400	0.8	9.91	9.80	0.86	0.86	0.21	0.20	0.84	0.83	3.43	3.46	5.09	5.05	6.14	6.05
24	Linear	normal	1000	0.8	13.99	13.69	1.43	1.43	0.20	0.20	1.92	1.91	7.98	8.01	8.08	8.04	6.14	6.09
25	Linear	right-skewed	400	0.4	11.58	10.26	1.00	0.86	0.28	0.18	0.83	0.82	3.11	3.11	8.18	6.06	6.82	5.39
26	Linear	right-skewed	1000	0.4	13.39	13.34	1.42	1.40	0.19	0.18	1.91	1.90	7.53	7.53	9.18	9.00	5.58	5.33
27	Linear	right-skewed	400	0.6	9.14	9.09	0.79	0.77	0.18	0.17	0.83	0.83	3.16	3.16	5.27	5.14	5.48	5.38
28	Linear	right-skewed	1000	0.6	13.14	13.12	1.36	1.35	0.17	0.17	1.91	1.90	7.53	7.54	8.45	8.37	5.30	5.30
29	Linear	right-skewed	400	0.8	10.24	9.97	0.91	0.90	0.22	0.21	0.83	0.82	3.17	3.17	5.67	5.60	6.34	6.34
30	Linear	right-skewed	1000	0.8	15.15	14.94	1.58	1.56	0.22	0.21	1.90	1.90	7.50	7.51	9.21	9.12	6.24	6.24
31	Linear	uniform	400	0.4	16.03	11.02	1.77	1.00	0.92	0.23	0.82	0.81	3.11	3.09	11.82	8.97	10.72	8.20
32	Linear	uniform	1000	0.4	15.45	14.09	1.45	1.22	0.29	0.19	1.91	1.90	7.71	7.70	11.10	8.54	6.93	5.64
33	Linear	uniform	400	0.6	9.11	9.09	0.72	0.71	0.17	0.16	0.84	0.83	3.49	3.53	4.86	4.72	5.64	5.49
34	Linear	uniform	1000	0.6	12.39	12.43	1.20	1.18	0.16	0.16	1.93	1.92	7.96	7.96	7.34	7.19	5.42	5.38
35	Linear	uniform	400	0.8	10.03	9.89	1.00	1.02	0.20	0.19	0.84	0.83	3.42	3.44	5.86	5.81	6.20	6.17
36	Linear	uniform	1000	0.8	14.58	14.40	1.72	1.76	0.20	0.19	1.90	1.89	7.88	7.84	9.36	9.35	6.11	6.10

Note. Values represent mean and median computation times across 1,000 replications.

Table S3

Simulation 1 - Condition 1: Full Model, Distribution = normal, N = 400, Reliability = 0.4

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.296	-0.020 ± 0.003	-0.063 ± 0.011	0.010 ± 0.001	0.102 ± 0.003	0.011 ± 0.001	0.104 ± 0.003	0.109 ± 0.005	0.330 ± 0.010	0.289	-0.027	0.0969	0.1005	0.974	93.399 ± 0.008	0.390 ± 0.002	87.239 ± 0.011
	beta32	0.316	0.302	-0.014 ± 0.003	-0.044 ± 0.010	0.010 ± 0.000	0.099 ± 0.002	0.010 ± 0.000	0.100 ± 0.003	0.101 ± 0.005	0.317 ± 0.009	0.298	-0.018	0.0982	0.0997	1.002	94.389 ± 0.008	0.390 ± 0.002	88.449 ± 0.011
	beta33	0.139	0.111	-0.028 ± 0.005	-0.199 ± 0.035	0.021 ± 0.001	0.145 ± 0.004	0.022 ± 0.001	0.147 ± 0.004	1.120 ± 0.056	1.058 ± 0.031	0.104	-0.035	0.1355	0.1400	0.980	95.160 ± 0.007	0.555 ± 0.004	13.201 ± 0.011
	beta34	0.101	0.107	0.006 ± 0.003	0.064 ± 0.032	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.098 ± 0.003	0.942 ± 0.055	0.971 ± 0.032	0.105	0.004	0.0915	0.0916	0.973	95.710 ± 0.007	0.373 ± 0.003	19.032 ± 0.013
	beta35	0.101	0.102	0.001 ± 0.003	0.012 ± 0.031	0.009 ± 0.000	0.094 ± 0.003	0.009 ± 0.000	0.094 ± 0.003	0.872 ± 0.047	0.934 ± 0.030	0.101	0.000	0.0940	0.0940	0.994	96.150 ± 0.006	0.368 ± 0.003	17.272 ± 0.013
LSAM	beta31	0.316	0.316	-0.000 ± 0.004	-0.000 ± 0.013	0.015 ± 0.001	0.121 ± 0.003	0.015 ± 0.001	0.121 ± 0.004	0.147 ± 0.008	0.384 ± 0.012	0.311	-0.005	0.1210	0.1211	1.176	97.360 ± 0.005	0.560 ± 0.011	69.747 ± 0.015
	beta32	0.316	0.324	0.008 ± 0.004	0.025 ± 0.012	0.013 ± 0.001	0.116 ± 0.003	0.013 ± 0.001	0.116 ± 0.004	0.135 ± 0.007	0.367 ± 0.012	0.320	0.004	0.1168	0.1169	1.207	98.680 ± 0.004	0.548 ± 0.009	72.937 ± 0.015
	beta33	0.139	0.152	0.013 ± 0.006	0.092 ± 0.043	0.033 ± 0.002	0.180 ± 0.005	0.033 ± 0.002	0.181 ± 0.006	1.689 ± 0.104	1.300 ± 0.046	0.142	0.003	0.1666	0.1666	1.150	98.570 ± 0.004	0.813 ± 0.036	10.891 ± 0.010
	beta34	0.101	0.105	0.004 ± 0.004	0.044 ± 0.042	0.016 ± 0.001	0.128 ± 0.004	0.016 ± 0.001	0.128 ± 0.004	1.595 ± 0.098	1.263 ± 0.044	0.104	0.003	0.1174	0.1175	1.117	98.460 ± 0.004	0.559 ± 0.023	8.141 ± 0.009
	beta35	0.101	0.104	0.003 ± 0.004	0.030 ± 0.043	0.017 ± 0.001	0.131 ± 0.004	0.017 ± 0.001	0.131 ± 0.005	1.682 ± 0.103	1.297 ± 0.045	0.099	-0.002	0.1138	0.1138	1.089	98.350 ± 0.004	0.559 ± 0.020	10.781 ± 0.010
QML	beta31	0.316	0.296	-0.020 ± 0.003	-0.062 ± 0.011	0.011 ± 0.001	0.102 ± 0.003	0.011 ± 0.001	0.104 ± 0.003	0.109 ± 0.005	0.330 ± 0.010	0.289	-0.027	0.0980	0.1015	0.972	93.179 ± 0.008	0.390 ± 0.002	87.349 ± 0.011
	beta32	0.316	0.302	-0.014 ± 0.003	-0.043 ± 0.010	0.010 ± 0.000	0.100 ± 0.002	0.010 ± 0.000	0.101 ± 0.003	0.101 ± 0.005	0.318 ± 0.009	0.299	-0.017	0.0976	0.0990	0.999	94.389 ± 0.008	0.390 ± 0.002	88.889 ± 0.010
	beta33	0.139	0.113	-0.026 ± 0.005	-0.184 ± 0.035	0.021 ± 0.001	0.146 ± 0.004	0.022 ± 0.001	0.148 ± 0.004	1.130 ± 0.058	1.063 ± 0.032	0.106	-0.033	0.1391	0.1429	0.975	95.160 ± 0.007	0.556 ± 0.004	13.531 ± 0.011
	beta34	0.101	0.106	0.005 ± 0.003	0.052 ± 0.032	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.098 ± 0.003	0.945 ± 0.053	0.972 ± 0.032	0.104	0.003	0.0923	0.0923	0.970	95.380 ± 0.007	0.373 ± 0.003	19.032 ± 0.013
	beta35	0.101	0.102	0.001 ± 0.003	0.005 ± 0.031	0.009 ± 0.001	0.095 ± 0.003	0.009 ± 0.001	0.095 ± 0.003	0.892 ± 0.049	0.944 ± 0.030	0.099	-0.002	0.0940	0.0941	0.981	96.040 ± 0.006	0.367 ± 0.003	17.382 ± 0.013
UPI	beta31	0.316	0.319	0.003 ± 0.004	0.009 ± 0.013	0.016 ± 0.001	0.127 ± 0.003	0.016 ± 0.001	0.127 ± 0.004	0.161 ± 0.009	0.402 ± 0.013	0.314	-0.002	0.1240	0.1240	1.656	95.160 ± 0.007	0.824 ± 0.339	77.558 ± 0.014
	beta32	0.316	0.329	0.013 ± 0.004	0.041 ± 0.013	0.015 ± 0.001	0.121 ± 0.003	0.015 ± 0.001	0.122 ± 0.004	0.148 ± 0.008	0.385 ± 0.012	0.324	0.008	0.1243	0.1246	1.755	96.920 ± 0.006	0.833 ± 0.351	79.648 ± 0.013
	beta33	0.139	0.168	0.029 ± 0.007	0.208 ± 0.050	0.043 ± 0.003	0.208 ± 0.007	0.044 ± 0.003	0.210 ± 0.008	2.275 ± 0.147	1.508 ± 0.055	0.145	0.006	0.1741	0.1743	1.567	95.930 ± 0.007	1.276 ± 0.492	15.622 ± 0.012
	beta34	0.101	0.107	0.006 ± 0.005	0.058 ± 0.049	0.022 ± 0.001	0.150 ± 0.005	0.022 ± 0.001	0.150 ± 0.005	2.203 ± 0.133	1.484 ± 0.051	0.100	-0.001	0.1282	0.1282	1.283	94.279 ± 0.008	0.754 ± 0.198	11.771 ± 0.011
	beta35	0.101	0.098	-0.003 ± 0.005	-0.027 ± 0.051	0.025 ± 0.002	0.157 ± 0.005	0.025 ± 0.002	0.157 ± 0.006	2.405 ± 0.149	1.551 ± 0.055	0.089	-0.012	0.1325	0.1331	3.226	95.270 ± 0.007	1.982 ± 1.425	13.861 ± 0.011

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S4

Simulation 1 - Condition 2: Full Model, Distribution = normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.297	-0.019 ± 0.002	-0.061 ± 0.006	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.042 ± 0.002	0.205 ± 0.006	0.296	-0.020	0.0599	0.0633	1.011	93.939 ± 0.008	0.245 ± 0.001	99.697 ± 0.002
	beta32	0.316	0.294	-0.022 ± 0.002	-0.068 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.041 ± 0.002	0.201 ± 0.005	0.292	-0.024	0.0638	0.0681	1.042	93.939 ± 0.008	0.245 ± 0.001	99.899 ± 0.001
	beta33	0.139	0.106	-0.033 ± 0.003	-0.236 ± 0.020	0.008 ± 0.000	0.089 ± 0.002	0.009 ± 0.000	0.095 ± 0.003	0.469 ± 0.022	0.685 ± 0.019	0.109	-0.030	0.0865	0.0914	0.993	93.636 ± 0.008	0.348 ± 0.002	21.616 ± 0.013
	beta34	0.101	0.109	0.008 ± 0.002	0.079 ± 0.018	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.341 ± 0.017	0.584 ± 0.017	0.108	0.007	0.0557	0.0560	1.005	96.263 ± 0.006	0.231 ± 0.001	46.869 ± 0.016
	beta35	0.101	0.108	0.007 ± 0.002	0.072 ± 0.019	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.352 ± 0.017	0.593 ± 0.017	0.107	0.006	0.0560	0.0564	0.991	95.455 ± 0.007	0.231 ± 0.001	44.949 ± 0.016
LSAM	beta31	0.316	0.317	0.001 ± 0.002	0.004 ± 0.007	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.049 ± 0.002	0.222 ± 0.006	0.315	-0.001	0.0688	0.0688	1.102	96.667 ± 0.006	0.303 ± 0.001	99.192 ± 0.003
	beta32	0.316	0.314	-0.002 ± 0.002	-0.006 ± 0.007	0.005 ± 0.000	0.067 ± 0.001	0.005 ± 0.000	0.067 ± 0.002	0.045 ± 0.002	0.213 ± 0.006	0.313	-0.003	0.0716	0.0717	1.122	97.475 ± 0.005	0.296 ± 0.001	99.596 ± 0.002
	beta33	0.139	0.138	-0.001 ± 0.003	-0.004 ± 0.022	0.009 ± 0.000	0.097 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	0.488 ± 0.025	0.698 ± 0.021	0.141	0.002	0.0929	0.0930	1.020	97.374 ± 0.005	0.388 ± 0.004	34.444 ± 0.015
	beta34	0.101	0.106	0.005 ± 0.002	0.052 ± 0.022	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.471 ± 0.025	0.686 ± 0.021	0.105	0.004	0.0643	0.0644	1.018	96.162 ± 0.006	0.276 ± 0.003	33.232 ± 0.015
	beta35	0.101	0.108	0.007 ± 0.002	0.065 ± 0.022	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.502 ± 0.027	0.709 ± 0.022	0.107	0.006	0.0694	0.0696	0.998	96.364 ± 0.006	0.279 ± 0.003	33.636 ± 0.015
QML	beta31	0.316	0.297	-0.019 ± 0.002	-0.060 ± 0.006	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.042 ± 0.002	0.205 ± 0.006	0.296	-0.020	0.0599	0.0632	1.009	93.838 ± 0.008	0.245 ± 0.001	99.697 ± 0.002
	beta32	0.316	0.294	-0.022 ± 0.002	-0.068 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.040 ± 0.002	0.201 ± 0.005	0.292	-0.024	0.0642	0.0685	1.042	94.141 ± 0.007	0.244 ± 0.001	99.899 ± 0.001
	beta33	0.139	0.108	-0.031 ± 0.003	-0.221 ± 0.020	0.008 ± 0.000	0.090 ± 0.002	0.009 ± 0.000	0.095 ± 0.003	0.463 ± 0.022	0.681 ± 0.019	0.112	-0.027	0.0855	0.0898	0.988	93.131 ± 0.008	0.347 ± 0.002	22.323 ± 0.013
	beta34	0.101	0.108	0.007 ± 0.002	0.066 ± 0.018	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.339 ± 0.017	0.582 ± 0.017	0.106	0.005	0.0564	0.0566	1.002	96.364 ± 0.006	0.229 ± 0.001	46.768 ± 0.016
	beta35	0.101	0.107	0.006 ± 0.002	0.059 ± 0.019	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.351 ± 0.017	0.593 ± 0.017	0.106	0.005	0.0566	0.0568	0.986	95.556 ± 0.007	0.230 ± 0.001	44.646 ± 0.016
UPI	beta31	0.316	0.319	0.003 ± 0.002	0.009 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.051 ± 0.002	0.225 ± 0.006	0.318	0.002	0.0693	0.0693	0.985	94.545 ± 0.007	0.275 ± 0.001	99.394 ± 0.002
	beta32	0.316	0.316	-0.000 ± 0.002	-0.000 ± 0.007	0.005 ± 0.000	0.068 ± 0.001	0.005 ± 0.000	0.068 ± 0.002	0.046 ± 0.002	0.215 ± 0.006	0.314	-0.002	0.0714	0.0714	1.029	96.263 ± 0.006	0.274 ± 0.001	99.798 ± 0.001
	beta33	0.139	0.142	0.003 ± 0.003	0.021 ± 0.024	0.011 ± 0.001	0.103 ± 0.003	0.011 ± 0.001	0.103 ± 0.003	0.553 ± 0.029	0.744 ± 0.023	0.140	0.001	0.0974	0.0974	0.935	94.949 ± 0.007	0.379 ± 0.004	36.263 ± 0.015
	beta34	0.101	0.108	0.007 ± 0.002	0.065 ± 0.024	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.581 ± 0.032	0.762 ± 0.024	0.104	0.003	0.0710	0.0710	0.902	93.838 ± 0.008	0.271 ± 0.003	37.071 ± 0.015
	beta35	0.101	0.106	0.005 ± 0.003	0.054 ± 0.025	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.609 ± 0.033	0.781 ± 0.025	0.102	0.001	0.0734	0.0734	0.887	93.232 ± 0.008	0.274 ± 0.003	36.869 ± 0.015

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S5

Simulation 1 - Condition 3: Full Model, Distribution = normal, N = 400, Reliability = 0.6

			Mean-based										Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.307	-0.009 ± 0.002	-0.029 ± 0.007	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.052 ± 0.002	0.229 ± 0.006	0.306	-0.010	0.0705	0.0713	1.041	95.486 ± 0.007	0.293 ± 0.001	98.796 ± 0.003
	beta32	0.316	0.307	-0.009 ± 0.002	-0.028 ± 0.007	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.056 ± 0.003	0.237 ± 0.007	0.304	-0.012	0.0758	0.0767	1.003	94.885 ± 0.007	0.292 ± 0.001	99.097 ± 0.003
	beta33	0.139	0.121	-0.018 ± 0.003	-0.127 ± 0.023	0.010 ± 0.001	0.101 ± 0.003	0.010 ± 0.001	0.102 ± 0.003	0.541 ± 0.028	0.736 ± 0.022	0.120	-0.019	0.0913	0.0933	0.927	92.678 ± 0.008	0.366 ± 0.002	26.780 ± 0.014
	beta34	0.101	0.105	0.004 ± 0.002	0.035 ± 0.021	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.424 ± 0.019	0.651 ± 0.018	0.104	0.003	0.0671	0.0672	0.964	94.985 ± 0.007	0.248 ± 0.001	40.522 ± 0.016
	beta35	0.101	0.104	0.003 ± 0.002	0.026 ± 0.021	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.428 ± 0.020	0.654 ± 0.019	0.102	0.001	0.0615	0.0615	0.959	94.383 ± 0.007	0.248 ± 0.001	38.215 ± 0.015
LSAM	beta31	0.316	0.318	0.002 ± 0.002	0.008 ± 0.008	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.059 ± 0.003	0.242 ± 0.007	0.318	0.002	0.0740	0.0741	1.074	96.690 ± 0.006	0.322 ± 0.001	98.195 ± 0.004
	beta32	0.316	0.319	0.003 ± 0.003	0.008 ± 0.008	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.064 ± 0.003	0.252 ± 0.007	0.314	-0.002	0.0809	0.0810	1.013	95.486 ± 0.007	0.316 ± 0.001	98.195 ± 0.004
	beta33	0.139	0.141	0.002 ± 0.003	0.016 ± 0.024	0.011 ± 0.001	0.107 ± 0.003	0.011 ± 0.001	0.107 ± 0.003	0.587 ± 0.032	0.766 ± 0.024	0.141	0.002	0.0967	0.0967	0.915	93.079 ± 0.008	0.382 ± 0.003	34.403 ± 0.015
	beta34	0.101	0.104	0.003 ± 0.002	0.031 ± 0.023	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.519 ± 0.026	0.720 ± 0.021	0.104	0.003	0.0713	0.0714	0.951	95.587 ± 0.007	0.271 ± 0.002	35.306 ± 0.015
	beta35	0.101	0.104	0.003 ± 0.002	0.028 ± 0.023	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.506 ± 0.025	0.711 ± 0.021	0.100	-0.001	0.0666	0.0666	0.951	94.383 ± 0.007	0.268 ± 0.002	36.409 ± 0.015
QML	beta31	0.316	0.307	-0.009 ± 0.002	-0.029 ± 0.007	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.052 ± 0.002	0.229 ± 0.006	0.305	-0.011	0.0703	0.0711	1.040	95.486 ± 0.007	0.293 ± 0.001	98.696 ± 0.004
	beta32	0.316	0.307	-0.009 ± 0.002	-0.028 ± 0.007	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.056 ± 0.003	0.237 ± 0.007	0.304	-0.012	0.0754	0.0764	1.003	94.885 ± 0.007	0.292 ± 0.001	99.097 ± 0.003
	beta33	0.139	0.122	-0.017 ± 0.003	-0.120 ± 0.023	0.010 ± 0.001	0.101 ± 0.003	0.010 ± 0.001	0.102 ± 0.003	0.538 ± 0.027	0.734 ± 0.022	0.121	-0.018	0.0913	0.0931	0.928	92.477 ± 0.008	0.366 ± 0.002	27.382 ± 0.014
	beta34	0.101	0.104	0.003 ± 0.002	0.026 ± 0.021	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.421 ± 0.019	0.649 ± 0.018	0.103	0.002	0.0666	0.0666	0.965	94.584 ± 0.007	0.248 ± 0.001	40.221 ± 0.016
	beta35	0.101	0.103	0.002 ± 0.002	0.017 ± 0.021	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.427 ± 0.020	0.654 ± 0.019	0.102	0.001	0.0614	0.0614	0.958	94.183 ± 0.007	0.248 ± 0.001	37.412 ± 0.015
UPI	beta31	0.316	0.319	0.003 ± 0.002	0.010 ± 0.008	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.060 ± 0.003	0.245 ± 0.007	0.318	0.002	0.0774	0.0774	1.004	95.186 ± 0.007	0.305 ± 0.001	98.395 ± 0.004
	beta32	0.316	0.320	0.004 ± 0.003	0.011 ± 0.008	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.063 ± 0.003	0.252 ± 0.007	0.316	0.000	0.0770	0.0770	0.978	95.085 ± 0.007	0.305 ± 0.001	98.395 ± 0.004
	beta33	0.139	0.143	0.004 ± 0.003	0.029 ± 0.025	0.012 ± 0.001	0.110 ± 0.003	0.012 ± 0.001	0.110 ± 0.003	0.628 ± 0.034	0.792 ± 0.025	0.142	0.003	0.1009	0.1010	0.883	92.477 ± 0.008	0.381 ± 0.003	35.707 ± 0.015
	beta34	0.101	0.104	0.003 ± 0.002	0.032 ± 0.024	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.578 ± 0.029	0.760 ± 0.023	0.104	0.003	0.0746	0.0747	0.891	94.183 ± 0.007	0.268 ± 0.002	36.911 ± 0.015
	beta35	0.101	0.103	0.002 ± 0.002	0.020 ± 0.023	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.547 ± 0.027	0.739 ± 0.022	0.100	-0.001	0.0683	0.0684	0.916	93.480 ± 0.008	0.268 ± 0.002	33.801 ± 0.015

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S6

Simulation 1 - Condition 4: Full Model, Distribution = normal, N = 1000, Reliability = 0.6

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.304	-0.012 ± 0.001	-0.038 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.149 ± 0.004	0.304	-0.012	0.0445	0.0462	1.025	93.788 ± 0.008	0.183 ± 0.000	100.000 ± 0.000
	beta32	0.316	0.305	-0.011 ± 0.002	-0.034 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.154 ± 0.004	0.304	-0.012	0.0484	0.0498	0.983	93.788 ± 0.008	0.183 ± 0.000	100.000 ± 0.000
	beta33	0.139	0.120	-0.019 ± 0.002	-0.137 ± 0.013	0.003 ± 0.000	0.056 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.183 ± 0.008	0.428 ± 0.011	0.118	-0.021	0.0570	0.0608	1.024	94.489 ± 0.007	0.226 ± 0.001	54.509 ± 0.016
	beta34	0.101	0.106	0.005 ± 0.001	0.047 ± 0.012	0.001 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.148 ± 0.007	0.385 ± 0.011	0.105	0.004	0.0370	0.0373	1.014	94.990 ± 0.007	0.154 ± 0.000	78.858 ± 0.013
	beta35	0.101	0.105	0.004 ± 0.001	0.038 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.145 ± 0.006	0.381 ± 0.010	0.105	0.004	0.0393	0.0396	1.022	95.691 ± 0.006	0.154 ± 0.001	76.453 ± 0.013
LSAM	beta31	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.152 ± 0.004	0.314	-0.002	0.0467	0.0467	1.056	95.691 ± 0.006	0.199 ± 0.000	100.000 ± 0.000
	beta32	0.316	0.317	0.001 ± 0.002	0.002 ± 0.005	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.025 ± 0.001	0.158 ± 0.004	0.315	-0.001	0.0520	0.0520	0.990	94.890 ± 0.007	0.194 ± 0.000	100.000 ± 0.000
	beta33	0.139	0.137	-0.002 ± 0.002	-0.011 ± 0.013	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.166 ± 0.008	0.408 ± 0.011	0.137	-0.002	0.0563	0.0563	1.024	95.992 ± 0.006	0.228 ± 0.001	67.335 ± 0.015
	beta34	0.101	0.103	0.002 ± 0.001	0.021 ± 0.013	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.165 ± 0.008	0.406 ± 0.012	0.102	0.001	0.0410	0.0410	1.009	95.090 ± 0.007	0.162 ± 0.001	72.044 ± 0.014
	beta35	0.101	0.102	0.001 ± 0.001	0.013 ± 0.013	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.163 ± 0.007	0.404 ± 0.011	0.102	0.001	0.0417	0.0417	1.009	95.291 ± 0.007	0.161 ± 0.001	70.842 ± 0.014
QML	beta31	0.316	0.304	-0.012 ± 0.001	-0.038 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.149 ± 0.004	0.304	-0.012	0.0442	0.0459	1.024	93.788 ± 0.008	0.183 ± 0.000	100.000 ± 0.000
	beta32	0.316	0.305	-0.011 ± 0.002	-0.033 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.154 ± 0.004	0.304	-0.012	0.0488	0.0502	0.981	93.788 ± 0.008	0.183 ± 0.000	100.000 ± 0.000
	beta33	0.139	0.121	-0.018 ± 0.002	-0.133 ± 0.013	0.003 ± 0.000	0.056 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.182 ± 0.008	0.427 ± 0.011	0.118	-0.021	0.0566	0.0602	1.023	94.589 ± 0.007	0.226 ± 0.001	54.810 ± 0.016
	beta34	0.101	0.105	0.004 ± 0.001	0.041 ± 0.012	0.001 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.148 ± 0.007	0.384 ± 0.011	0.104	0.003	0.0371	0.0373	1.013	94.790 ± 0.007	0.153 ± 0.000	78.457 ± 0.013
	beta35	0.101	0.104	0.003 ± 0.001	0.032 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.145 ± 0.006	0.381 ± 0.010	0.104	0.003	0.0395	0.0396	1.021	95.591 ± 0.006	0.153 ± 0.001	76.353 ± 0.013
UPI	beta31	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.152 ± 0.004	0.315	-0.001	0.0454	0.0454	1.003	94.589 ± 0.007	0.189 ± 0.000	100.000 ± 0.000
	beta32	0.316	0.317	0.001 ± 0.002	0.003 ± 0.005	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.025 ± 0.001	0.159 ± 0.004	0.316	0.000	0.0517	0.0517	0.961	95.090 ± 0.007	0.189 ± 0.000	100.000 ± 0.000
	beta33	0.139	0.138	-0.001 ± 0.002	-0.005 ± 0.013	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.177 ± 0.008	0.421 ± 0.012	0.138	-0.001	0.0559	0.0559	0.978	94.890 ± 0.007	0.224 ± 0.001	68.236 ± 0.015
	beta34	0.101	0.102	0.001 ± 0.001	0.013 ± 0.013	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.172 ± 0.008	0.415 ± 0.012	0.101	0.000	0.0410	0.0410	0.966	94.389 ± 0.007	0.159 ± 0.001	72.745 ± 0.014
	beta35	0.101	0.102	0.001 ± 0.001	0.009 ± 0.013	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.169 ± 0.008	0.411 ± 0.012	0.102	0.001	0.0414	0.0414	0.981	94.790 ± 0.007	0.160 ± 0.001	71.643 ± 0.014

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S7

Simulation 1 - Condition 5: Full Model, Distribution = normal, N = 400, Reliability = 0.8

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.006	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.041 ± 0.002	0.202 ± 0.006	0.313	-0.003	0.0618	0.0619	0.975	94.094 ± 0.007	0.244 ± 0.000	99.900 ± 0.001
	beta32	0.316	0.310	-0.006 ± 0.002	-0.018 ± 0.006	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.040 ± 0.002	0.200 ± 0.005	0.311	-0.005	0.0634	0.0636	0.986	93.994 ± 0.008	0.244 ± 0.001	99.800 ± 0.001
	beta33	0.139	0.135	-0.004 ± 0.002	-0.031 ± 0.017	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.278 ± 0.012	0.527 ± 0.014	0.135	-0.004	0.0728	0.0729	0.962	94.995 ± 0.007	0.276 ± 0.001	48.849 ± 0.016
	beta34	0.101	0.102	0.001 ± 0.002	0.014 ± 0.015	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.239 ± 0.010	0.488 ± 0.013	0.101	0.000	0.0493	0.0493	0.984	94.294 ± 0.007	0.190 ± 0.001	56.156 ± 0.016
	beta35	0.101	0.101	-0.000 ± 0.002	-0.001 ± 0.016	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.242 ± 0.011	0.492 ± 0.013	0.101	0.000	0.0490	0.0490	0.980	94.995 ± 0.007	0.191 ± 0.001	56.757 ± 0.016
LSAM	beta31	0.316	0.320	0.004 ± 0.002	0.012 ± 0.007	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.043 ± 0.002	0.207 ± 0.006	0.317	0.001	0.0654	0.0654	0.968	94.194 ± 0.007	0.248 ± 0.001	99.800 ± 0.001
	beta32	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.006	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.041 ± 0.002	0.203 ± 0.006	0.316	0.000	0.0631	0.0631	0.970	93.894 ± 0.008	0.244 ± 0.001	99.900 ± 0.001
	beta33	0.139	0.142	0.003 ± 0.002	0.022 ± 0.017	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.285 ± 0.013	0.534 ± 0.015	0.142	0.003	0.0753	0.0754	0.933	93.193 ± 0.008	0.271 ± 0.001	54.354 ± 0.016
	beta34	0.101	0.102	0.001 ± 0.002	0.013 ± 0.016	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.256 ± 0.012	0.506 ± 0.014	0.100	-0.001	0.0515	0.0515	0.961	94.394 ± 0.007	0.192 ± 0.001	54.855 ± 0.016
	beta35	0.101	0.101	-0.000 ± 0.002	-0.003 ± 0.016	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.260 ± 0.012	0.510 ± 0.014	0.101	0.000	0.0522	0.0522	0.943	93.594 ± 0.008	0.190 ± 0.001	55.956 ± 0.016
QML	beta31	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.006	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.041 ± 0.002	0.202 ± 0.006	0.313	-0.003	0.0621	0.0622	0.975	94.094 ± 0.007	0.244 ± 0.000	99.900 ± 0.001
	beta32	0.316	0.310	-0.006 ± 0.002	-0.019 ± 0.006	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.040 ± 0.002	0.201 ± 0.005	0.311	-0.005	0.0631	0.0633	0.986	93.794 ± 0.008	0.244 ± 0.000	99.800 ± 0.001
	beta33	0.139	0.134	-0.005 ± 0.002	-0.033 ± 0.017	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.278 ± 0.012	0.527 ± 0.014	0.135	-0.004	0.0720	0.0722	0.962	94.995 ± 0.007	0.276 ± 0.001	48.949 ± 0.016
	beta34	0.101	0.102	0.001 ± 0.002	0.013 ± 0.015	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.239 ± 0.010	0.489 ± 0.013	0.101	0.000	0.0495	0.0495	0.983	94.294 ± 0.007	0.190 ± 0.001	56.056 ± 0.016
	beta35	0.101	0.101	-0.000 ± 0.002	-0.002 ± 0.016	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.242 ± 0.011	0.492 ± 0.013	0.101	0.000	0.0493	0.0493	0.979	94.995 ± 0.007	0.191 ± 0.001	56.757 ± 0.016
UPI	beta31	0.316	0.320	0.004 ± 0.002	0.013 ± 0.007	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.043 ± 0.002	0.207 ± 0.006	0.317	0.001	0.0649	0.0649	0.946	93.493 ± 0.008	0.242 ± 0.001	99.800 ± 0.001
	beta32	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.006	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.042 ± 0.002	0.204 ± 0.006	0.316	0.000	0.0630	0.0630	0.956	93.694 ± 0.008	0.242 ± 0.001	99.900 ± 0.001
	beta33	0.139	0.143	0.004 ± 0.002	0.026 ± 0.017	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.292 ± 0.013	0.541 ± 0.015	0.142	0.003	0.0757	0.0758	0.936	93.894 ± 0.008	0.275 ± 0.001	53.453 ± 0.016
	beta34	0.101	0.102	0.001 ± 0.002	0.006 ± 0.016	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.258 ± 0.012	0.508 ± 0.014	0.099	-0.002	0.0496	0.0497	0.956	93.594 ± 0.008	0.192 ± 0.001	53.854 ± 0.016
	beta35	0.101	0.101	-0.000 ± 0.002	-0.003 ± 0.017	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.276 ± 0.013	0.525 ± 0.015	0.100	-0.001	0.0525	0.0525	0.932	93.894 ± 0.008	0.194 ± 0.001	54.454 ± 0.016

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S8

Simulation 1 - Condition 6: Full Model, Distribution = normal, N = 1000, Reliability = 0.8

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.311	-0.005 ± 0.001	-0.016 ± 0.004	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.017 ± 0.001	0.132 ± 0.004	0.312	-0.004	0.0418	0.0420	0.948	93.400 ± 0.008	0.154 ± 0.000	100.000 ± 0.000
	beta32	0.316	0.311	-0.005 ± 0.001	-0.016 ± 0.004	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.017 ± 0.001	0.129 ± 0.004	0.310	-0.006	0.0398	0.0403	0.971	94.400 ± 0.007	0.154 ± 0.000	100.000 ± 0.000
	beta33	0.139	0.134	-0.005 ± 0.001	-0.035 ± 0.010	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.096 ± 0.004	0.310 ± 0.009	0.131	-0.008	0.0415	0.0423	1.022	95.500 ± 0.007	0.172 ± 0.000	87.700 ± 0.010
	beta34	0.101	0.102	0.001 ± 0.001	0.008 ± 0.010	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.095 ± 0.004	0.309 ± 0.009	0.101	0.000	0.0298	0.0298	0.973	94.400 ± 0.007	0.119 ± 0.000	91.000 ± 0.009
	beta35	0.101	0.101	0.000 ± 0.001	0.002 ± 0.009	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.090 ± 0.004	0.299 ± 0.008	0.102	0.001	0.0302	0.0302	1.004	95.400 ± 0.007	0.119 ± 0.000	91.800 ± 0.009
LSAM	beta31	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.004	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.018 ± 0.001	0.133 ± 0.004	0.317	0.001	0.0425	0.0426	0.948	93.300 ± 0.008	0.157 ± 0.000	100.000 ± 0.000
	beta32	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.004	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.017 ± 0.001	0.130 ± 0.004	0.315	-0.001	0.0405	0.0406	0.959	94.100 ± 0.007	0.154 ± 0.000	100.000 ± 0.000
	beta33	0.139	0.141	0.002 ± 0.001	0.016 ± 0.010	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.094 ± 0.005	0.306 ± 0.009	0.139	0.000	0.0407	0.0407	1.020	95.700 ± 0.006	0.170 ± 0.001	91.600 ± 0.009
	beta34	0.101	0.101	-0.000 ± 0.001	-0.003 ± 0.010	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.101 ± 0.005	0.318 ± 0.009	0.100	-0.001	0.0309	0.0309	0.962	93.300 ± 0.008	0.121 ± 0.000	89.500 ± 0.010
	beta35	0.101	0.100	-0.001 ± 0.001	-0.008 ± 0.010	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.094 ± 0.004	0.307 ± 0.009	0.100	-0.001	0.0310	0.0310	0.984	95.100 ± 0.007	0.119 ± 0.000	90.300 ± 0.009
QML	beta31	0.316	0.311	-0.005 ± 0.001	-0.016 ± 0.004	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.017 ± 0.001	0.132 ± 0.004	0.312	-0.004	0.0422	0.0424	0.948	93.400 ± 0.008	0.154 ± 0.000	100.000 ± 0.000
	beta32	0.316	0.311	-0.005 ± 0.001	-0.017 ± 0.004	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.017 ± 0.001	0.129 ± 0.004	0.310	-0.006	0.0400	0.0405	0.971	94.400 ± 0.007	0.154 ± 0.000	100.000 ± 0.000
	beta33	0.139	0.134	-0.005 ± 0.001	-0.037 ± 0.010	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.096 ± 0.004	0.310 ± 0.009	0.131	-0.008	0.0412	0.0420	1.022	95.400 ± 0.007	0.172 ± 0.000	87.800 ± 0.010
	beta34	0.101	0.102	0.001 ± 0.001	0.007 ± 0.010	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.095 ± 0.004	0.309 ± 0.009	0.101	0.000	0.0298	0.0298	0.973	94.200 ± 0.007	0.119 ± 0.000	91.000 ± 0.009
	beta35	0.101	0.101	0.000 ± 0.001	0.001 ± 0.009	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.090 ± 0.004	0.299 ± 0.008	0.102	0.001	0.0302	0.0302	1.004	95.200 ± 0.007	0.119 ± 0.000	91.800 ± 0.009
UPI	beta31	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.004	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.018 ± 0.001	0.134 ± 0.004	0.317	0.001	0.0436	0.0436	0.922	92.800 ± 0.008	0.153 ± 0.000	100.000 ± 0.000
	beta32	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.004	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.017 ± 0.001	0.130 ± 0.004	0.315	-0.001	0.0409	0.0410	0.949	93.800 ± 0.008	0.153 ± 0.000	100.000 ± 0.000
	beta33	0.139	0.142	0.003 ± 0.001	0.018 ± 0.010	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.096 ± 0.005	0.309 ± 0.009	0.138	-0.001	0.0413	0.0413	1.009	94.600 ± 0.007	0.170 ± 0.000	91.900 ± 0.009
	beta34	0.101	0.101	-0.000 ± 0.001	-0.002 ± 0.010	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.103 ± 0.005	0.322 ± 0.009	0.100	-0.001	0.0310	0.0310	0.941	93.200 ± 0.008	0.120 ± 0.000	90.200 ± 0.009
	beta35	0.101	0.100	-0.001 ± 0.001	-0.008 ± 0.010	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.096 ± 0.004	0.309 ± 0.009	0.101	0.000	0.0317	0.0317	0.981	95.400 ± 0.007	0.120 ± 0.000	90.300 ± 0.009

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S9

Simulation 1 - Condition 7: Full Model, Distribution = right-skewed, N = 400, Reliability = 0.4

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.345	0.029 ± 0.004	0.091 ± 0.012	0.013 ± 0.001	0.113 ± 0.003	0.014 ± 0.001	0.116 ± 0.004	0.136 ± 0.007	0.368 ± 0.011	0.339	0.023	0.1097	0.1122	1.014	95.986 ± 0.007	0.449 ± 0.002	87.603 ± 0.011
	beta32	0.316	0.340	0.024 ± 0.004	0.075 ± 0.012	0.012 ± 0.001	0.111 ± 0.003	0.013 ± 0.001	0.113 ± 0.003	0.129 ± 0.006	0.358 ± 0.011	0.333	0.017	0.1146	0.1160	1.028	95.750 ± 0.007	0.447 ± 0.002	86.895 ± 0.012
	beta33	0.139	0.014	-0.125 ± 0.005	-0.900 ± 0.035	0.020 ± 0.001	0.140 ± 0.004	0.035 ± 0.002	0.188 ± 0.005	1.826 ± 0.084	1.351 ± 0.034	0.012	-0.127	0.1438	0.1920	1.080	89.610 ± 0.010	0.594 ± 0.006	2.715 ± 0.006
	beta34	0.101	0.220	0.119 ± 0.003	1.181 ± 0.032	0.009 ± 0.000	0.093 ± 0.002	0.023 ± 0.001	0.151 ± 0.003	2.234 ± 0.096	1.495 ± 0.033	0.215	0.114	0.0851	0.1420	1.028	81.936 ± 0.013	0.373 ± 0.005	73.554 ± 0.015
	beta35	0.101	0.219	0.118 ± 0.003	1.164 ± 0.031	0.008 ± 0.000	0.092 ± 0.003	0.022 ± 0.001	0.149 ± 0.004	2.182 ± 0.100	1.477 ± 0.035	0.212	0.111	0.0865	0.1408	1.029	84.061 ± 0.013	0.370 ± 0.005	72.373 ± 0.015
LSAM	beta31	0.316	0.311	-0.005 ± 0.008	-0.014 ± 0.025	0.054 ± 0.003	0.233 ± 0.007	0.054 ± 0.003	0.233 ± 0.008	0.545 ± 0.030	0.738 ± 0.024	0.309	-0.007	0.2369	0.2370	1.153	99.292 ± 0.003	1.055 ± 0.021	25.148 ± 0.015
	beta32	0.316	0.312	-0.004 ± 0.008	-0.014 ± 0.026	0.055 ± 0.003	0.236 ± 0.006	0.055 ± 0.003	0.235 ± 0.008	0.555 ± 0.030	0.745 ± 0.024	0.301	-0.015	0.2278	0.2283	1.200	99.292 ± 0.003	1.108 ± 0.029	24.439 ± 0.015
	beta33	0.139	0.169	0.030 ± 0.007	0.214 ± 0.047	0.037 ± 0.002	0.192 ± 0.006	0.038 ± 0.002	0.194 ± 0.007	1.950 ± 0.129	1.397 ± 0.052	0.160	0.021	0.1768	0.1781	1.136	96.340 ± 0.006	0.855 ± 0.037	8.737 ± 0.010
	beta34	0.101	0.109	0.008 ± 0.004	0.077 ± 0.038	0.012 ± 0.001	0.111 ± 0.004	0.012 ± 0.001	0.111 ± 0.004	1.205 ± 0.078	1.098 ± 0.040	0.102	0.001	0.0934	0.0934	1.044	97.993 ± 0.005	0.453 ± 0.014	20.189 ± 0.014
	beta35	0.101	0.105	0.004 ± 0.004	0.037 ± 0.037	0.012 ± 0.001	0.107 ± 0.003	0.012 ± 0.001	0.107 ± 0.004	1.132 ± 0.069	1.064 ± 0.037	0.103	0.002	0.0951	0.0951	1.148	97.993 ± 0.005	0.484 ± 0.020	22.668 ± 0.014
QML	beta31	0.316	0.345	0.029 ± 0.004	0.091 ± 0.012	0.013 ± 0.001	0.113 ± 0.003	0.014 ± 0.001	0.117 ± 0.004	0.137 ± 0.007	0.370 ± 0.011	0.340	0.024	0.1118	0.1144	1.009	95.868 ± 0.007	0.448 ± 0.002	87.367 ± 0.011
	beta32	0.316	0.340	0.024 ± 0.004	0.075 ± 0.012	0.012 ± 0.001	0.111 ± 0.003	0.013 ± 0.001	0.114 ± 0.003	0.130 ± 0.006	0.360 ± 0.011	0.336	0.020	0.1144	0.1162	1.022	95.868 ± 0.007	0.447 ± 0.002	87.485 ± 0.011
	beta33	0.139	0.016	-0.123 ± 0.005	-0.886 ± 0.035	0.020 ± 0.001	0.143 ± 0.004	0.036 ± 0.002	0.188 ± 0.005	1.838 ± 0.091	1.356 ± 0.036	0.017	-0.122	0.1435	0.1886	1.058	88.666 ± 0.011	0.591 ± 0.006	2.952 ± 0.006
	beta34	0.101	0.220	0.119 ± 0.003	1.181 ± 0.032	0.009 ± 0.001	0.095 ± 0.003	0.023 ± 0.001	0.153 ± 0.004	2.281 ± 0.104	1.510 ± 0.036	0.213	0.112	0.0870	0.1422	0.990	81.346 ± 0.013	0.369 ± 0.005	73.081 ± 0.015
	beta35	0.101	0.217	0.116 ± 0.003	1.152 ± 0.031	0.008 ± 0.000	0.092 ± 0.003	0.022 ± 0.001	0.148 ± 0.004	2.156 ± 0.100	1.468 ± 0.036	0.210	0.109	0.0869	0.1392	1.013	83.235 ± 0.013	0.366 ± 0.005	71.783 ± 0.015
UPI	beta31	0.316	0.339	0.023 ± 0.008	0.073 ± 0.027	0.061 ± 0.004	0.247 ± 0.008	0.061 ± 0.004	0.248 ± 0.009	0.614 ± 0.038	0.784 ± 0.028	0.328	0.012	0.2412	0.2415	1.074	98.347 ± 0.004	1.039 ± 0.026	32.940 ± 0.016
	beta32	0.316	0.323	0.007 ± 0.009	0.021 ± 0.027	0.062 ± 0.004	0.248 ± 0.007	0.062 ± 0.004	0.248 ± 0.008	0.618 ± 0.036	0.786 ± 0.026	0.320	0.004	0.2384	0.2385	1.107	97.757 ± 0.005	1.078 ± 0.032	31.169 ± 0.016
	beta33	0.139	0.143	0.004 ± 0.007	0.030 ± 0.051	0.043 ± 0.003	0.206 ± 0.007	0.043 ± 0.003	0.206 ± 0.008	2.204 ± 0.158	1.485 ± 0.059	0.137	-0.002	0.1796	0.1796	0.937	90.791 ± 0.010	0.759 ± 0.026	15.939 ± 0.013
	beta34	0.101	0.100	-0.001 ± 0.004	-0.009 ± 0.038	0.012 ± 0.001	0.111 ± 0.004	0.012 ± 0.001	0.111 ± 0.004	1.216 ± 0.078	1.103 ± 0.040	0.096	-0.005	0.1014	0.1015	0.947	93.979 ± 0.008	0.414 ± 0.012	27.037 ± 0.015
	beta35	0.101	0.104	0.003 ± 0.004	0.027 ± 0.039	0.013 ± 0.001	0.113 ± 0.004	0.013 ± 0.001	0.113 ± 0.004	1.260 ± 0.083	1.122 ± 0.042	0.092	-0.009	0.0946	0.0950	1.001	94.569 ± 0.008	0.445 ± 0.019	27.745 ± 0.015

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S10

Simulation 1 - Condition 8: Full Model, Distribution = right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.340	0.024 ± 0.002	0.075 ± 0.007	0.005 ± 0.000	0.070 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.055 ± 0.003	0.235 ± 0.007	0.339	0.023	0.0664	0.0704	1.010	94.280 ± 0.007	0.279 ± 0.001	99.796 ± 0.001
	beta32	0.316	0.339	0.023 ± 0.002	0.072 ± 0.007	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.052 ± 0.002	0.228 ± 0.006	0.339	0.023	0.0711	0.0746	1.040	96.118 ± 0.006	0.279 ± 0.001	99.898 ± 0.001
	beta33	0.139	0.010	-0.129 ± 0.003	-0.927 ± 0.021	0.008 ± 0.000	0.091 ± 0.002	0.025 ± 0.001	0.158 ± 0.003	1.286 ± 0.045	1.134 ± 0.021	0.011	-0.128	0.0892	0.1561	1.001	69.663 ± 0.015	0.357 ± 0.002	5.005 ± 0.007
	beta34	0.101	0.224	0.123 ± 0.002	1.214 ± 0.018	0.003 ± 0.000	0.058 ± 0.001	0.018 ± 0.001	0.136 ± 0.002	1.807 ± 0.052	1.344 ± 0.020	0.219	0.118	0.0566	0.1305	0.966	38.611 ± 0.016	0.221 ± 0.002	99.081 ± 0.003
	beta35	0.101	0.226	0.125 ± 0.002	1.234 ± 0.017	0.003 ± 0.000	0.054 ± 0.001	0.018 ± 0.000	0.136 ± 0.002	1.810 ± 0.047	1.346 ± 0.018	0.224	0.123	0.0553	0.1350	1.039	36.670 ± 0.015	0.220 ± 0.002	99.387 ± 0.002
LSAM	beta31	0.316	0.314	-0.002 ± 0.004	-0.007 ± 0.014	0.018 ± 0.001	0.135 ± 0.003	0.018 ± 0.001	0.135 ± 0.004	0.184 ± 0.009	0.428 ± 0.013	0.314	-0.002	0.1283	0.1283	1.035	96.527 ± 0.006	0.550 ± 0.004	64.760 ± 0.015
	beta32	0.316	0.311	-0.005 ± 0.004	-0.017 ± 0.014	0.018 ± 0.001	0.136 ± 0.003	0.018 ± 0.001	0.136 ± 0.004	0.185 ± 0.009	0.430 ± 0.013	0.306	-0.010	0.1238	0.1242	1.031	97.038 ± 0.005	0.549 ± 0.004	65.169 ± 0.015
	beta33	0.139	0.150	0.011 ± 0.003	0.077 ± 0.022	0.009 ± 0.000	0.096 ± 0.002	0.009 ± 0.000	0.096 ± 0.003	0.480 ± 0.025	0.693 ± 0.021	0.142	0.003	0.0883	0.0884	0.991	96.118 ± 0.006	0.372 ± 0.004	34.627 ± 0.015
	beta34	0.101	0.104	0.003 ± 0.002	0.029 ± 0.017	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.275 ± 0.014	0.524 ± 0.016	0.101	0.000	0.0510	0.0510	0.987	95.608 ± 0.007	0.205 ± 0.002	58.018 ± 0.016
	beta35	0.101	0.106	0.005 ± 0.002	0.045 ± 0.017	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.278 ± 0.015	0.527 ± 0.016	0.103	0.002	0.0484	0.0484	0.957	95.301 ± 0.007	0.199 ± 0.002	61.185 ± 0.016
QML	beta31	0.316	0.340	0.024 ± 0.002	0.076 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.056 ± 0.003	0.236 ± 0.007	0.340	0.024	0.0671	0.0712	1.006	94.688 ± 0.007	0.278 ± 0.001	99.796 ± 0.001
	beta32	0.316	0.339	0.023 ± 0.002	0.073 ± 0.007	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.053 ± 0.002	0.229 ± 0.006	0.339	0.023	0.0716	0.0751	1.035	95.914 ± 0.006	0.279 ± 0.001	99.898 ± 0.001
	beta33	0.139	0.014	-0.125 ± 0.003	-0.902 ± 0.021	0.008 ± 0.000	0.091 ± 0.002	0.024 ± 0.001	0.155 ± 0.003	1.243 ± 0.044	1.115 ± 0.021	0.014	-0.125	0.0890	0.1538	0.989	70.276 ± 0.015	0.353 ± 0.002	5.209 ± 0.007
	beta34	0.101	0.222	0.121 ± 0.002	1.197 ± 0.019	0.003 ± 0.000	0.059 ± 0.001	0.018 ± 0.001	0.134 ± 0.002	1.767 ± 0.052	1.329 ± 0.020	0.216	0.115	0.0579	0.1287	0.947	38.611 ± 0.016	0.217 ± 0.002	99.081 ± 0.003
	beta35	0.101	0.224	0.123 ± 0.002	1.216 ± 0.017	0.003 ± 0.000	0.054 ± 0.001	0.018 ± 0.000	0.134 ± 0.002	1.765 ± 0.046	1.328 ± 0.017	0.223	0.122	0.0544	0.1335	1.024	36.568 ± 0.015	0.217 ± 0.002	99.387 ± 0.002
UPI	beta31	0.316	0.321	0.005 ± 0.004	0.016 ± 0.014	0.020 ± 0.001	0.140 ± 0.003	0.020 ± 0.001	0.140 ± 0.004	0.197 ± 0.010	0.444 ± 0.013	0.320	0.004	0.1326	0.1327	0.984	95.403 ± 0.007	0.540 ± 0.005	66.496 ± 0.015
	beta32	0.316	0.318	0.002 ± 0.005	0.008 ± 0.014	0.020 ± 0.001	0.141 ± 0.003	0.020 ± 0.001	0.141 ± 0.004	0.200 ± 0.010	0.447 ± 0.013	0.314	-0.002	0.1373	0.1373	0.977	96.221 ± 0.006	0.541 ± 0.005	66.394 ± 0.015
	beta33	0.139	0.146	0.007 ± 0.004	0.054 ± 0.026	0.013 ± 0.001	0.112 ± 0.004	0.013 ± 0.001	0.113 ± 0.004	0.657 ± 0.042	0.811 ± 0.029	0.133	-0.006	0.0943	0.0945	0.761	88.560 ± 0.010	0.335 ± 0.006	45.761 ± 0.016
	beta34	0.101	0.102	0.001 ± 0.002	0.008 ± 0.018	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.302 ± 0.015	0.550 ± 0.016	0.099	-0.002	0.0546	0.0546	0.851	91.522 ± 0.009	0.185 ± 0.002	62.513 ± 0.015
	beta35	0.101	0.103	0.002 ± 0.002	0.024 ± 0.018	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	0.310 ± 0.017	0.557 ± 0.017	0.098	-0.003	0.0489	0.0490	0.835	91.420 ± 0.009	0.184 ± 0.002	64.658 ± 0.015

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S11

Simulation 1 - Condition 9: Full Model, Distribution = right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.325	0.009 ± 0.003	0.028 ± 0.009	0.009 ± 0.000	0.092 ± 0.002	0.009 ± 0.000	0.093 ± 0.003	0.086 ± 0.004	0.294 ± 0.008	0.324	0.008	0.0915	0.0919	1.004	95.635 ± 0.007	0.364 ± 0.001	93.909 ± 0.008
	beta32	0.316	0.327	0.011 ± 0.003	0.036 ± 0.009	0.008 ± 0.000	0.091 ± 0.002	0.008 ± 0.000	0.092 ± 0.003	0.085 ± 0.004	0.292 ± 0.008	0.325	0.009	0.0898	0.0902	1.017	95.330 ± 0.007	0.365 ± 0.001	94.315 ± 0.007
	beta33	0.139	0.083	-0.056 ± 0.003	-0.402 ± 0.022	0.009 ± 0.000	0.096 ± 0.002	0.012 ± 0.001	0.111 ± 0.003	0.636 ± 0.028	0.798 ± 0.020	0.080	-0.059	0.0942	0.1109	1.001	90.660 ± 0.009	0.376 ± 0.002	15.127 ± 0.011
	beta34	0.101	0.155	0.054 ± 0.002	0.533 ± 0.017	0.003 ± 0.000	0.055 ± 0.001	0.006 ± 0.000	0.077 ± 0.002	0.578 ± 0.024	0.761 ± 0.017	0.152	0.051	0.0538	0.0743	0.995	85.990 ± 0.011	0.214 ± 0.002	82.741 ± 0.012
	beta35	0.101	0.154	0.053 ± 0.002	0.524 ± 0.018	0.003 ± 0.000	0.056 ± 0.001	0.006 ± 0.000	0.077 ± 0.002	0.578 ± 0.025	0.760 ± 0.018	0.153	0.052	0.0515	0.0735	0.992	86.396 ± 0.011	0.217 ± 0.002	82.335 ± 0.012
LSAM	beta31	0.316	0.318	0.002 ± 0.005	0.007 ± 0.015	0.021 ± 0.001	0.144 ± 0.003	0.021 ± 0.001	0.144 ± 0.004	0.209 ± 0.010	0.457 ± 0.013	0.316	0.000	0.1400	0.1400	1.027	96.142 ± 0.006	0.582 ± 0.004	61.320 ± 0.016
	beta32	0.316	0.320	0.004 ± 0.005	0.014 ± 0.015	0.021 ± 0.001	0.145 ± 0.003	0.021 ± 0.001	0.145 ± 0.004	0.212 ± 0.010	0.460 ± 0.013	0.315	-0.001	0.1343	0.1343	1.017	95.838 ± 0.006	0.580 ± 0.004	61.624 ± 0.015
	beta33	0.139	0.152	0.013 ± 0.003	0.097 ± 0.024	0.011 ± 0.001	0.103 ± 0.003	0.011 ± 0.001	0.103 ± 0.003	0.554 ± 0.028	0.744 ± 0.022	0.149	0.010	0.1011	0.1016	0.956	95.228 ± 0.007	0.385 ± 0.004	36.650 ± 0.015
	beta34	0.101	0.101	-0.000 ± 0.002	-0.002 ± 0.018	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	0.336 ± 0.018	0.580 ± 0.018	0.102	0.001	0.0530	0.0530	0.950	94.721 ± 0.007	0.218 ± 0.003	53.807 ± 0.016
	beta35	0.101	0.100	-0.001 ± 0.002	-0.014 ± 0.019	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.369 ± 0.020	0.608 ± 0.019	0.100	-0.001	0.0534	0.0535	0.904	91.878 ± 0.009	0.218 ± 0.003	50.863 ± 0.016
QML	beta31	0.316	0.325	0.009 ± 0.003	0.030 ± 0.009	0.009 ± 0.000	0.093 ± 0.002	0.009 ± 0.000	0.093 ± 0.003	0.087 ± 0.004	0.295 ± 0.008	0.324	0.008	0.0915	0.0919	1.001	95.635 ± 0.007	0.364 ± 0.001	93.909 ± 0.008
	beta32	0.316	0.328	0.012 ± 0.003	0.038 ± 0.009	0.008 ± 0.000	0.092 ± 0.002	0.009 ± 0.000	0.092 ± 0.003	0.085 ± 0.004	0.292 ± 0.008	0.325	0.009	0.0903	0.0908	1.016	95.127 ± 0.007	0.365 ± 0.001	94.416 ± 0.007
	beta33	0.139	0.084	-0.055 ± 0.003	-0.395 ± 0.022	0.009 ± 0.000	0.096 ± 0.002	0.012 ± 0.001	0.111 ± 0.003	0.633 ± 0.027	0.795 ± 0.020	0.082	-0.057	0.0950	0.1108	0.996	90.254 ± 0.009	0.375 ± 0.002	15.228 ± 0.011
	beta34	0.101	0.154	0.053 ± 0.002	0.523 ± 0.017	0.003 ± 0.000	0.055 ± 0.001	0.006 ± 0.000	0.076 ± 0.002	0.569 ± 0.024	0.754 ± 0.017	0.151	0.050	0.0541	0.0740	0.991	85.888 ± 0.011	0.213 ± 0.002	82.234 ± 0.012
	beta35	0.101	0.153	0.052 ± 0.002	0.514 ± 0.018	0.003 ± 0.000	0.056 ± 0.001	0.006 ± 0.000	0.076 ± 0.002	0.566 ± 0.024	0.752 ± 0.017	0.153	0.052	0.0516	0.0731	0.990	86.091 ± 0.011	0.216 ± 0.002	81.929 ± 0.012
UPI	beta31	0.316	0.328	0.012 ± 0.005	0.037 ± 0.015	0.021 ± 0.001	0.145 ± 0.003	0.021 ± 0.001	0.146 ± 0.004	0.213 ± 0.010	0.462 ± 0.013	0.323	0.007	0.1422	0.1424	0.988	95.939 ± 0.006	0.563 ± 0.004	65.076 ± 0.015
	beta32	0.316	0.329	0.013 ± 0.005	0.040 ± 0.015	0.021 ± 0.001	0.146 ± 0.004	0.022 ± 0.001	0.147 ± 0.004	0.216 ± 0.011	0.465 ± 0.014	0.324	0.008	0.1314	0.1316	0.987	94.924 ± 0.007	0.567 ± 0.004	65.279 ± 0.015
	beta33	0.139	0.148	0.009 ± 0.003	0.067 ± 0.025	0.011 ± 0.001	0.107 ± 0.003	0.012 ± 0.001	0.107 ± 0.003	0.597 ± 0.032	0.773 ± 0.024	0.141	0.002	0.1041	0.1041	0.867	92.386 ± 0.008	0.364 ± 0.004	39.188 ± 0.016
	beta34	0.101	0.097	-0.004 ± 0.002	-0.035 ± 0.019	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.352 ± 0.019	0.593 ± 0.019	0.098	-0.003	0.0532	0.0533	0.887	92.690 ± 0.008	0.208 ± 0.003	53.909 ± 0.016
	beta35	0.101	0.097	-0.004 ± 0.002	-0.039 ± 0.020	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.399 ± 0.024	0.631 ± 0.021	0.096	-0.005	0.0544	0.0546	0.856	91.777 ± 0.009	0.214 ± 0.003	51.574 ± 0.016

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S12

Simulation 1 - Condition 10: Full Model, Distribution = right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.321	0.005 ± 0.002	0.017 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.002	0.179 ± 0.005	0.319	0.003	0.0539	0.0540	1.031	94.874 ± 0.007	0.228 ± 0.001	100.000 ± 0.000
	beta32	0.316	0.321	0.005 ± 0.002	0.016 ± 0.006	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.188 ± 0.005	0.322	0.006	0.0597	0.0600	0.982	95.176 ± 0.007	0.228 ± 0.001	99.899 ± 0.001
	beta33	0.139	0.085	-0.054 ± 0.002	-0.391 ± 0.014	0.004 ± 0.000	0.059 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.335 ± 0.015	0.579 ± 0.014	0.087	-0.052	0.0566	0.0769	0.977	84.623 ± 0.011	0.227 ± 0.001	31.658 ± 0.015
	beta34	0.101	0.158	0.057 ± 0.001	0.563 ± 0.011	0.001 ± 0.000	0.034 ± 0.001	0.004 ± 0.000	0.066 ± 0.001	0.431 ± 0.014	0.656 ± 0.011	0.157	0.056	0.0339	0.0653	0.966	57.588 ± 0.016	0.129 ± 0.001	100.000 ± 0.000
	beta35	0.101	0.157	0.056 ± 0.001	0.555 ± 0.011	0.001 ± 0.000	0.034 ± 0.001	0.004 ± 0.000	0.066 ± 0.001	0.423 ± 0.014	0.650 ± 0.011	0.156	0.055	0.0328	0.0638	0.963	60.302 ± 0.016	0.129 ± 0.001	99.598 ± 0.002
LSAM	beta31	0.316	0.315	-0.001 ± 0.003	-0.003 ± 0.009	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.002	0.074 ± 0.003	0.271 ± 0.008	0.313	-0.003	0.0801	0.0801	1.025	95.678 ± 0.006	0.345 ± 0.001	94.673 ± 0.007
	beta32	0.316	0.313	-0.003 ± 0.003	-0.008 ± 0.009	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.002	0.076 ± 0.003	0.276 ± 0.008	0.314	-0.002	0.0858	0.0858	1.001	94.774 ± 0.007	0.342 ± 0.001	93.367 ± 0.008
	beta33	0.139	0.144	0.005 ± 0.002	0.039 ± 0.013	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.168 ± 0.008	0.410 ± 0.012	0.145	0.006	0.0542	0.0545	0.964	94.070 ± 0.007	0.214 ± 0.001	76.080 ± 0.014
	beta34	0.101	0.102	0.001 ± 0.001	0.012 ± 0.010	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.107 ± 0.005	0.327 ± 0.009	0.101	0.000	0.0330	0.0330	0.949	93.769 ± 0.008	0.123 ± 0.001	88.643 ± 0.010
	beta35	0.101	0.103	0.002 ± 0.001	0.019 ± 0.011	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.112 ± 0.006	0.334 ± 0.010	0.103	0.002	0.0309	0.0309	0.912	92.060 ± 0.009	0.121 ± 0.001	89.347 ± 0.010
QML	beta31	0.316	0.322	0.006 ± 0.002	0.018 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.002	0.180 ± 0.005	0.320	0.004	0.0538	0.0540	1.027	94.874 ± 0.007	0.228 ± 0.000	100.000 ± 0.000
	beta32	0.316	0.321	0.005 ± 0.002	0.017 ± 0.006	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.188 ± 0.005	0.322	0.006	0.0598	0.0601	0.979	95.075 ± 0.007	0.227 ± 0.000	100.000 ± 0.000
	beta33	0.139	0.085	-0.054 ± 0.002	-0.388 ± 0.014	0.004 ± 0.000	0.060 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.334 ± 0.015	0.578 ± 0.014	0.088	-0.051	0.0565	0.0764	0.969	83.819 ± 0.012	0.226 ± 0.001	32.261 ± 0.015
	beta34	0.101	0.157	0.056 ± 0.001	0.558 ± 0.011	0.001 ± 0.000	0.034 ± 0.001	0.004 ± 0.000	0.066 ± 0.001	0.425 ± 0.014	0.652 ± 0.011	0.156	0.055	0.0342	0.0648	0.961	57.990 ± 0.016	0.128 ± 0.001	100.000 ± 0.000
	beta35	0.101	0.157	0.056 ± 0.001	0.550 ± 0.011	0.001 ± 0.000	0.034 ± 0.001	0.004 ± 0.000	0.065 ± 0.001	0.417 ± 0.014	0.645 ± 0.011	0.155	0.054	0.0328	0.0631	0.958	60.201 ± 0.016	0.128 ± 0.001	99.598 ± 0.002
UPI	beta31	0.316	0.319	0.003 ± 0.003	0.010 ± 0.009	0.007 ± 0.000	0.087 ± 0.002	0.007 ± 0.000	0.087 ± 0.003	0.075 ± 0.004	0.274 ± 0.008	0.318	0.002	0.0810	0.0810	0.989	94.472 ± 0.007	0.335 ± 0.001	95.578 ± 0.007
	beta32	0.316	0.317	0.001 ± 0.003	0.003 ± 0.009	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.002	0.076 ± 0.003	0.276 ± 0.008	0.319	0.003	0.0863	0.0864	0.981	94.070 ± 0.007	0.335 ± 0.001	94.171 ± 0.007
	beta33	0.139	0.145	0.006 ± 0.002	0.042 ± 0.014	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.189 ± 0.009	0.435 ± 0.013	0.144	0.005	0.0588	0.0590	0.836	90.854 ± 0.009	0.197 ± 0.001	80.804 ± 0.012
	beta34	0.101	0.101	0.000 ± 0.001	0.002 ± 0.011	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.111 ± 0.005	0.333 ± 0.010	0.101	0.000	0.0335	0.0335	0.862	91.256 ± 0.009	0.114 ± 0.001	89.749 ± 0.010
	beta35	0.101	0.101	0.000 ± 0.001	0.004 ± 0.011	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.112 ± 0.006	0.334 ± 0.010	0.101	0.000	0.0322	0.0322	0.866	91.055 ± 0.009	0.115 ± 0.001	89.146 ± 0.010

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S13

Simulation 1 - Condition 11: Full Model, Distribution = right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.318	0.002 ± 0.003	0.007 ± 0.009	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.073 ± 0.003	0.270 ± 0.008	0.322	0.006	0.0827	0.0829	0.995	94.472 ± 0.007	0.332 ± 0.001	95.477 ± 0.007
	beta32	0.316	0.321	0.005 ± 0.003	0.015 ± 0.009	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	0.074 ± 0.004	0.273 ± 0.008	0.321	0.005	0.0799	0.0800	0.982	94.673 ± 0.007	0.332 ± 0.001	96.181 ± 0.006
	beta33	0.139	0.116	-0.023 ± 0.002	-0.164 ± 0.016	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.270 ± 0.013	0.520 ± 0.014	0.116	-0.023	0.0678	0.0717	1.008	94.573 ± 0.007	0.271 ± 0.001	40.704 ± 0.016
	beta34	0.101	0.121	0.020 ± 0.001	0.200 ± 0.013	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.195 ± 0.010	0.442 ± 0.013	0.121	0.020	0.0363	0.0414	0.974	92.261 ± 0.008	0.152 ± 0.001	86.332 ± 0.011
	beta35	0.101	0.120	0.019 ± 0.001	0.186 ± 0.013	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.196 ± 0.010	0.442 ± 0.013	0.118	0.017	0.0390	0.0427	0.952	91.558 ± 0.009	0.151 ± 0.001	85.528 ± 0.011
LSAM	beta31	0.316	0.316	-0.000 ± 0.003	-0.000 ± 0.011	0.011 ± 0.001	0.107 ± 0.003	0.011 ± 0.001	0.107 ± 0.003	0.114 ± 0.006	0.338 ± 0.010	0.322	0.006	0.1039	0.1041	0.981	94.774 ± 0.007	0.411 ± 0.002	84.523 ± 0.011
	beta32	0.316	0.320	0.004 ± 0.003	0.013 ± 0.011	0.012 ± 0.001	0.108 ± 0.003	0.012 ± 0.001	0.108 ± 0.003	0.117 ± 0.006	0.342 ± 0.010	0.320	0.004	0.1019	0.1020	0.966	93.769 ± 0.008	0.408 ± 0.002	84.322 ± 0.012
	beta33	0.139	0.139	0.000 ± 0.002	0.003 ± 0.016	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.248 ± 0.012	0.498 ± 0.015	0.137	-0.002	0.0663	0.0663	0.938	92.764 ± 0.008	0.255 ± 0.002	59.196 ± 0.016
	beta34	0.101	0.102	0.001 ± 0.001	0.008 ± 0.013	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.179 ± 0.009	0.423 ± 0.013	0.101	0.000	0.0397	0.0397	0.894	92.864 ± 0.008	0.150 ± 0.002	75.578 ± 0.014
	beta35	0.101	0.100	-0.001 ± 0.001	-0.011 ± 0.014	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.183 ± 0.010	0.427 ± 0.013	0.098	-0.003	0.0407	0.0408	0.867	90.955 ± 0.009	0.147 ± 0.002	72.261 ± 0.014
QML	beta31	0.316	0.318	0.002 ± 0.003	0.008 ± 0.009	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.073 ± 0.003	0.270 ± 0.008	0.322	0.006	0.0834	0.0837	0.993	94.472 ± 0.007	0.332 ± 0.001	95.578 ± 0.007
	beta32	0.316	0.321	0.005 ± 0.003	0.014 ± 0.009	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	0.075 ± 0.004	0.273 ± 0.008	0.320	0.004	0.0805	0.0806	0.981	94.673 ± 0.007	0.332 ± 0.001	96.181 ± 0.006
	beta33	0.139	0.116	-0.023 ± 0.002	-0.165 ± 0.016	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.271 ± 0.013	0.520 ± 0.015	0.116	-0.023	0.0670	0.0708	1.007	94.472 ± 0.007	0.271 ± 0.001	40.704 ± 0.016
	beta34	0.101	0.121	0.020 ± 0.001	0.197 ± 0.012	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.194 ± 0.010	0.441 ± 0.013	0.120	0.019	0.0365	0.0413	0.973	92.362 ± 0.008	0.152 ± 0.001	86.332 ± 0.011
	beta35	0.101	0.120	0.019 ± 0.001	0.185 ± 0.013	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.196 ± 0.010	0.442 ± 0.013	0.118	0.017	0.0389	0.0425	0.950	91.457 ± 0.009	0.151 ± 0.001	85.427 ± 0.011
UPI	beta31	0.316	0.320	0.004 ± 0.003	0.013 ± 0.011	0.011 ± 0.001	0.107 ± 0.003	0.011 ± 0.001	0.107 ± 0.003	0.115 ± 0.006	0.339 ± 0.010	0.323	0.007	0.1035	0.1037	0.959	94.070 ± 0.007	0.403 ± 0.002	85.628 ± 0.011
	beta32	0.316	0.323	0.007 ± 0.003	0.023 ± 0.011	0.012 ± 0.001	0.108 ± 0.003	0.012 ± 0.001	0.108 ± 0.003	0.117 ± 0.006	0.343 ± 0.010	0.322	0.006	0.1041	0.1042	0.948	93.467 ± 0.008	0.402 ± 0.002	86.131 ± 0.011
	beta33	0.139	0.140	0.001 ± 0.002	0.006 ± 0.016	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.262 ± 0.013	0.512 ± 0.015	0.139	0.000	0.0677	0.0677	0.923	93.869 ± 0.008	0.258 ± 0.002	57.588 ± 0.016
	beta34	0.101	0.100	-0.001 ± 0.001	-0.008 ± 0.014	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.183 ± 0.010	0.428 ± 0.013	0.100	-0.001	0.0398	0.0398	0.890	92.965 ± 0.008	0.151 ± 0.002	73.869 ± 0.014
	beta35	0.101	0.099	-0.002 ± 0.001	-0.018 ± 0.014	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.186 ± 0.010	0.431 ± 0.013	0.097	-0.004	0.0392	0.0394	0.880	92.060 ± 0.009	0.150 ± 0.002	71.859 ± 0.014

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S14

Simulation 1 - Condition 12: Full Model, Distribution = right-skewed, N = 1000, Reliability = 0.8

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.316	0.000 ± 0.002	0.001 ± 0.005	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.026 ± 0.001	0.162 ± 0.005	0.317	0.001	0.0505	0.0506	1.035	96.096 ± 0.006	0.208 ± 0.000	100.000 ± 0.000
	beta32	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.005	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.029 ± 0.001	0.170 ± 0.005	0.314	-0.002	0.0510	0.0510	0.986	94.194 ± 0.007	0.208 ± 0.000	100.000 ± 0.000
	beta33	0.139	0.120	-0.019 ± 0.001	-0.138 ± 0.010	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.116 ± 0.005	0.341 ± 0.009	0.120	-0.019	0.0429	0.0470	0.963	91.592 ± 0.009	0.164 ± 0.001	80.881 ± 0.012
	beta34	0.101	0.122	0.021 ± 0.001	0.209 ± 0.007	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.091 ± 0.004	0.302 ± 0.007	0.122	0.021	0.0221	0.0303	1.043	87.187 ± 0.011	0.090 ± 0.001	100.000 ± 0.000
	beta35	0.101	0.122	0.021 ± 0.001	0.208 ± 0.008	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.105 ± 0.005	0.323 ± 0.008	0.122	0.021	0.0244	0.0319	0.920	82.883 ± 0.012	0.090 ± 0.001	99.700 ± 0.002
LSAM	beta31	0.316	0.316	-0.000 ± 0.002	-0.000 ± 0.006	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.039 ± 0.002	0.197 ± 0.006	0.314	-0.002	0.0607	0.0607	1.031	96.096 ± 0.006	0.252 ± 0.001	99.700 ± 0.002
	beta32	0.316	0.312	-0.004 ± 0.002	-0.013 ± 0.007	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.044 ± 0.002	0.209 ± 0.006	0.314	-0.002	0.0619	0.0620	0.970	93.694 ± 0.008	0.251 ± 0.001	99.499 ± 0.002
	beta33	0.139	0.140	0.001 ± 0.001	0.005 ± 0.010	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.093 ± 0.004	0.306 ± 0.008	0.140	0.001	0.0425	0.0425	0.928	92.192 ± 0.008	0.155 ± 0.001	93.193 ± 0.008
	beta34	0.101	0.102	0.001 ± 0.001	0.010 ± 0.007	0.001 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.022 ± 0.001	0.049 ± 0.002	0.222 ± 0.006	0.101	0.000	0.0221	0.0221	0.993	94.695 ± 0.007	0.087 ± 0.001	98.098 ± 0.004
	beta35	0.101	0.103	0.002 ± 0.001	0.016 ± 0.008	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	0.063 ± 0.003	0.252 ± 0.007	0.102	0.001	0.0243	0.0243	0.869	90.390 ± 0.009	0.087 ± 0.001	97.598 ± 0.005
QML	beta31	0.316	0.317	0.001 ± 0.002	0.002 ± 0.005	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.026 ± 0.001	0.162 ± 0.005	0.317	0.001	0.0503	0.0503	1.034	95.696 ± 0.006	0.208 ± 0.000	100.000 ± 0.000
	beta32	0.316	0.314	-0.002 ± 0.002	-0.008 ± 0.005	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.029 ± 0.001	0.171 ± 0.005	0.314	-0.002	0.0506	0.0507	0.983	94.394 ± 0.007	0.208 ± 0.000	100.000 ± 0.000
	beta33	0.139	0.120	-0.019 ± 0.001	-0.137 ± 0.010	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.117 ± 0.005	0.341 ± 0.009	0.120	-0.019	0.0433	0.0474	0.960	91.592 ± 0.009	0.164 ± 0.001	80.981 ± 0.012
	beta34	0.101	0.122	0.021 ± 0.001	0.205 ± 0.007	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.089 ± 0.004	0.299 ± 0.007	0.121	0.020	0.0221	0.0298	1.042	87.387 ± 0.011	0.090 ± 0.001	100.000 ± 0.000
	beta35	0.101	0.122	0.021 ± 0.001	0.208 ± 0.008	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.105 ± 0.005	0.324 ± 0.008	0.122	0.021	0.0242	0.0318	0.917	82.583 ± 0.012	0.090 ± 0.001	99.700 ± 0.002
UPI	beta31	0.316	0.318	0.002 ± 0.002	0.005 ± 0.006	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.039 ± 0.002	0.198 ± 0.006	0.316	0.000	0.0606	0.0606	1.009	95.495 ± 0.007	0.247 ± 0.001	99.600 ± 0.002
	beta32	0.316	0.313	-0.003 ± 0.002	-0.008 ± 0.007	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.044 ± 0.002	0.209 ± 0.006	0.315	-0.001	0.0619	0.0619	0.956	93.694 ± 0.008	0.248 ± 0.001	99.499 ± 0.002
	beta33	0.139	0.140	0.001 ± 0.001	0.005 ± 0.010	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.097 ± 0.004	0.311 ± 0.009	0.139	0.000	0.0431	0.0431	0.883	90.991 ± 0.009	0.150 ± 0.001	93.694 ± 0.008
	beta34	0.101	0.101	0.000 ± 0.001	0.003 ± 0.007	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.051 ± 0.002	0.225 ± 0.006	0.101	0.000	0.0221	0.0221	0.962	93.493 ± 0.008	0.086 ± 0.001	98.699 ± 0.004
	beta35	0.101	0.102	0.001 ± 0.001	0.012 ± 0.008	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	0.064 ± 0.003	0.254 ± 0.008	0.101	0.000	0.0245	0.0245	0.859	90.891 ± 0.009	0.086 ± 0.001	98.298 ± 0.004

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S15

Simulation 1 - Condition 13: Full Model, Distribution = uniform, N = 400, Reliability = 0.4

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.304	-0.012 ± 0.004	-0.037 ± 0.013	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	0.097 ± 0.006	0.312 ± 0.012	0.306	-0.010	0.0979	0.0984	0.993	94.000 ± 0.010	0.382 ± 0.002	89.455 ± 0.013
	beta32	0.316	0.308	-0.008 ± 0.004	-0.026 ± 0.012	0.008 ± 0.001	0.090 ± 0.003	0.008 ± 0.001	0.090 ± 0.003	0.082 ± 0.005	0.286 ± 0.011	0.300	-0.016	0.0975	0.0988	1.088	97.091 ± 0.007	0.383 ± 0.002	93.091 ± 0.011
	beta33	0.139	0.130	-0.009 ± 0.006	-0.065 ± 0.042	0.018 ± 0.001	0.135 ± 0.005	0.018 ± 0.001	0.136 ± 0.005	0.950 ± 0.064	0.975 ± 0.039	0.124	-0.015	0.1245	0.1253	1.016	96.182 ± 0.008	0.539 ± 0.005	12.727 ± 0.014
	beta34	0.101	0.045	-0.056 ± 0.004	-0.554 ± 0.042	0.010 ± 0.001	0.099 ± 0.003	0.013 ± 0.001	0.114 ± 0.004	1.267 ± 0.072	1.126 ± 0.037	0.050	-0.051	0.0973	0.1099	0.989	89.455 ± 0.013	0.384 ± 0.004	6.727 ± 0.011
	beta35	0.101	0.044	-0.057 ± 0.004	-0.564 ± 0.042	0.010 ± 0.001	0.099 ± 0.004	0.013 ± 0.001	0.114 ± 0.004	1.285 ± 0.082	1.134 ± 0.041	0.041	-0.060	0.0991	0.1157	0.992	90.545 ± 0.012	0.386 ± 0.004	6.364 ± 0.010
LSAM	beta31	0.316	0.321	0.005 ± 0.005	0.017 ± 0.016	0.015 ± 0.001	0.121 ± 0.004	0.015 ± 0.001	0.121 ± 0.005	0.147 ± 0.010	0.383 ± 0.015	0.323	0.007	0.1256	0.1258	1.471	97.818 ± 0.006	0.699 ± 0.026	61.091 ± 0.021
	beta32	0.316	0.323	0.007 ± 0.005	0.021 ± 0.016	0.014 ± 0.001	0.117 ± 0.004	0.014 ± 0.001	0.117 ± 0.005	0.137 ± 0.009	0.370 ± 0.015	0.320	0.004	0.1166	0.1167	1.556	99.091 ± 0.004	0.712 ± 0.030	62.182 ± 0.021
	beta33	0.139	0.141	0.002 ± 0.008	0.018 ± 0.060	0.038 ± 0.003	0.196 ± 0.008	0.038 ± 0.003	0.196 ± 0.009	1.982 ± 0.164	1.408 ± 0.066	0.131	-0.008	0.1761	0.1763	1.329	99.091 ± 0.004	1.021 ± 0.049	6.182 ± 0.010
	beta34	0.101	0.110	0.009 ± 0.010	0.087 ± 0.103	0.059 ± 0.005	0.243 ± 0.010	0.059 ± 0.005	0.243 ± 0.011	5.782 ± 0.471	2.404 ± 0.111	0.112	0.011	0.2125	0.2128	1.432	98.727 ± 0.005	1.364 ± 0.101	1.091 ± 0.004
	beta35	0.101	0.100	-0.001 ± 0.011	-0.013 ± 0.108	0.066 ± 0.006	0.257 ± 0.012	0.066 ± 0.006	0.257 ± 0.013	6.453 ± 0.574	2.540 ± 0.126	0.095	-0.006	0.2285	0.2286	1.449	99.091 ± 0.004	1.458 ± 0.129	0.545 ± 0.003
QML	beta31	0.316	0.305	-0.011 ± 0.004	-0.035 ± 0.013	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	0.098 ± 0.007	0.313 ± 0.012	0.307	-0.009	0.0966	0.0971	0.991	93.818 ± 0.010	0.382 ± 0.002	89.455 ± 0.013
	beta32	0.316	0.308	-0.008 ± 0.004	-0.025 ± 0.012	0.008 ± 0.001	0.090 ± 0.003	0.008 ± 0.001	0.091 ± 0.003	0.082 ± 0.005	0.286 ± 0.011	0.300	-0.016	0.0967	0.0979	1.085	97.091 ± 0.007	0.384 ± 0.002	93.273 ± 0.011
	beta33	0.139	0.132	-0.007 ± 0.006	-0.049 ± 0.043	0.019 ± 0.001	0.139 ± 0.005	0.019 ± 0.001	0.139 ± 0.006	1.003 ± 0.072	1.002 ± 0.042	0.125	-0.014	0.1208	0.1216	0.998	96.182 ± 0.008	0.545 ± 0.006	13.091 ± 0.014
	beta34	0.101	0.044	-0.057 ± 0.004	-0.563 ± 0.043	0.010 ± 0.001	0.102 ± 0.004	0.014 ± 0.001	0.116 ± 0.004	1.328 ± 0.080	1.153 ± 0.040	0.049	-0.052	0.0954	0.1089	0.970	89.636 ± 0.013	0.387 ± 0.004	6.545 ± 0.011
	beta35	0.101	0.043	-0.058 ± 0.004	-0.571 ± 0.043	0.010 ± 0.001	0.102 ± 0.004	0.014 ± 0.001	0.117 ± 0.004	1.345 ± 0.090	1.160 ± 0.043	0.041	-0.060	0.0988	0.1155	0.975	90.000 ± 0.013	0.390 ± 0.004	6.727 ± 0.011
UPI	beta31	0.316	0.324	0.008 ± 0.006	0.026 ± 0.017	0.017 ± 0.001	0.129 ± 0.004	0.017 ± 0.001	0.129 ± 0.005	0.167 ± 0.011	0.409 ± 0.017	0.318	0.002	0.1338	0.1338	1.111	93.818 ± 0.010	0.562 ± 0.025	75.455 ± 0.018
	beta32	0.316	0.326	0.010 ± 0.005	0.031 ± 0.016	0.014 ± 0.001	0.120 ± 0.004	0.015 ± 0.001	0.121 ± 0.005	0.146 ± 0.011	0.382 ± 0.017	0.323	0.007	0.1229	0.1231	1.186	97.818 ± 0.006	0.560 ± 0.021	77.091 ± 0.018
	beta33	0.139	0.159	0.020 ± 0.009	0.147 ± 0.062	0.041 ± 0.003	0.203 ± 0.008	0.042 ± 0.003	0.204 ± 0.009	2.160 ± 0.170	1.470 ± 0.066	0.147	0.008	0.1957	0.1959	1.137	97.455 ± 0.007	0.907 ± 0.041	12.364 ± 0.014
	beta34	0.101	0.111	0.010 ± 0.013	0.100 ± 0.128	0.092 ± 0.008	0.304 ± 0.013	0.092 ± 0.008	0.304 ± 0.014	9.045 ± 0.764	3.008 ± 0.143	0.075	-0.026	0.2519	0.2532	1.178	93.091 ± 0.011	1.404 ± 0.096	1.636 ± 0.005
	beta35	0.101	0.095	-0.006 ± 0.012	-0.059 ± 0.122	0.084 ± 0.007	0.289 ± 0.013	0.084 ± 0.007	0.289 ± 0.014	8.187 ± 0.730	2.861 ± 0.142	0.064	-0.037	0.2410	0.2438	1.192	93.273 ± 0.011	1.352 ± 0.093	1.636 ± 0.005

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S16

Simulation 1 - Condition 14: Full Model, Distribution = uniform, N = 1000, Reliability = 0.4

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.306	-0.010 ± 0.002	-0.030 ± 0.007	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.040 ± 0.002	0.200 ± 0.006	0.308	-0.008	0.0620	0.0626	0.987	93.798 ± 0.008	0.241 ± 0.001	99.889 ± 0.001
	beta32	0.316	0.302	-0.014 ± 0.002	-0.045 ± 0.007	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.043 ± 0.002	0.207 ± 0.006	0.298	-0.018	0.0673	0.0695	0.959	92.580 ± 0.009	0.241 ± 0.001	99.889 ± 0.001
	beta33	0.139	0.139	-0.000 ± 0.003	-0.003 ± 0.021	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	0.385 ± 0.019	0.621 ± 0.018	0.140	0.001	0.0843	0.0843	1.010	94.795 ± 0.007	0.342 ± 0.002	36.988 ± 0.016
	beta34	0.101	0.041	-0.060 ± 0.002	-0.593 ± 0.020	0.004 ± 0.000	0.062 ± 0.001	0.007 ± 0.000	0.086 ± 0.002	0.722 ± 0.028	0.850 ± 0.018	0.039	-0.062	0.0648	0.0899	1.013	80.731 ± 0.013	0.244 ± 0.001	9.635 ± 0.010
	beta35	0.101	0.039	-0.062 ± 0.002	-0.617 ± 0.020	0.004 ± 0.000	0.062 ± 0.002	0.008 ± 0.000	0.088 ± 0.002	0.755 ± 0.031	0.869 ± 0.019	0.041	-0.060	0.0575	0.0832	0.999	81.949 ± 0.013	0.242 ± 0.001	8.527 ± 0.009
LSAM	beta31	0.316	0.319	0.003 ± 0.002	0.010 ± 0.008	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.051 ± 0.003	0.226 ± 0.007	0.320	0.004	0.0702	0.0704	1.144	97.674 ± 0.005	0.320 ± 0.002	98.782 ± 0.004
	beta32	0.316	0.316	-0.000 ± 0.002	-0.000 ± 0.008	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.052 ± 0.002	0.228 ± 0.007	0.316	0.000	0.0714	0.0714	1.113	97.453 ± 0.005	0.314 ± 0.002	97.674 ± 0.005
	beta33	0.139	0.139	0.000 ± 0.003	0.002 ± 0.024	0.010 ± 0.001	0.101 ± 0.003	0.010 ± 0.001	0.101 ± 0.003	0.526 ± 0.028	0.725 ± 0.023	0.139	0.000	0.0965	0.0965	1.065	96.788 ± 0.006	0.421 ± 0.005	31.561 ± 0.015
	beta34	0.101	0.110	0.009 ± 0.004	0.090 ± 0.044	0.017 ± 0.001	0.132 ± 0.004	0.018 ± 0.001	0.132 ± 0.005	1.717 ± 0.102	1.310 ± 0.045	0.097	-0.004	0.1256	0.1257	1.069	97.896 ± 0.005	0.553 ± 0.008	7.087 ± 0.009
	beta35	0.101	0.101	0.000 ± 0.004	0.003 ± 0.043	0.017 ± 0.001	0.131 ± 0.004	0.017 ± 0.001	0.131 ± 0.005	1.678 ± 0.101	1.295 ± 0.045	0.095	-0.006	0.1190	0.1192	1.053	97.564 ± 0.005	0.541 ± 0.009	5.648 ± 0.008
QML	beta31	0.316	0.307	-0.009 ± 0.002	-0.029 ± 0.007	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.040 ± 0.002	0.200 ± 0.006	0.308	-0.008	0.0612	0.0617	0.987	93.909 ± 0.008	0.242 ± 0.001	99.889 ± 0.001
	beta32	0.316	0.302	-0.014 ± 0.002	-0.044 ± 0.007	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.043 ± 0.002	0.207 ± 0.006	0.298	-0.018	0.0674	0.0697	0.958	92.470 ± 0.009	0.241 ± 0.001	99.889 ± 0.001
	beta33	0.139	0.140	0.001 ± 0.003	0.005 ± 0.021	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	0.385 ± 0.019	0.621 ± 0.018	0.140	0.001	0.0849	0.0849	1.012	94.906 ± 0.007	0.342 ± 0.002	36.877 ± 0.016
	beta34	0.101	0.040	-0.061 ± 0.002	-0.601 ± 0.020	0.004 ± 0.000	0.062 ± 0.001	0.007 ± 0.000	0.086 ± 0.002	0.731 ± 0.028	0.855 ± 0.018	0.038	-0.063	0.0644	0.0901	1.013	80.399 ± 0.013	0.244 ± 0.001	9.192 ± 0.010
	beta35	0.101	0.038	-0.063 ± 0.002	-0.624 ± 0.020	0.004 ± 0.000	0.062 ± 0.002	0.008 ± 0.000	0.088 ± 0.002	0.766 ± 0.031	0.875 ± 0.019	0.040	-0.061	0.0574	0.0834	0.996	81.506 ± 0.013	0.242 ± 0.001	8.749 ± 0.009
UPI	beta31	0.316	0.321	0.005 ± 0.002	0.016 ± 0.008	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.055 ± 0.003	0.234 ± 0.007	0.321	0.005	0.0745	0.0746	0.993	95.127 ± 0.007	0.288 ± 0.003	98.450 ± 0.004
	beta32	0.316	0.316	0.000 ± 0.002	0.000 ± 0.008	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.056 ± 0.003	0.236 ± 0.007	0.314	-0.002	0.0729	0.0729	0.988	94.574 ± 0.008	0.289 ± 0.003	98.339 ± 0.004
	beta33	0.139	0.147	0.008 ± 0.004	0.056 ± 0.025	0.011 ± 0.001	0.106 ± 0.003	0.011 ± 0.001	0.107 ± 0.003	0.590 ± 0.033	0.768 ± 0.025	0.142	0.003	0.0965	0.0966	1.002	95.127 ± 0.007	0.418 ± 0.008	35.770 ± 0.016
	beta34	0.101	0.117	0.016 ± 0.006	0.161 ± 0.055	0.027 ± 0.002	0.165 ± 0.006	0.028 ± 0.002	0.166 ± 0.007	2.706 ± 0.192	1.645 ± 0.065	0.097	-0.004	0.1416	0.1416	0.920	94.131 ± 0.008	0.597 ± 0.014	6.645 ± 0.008
	beta35	0.101	0.106	0.005 ± 0.005	0.053 ± 0.053	0.026 ± 0.002	0.162 ± 0.005	0.026 ± 0.002	0.162 ± 0.006	2.559 ± 0.168	1.600 ± 0.059	0.090	-0.011	0.1404	0.1409	0.922	93.909 ± 0.008	0.584 ± 0.015	5.869 ± 0.008

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S17

Simulation 1 - Condition 15: Full Model, Distribution = uniform, N = 400, Reliability = 0.6

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.309	-0.007 ± 0.002	-0.024 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.051 ± 0.002	0.227 ± 0.007	0.307	-0.009	0.0697	0.0702	1.028	95.935 ± 0.006	0.287 ± 0.001	99.187 ± 0.003
	beta32	0.316	0.307	-0.009 ± 0.002	-0.030 ± 0.007	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.055 ± 0.003	0.236 ± 0.007	0.305	-0.011	0.0741	0.0750	0.990	94.512 ± 0.007	0.287 ± 0.001	98.780 ± 0.003
	beta33	0.139	0.144	0.005 ± 0.003	0.033 ± 0.021	0.008 ± 0.000	0.092 ± 0.002	0.008 ± 0.000	0.092 ± 0.003	0.439 ± 0.020	0.663 ± 0.019	0.141	0.002	0.0904	0.0904	1.000	96.138 ± 0.006	0.361 ± 0.002	34.654 ± 0.015
	beta34	0.101	0.058	-0.043 ± 0.002	-0.425 ± 0.024	0.006 ± 0.000	0.076 ± 0.002	0.008 ± 0.000	0.087 ± 0.002	0.746 ± 0.033	0.864 ± 0.022	0.058	-0.043	0.0723	0.0839	0.975	89.634 ± 0.010	0.290 ± 0.001	11.687 ± 0.010
	beta35	0.101	0.057	-0.044 ± 0.002	-0.436 ± 0.024	0.006 ± 0.000	0.075 ± 0.002	0.008 ± 0.000	0.087 ± 0.002	0.747 ± 0.032	0.865 ± 0.021	0.056	-0.045	0.0708	0.0838	0.977	89.939 ± 0.010	0.289 ± 0.001	11.585 ± 0.010
LSAM	beta31	0.316	0.316	0.000 ± 0.002	0.001 ± 0.008	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.060 ± 0.003	0.245 ± 0.007	0.316	0.000	0.0785	0.0785	1.077	96.646 ± 0.006	0.327 ± 0.001	97.358 ± 0.005
	beta32	0.316	0.315	-0.001 ± 0.003	-0.002 ± 0.008	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.062 ± 0.003	0.250 ± 0.007	0.314	-0.002	0.0791	0.0792	1.038	95.833 ± 0.006	0.321 ± 0.001	97.154 ± 0.005
	beta33	0.139	0.142	0.003 ± 0.003	0.023 ± 0.023	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	0.522 ± 0.028	0.722 ± 0.022	0.137	-0.002	0.0988	0.0988	0.984	96.341 ± 0.006	0.387 ± 0.003	33.435 ± 0.015
	beta34	0.101	0.103	0.002 ± 0.004	0.023 ± 0.039	0.015 ± 0.001	0.123 ± 0.003	0.015 ± 0.001	0.123 ± 0.004	1.487 ± 0.078	1.219 ± 0.037	0.102	0.001	0.1169	0.1169	0.995	95.935 ± 0.006	0.481 ± 0.005	11.382 ± 0.010
	beta35	0.101	0.103	0.002 ± 0.004	0.017 ± 0.039	0.015 ± 0.001	0.124 ± 0.003	0.015 ± 0.001	0.124 ± 0.004	1.512 ± 0.081	1.230 ± 0.038	0.098	-0.003	0.1149	0.1149	0.977	95.935 ± 0.006	0.476 ± 0.004	10.976 ± 0.010
QML	beta31	0.316	0.309	-0.007 ± 0.002	-0.023 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.051 ± 0.002	0.227 ± 0.007	0.307	-0.009	0.0695	0.0700	1.027	95.833 ± 0.006	0.287 ± 0.001	99.187 ± 0.003
	beta32	0.316	0.307	-0.009 ± 0.002	-0.029 ± 0.007	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.056 ± 0.003	0.236 ± 0.007	0.304	-0.012	0.0734	0.0743	0.989	94.614 ± 0.007	0.287 ± 0.001	98.780 ± 0.003
	beta33	0.139	0.144	0.005 ± 0.003	0.038 ± 0.021	0.009 ± 0.000	0.092 ± 0.002	0.009 ± 0.000	0.092 ± 0.003	0.442 ± 0.020	0.665 ± 0.019	0.142	0.003	0.0908	0.0908	0.999	96.240 ± 0.006	0.361 ± 0.002	34.756 ± 0.015
	beta34	0.101	0.057	-0.044 ± 0.002	-0.436 ± 0.024	0.006 ± 0.000	0.076 ± 0.002	0.008 ± 0.000	0.088 ± 0.002	0.754 ± 0.033	0.868 ± 0.022	0.058	-0.043	0.0712	0.0834	0.976	89.329 ± 0.010	0.290 ± 0.001	11.687 ± 0.010
	beta35	0.101	0.056	-0.045 ± 0.002	-0.445 ± 0.024	0.006 ± 0.000	0.076 ± 0.002	0.008 ± 0.000	0.088 ± 0.002	0.757 ± 0.032	0.870 ± 0.021	0.055	-0.046	0.0711	0.0847	0.978	90.346 ± 0.009	0.290 ± 0.001	11.179 ± 0.010
UPI	beta31	0.316	0.317	0.001 ± 0.003	0.004 ± 0.008	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.062 ± 0.003	0.249 ± 0.007	0.317	0.001	0.0784	0.0784	0.998	95.732 ± 0.006	0.308 ± 0.001	97.967 ± 0.004
	beta32	0.316	0.317	0.001 ± 0.003	0.002 ± 0.008	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.064 ± 0.003	0.253 ± 0.007	0.317	0.001	0.0786	0.0786	0.976	94.309 ± 0.007	0.306 ± 0.001	98.171 ± 0.004
	beta33	0.139	0.148	0.009 ± 0.003	0.065 ± 0.024	0.011 ± 0.001	0.107 ± 0.003	0.011 ± 0.001	0.107 ± 0.004	0.592 ± 0.034	0.769 ± 0.025	0.147	0.008	0.1014	0.1018	0.924	95.427 ± 0.007	0.386 ± 0.003	35.061 ± 0.015
	beta34	0.101	0.104	0.003 ± 0.004	0.033 ± 0.044	0.019 ± 0.001	0.139 ± 0.004	0.019 ± 0.001	0.139 ± 0.005	1.888 ± 0.107	1.374 ± 0.045	0.102	0.001	0.1301	0.1301	0.890	93.293 ± 0.008	0.484 ± 0.006	12.602 ± 0.011
	beta35	0.101	0.106	0.005 ± 0.004	0.046 ± 0.044	0.019 ± 0.001	0.139 ± 0.004	0.019 ± 0.001	0.139 ± 0.004	1.886 ± 0.106	1.373 ± 0.044	0.099	-0.002	0.1216	0.1216	0.890	94.309 ± 0.007	0.484 ± 0.006	13.211 ± 0.011

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S18

Simulation 1 - Condition 16: Full Model, Distribution = uniform, N = 1000, Reliability = 0.6

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.309	-0.007 ± 0.001	-0.021 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.310	-0.006	0.0449	0.0453	1.008	94.573 ± 0.007	0.180 ± 0.000	100.000 ± 0.000
	beta32	0.316	0.308	-0.008 ± 0.001	-0.025 ± 0.004	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.309	-0.007	0.0445	0.0450	1.025	94.975 ± 0.007	0.180 ± 0.000	100.000 ± 0.000
	beta33	0.139	0.142	0.003 ± 0.002	0.018 ± 0.013	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.167 ± 0.008	0.408 ± 0.012	0.141	0.002	0.0559	0.0559	1.010	95.176 ± 0.007	0.225 ± 0.001	70.854 ± 0.014
	beta34	0.101	0.059	-0.042 ± 0.001	-0.418 ± 0.014	0.002 ± 0.000	0.046 ± 0.001	0.004 ± 0.000	0.062 ± 0.001	0.381 ± 0.015	0.617 ± 0.013	0.058	-0.043	0.0433	0.0609	1.004	85.729 ± 0.011	0.181 ± 0.001	23.116 ± 0.013
	beta35	0.101	0.060	-0.041 ± 0.001	-0.407 ± 0.014	0.002 ± 0.000	0.045 ± 0.001	0.004 ± 0.000	0.061 ± 0.001	0.365 ± 0.015	0.605 ± 0.013	0.060	-0.041	0.0456	0.0615	1.020	85.427 ± 0.011	0.181 ± 0.001	25.226 ± 0.014
LSAM	beta31	0.316	0.317	0.001 ± 0.002	0.003 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.151 ± 0.004	0.317	0.001	0.0482	0.0483	1.050	96.080 ± 0.006	0.197 ± 0.000	100.000 ± 0.000
	beta32	0.316	0.316	0.000 ± 0.001	0.001 ± 0.005	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.149 ± 0.004	0.316	0.000	0.0466	0.0466	1.042	95.980 ± 0.006	0.193 ± 0.000	100.000 ± 0.000
	beta33	0.139	0.140	0.001 ± 0.002	0.008 ± 0.013	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.170 ± 0.008	0.412 ± 0.012	0.140	0.001	0.0558	0.0558	1.022	95.477 ± 0.007	0.229 ± 0.001	67.437 ± 0.015
	beta34	0.101	0.103	0.002 ± 0.002	0.017 ± 0.022	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.483 ± 0.023	0.695 ± 0.020	0.098	-0.003	0.0670	0.0671	1.029	95.779 ± 0.006	0.283 ± 0.001	27.638 ± 0.014
	beta35	0.101	0.104	0.003 ± 0.002	0.027 ± 0.022	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.479 ± 0.022	0.692 ± 0.019	0.101	0.000	0.0704	0.0704	1.031	95.176 ± 0.007	0.282 ± 0.001	29.548 ± 0.014
QML	beta31	0.316	0.309	-0.007 ± 0.001	-0.021 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.310	-0.006	0.0448	0.0452	1.008	94.372 ± 0.007	0.180 ± 0.000	100.000 ± 0.000
	beta32	0.316	0.308	-0.008 ± 0.001	-0.024 ± 0.004	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.309	-0.007	0.0446	0.0451	1.025	94.975 ± 0.007	0.180 ± 0.000	100.000 ± 0.000
	beta33	0.139	0.142	0.003 ± 0.002	0.023 ± 0.013	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.167 ± 0.008	0.408 ± 0.012	0.141	0.002	0.0555	0.0556	1.011	95.276 ± 0.007	0.225 ± 0.001	71.156 ± 0.014
	beta34	0.101	0.058	-0.043 ± 0.001	-0.425 ± 0.014	0.002 ± 0.000	0.046 ± 0.001	0.004 ± 0.000	0.063 ± 0.001	0.387 ± 0.015	0.622 ± 0.013	0.057	-0.044	0.0434	0.0614	1.004	85.528 ± 0.011	0.181 ± 0.001	22.613 ± 0.013
	beta35	0.101	0.059	-0.042 ± 0.001	-0.415 ± 0.014	0.002 ± 0.000	0.045 ± 0.001	0.004 ± 0.000	0.062 ± 0.001	0.371 ± 0.015	0.609 ± 0.013	0.059	-0.042	0.0456	0.0623	1.022	85.025 ± 0.011	0.181 ± 0.001	24.724 ± 0.014
UPI	beta31	0.316	0.318	0.002 ± 0.002	0.005 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.152 ± 0.004	0.319	0.003	0.0484	0.0485	0.996	95.176 ± 0.007	0.188 ± 0.000	100.000 ± 0.000
	beta32	0.316	0.317	0.001 ± 0.001	0.003 ± 0.005	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.150 ± 0.004	0.316	0.000	0.0467	0.0467	1.013	95.779 ± 0.006	0.188 ± 0.000	100.000 ± 0.000
	beta33	0.139	0.141	0.002 ± 0.002	0.017 ± 0.013	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.176 ± 0.008	0.419 ± 0.012	0.142	0.003	0.0551	0.0551	1.000	95.377 ± 0.007	0.228 ± 0.001	69.146 ± 0.015
	beta34	0.101	0.104	0.003 ± 0.002	0.026 ± 0.023	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.534 ± 0.026	0.731 ± 0.021	0.099	-0.002	0.0701	0.0702	0.969	94.171 ± 0.007	0.280 ± 0.002	29.146 ± 0.014
	beta35	0.101	0.105	0.004 ± 0.002	0.040 ± 0.024	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.568 ± 0.029	0.754 ± 0.023	0.102	0.001	0.0727	0.0727	0.945	94.673 ± 0.007	0.282 ± 0.002	31.759 ± 0.015

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S19

Simulation 1 - Condition 17: Full Model, Distribution = uniform, N = 400, Reliability = 0.8

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.310	-0.006 ± 0.002	-0.018 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.037 ± 0.002	0.191 ± 0.005	0.310	-0.006	0.0615	0.0618	1.010	95.687 ± 0.006	0.238 ± 0.000	100.000 ± 0.000
	beta32	0.316	0.314	-0.002 ± 0.002	-0.006 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.193 ± 0.005	0.313	-0.003	0.0634	0.0635	0.998	95.186 ± 0.007	0.239 ± 0.000	100.000 ± 0.000
	beta33	0.139	0.146	0.007 ± 0.002	0.049 ± 0.016	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.249 ± 0.011	0.499 ± 0.013	0.148	0.009	0.0686	0.0692	0.996	95.286 ± 0.007	0.270 ± 0.001	58.074 ± 0.016
	beta34	0.101	0.076	-0.025 ± 0.002	-0.251 ± 0.020	0.004 ± 0.000	0.062 ± 0.001	0.005 ± 0.000	0.067 ± 0.002	0.444 ± 0.020	0.666 ± 0.017	0.077	-0.024	0.0615	0.0662	1.007	92.778 ± 0.008	0.246 ± 0.001	21.866 ± 0.013
	beta35	0.101	0.077	-0.024 ± 0.002	-0.240 ± 0.019	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.067 ± 0.002	0.435 ± 0.018	0.659 ± 0.017	0.074	-0.027	0.0615	0.0671	1.013	93.280 ± 0.008	0.246 ± 0.001	22.267 ± 0.013
LSAM	beta31	0.316	0.314	-0.002 ± 0.002	-0.006 ± 0.006	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.038 ± 0.002	0.196 ± 0.005	0.314	-0.002	0.0623	0.0623	1.009	95.988 ± 0.006	0.245 ± 0.001	100.000 ± 0.000
	beta32	0.316	0.318	0.002 ± 0.002	0.005 ± 0.006	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.039 ± 0.002	0.198 ± 0.005	0.317	0.001	0.0641	0.0641	0.983	94.483 ± 0.007	0.242 ± 0.001	99.900 ± 0.001
	beta33	0.139	0.144	0.005 ± 0.002	0.034 ± 0.016	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.251 ± 0.011	0.501 ± 0.013	0.144	0.005	0.0686	0.0688	0.984	94.584 ± 0.007	0.268 ± 0.001	56.469 ± 0.016
	beta34	0.101	0.098	-0.003 ± 0.002	-0.027 ± 0.024	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.593 ± 0.026	0.770 ± 0.021	0.100	-0.001	0.0778	0.0778	1.017	95.286 ± 0.007	0.310 ± 0.001	22.267 ± 0.013
	beta35	0.101	0.099	-0.002 ± 0.002	-0.020 ± 0.024	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.581 ± 0.027	0.763 ± 0.021	0.096	-0.005	0.0762	0.0764	1.013	95.787 ± 0.006	0.306 ± 0.001	23.872 ± 0.014
QML	beta31	0.316	0.310	-0.006 ± 0.002	-0.018 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.037 ± 0.002	0.191 ± 0.005	0.310	-0.006	0.0616	0.0619	1.009	95.787 ± 0.006	0.238 ± 0.000	100.000 ± 0.000
	beta32	0.316	0.314	-0.002 ± 0.002	-0.006 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.193 ± 0.005	0.313	-0.003	0.0634	0.0634	0.998	95.085 ± 0.007	0.239 ± 0.000	100.000 ± 0.000
	beta33	0.139	0.146	0.007 ± 0.002	0.048 ± 0.016	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.249 ± 0.011	0.499 ± 0.013	0.148	0.009	0.0686	0.0692	0.996	95.186 ± 0.007	0.270 ± 0.001	57.874 ± 0.016
	beta34	0.101	0.075	-0.026 ± 0.002	-0.256 ± 0.020	0.004 ± 0.000	0.062 ± 0.001	0.005 ± 0.000	0.068 ± 0.002	0.447 ± 0.020	0.668 ± 0.017	0.076	-0.025	0.0617	0.0666	1.005	92.477 ± 0.008	0.246 ± 0.001	21.866 ± 0.013
	beta35	0.101	0.076	-0.025 ± 0.002	-0.245 ± 0.019	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.067 ± 0.002	0.437 ± 0.019	0.661 ± 0.017	0.074	-0.027	0.0614	0.0673	1.011	93.380 ± 0.008	0.246 ± 0.001	22.166 ± 0.013
UPI	beta31	0.316	0.314	-0.002 ± 0.002	-0.005 ± 0.006	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.039 ± 0.002	0.197 ± 0.005	0.314	-0.002	0.0625	0.0625	0.978	95.085 ± 0.007	0.238 ± 0.000	100.000 ± 0.000
	beta32	0.316	0.318	0.002 ± 0.002	0.006 ± 0.006	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.040 ± 0.002	0.199 ± 0.005	0.316	0.000	0.0642	0.0642	0.969	94.383 ± 0.007	0.239 ± 0.001	99.900 ± 0.001
	beta33	0.139	0.145	0.006 ± 0.002	0.041 ± 0.016	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.258 ± 0.011	0.508 ± 0.014	0.145	0.006	0.0699	0.0701	0.978	94.584 ± 0.007	0.270 ± 0.001	56.068 ± 0.016
	beta34	0.101	0.099	-0.002 ± 0.003	-0.020 ± 0.025	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.627 ± 0.028	0.792 ± 0.021	0.100	-0.001	0.0796	0.0796	0.974	94.684 ± 0.007	0.305 ± 0.001	24.373 ± 0.014
	beta35	0.101	0.100	-0.001 ± 0.003	-0.015 ± 0.025	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.612 ± 0.029	0.782 ± 0.022	0.097	-0.004	0.0736	0.0737	0.989	94.784 ± 0.007	0.306 ± 0.001	24.173 ± 0.014

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S20

Simulation 1 - Condition 18: Full Model, Distribution = uniform, N = 1000, Reliability = 0.8

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.313	-0.003 ± 0.001	-0.010 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.015 ± 0.001	0.121 ± 0.003	0.314	-0.002	0.0385	0.0385	1.006	95.000 ± 0.007	0.150 ± 0.000	100.000 ± 0.000
	beta32	0.316	0.312	-0.004 ± 0.001	-0.014 ± 0.004	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.016 ± 0.001	0.125 ± 0.003	0.311	-0.005	0.0387	0.0390	0.977	94.700 ± 0.007	0.150 ± 0.000	100.000 ± 0.000
	beta33	0.139	0.142	0.003 ± 0.001	0.020 ± 0.010	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.094 ± 0.004	0.306 ± 0.008	0.140	0.001	0.0422	0.0422	1.024	95.900 ± 0.006	0.170 ± 0.000	91.600 ± 0.009
	beta34	0.101	0.076	-0.025 ± 0.001	-0.246 ± 0.012	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.214 ± 0.009	0.462 ± 0.011	0.075	-0.026	0.0397	0.0472	1.002	90.900 ± 0.009	0.155 ± 0.000	48.800 ± 0.016
	beta35	0.101	0.078	-0.023 ± 0.001	-0.228 ± 0.013	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.212 ± 0.009	0.460 ± 0.011	0.078	-0.023	0.0398	0.0460	0.980	90.800 ± 0.009	0.155 ± 0.000	50.400 ± 0.016
LSAM	beta31	0.316	0.317	0.001 ± 0.001	0.003 ± 0.004	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.015 ± 0.001	0.123 ± 0.003	0.318	0.002	0.0395	0.0396	1.005	95.500 ± 0.007	0.154 ± 0.000	100.000 ± 0.000
	beta32	0.316	0.315	-0.001 ± 0.001	-0.003 ± 0.004	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.016 ± 0.001	0.127 ± 0.003	0.314	-0.002	0.0405	0.0406	0.962	94.500 ± 0.007	0.152 ± 0.000	100.000 ± 0.000
	beta33	0.139	0.140	0.001 ± 0.001	0.005 ± 0.010	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.092 ± 0.004	0.303 ± 0.008	0.138	-0.001	0.0424	0.0424	1.026	95.600 ± 0.006	0.169 ± 0.000	91.200 ± 0.009
	beta34	0.101	0.098	-0.003 ± 0.002	-0.027 ± 0.015	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.239 ± 0.011	0.489 ± 0.014	0.099	-0.002	0.0489	0.0490	1.005	94.900 ± 0.007	0.194 ± 0.000	52.200 ± 0.016
	beta35	0.101	0.101	-0.000 ± 0.002	-0.005 ± 0.016	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.244 ± 0.011	0.494 ± 0.013	0.100	-0.001	0.0497	0.0497	0.978	94.100 ± 0.007	0.191 ± 0.000	53.100 ± 0.016
QML	beta31	0.316	0.313	-0.003 ± 0.001	-0.010 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.015 ± 0.001	0.121 ± 0.003	0.314	-0.002	0.0384	0.0385	1.006	95.000 ± 0.007	0.150 ± 0.000	100.000 ± 0.000
	beta32	0.316	0.311	-0.005 ± 0.001	-0.014 ± 0.004	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.016 ± 0.001	0.125 ± 0.003	0.311	-0.005	0.0388	0.0391	0.976	94.700 ± 0.007	0.150 ± 0.000	100.000 ± 0.000
	beta33	0.139	0.142	0.003 ± 0.001	0.018 ± 0.010	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.093 ± 0.004	0.305 ± 0.008	0.140	0.001	0.0418	0.0418	1.024	96.200 ± 0.006	0.170 ± 0.000	91.700 ± 0.009
	beta34	0.101	0.076	-0.025 ± 0.001	-0.250 ± 0.012	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.216 ± 0.009	0.464 ± 0.011	0.075	-0.026	0.0395	0.0472	1.001	90.700 ± 0.009	0.155 ± 0.000	48.400 ± 0.016
	beta35	0.101	0.078	-0.023 ± 0.001	-0.231 ± 0.013	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.213 ± 0.009	0.462 ± 0.011	0.077	-0.024	0.0397	0.0463	0.979	90.600 ± 0.009	0.155 ± 0.000	50.200 ± 0.016
UPI	beta31	0.316	0.317	0.001 ± 0.001	0.003 ± 0.004	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.015 ± 0.001	0.124 ± 0.003	0.318	0.002	0.0399	0.0399	0.978	94.800 ± 0.007	0.150 ± 0.000	100.000 ± 0.000
	beta32	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.004	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.016 ± 0.001	0.127 ± 0.003	0.314	-0.002	0.0402	0.0402	0.950	94.400 ± 0.007	0.150 ± 0.000	100.000 ± 0.000
	beta33	0.139	0.140	0.001 ± 0.001	0.006 ± 0.010	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.093 ± 0.004	0.305 ± 0.008	0.138	-0.001	0.0430	0.0430	1.021	95.500 ± 0.007	0.169 ± 0.000	90.900 ± 0.009
	beta34	0.101	0.099	-0.002 ± 0.002	-0.018 ± 0.016	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.250 ± 0.012	0.500 ± 0.014	0.099	-0.002	0.0506	0.0507	0.972	94.400 ± 0.007	0.192 ± 0.001	53.200 ± 0.016
	beta35	0.101	0.101	0.000 ± 0.002	0.001 ± 0.016	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.253 ± 0.011	0.503 ± 0.014	0.100	-0.001	0.0495	0.0495	0.959	94.100 ± 0.007	0.191 ± 0.000	54.400 ± 0.016

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S21

Simulation 1 - Condition 19: Linear Model, Distribution = normal, N = 400, Reliability = 0.4

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.312	-0.004 ± 0.003	-0.013 ± 0.011	0.010 ± 0.000	0.100 ± 0.002	0.010 ± 0.000	0.100 ± 0.003	0.099 ± 0.005	0.315 ± 0.009	0.307	-0.009	0.0982	0.0986	0.985	95.398 ± 0.007	0.385 ± 0.002	—
	beta32	0.316	0.314	-0.002 ± 0.003	-0.007 ± 0.011	0.010 ± 0.000	0.100 ± 0.002	0.010 ± 0.000	0.100 ± 0.003	0.100 ± 0.005	0.316 ± 0.009	0.305	-0.011	0.1000	0.1006	0.982	95.398 ± 0.007	0.384 ± 0.002	—
	beta33	0.000	-0.040	-0.040 ± 0.004	—	0.018 ± 0.001	0.133 ± 0.003	0.019 ± 0.001	0.139 ± 0.004	—	—	-0.041	-0.041	0.1344	0.1404	1.048	95.960 ± 0.007	0.545 ± 0.006	4.040 ± 0.007
	beta34	0.000	0.008	0.008 ± 0.003	—	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	—	—	0.002	0.002	0.0800	0.0801	1.074	97.082 ± 0.006	0.358 ± 0.005	2.918 ± 0.006
	beta35	0.000	0.004	0.004 ± 0.003	—	0.008 ± 0.000	0.090 ± 0.002	0.008 ± 0.000	0.090 ± 0.003	—	—	0.005	0.005	0.0888	0.0889	1.004	96.521 ± 0.006	0.353 ± 0.003	3.479 ± 0.006
LSAM	beta31	0.316	0.315	-0.001 ± 0.004	-0.002 ± 0.012	0.013 ± 0.001	0.115 ± 0.003	0.013 ± 0.001	0.115 ± 0.004	0.133 ± 0.007	0.364 ± 0.011	0.310	-0.006	0.1121	0.1123	1.129	96.857 ± 0.006	0.510 ± 0.005	—
	beta32	0.316	0.316	-0.000 ± 0.004	-0.001 ± 0.012	0.013 ± 0.001	0.114 ± 0.003	0.013 ± 0.001	0.114 ± 0.004	0.130 ± 0.007	0.360 ± 0.011	0.308	-0.008	0.1128	0.1131	1.122	97.194 ± 0.006	0.501 ± 0.005	—
	beta33	0.000	-0.001	-0.001 ± 0.005	—	0.025 ± 0.001	0.157 ± 0.005	0.025 ± 0.001	0.157 ± 0.005	—	—	-0.002	-0.002	0.1494	0.1495	1.125	98.878 ± 0.004	0.695 ± 0.012	1.122 ± 0.004
	beta34	0.000	0.003	0.003 ± 0.004	—	0.012 ± 0.001	0.110 ± 0.003	0.012 ± 0.001	0.110 ± 0.004	—	—	0.000	0.000	0.1048	0.1048	1.123	98.653 ± 0.004	0.485 ± 0.009	1.347 ± 0.004
	beta35	0.000	0.000	-0.000 ± 0.004	—	0.013 ± 0.001	0.112 ± 0.003	0.013 ± 0.001	0.112 ± 0.004	—	—	-0.001	-0.001	0.1103	0.1103	1.088	98.316 ± 0.004	0.478 ± 0.008	1.684 ± 0.004
QML	beta31	0.316	0.312	-0.004 ± 0.003	-0.012 ± 0.011	0.010 ± 0.000	0.100 ± 0.002	0.010 ± 0.000	0.100 ± 0.003	0.099 ± 0.005	0.315 ± 0.009	0.308	-0.008	0.0992	0.0995	0.983	95.286 ± 0.007	0.384 ± 0.002	—
	beta32	0.316	0.314	-0.002 ± 0.003	-0.007 ± 0.011	0.010 ± 0.000	0.100 ± 0.002	0.010 ± 0.000	0.100 ± 0.003	0.100 ± 0.005	0.316 ± 0.009	0.306	-0.010	0.1000	0.1006	0.981	95.398 ± 0.007	0.384 ± 0.002	—
	beta33	0.000	-0.040	-0.040 ± 0.004	—	0.017 ± 0.001	0.132 ± 0.003	0.019 ± 0.001	0.138 ± 0.004	—	—	-0.041	-0.041	0.1334	0.1397	1.044	96.184 ± 0.006	0.542 ± 0.004	3.816 ± 0.006
	beta34	0.000	0.008	0.008 ± 0.003	—	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	—	—	0.005	0.005	0.0804	0.0806	1.069	97.194 ± 0.006	0.354 ± 0.003	2.806 ± 0.006
	beta35	0.000	0.004	0.004 ± 0.003	—	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	—	—	0.004	0.004	0.0880	0.0881	1.008	96.409 ± 0.006	0.353 ± 0.003	3.591 ± 0.006
UPI	beta31	0.316	0.321	0.005 ± 0.004	0.016 ± 0.012	0.014 ± 0.001	0.117 ± 0.003	0.014 ± 0.001	0.117 ± 0.004	0.138 ± 0.008	0.372 ± 0.012	0.312	-0.004	0.1154	0.1155	0.983	95.398 ± 0.007	0.452 ± 0.004	—
	beta32	0.316	0.321	0.005 ± 0.004	0.017 ± 0.013	0.014 ± 0.001	0.119 ± 0.003	0.014 ± 0.001	0.119 ± 0.004	0.142 ± 0.007	0.377 ± 0.012	0.312	-0.004	0.1182	0.1183	0.962	95.062 ± 0.007	0.449 ± 0.004	—
	beta33	0.000	0.002	0.002 ± 0.006	—	0.029 ± 0.002	0.169 ± 0.005	0.029 ± 0.002	0.169 ± 0.006	—	—	-0.002	-0.002	0.1420	0.1420	1.022	97.755 ± 0.005	0.678 ± 0.017	2.245 ± 0.005
	beta34	0.000	0.001	0.001 ± 0.004	—	0.016 ± 0.001	0.127 ± 0.004	0.016 ± 0.001	0.127 ± 0.005	—	—	0.001	0.001	0.1077	0.1077	0.977	96.633 ± 0.006	0.485 ± 0.011	3.367 ± 0.006
	beta35	0.000	0.000	-0.000 ± 0.004	—	0.016 ± 0.001	0.128 ± 0.004	0.016 ± 0.001	0.128 ± 0.005	—	—	0.005	0.005	0.1156	0.1157	0.977	96.970 ± 0.006	0.490 ± 0.015	3.030 ± 0.006

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S22

Simulation 1 - Condition 20: Linear Model, Distribution = normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.007	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.042 ± 0.002	0.205 ± 0.006	0.313	-0.003	0.0656	0.0657	0.951	94.315 ± 0.007	0.242 ± 0.001	—
	beta32	0.316	0.312	-0.004 ± 0.002	-0.011 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.192 ± 0.005	0.312	-0.004	0.0631	0.0632	1.016	94.924 ± 0.007	0.241 ± 0.001	—
	beta33	0.000	-0.041	-0.041 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.002	0.009 ± 0.000	0.095 ± 0.003	—	—	-0.044	-0.044	0.0843	0.0953	1.003	93.299 ± 0.008	0.338 ± 0.002	6.701 ± 0.008
	beta34	0.000	0.006	0.006 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	0.006	0.006	0.0584	0.0587	0.965	95.635 ± 0.007	0.220 ± 0.001	4.365 ± 0.007
	beta35	0.000	0.004	0.004 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	0.003	0.003	0.0580	0.0581	0.953	94.416 ± 0.007	0.217 ± 0.001	5.584 ± 0.007
LSAM	beta31	0.316	0.318	0.002 ± 0.002	0.006 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.051 ± 0.002	0.226 ± 0.006	0.316	0.000	0.0690	0.0690	1.038	96.041 ± 0.006	0.291 ± 0.001	—
	beta32	0.316	0.316	-0.000 ± 0.002	-0.000 ± 0.007	0.004 ± 0.000	0.067 ± 0.001	0.004 ± 0.000	0.067 ± 0.002	0.044 ± 0.002	0.211 ± 0.005	0.317	0.001	0.0693	0.0694	1.090	96.447 ± 0.006	0.284 ± 0.001	—
	beta33	0.000	-0.001	-0.001 ± 0.003	—	0.008 ± 0.000	0.092 ± 0.002	0.008 ± 0.000	0.092 ± 0.003	—	—	-0.005	-0.005	0.0869	0.0870	1.009	97.259 ± 0.005	0.363 ± 0.003	2.741 ± 0.005
	beta34	0.000	0.002	0.002 ± 0.002	—	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	—	—	0.004	0.004	0.0635	0.0637	0.971	97.360 ± 0.005	0.257 ± 0.002	2.640 ± 0.005
	beta35	0.000	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.002	-0.002	0.0621	0.0621	0.960	95.838 ± 0.006	0.250 ± 0.002	4.162 ± 0.006
QML	beta31	0.316	0.316	-0.000 ± 0.002	-0.000 ± 0.007	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.042 ± 0.002	0.205 ± 0.006	0.313	-0.003	0.0656	0.0657	0.950	94.315 ± 0.007	0.242 ± 0.001	—
	beta32	0.316	0.312	-0.004 ± 0.002	-0.011 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.192 ± 0.005	0.312	-0.004	0.0632	0.0634	1.014	94.924 ± 0.007	0.241 ± 0.001	—
	beta33	0.000	-0.041	-0.041 ± 0.003	—	0.007 ± 0.000	0.085 ± 0.002	0.009 ± 0.000	0.094 ± 0.002	—	—	-0.045	-0.045	0.0826	0.0939	1.012	93.198 ± 0.008	0.339 ± 0.002	6.802 ± 0.008
	beta34	0.000	0.005	0.005 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	0.005	0.005	0.0573	0.0576	0.974	95.431 ± 0.007	0.220 ± 0.001	4.569 ± 0.007
	beta35	0.000	0.004	0.004 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	0.003	0.003	0.0568	0.0569	0.964	94.721 ± 0.007	0.218 ± 0.001	5.279 ± 0.007
UPI	beta31	0.316	0.320	0.004 ± 0.002	0.013 ± 0.007	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.052 ± 0.002	0.227 ± 0.006	0.318	0.002	0.0691	0.0692	0.948	94.416 ± 0.007	0.266 ± 0.001	—
	beta32	0.316	0.318	0.002 ± 0.002	0.006 ± 0.007	0.004 ± 0.000	0.067 ± 0.001	0.004 ± 0.000	0.067 ± 0.002	0.045 ± 0.002	0.212 ± 0.005	0.318	0.002	0.0701	0.0701	1.014	95.635 ± 0.007	0.266 ± 0.001	—
	beta33	0.000	-0.002	-0.002 ± 0.003	—	0.009 ± 0.000	0.096 ± 0.003	0.009 ± 0.000	0.096 ± 0.003	—	—	-0.006	-0.006	0.0884	0.0886	0.957	96.244 ± 0.006	0.362 ± 0.004	3.756 ± 0.006
	beta34	0.000	0.002	0.002 ± 0.002	—	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	—	—	0.002	0.002	0.0671	0.0672	0.897	95.736 ± 0.006	0.258 ± 0.003	4.264 ± 0.006
	beta35	0.000	-0.001	-0.001 ± 0.002	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	-0.001	-0.001	0.0665	0.0665	0.898	94.112 ± 0.008	0.249 ± 0.003	5.888 ± 0.008

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S23

Simulation 1 - Condition 21: Linear Model, Distribution = normal, N = 400, Reliability = 0.6

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.318	0.002 ± 0.002	0.005 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.053 ± 0.002	0.230 ± 0.006	0.318	0.002	0.0714	0.0714	1.004	94.625 ± 0.007	0.286 ± 0.001	—
	beta32	0.316	0.313	-0.003 ± 0.002	-0.010 ± 0.008	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.056 ± 0.003	0.237 ± 0.007	0.309	-0.007	0.0779	0.0782	0.975	94.625 ± 0.007	0.286 ± 0.001	—
	beta33	0.000	-0.019	-0.019 ± 0.003	—	0.008 ± 0.000	0.091 ± 0.002	0.009 ± 0.000	0.093 ± 0.003	—	—	-0.019	-0.019	0.0857	0.0877	1.004	95.132 ± 0.007	0.358 ± 0.002	4.868 ± 0.007
	beta34	0.000	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	0.001	0.001	0.0584	0.0584	1.015	96.247 ± 0.006	0.239 ± 0.001	3.753 ± 0.006
	beta35	0.000	0.004	0.004 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	0.005	0.005	0.0575	0.0577	1.014	95.943 ± 0.006	0.242 ± 0.001	4.057 ± 0.006
LSAM	beta31	0.316	0.320	0.004 ± 0.002	0.013 ± 0.008	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.059 ± 0.003	0.243 ± 0.007	0.321	0.005	0.0756	0.0757	1.048	96.247 ± 0.006	0.315 ± 0.001	—
	beta32	0.316	0.315	-0.001 ± 0.003	-0.005 ± 0.008	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.063 ± 0.003	0.250 ± 0.007	0.311	-0.005	0.0816	0.0817	0.995	95.538 ± 0.007	0.309 ± 0.001	—
	beta33	0.000	0.001	0.001 ± 0.003	—	0.009 ± 0.000	0.097 ± 0.003	0.009 ± 0.000	0.097 ± 0.003	—	—	0.002	0.002	0.0856	0.0856	0.960	94.726 ± 0.007	0.365 ± 0.003	5.274 ± 0.007
	beta34	0.000	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.002	-0.002	0.0624	0.0624	0.977	95.538 ± 0.007	0.254 ± 0.002	4.462 ± 0.007
	beta35	0.000	0.002	0.002 ± 0.002	—	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	—	—	0.003	0.003	0.0624	0.0624	0.984	95.538 ± 0.007	0.260 ± 0.002	4.462 ± 0.007
QML	beta31	0.316	0.318	0.002 ± 0.002	0.005 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.053 ± 0.002	0.230 ± 0.006	0.319	0.003	0.0721	0.0721	1.004	94.625 ± 0.007	0.286 ± 0.001	—
	beta32	0.316	0.313	-0.003 ± 0.002	-0.010 ± 0.008	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.056 ± 0.003	0.237 ± 0.007	0.309	-0.007	0.0778	0.0782	0.974	94.726 ± 0.007	0.286 ± 0.001	—
	beta33	0.000	-0.019	-0.019 ± 0.003	—	0.008 ± 0.000	0.091 ± 0.002	0.009 ± 0.000	0.093 ± 0.003	—	—	-0.019	-0.019	0.0847	0.0868	1.006	95.132 ± 0.007	0.358 ± 0.002	4.868 ± 0.007
	beta34	0.000	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	0.000	0.000	0.0578	0.0578	1.021	96.349 ± 0.006	0.239 ± 0.001	3.651 ± 0.006
	beta35	0.000	0.004	0.004 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	0.005	0.005	0.0574	0.0576	1.017	96.146 ± 0.006	0.242 ± 0.001	3.854 ± 0.006
UPI	beta31	0.316	0.322	0.006 ± 0.002	0.018 ± 0.008	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.060 ± 0.003	0.245 ± 0.007	0.324	0.008	0.0769	0.0773	0.996	94.219 ± 0.007	0.302 ± 0.001	—
	beta32	0.316	0.316	-0.000 ± 0.003	-0.000 ± 0.008	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.064 ± 0.003	0.252 ± 0.007	0.313	-0.003	0.0806	0.0807	0.964	94.219 ± 0.007	0.301 ± 0.001	—
	beta33	0.000	0.001	0.001 ± 0.003	—	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	—	—	0.002	0.002	0.0878	0.0878	0.941	94.118 ± 0.007	0.368 ± 0.003	5.882 ± 0.007
	beta34	0.000	-0.003	-0.003 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	-0.002	-0.002	0.0638	0.0639	0.948	94.625 ± 0.007	0.257 ± 0.002	5.375 ± 0.007
	beta35	0.000	0.002	0.002 ± 0.002	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	0.002	0.002	0.0654	0.0654	0.946	95.335 ± 0.007	0.262 ± 0.002	4.665 ± 0.007

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S24

Simulation 1 - Condition 22: Linear Model, Distribution = normal, N = 1000, Reliability = 0.6

			Mean-based										Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.317	0.001 ± 0.001	0.002 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.317	0.001	0.0450	0.0450	0.992	94.562 ± 0.007	0.179 ± 0.000	—
	beta32	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.005	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.149 ± 0.004	0.316	0.000	0.0447	0.0447	0.971	94.562 ± 0.007	0.179 ± 0.000	—
	beta33	0.000	-0.017	-0.017 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.019	-0.019	0.0550	0.0580	0.999	93.555 ± 0.008	0.221 ± 0.001	6.445 ± 0.008
	beta34	0.000	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.001	0.001	0.0384	0.0384	0.960	94.663 ± 0.007	0.149 ± 0.001	5.337 ± 0.007
	beta35	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.001	-0.001	0.0375	0.0375	1.003	95.267 ± 0.007	0.148 ± 0.001	4.733 ± 0.007
LSAM	beta31	0.316	0.319	0.003 ± 0.002	0.009 ± 0.005	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.154 ± 0.004	0.320	0.004	0.0480	0.0481	1.021	95.670 ± 0.006	0.195 ± 0.000	—
	beta32	0.316	0.318	0.002 ± 0.002	0.006 ± 0.005	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.155 ± 0.004	0.318	0.002	0.0474	0.0475	0.987	94.361 ± 0.007	0.190 ± 0.000	—
	beta33	0.000	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	0.002	0.002	0.0549	0.0549	0.988	94.058 ± 0.008	0.218 ± 0.001	5.942 ± 0.008
	beta34	0.000	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.000	0.000	0.0392	0.0392	0.951	94.562 ± 0.007	0.155 ± 0.001	5.438 ± 0.007
	beta35	0.000	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.003	-0.003	0.0386	0.0387	0.984	94.965 ± 0.007	0.155 ± 0.001	5.035 ± 0.007
QML	beta31	0.316	0.317	0.001 ± 0.001	0.002 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.317	0.001	0.0451	0.0452	0.991	94.562 ± 0.007	0.179 ± 0.000	—
	beta32	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.005	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.149 ± 0.004	0.316	0.000	0.0445	0.0445	0.971	94.562 ± 0.007	0.179 ± 0.000	—
	beta33	0.000	-0.017	-0.017 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.019	-0.019	0.0547	0.0578	1.002	93.656 ± 0.008	0.221 ± 0.001	6.344 ± 0.008
	beta34	0.000	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.001	0.001	0.0383	0.0383	0.966	94.663 ± 0.007	0.149 ± 0.001	5.337 ± 0.007
	beta35	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.001	-0.001	0.0372	0.0372	1.009	95.468 ± 0.007	0.149 ± 0.001	4.532 ± 0.007
UPI	beta31	0.316	0.319	0.003 ± 0.002	0.011 ± 0.005	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.154 ± 0.004	0.319	0.003	0.0473	0.0475	0.980	94.663 ± 0.007	0.187 ± 0.000	—
	beta32	0.316	0.318	0.002 ± 0.002	0.008 ± 0.005	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.155 ± 0.005	0.318	0.002	0.0461	0.0461	0.971	94.461 ± 0.007	0.187 ± 0.000	—
	beta33	0.000	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	0.002	0.002	0.0535	0.0535	0.984	94.159 ± 0.007	0.219 ± 0.001	5.841 ± 0.007
	beta34	0.000	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.001	-0.001	0.0396	0.0396	0.948	94.461 ± 0.007	0.155 ± 0.001	5.539 ± 0.007
	beta35	0.000	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.002	-0.002	0.0400	0.0401	0.986	95.065 ± 0.007	0.155 ± 0.001	4.935 ± 0.007

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S25

Simulation 1 - Condition 23: Linear Model, Distribution = normal, N = 400, Reliability = 0.8

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.314	-0.002 ± 0.002	-0.006 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.038 ± 0.002	0.194 ± 0.005	0.317	0.001	0.0600	0.0600	0.986	94.684 ± 0.007	0.237 ± 0.001	—
	beta32	0.316	0.314	-0.002 ± 0.002	-0.006 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.191 ± 0.005	0.315	-0.001	0.0570	0.0570	0.997	94.383 ± 0.007	0.237 ± 0.001	—
	beta33	0.000	-0.008	-0.008 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	-0.008	-0.008	0.0684	0.0689	1.032	96.389 ± 0.006	0.273 ± 0.001	3.611 ± 0.006
	beta34	0.000	0.002	0.002 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.002	0.002	0.0487	0.0487	0.993	96.389 ± 0.006	0.188 ± 0.001	3.611 ± 0.006
	beta35	0.000	0.000	0.000 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.002	0.002	0.0484	0.0484	0.989	94.183 ± 0.007	0.188 ± 0.001	5.817 ± 0.007
LSAM	beta31	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.006	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.039 ± 0.002	0.197 ± 0.005	0.318	0.002	0.0618	0.0619	1.008	95.286 ± 0.007	0.246 ± 0.001	—
	beta32	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.006	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.038 ± 0.002	0.196 ± 0.005	0.316	0.000	0.0581	0.0581	1.000	94.684 ± 0.007	0.242 ± 0.001	—
	beta33	0.000	0.001	0.001 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	0.000	0.000	0.0685	0.0685	1.007	95.186 ± 0.007	0.269 ± 0.001	4.814 ± 0.007
	beta34	0.000	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	—	—	0.001	0.001	0.0503	0.0503	0.966	94.885 ± 0.007	0.190 ± 0.001	5.115 ± 0.007
	beta35	0.000	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	—	—	0.002	0.002	0.0495	0.0496	0.966	93.581 ± 0.008	0.189 ± 0.001	6.419 ± 0.008
QML	beta31	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.038 ± 0.002	0.194 ± 0.005	0.317	0.001	0.0597	0.0597	0.986	94.684 ± 0.007	0.237 ± 0.001	—
	beta32	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.192 ± 0.005	0.314	-0.002	0.0573	0.0573	0.997	94.283 ± 0.007	0.237 ± 0.001	—
	beta33	0.000	-0.008	-0.008 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	-0.008	-0.008	0.0686	0.0691	1.031	96.489 ± 0.006	0.273 ± 0.001	3.511 ± 0.006
	beta34	0.000	0.002	0.002 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.002	0.002	0.0486	0.0486	0.994	96.489 ± 0.006	0.189 ± 0.001	3.511 ± 0.006
	beta35	0.000	0.000	0.000 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.002	0.002	0.0482	0.0482	0.989	94.283 ± 0.007	0.188 ± 0.001	5.717 ± 0.007
UPI	beta31	0.316	0.315	-0.001 ± 0.002	-0.002 ± 0.006	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.039 ± 0.002	0.198 ± 0.005	0.318	0.002	0.0620	0.0621	0.984	94.584 ± 0.007	0.241 ± 0.001	—
	beta32	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.006	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.039 ± 0.002	0.197 ± 0.005	0.316	0.000	0.0587	0.0587	0.988	94.082 ± 0.007	0.241 ± 0.001	—
	beta33	0.000	0.000	0.000 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.002	0.002	0.0690	0.0690	1.014	95.988 ± 0.006	0.273 ± 0.001	4.012 ± 0.006
	beta34	0.000	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	0.000	0.000	0.0500	0.0500	0.968	95.286 ± 0.007	0.193 ± 0.001	4.714 ± 0.007
	beta35	0.000	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	—	—	0.002	0.002	0.0483	0.0483	0.970	93.882 ± 0.008	0.192 ± 0.001	6.118 ± 0.008

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S26

Simulation 1 - Condition 24: Linear Model, Distribution = normal, N = 1000, Reliability = 0.8

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.314	-0.002 ± 0.001	-0.006 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.116 ± 0.003	0.315	-0.001	0.0365	0.0365	1.034	94.864 ± 0.007	0.149 ± 0.000	—
	beta32	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.317	0.001	0.0353	0.0353	1.033	95.065 ± 0.007	0.149 ± 0.000	—
	beta33	0.000	-0.009	-0.009 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.010	-0.010	0.0444	0.0454	0.955	94.461 ± 0.007	0.170 ± 0.000	5.539 ± 0.007
	beta34	0.000	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.003	0.003	0.0267	0.0269	1.043	95.871 ± 0.006	0.117 ± 0.000	4.129 ± 0.006
	beta35	0.000	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.001	0.001	0.0303	0.0304	0.983	95.770 ± 0.006	0.117 ± 0.000	4.230 ± 0.006
LSAM	beta31	0.316	0.315	-0.001 ± 0.001	-0.004 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.316	0.000	0.0368	0.0368	1.051	95.368 ± 0.007	0.155 ± 0.000	—
	beta32	0.316	0.317	0.001 ± 0.001	0.003 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.318	0.002	0.0363	0.0363	1.036	94.562 ± 0.007	0.152 ± 0.000	—
	beta33	0.000	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.002	-0.002	0.0444	0.0444	0.944	93.555 ± 0.008	0.167 ± 0.001	6.445 ± 0.008
	beta34	0.000	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.002	0.002	0.0275	0.0276	1.029	95.468 ± 0.007	0.118 ± 0.000	4.532 ± 0.007
	beta35	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	0.001	0.001	0.0316	0.0316	0.966	95.065 ± 0.007	0.118 ± 0.000	4.935 ± 0.007
QML	beta31	0.316	0.314	-0.002 ± 0.001	-0.006 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.116 ± 0.003	0.315	-0.001	0.0367	0.0367	1.033	94.965 ± 0.007	0.149 ± 0.000	—
	beta32	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.317	0.001	0.0355	0.0355	1.032	94.965 ± 0.007	0.149 ± 0.000	—
	beta33	0.000	-0.009	-0.009 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.010	-0.010	0.0444	0.0454	0.954	94.260 ± 0.007	0.170 ± 0.000	5.740 ± 0.007
	beta34	0.000	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.003	0.003	0.0266	0.0268	1.041	95.770 ± 0.006	0.117 ± 0.000	4.230 ± 0.006
	beta35	0.000	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.001	0.001	0.0302	0.0303	0.984	95.770 ± 0.006	0.117 ± 0.000	4.230 ± 0.006
UPI	beta31	0.316	0.315	-0.001 ± 0.001	-0.003 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.316	0.000	0.0372	0.0372	1.027	94.965 ± 0.007	0.152 ± 0.000	—
	beta32	0.316	0.317	0.001 ± 0.001	0.003 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.318	0.002	0.0364	0.0365	1.033	94.763 ± 0.007	0.152 ± 0.000	—
	beta33	0.000	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.002	-0.002	0.0444	0.0444	0.949	93.958 ± 0.008	0.168 ± 0.000	6.042 ± 0.008
	beta34	0.000	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.002	0.002	0.0279	0.0280	1.036	95.871 ± 0.006	0.119 ± 0.000	4.129 ± 0.006
	beta35	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	0.000	0.000	0.0314	0.0314	0.975	95.166 ± 0.007	0.119 ± 0.000	4.834 ± 0.007

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S27

Simulation 1 - Condition 25: Linear Model, Distribution = right-skewed, N = 400, Reliability = 0.4

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.276	-0.040 ± 0.004	-0.125 ± 0.012	0.012 ± 0.001	0.111 ± 0.003	0.014 ± 0.001	0.118 ± 0.003	0.140 ± 0.007	0.374 ± 0.011	0.267	-0.049	0.1109	0.1212	0.989	92.857 ± 0.009	0.431 ± 0.003	—
	beta32	0.316	0.272	-0.044 ± 0.004	-0.138 ± 0.012	0.013 ± 0.001	0.114 ± 0.003	0.015 ± 0.001	0.122 ± 0.003	0.149 ± 0.007	0.386 ± 0.010	0.267	-0.049	0.1179	0.1278	0.958	91.101 ± 0.010	0.428 ± 0.002	—
	beta33	0.000	-0.043	-0.043 ± 0.004	—	0.016 ± 0.001	0.127 ± 0.003	0.018 ± 0.001	0.134 ± 0.004	—	—	-0.044	-0.044	0.1285	0.1359	1.056	96.838 ± 0.006	0.528 ± 0.006	3.162 ± 0.006
	beta34	0.000	0.074	0.074 ± 0.003	—	0.007 ± 0.000	0.082 ± 0.002	0.012 ± 0.001	0.111 ± 0.003	—	—	0.070	0.070	0.0757	0.1029	0.967	89.110 ± 0.011	0.312 ± 0.004	10.890 ± 0.011
	beta35	0.000	0.068	0.068 ± 0.003	—	0.006 ± 0.000	0.074 ± 0.002	0.010 ± 0.001	0.101 ± 0.003	—	—	0.066	0.066	0.0716	0.0975	1.039	89.461 ± 0.011	0.303 ± 0.004	10.539 ± 0.011
LSAM	beta31	0.316	0.312	-0.004 ± 0.008	-0.012 ± 0.026	0.059 ± 0.003	0.242 ± 0.007	0.058 ± 0.003	0.242 ± 0.008	0.585 ± 0.034	0.765 ± 0.026	0.300	-0.016	0.2255	0.2260	1.100	98.361 ± 0.004	1.043 ± 0.022	—
	beta32	0.316	0.313	-0.003 ± 0.008	-0.011 ± 0.025	0.055 ± 0.003	0.234 ± 0.007	0.055 ± 0.003	0.234 ± 0.008	0.547 ± 0.031	0.740 ± 0.025	0.298	-0.018	0.2271	0.2278	1.114	98.009 ± 0.005	1.021 ± 0.020	—
	beta33	0.000	0.008	0.008 ± 0.005	—	0.025 ± 0.002	0.157 ± 0.005	0.025 ± 0.002	0.157 ± 0.006	—	—	0.005	0.005	0.1446	0.1447	1.118	98.244 ± 0.004	0.688 ± 0.018	1.756 ± 0.004
	beta34	0.000	0.005	0.005 ± 0.003	—	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.004	—	—	0.007	0.007	0.0824	0.0827	1.069	99.297 ± 0.003	0.417 ± 0.012	0.703 ± 0.003
	beta35	0.000	-0.001	-0.001 ± 0.003	—	0.008 ± 0.001	0.091 ± 0.003	0.008 ± 0.001	0.091 ± 0.003	—	—	0.002	0.002	0.0794	0.0795	1.092	98.361 ± 0.004	0.390 ± 0.010	1.639 ± 0.004
QML	beta31	0.316	0.277	-0.039 ± 0.004	-0.125 ± 0.012	0.012 ± 0.001	0.111 ± 0.003	0.014 ± 0.001	0.118 ± 0.003	0.140 ± 0.007	0.374 ± 0.011	0.268	-0.048	0.1110	0.1209	0.987	92.740 ± 0.009	0.431 ± 0.003	—
	beta32	0.316	0.273	-0.043 ± 0.004	-0.137 ± 0.012	0.013 ± 0.001	0.114 ± 0.003	0.015 ± 0.001	0.122 ± 0.003	0.150 ± 0.007	0.387 ± 0.011	0.269	-0.047	0.1189	0.1278	0.954	91.101 ± 0.010	0.428 ± 0.002	—
	beta33	0.000	-0.042	-0.042 ± 0.004	—	0.016 ± 0.001	0.127 ± 0.003	0.018 ± 0.001	0.134 ± 0.004	—	—	-0.043	-0.043	0.1291	0.1360	1.057	96.721 ± 0.006	0.527 ± 0.006	3.279 ± 0.006
	beta34	0.000	0.074	0.074 ± 0.003	—	0.007 ± 0.000	0.082 ± 0.002	0.012 ± 0.001	0.110 ± 0.003	—	—	0.069	0.069	0.0764	0.1032	0.966	89.227 ± 0.011	0.311 ± 0.004	10.773 ± 0.011
	beta35	0.000	0.067	0.067 ± 0.003	—	0.006 ± 0.000	0.074 ± 0.002	0.010 ± 0.001	0.100 ± 0.003	—	—	0.065	0.065	0.0716	0.0968	1.037	89.461 ± 0.011	0.302 ± 0.004	10.539 ± 0.011
UPI	beta31	0.316	0.327	0.011 ± 0.008	0.036 ± 0.026	0.059 ± 0.003	0.244 ± 0.007	0.059 ± 0.003	0.244 ± 0.008	0.596 ± 0.034	0.772 ± 0.026	0.309	-0.007	0.2357	0.2358	1.078	97.775 ± 0.005	1.030 ± 0.025	—
	beta32	0.316	0.318	0.002 ± 0.008	0.007 ± 0.026	0.057 ± 0.003	0.238 ± 0.007	0.057 ± 0.003	0.238 ± 0.008	0.567 ± 0.033	0.753 ± 0.025	0.307	-0.009	0.2327	0.2328	1.078	96.604 ± 0.006	1.006 ± 0.025	—
	beta33	0.000	0.005	0.005 ± 0.006	—	0.028 ± 0.002	0.166 ± 0.005	0.028 ± 0.002	0.166 ± 0.006	—	—	-0.002	-0.002	0.1433	0.1433	1.443	96.721 ± 0.006	0.939 ± 0.231	3.279 ± 0.006
	beta34	0.000	-0.002	-0.002 ± 0.003	—	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	—	—	0.001	0.001	0.0828	0.0828	1.071	96.956 ± 0.006	0.417 ± 0.017	3.044 ± 0.006
	beta35	0.000	-0.003	-0.003 ± 0.003	—	0.008 ± 0.000	0.090 ± 0.003	0.008 ± 0.000	0.090 ± 0.003	—	—	0.000	0.000	0.0824	0.0824	1.106	97.658 ± 0.005	0.391 ± 0.014	2.342 ± 0.005

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S28

Simulation 1 - Condition 26: Linear Model, Distribution = right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.273	-0.043 ± 0.002	-0.137 ± 0.007	0.005 ± 0.000	0.068 ± 0.002	0.006 ± 0.000	0.081 ± 0.002	0.065 ± 0.003	0.255 ± 0.006	0.270	-0.046	0.0663	0.0805	1.008	88.115 ± 0.010	0.268 ± 0.001	—
	beta32	0.316	0.276	-0.040 ± 0.002	-0.126 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.069 ± 0.003	0.263 ± 0.007	0.274	-0.042	0.0734	0.0845	0.944	87.910 ± 0.010	0.270 ± 0.001	—
	beta33	0.000	-0.045	-0.045 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.002	0.008 ± 0.000	0.092 ± 0.002	—	—	-0.047	-0.047	0.0807	0.0933	1.032	93.750 ± 0.008	0.324 ± 0.002	6.250 ± 0.008
	beta34	0.000	0.071	0.071 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.007 ± 0.000	0.086 ± 0.002	—	—	0.066	0.066	0.0441	0.0796	0.960	69.570 ± 0.015	0.182 ± 0.001	30.430 ± 0.015
	beta35	0.000	0.069	0.069 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.007 ± 0.000	0.083 ± 0.002	—	—	0.067	0.067	0.0422	0.0792	1.019	70.389 ± 0.015	0.183 ± 0.001	29.611 ± 0.015
LSAM	beta31	0.316	0.311	-0.005 ± 0.004	-0.015 ± 0.013	0.017 ± 0.001	0.131 ± 0.003	0.017 ± 0.001	0.131 ± 0.004	0.173 ± 0.009	0.416 ± 0.012	0.305	-0.011	0.1257	0.1261	1.044	96.824 ± 0.006	0.537 ± 0.004	—
	beta32	0.316	0.319	0.003 ± 0.004	0.009 ± 0.014	0.019 ± 0.001	0.137 ± 0.003	0.019 ± 0.001	0.137 ± 0.004	0.187 ± 0.009	0.432 ± 0.013	0.315	-0.001	0.1350	0.1350	1.019	96.107 ± 0.006	0.546 ± 0.005	—
	beta33	0.000	0.003	0.003 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	—	—	-0.001	-0.001	0.0740	0.0740	1.047	97.029 ± 0.005	0.306 ± 0.003	2.971 ± 0.005
	beta34	0.000	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.002	0.002	0.0390	0.0391	1.015	96.926 ± 0.006	0.179 ± 0.002	3.074 ± 0.006
	beta35	0.000	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.002	—	—	0.001	0.001	0.0381	0.0381	1.013	96.516 ± 0.006	0.183 ± 0.003	3.484 ± 0.006
QML	beta31	0.316	0.273	-0.043 ± 0.002	-0.136 ± 0.007	0.005 ± 0.000	0.068 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.065 ± 0.003	0.255 ± 0.006	0.271	-0.045	0.0664	0.0803	1.005	88.115 ± 0.010	0.268 ± 0.001	—
	beta32	0.316	0.276	-0.040 ± 0.002	-0.125 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.069 ± 0.003	0.263 ± 0.007	0.275	-0.041	0.0736	0.0845	0.941	88.012 ± 0.010	0.269 ± 0.001	—
	beta33	0.000	-0.044	-0.044 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.002	0.008 ± 0.000	0.091 ± 0.002	—	—	-0.045	-0.045	0.0796	0.0915	1.032	93.545 ± 0.008	0.324 ± 0.002	6.455 ± 0.008
	beta34	0.000	0.070	0.070 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.007 ± 0.000	0.085 ± 0.002	—	—	0.066	0.066	0.0444	0.0793	0.958	70.594 ± 0.015	0.181 ± 0.001	29.406 ± 0.015
	beta35	0.000	0.068	0.068 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.007 ± 0.000	0.082 ± 0.002	—	—	0.066	0.066	0.0424	0.0784	1.016	70.799 ± 0.015	0.183 ± 0.001	29.201 ± 0.015
UPI	beta31	0.316	0.315	-0.001 ± 0.004	-0.002 ± 0.014	0.018 ± 0.001	0.133 ± 0.003	0.018 ± 0.001	0.133 ± 0.004	0.178 ± 0.009	0.422 ± 0.013	0.311	-0.005	0.1291	0.1292	1.002	95.594 ± 0.007	0.524 ± 0.004	—
	beta32	0.316	0.323	0.007 ± 0.004	0.022 ± 0.014	0.019 ± 0.001	0.137 ± 0.003	0.019 ± 0.001	0.137 ± 0.004	0.188 ± 0.009	0.434 ± 0.013	0.318	0.002	0.1388	0.1388	0.982	95.697 ± 0.006	0.527 ± 0.004	—
	beta33	0.000	0.000	0.000 ± 0.003	—	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	—	—	-0.002	-0.002	0.0767	0.0767	0.970	96.209 ± 0.006	0.312 ± 0.005	3.791 ± 0.006
	beta34	0.000	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.001	0.001	0.0416	0.0416	0.972	95.594 ± 0.007	0.178 ± 0.002	4.406 ± 0.007
	beta35	0.000	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.001	-0.001	0.0394	0.0394	1.006	95.287 ± 0.007	0.179 ± 0.002	4.713 ± 0.007

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S29

Simulation 1 - Condition 27: Linear Model, Distribution = right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.286	-0.030 ± 0.003	-0.095 ± 0.009	0.008 ± 0.000	0.091 ± 0.002	0.009 ± 0.000	0.096 ± 0.003	0.092 ± 0.004	0.304 ± 0.009	0.286	-0.030	0.0874	0.0924	0.992	93.011 ± 0.008	0.354 ± 0.001	—
	beta32	0.316	0.284	-0.032 ± 0.003	-0.101 ± 0.009	0.008 ± 0.000	0.088 ± 0.002	0.009 ± 0.000	0.094 ± 0.002	0.088 ± 0.004	0.297 ± 0.008	0.286	-0.030	0.0857	0.0909	1.026	93.834 ± 0.008	0.355 ± 0.001	—
	beta33	0.000	-0.019	-0.019 ± 0.003	—	0.008 ± 0.000	0.090 ± 0.002	0.008 ± 0.000	0.092 ± 0.003	—	—	-0.015	-0.015	0.0890	0.0903	0.976	94.656 ± 0.007	0.345 ± 0.002	5.344 ± 0.007
	beta34	0.000	0.035	0.035 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	0.031	0.031	0.0448	0.0547	0.940	89.825 ± 0.010	0.190 ± 0.002	10.175 ± 0.010
	beta35	0.000	0.036	0.036 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	0.033	0.033	0.0455	0.0564	0.980	90.031 ± 0.010	0.192 ± 0.002	9.969 ± 0.010
LSAM	beta31	0.316	0.316	-0.000 ± 0.005	-0.000 ± 0.015	0.021 ± 0.001	0.146 ± 0.004	0.021 ± 0.001	0.145 ± 0.004	0.212 ± 0.011	0.460 ± 0.014	0.313	-0.003	0.1335	0.1335	1.004	96.403 ± 0.006	0.573 ± 0.004	—
	beta32	0.316	0.310	-0.006 ± 0.005	-0.019 ± 0.014	0.020 ± 0.001	0.141 ± 0.003	0.020 ± 0.001	0.142 ± 0.004	0.201 ± 0.010	0.448 ± 0.013	0.305	-0.011	0.1356	0.1360	1.034	96.095 ± 0.006	0.573 ± 0.004	—
	beta33	0.000	0.002	0.002 ± 0.003	—	0.009 ± 0.000	0.094 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	—	—	0.002	0.002	0.0874	0.0874	0.922	93.320 ± 0.008	0.338 ± 0.004	6.680 ± 0.008
	beta34	0.000	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	—	—	0.001	0.001	0.0475	0.0475	0.915	94.758 ± 0.007	0.202 ± 0.003	5.242 ± 0.007
	beta35	0.000	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	—	—	0.003	0.003	0.0468	0.0470	0.947	95.375 ± 0.007	0.205 ± 0.003	4.625 ± 0.007
QML	beta31	0.316	0.286	-0.030 ± 0.003	-0.094 ± 0.009	0.008 ± 0.000	0.092 ± 0.002	0.009 ± 0.000	0.096 ± 0.003	0.093 ± 0.005	0.305 ± 0.009	0.286	-0.030	0.0876	0.0925	0.987	92.806 ± 0.008	0.354 ± 0.001	—
	beta32	0.316	0.284	-0.032 ± 0.003	-0.101 ± 0.009	0.008 ± 0.000	0.089 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	0.089 ± 0.004	0.298 ± 0.008	0.285	-0.031	0.0867	0.0920	1.021	93.731 ± 0.008	0.355 ± 0.001	—
	beta33	0.000	-0.019	-0.019 ± 0.003	—	0.008 ± 0.000	0.090 ± 0.002	0.008 ± 0.000	0.092 ± 0.003	—	—	-0.015	-0.015	0.0889	0.0902	0.977	94.553 ± 0.007	0.345 ± 0.002	5.447 ± 0.007
	beta34	0.000	0.035	0.035 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	0.031	0.031	0.0445	0.0543	0.941	89.723 ± 0.010	0.190 ± 0.002	10.277 ± 0.010
	beta35	0.000	0.036	0.036 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	0.033	0.033	0.0450	0.0558	0.981	90.134 ± 0.010	0.191 ± 0.002	9.866 ± 0.010
UPI	beta31	0.316	0.319	0.003 ± 0.005	0.011 ± 0.015	0.021 ± 0.001	0.146 ± 0.004	0.021 ± 0.001	0.146 ± 0.004	0.214 ± 0.011	0.462 ± 0.014	0.317	0.001	0.1355	0.1355	0.971	95.375 ± 0.007	0.556 ± 0.004	—
	beta32	0.316	0.312	-0.004 ± 0.005	-0.013 ± 0.014	0.020 ± 0.001	0.141 ± 0.003	0.020 ± 0.001	0.141 ± 0.004	0.199 ± 0.010	0.447 ± 0.013	0.312	-0.004	0.1418	0.1419	1.012	95.786 ± 0.006	0.560 ± 0.004	—
	beta33	0.000	0.001	0.001 ± 0.003	—	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	—	—	0.002	0.002	0.0924	0.0925	0.911	94.039 ± 0.008	0.356 ± 0.004	5.961 ± 0.008
	beta34	0.000	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	—	—	0.001	0.001	0.0455	0.0455	0.918	94.861 ± 0.007	0.206 ± 0.003	5.139 ± 0.007
	beta35	0.000	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	—	—	0.004	0.004	0.0495	0.0497	0.944	95.581 ± 0.007	0.208 ± 0.003	4.419 ± 0.007

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S30

Simulation 1 - Condition 28: Linear Model, Distribution = right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.286	-0.030 ± 0.002	-0.094 ± 0.005	0.003 ± 0.000	0.055 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.039 ± 0.002	0.197 ± 0.005	0.287	-0.029	0.0559	0.0631	1.030	92.485 ± 0.008	0.220 ± 0.000	—
	beta32	0.316	0.284	-0.032 ± 0.002	-0.101 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.042 ± 0.002	0.206 ± 0.005	0.284	-0.032	0.0572	0.0656	0.989	90.481 ± 0.009	0.220 ± 0.000	—
	beta33	0.000	-0.012	-0.012 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.011	-0.011	0.0524	0.0535	0.958	93.487 ± 0.008	0.205 ± 0.001	6.513 ± 0.008
	beta34	0.000	0.031	0.031 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.030	0.030	0.0281	0.0409	0.938	82.365 ± 0.012	0.111 ± 0.001	17.635 ± 0.012
	beta35	0.000	0.030	0.030 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.029	0.029	0.0294	0.0410	0.937	82.064 ± 0.012	0.111 ± 0.001	17.936 ± 0.012
LSAM	beta31	0.316	0.316	0.000 ± 0.003	0.000 ± 0.008	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.069 ± 0.003	0.263 ± 0.007	0.317	0.001	0.0829	0.0829	1.034	95.892 ± 0.006	0.338 ± 0.001	—
	beta32	0.316	0.313	-0.003 ± 0.003	-0.010 ± 0.009	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.002	0.074 ± 0.003	0.271 ± 0.008	0.312	-0.004	0.0834	0.0834	0.992	95.291 ± 0.007	0.333 ± 0.001	—
	beta33	0.000	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	—	—	0.002	0.002	0.0478	0.0478	0.946	93.086 ± 0.008	0.186 ± 0.001	6.914 ± 0.008
	beta34	0.000	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.000	0.000	0.0272	0.0272	0.948	93.687 ± 0.008	0.110 ± 0.001	6.313 ± 0.008
	beta35	0.000	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.000	0.000	0.0279	0.0279	0.927	93.888 ± 0.008	0.109 ± 0.001	6.112 ± 0.008
QML	beta31	0.316	0.286	-0.030 ± 0.002	-0.094 ± 0.005	0.003 ± 0.000	0.055 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.039 ± 0.002	0.197 ± 0.005	0.287	-0.029	0.0564	0.0634	1.024	92.485 ± 0.008	0.220 ± 0.000	—
	beta32	0.316	0.284	-0.032 ± 0.002	-0.100 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.043 ± 0.002	0.207 ± 0.005	0.284	-0.032	0.0575	0.0657	0.983	90.080 ± 0.009	0.220 ± 0.000	—
	beta33	0.000	-0.012	-0.012 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.011	-0.011	0.0521	0.0531	0.961	93.387 ± 0.008	0.205 ± 0.001	6.613 ± 0.008
	beta34	0.000	0.031	0.031 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.029	0.029	0.0279	0.0406	0.940	82.866 ± 0.012	0.111 ± 0.001	17.134 ± 0.012
	beta35	0.000	0.030	0.030 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.028	0.028	0.0294	0.0408	0.938	82.164 ± 0.012	0.111 ± 0.001	17.836 ± 0.012
UPI	beta31	0.316	0.317	0.001 ± 0.003	0.004 ± 0.008	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.068 ± 0.003	0.262 ± 0.007	0.318	0.002	0.0831	0.0832	1.018	95.591 ± 0.006	0.330 ± 0.001	—
	beta32	0.316	0.314	-0.002 ± 0.003	-0.007 ± 0.009	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.073 ± 0.003	0.271 ± 0.008	0.312	-0.004	0.0841	0.0842	0.980	94.790 ± 0.007	0.328 ± 0.001	—
	beta33	0.000	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	—	—	0.002	0.002	0.0498	0.0499	0.968	94.589 ± 0.007	0.191 ± 0.001	5.411 ± 0.007
	beta34	0.000	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.001	-0.001	0.0273	0.0273	0.962	94.088 ± 0.007	0.112 ± 0.001	5.912 ± 0.007
	beta35	0.000	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.000	0.000	0.0280	0.0280	0.945	94.188 ± 0.007	0.111 ± 0.001	5.812 ± 0.007

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S31

Simulation 1 - Condition 29: Linear Model, Distribution = right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.300	-0.016 ± 0.003	-0.050 ± 0.008	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.068 ± 0.003	0.261 ± 0.007	0.297	-0.019	0.0805	0.0827	1.027	95.678 ± 0.006	0.326 ± 0.001	—
	beta32	0.316	0.302	-0.014 ± 0.003	-0.045 ± 0.008	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	0.071 ± 0.003	0.266 ± 0.007	0.299	-0.017	0.0775	0.0793	1.003	94.372 ± 0.007	0.326 ± 0.001	—
	beta33	0.000	-0.006	-0.006 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.006	-0.006	0.0617	0.0621	1.006	95.377 ± 0.007	0.261 ± 0.001	4.623 ± 0.007
	beta34	0.000	0.013	0.013 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	0.014	0.014	0.0334	0.0363	1.014	95.075 ± 0.007	0.144 ± 0.001	4.925 ± 0.007
	beta35	0.000	0.013	0.013 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.014	0.014	0.0355	0.0380	0.979	94.975 ± 0.007	0.143 ± 0.001	5.025 ± 0.007
LSAM	beta31	0.316	0.315	-0.001 ± 0.003	-0.003 ± 0.010	0.010 ± 0.001	0.102 ± 0.002	0.010 ± 0.001	0.102 ± 0.003	0.104 ± 0.005	0.323 ± 0.009	0.312	-0.004	0.0976	0.0977	1.034	95.980 ± 0.006	0.414 ± 0.002	—
	beta32	0.316	0.316	-0.000 ± 0.003	-0.000 ± 0.010	0.011 ± 0.001	0.104 ± 0.003	0.011 ± 0.001	0.104 ± 0.003	0.108 ± 0.005	0.329 ± 0.010	0.311	-0.005	0.0974	0.0975	1.000	94.975 ± 0.007	0.407 ± 0.002	—
	beta33	0.000	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	—	—	0.000	0.000	0.0612	0.0612	0.932	92.864 ± 0.008	0.244 ± 0.002	7.136 ± 0.008
	beta34	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.003	0.003	0.0347	0.0348	0.944	93.668 ± 0.008	0.145 ± 0.002	6.332 ± 0.008
	beta35	0.000	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.003	0.003	0.0369	0.0370	0.901	93.065 ± 0.008	0.142 ± 0.002	6.935 ± 0.008
QML	beta31	0.316	0.300	-0.016 ± 0.003	-0.050 ± 0.008	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.069 ± 0.003	0.262 ± 0.007	0.297	-0.019	0.0812	0.0833	1.021	95.377 ± 0.007	0.326 ± 0.001	—
	beta32	0.316	0.302	-0.014 ± 0.003	-0.045 ± 0.008	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	0.071 ± 0.003	0.267 ± 0.007	0.300	-0.016	0.0776	0.0793	0.999	94.372 ± 0.007	0.326 ± 0.001	—
	beta33	0.000	-0.006	-0.006 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.006	-0.006	0.0621	0.0624	1.004	95.276 ± 0.007	0.260 ± 0.001	4.724 ± 0.007
	beta34	0.000	0.013	0.013 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	0.014	0.014	0.0333	0.0363	1.010	95.075 ± 0.007	0.144 ± 0.001	4.925 ± 0.007
	beta35	0.000	0.013	0.013 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.014	0.014	0.0355	0.0380	0.979	94.874 ± 0.007	0.143 ± 0.001	5.126 ± 0.007
UPI	beta31	0.316	0.316	-0.000 ± 0.003	-0.000 ± 0.010	0.010 ± 0.000	0.102 ± 0.002	0.010 ± 0.000	0.102 ± 0.003	0.104 ± 0.005	0.322 ± 0.009	0.314	-0.002	0.0991	0.0991	1.015	95.477 ± 0.007	0.405 ± 0.002	—
	beta32	0.316	0.317	0.001 ± 0.003	0.003 ± 0.010	0.011 ± 0.001	0.104 ± 0.002	0.011 ± 0.001	0.103 ± 0.003	0.107 ± 0.005	0.328 ± 0.009	0.313	-0.003	0.0998	0.0999	0.995	95.377 ± 0.007	0.404 ± 0.002	—
	beta33	0.000	0.000	0.000 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	0.000	0.000	0.0635	0.0635	0.974	94.472 ± 0.007	0.260 ± 0.002	5.528 ± 0.007
	beta34	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.003	0.003	0.0355	0.0355	0.983	95.176 ± 0.007	0.151 ± 0.002	4.824 ± 0.007
	beta35	0.000	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.002	0.002	0.0357	0.0358	0.960	95.879 ± 0.006	0.150 ± 0.002	4.121 ± 0.006

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S32

Simulation 1 - Condition 30: Linear Model, Distribution = right-skewed, N = 1000, Reliability = 0.8

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.300	-0.016 ± 0.002	-0.050 ± 0.005	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.029 ± 0.001	0.170 ± 0.005	0.300	-0.016	0.0512	0.0536	1.004	94.034 ± 0.008	0.202 ± 0.000	—
	beta32	0.316	0.301	-0.015 ± 0.002	-0.049 ± 0.005	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.029 ± 0.001	0.171 ± 0.005	0.299	-0.017	0.0523	0.0548	0.992	94.034 ± 0.008	0.202 ± 0.000	—
	beta33	0.000	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.002	-0.002	0.0393	0.0394	0.979	94.944 ± 0.007	0.154 ± 0.001	5.056 ± 0.007
	beta34	0.000	0.011	0.011 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	0.011	0.011	0.0192	0.0220	1.005	91.911 ± 0.009	0.083 ± 0.001	8.089 ± 0.009
	beta35	0.000	0.012	0.012 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	—	—	0.012	0.012	0.0204	0.0238	0.963	91.405 ± 0.009	0.082 ± 0.001	8.595 ± 0.009
LSAM	beta31	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.006	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.040 ± 0.002	0.199 ± 0.006	0.315	-0.001	0.0627	0.0628	1.013	95.854 ± 0.006	0.250 ± 0.001	—
	beta32	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.006	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.041 ± 0.002	0.201 ± 0.006	0.314	-0.002	0.0625	0.0626	0.987	95.248 ± 0.007	0.246 ± 0.001	—
	beta33	0.000	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.002	0.002	0.0364	0.0365	0.946	93.023 ± 0.008	0.145 ± 0.001	6.977 ± 0.008
	beta34	0.000	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	0.000	0.000	0.0192	0.0192	0.968	93.832 ± 0.008	0.082 ± 0.001	6.168 ± 0.008
	beta35	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.022 ± 0.001	—	—	0.001	0.001	0.0212	0.0212	0.918	92.619 ± 0.008	0.081 ± 0.001	7.381 ± 0.008
QML	beta31	0.316	0.300	-0.016 ± 0.002	-0.050 ± 0.005	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.029 ± 0.001	0.171 ± 0.005	0.300	-0.016	0.0514	0.0539	0.999	94.034 ± 0.008	0.202 ± 0.000	—
	beta32	0.316	0.301	-0.015 ± 0.002	-0.049 ± 0.005	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.030 ± 0.001	0.172 ± 0.005	0.300	-0.016	0.0522	0.0547	0.986	93.731 ± 0.008	0.202 ± 0.000	—
	beta33	0.000	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.002	-0.002	0.0389	0.0390	0.982	95.046 ± 0.007	0.155 ± 0.001	4.954 ± 0.007
	beta34	0.000	0.011	0.011 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	0.011	0.011	0.0194	0.0222	1.004	92.012 ± 0.009	0.083 ± 0.001	7.988 ± 0.009
	beta35	0.000	0.012	0.012 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	—	—	0.012	0.012	0.0205	0.0239	0.961	91.304 ± 0.009	0.082 ± 0.001	8.696 ± 0.009
UPI	beta31	0.316	0.315	-0.001 ± 0.002	-0.002 ± 0.006	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.040 ± 0.002	0.199 ± 0.006	0.314	-0.002	0.0612	0.0612	0.997	95.450 ± 0.007	0.246 ± 0.001	—
	beta32	0.316	0.315	-0.001 ± 0.002	-0.002 ± 0.006	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.040 ± 0.002	0.201 ± 0.006	0.313	-0.003	0.0616	0.0616	0.985	94.843 ± 0.007	0.245 ± 0.001	—
	beta33	0.000	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.001	0.001	0.0369	0.0370	0.972	94.439 ± 0.007	0.150 ± 0.001	5.561 ± 0.007
	beta34	0.000	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	0.000	0.000	0.0191	0.0191	1.003	95.450 ± 0.007	0.085 ± 0.001	4.550 ± 0.007
	beta35	0.000	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.022 ± 0.001	—	—	0.001	0.001	0.0211	0.0211	0.957	94.439 ± 0.007	0.084 ± 0.001	5.561 ± 0.007

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S33

Simulation 1 - Condition 31: Linear Model, Distribution = uniform, N = 400, Reliability = 0.4

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.317	0.001 ± 0.004	0.002 ± 0.013	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.093 ± 0.004	0.087 ± 0.006	0.296 ± 0.012	0.315	-0.001	0.0927	0.0927	1.039	94.717 ± 0.010	0.381 ± 0.002	—
	beta32	0.316	0.306	-0.010 ± 0.004	-0.031 ± 0.014	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	0.099 ± 0.006	0.314 ± 0.012	0.300	-0.016	0.0957	0.0971	0.979	93.962 ± 0.010	0.379 ± 0.002	—
	beta33	0.000	-0.028	-0.028 ± 0.005	—	0.015 ± 0.001	0.124 ± 0.004	0.016 ± 0.001	0.127 ± 0.005	—	—	-0.026	-0.026	0.1307	0.1334	1.074	96.792 ± 0.008	0.524 ± 0.005	3.208 ± 0.008
	beta34	0.000	0.000	-0.000 ± 0.004	—	0.008 ± 0.001	0.089 ± 0.003	0.008 ± 0.001	0.089 ± 0.003	—	—	0.002	0.002	0.1007	0.1007	1.067	97.547 ± 0.007	0.374 ± 0.004	2.453 ± 0.007
	beta35	0.000	0.007	0.007 ± 0.004	—	0.009 ± 0.001	0.093 ± 0.003	0.009 ± 0.001	0.093 ± 0.004	—	—	0.004	0.004	0.0943	0.0943	1.033	96.415 ± 0.008	0.378 ± 0.004	3.585 ± 0.008
LSAM	beta31	0.316	0.319	0.003 ± 0.005	0.010 ± 0.016	0.014 ± 0.001	0.118 ± 0.005	0.014 ± 0.001	0.118 ± 0.005	0.139 ± 0.011	0.373 ± 0.017	0.314	-0.002	0.1222	0.1222	1.494	97.736 ± 0.006	0.690 ± 0.032	—
	beta32	0.316	0.310	-0.006 ± 0.005	-0.020 ± 0.017	0.015 ± 0.001	0.122 ± 0.004	0.015 ± 0.001	0.122 ± 0.005	0.150 ± 0.010	0.387 ± 0.015	0.306	-0.010	0.1296	0.1300	1.410	97.547 ± 0.007	0.676 ± 0.030	—
	beta33	0.000	-0.009	-0.009 ± 0.008	—	0.035 ± 0.003	0.188 ± 0.009	0.035 ± 0.003	0.188 ± 0.010	—	—	-0.002	-0.002	0.1701	0.1701	1.502	98.679 ± 0.005	1.108 ± 0.089	1.321 ± 0.005
	beta34	0.000	0.001	0.001 ± 0.010	—	0.054 ± 0.005	0.232 ± 0.010	0.054 ± 0.005	0.232 ± 0.012	—	—	0.009	0.009	0.2014	0.2016	1.497	99.811 ± 0.002	1.360 ± 0.107	0.189 ± 0.002
	beta35	0.000	0.012	0.012 ± 0.010	—	0.055 ± 0.005	0.235 ± 0.011	0.055 ± 0.005	0.235 ± 0.012	—	—	-0.002	-0.002	0.2059	0.2059	1.376	99.623 ± 0.003	1.267 ± 0.096	0.377 ± 0.003
QML	beta31	0.316	0.317	0.001 ± 0.004	0.004 ± 0.013	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.094 ± 0.004	0.088 ± 0.006	0.297 ± 0.012	0.315	-0.001	0.0943	0.0943	1.037	94.906 ± 0.010	0.382 ± 0.002	—
	beta32	0.316	0.307	-0.009 ± 0.004	-0.030 ± 0.014	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	0.099 ± 0.006	0.315 ± 0.012	0.301	-0.015	0.0949	0.0961	0.978	93.962 ± 0.010	0.380 ± 0.002	—
	beta33	0.000	-0.028	-0.028 ± 0.005	—	0.016 ± 0.001	0.126 ± 0.004	0.017 ± 0.001	0.129 ± 0.005	—	—	-0.026	-0.026	0.1325	0.1350	1.069	96.792 ± 0.008	0.528 ± 0.005	3.208 ± 0.008
	beta34	0.000	0.000	-0.000 ± 0.004	—	0.008 ± 0.001	0.090 ± 0.003	0.008 ± 0.001	0.090 ± 0.004	—	—	0.002	0.002	0.1006	0.1007	1.064	97.547 ± 0.007	0.377 ± 0.004	2.453 ± 0.007
	beta35	0.000	0.008	0.008 ± 0.004	—	0.009 ± 0.001	0.095 ± 0.003	0.009 ± 0.001	0.095 ± 0.004	—	—	0.002	0.002	0.0949	0.0949	1.023	96.415 ± 0.008	0.379 ± 0.004	3.585 ± 0.008
UPI	beta31	0.316	0.323	0.007 ± 0.005	0.022 ± 0.017	0.015 ± 0.001	0.121 ± 0.005	0.015 ± 0.001	0.121 ± 0.005	0.147 ± 0.012	0.383 ± 0.017	0.318	0.002	0.1180	0.1180	1.165	96.604 ± 0.008	0.553 ± 0.028	—
	beta32	0.316	0.315	-0.001 ± 0.005	-0.004 ± 0.017	0.015 ± 0.001	0.121 ± 0.005	0.015 ± 0.001	0.121 ± 0.005	0.147 ± 0.011	0.384 ± 0.017	0.312	-0.004	0.1237	0.1238	1.236	96.038 ± 0.008	0.588 ± 0.071	—
	beta33	0.000	0.006	0.006 ± 0.009	—	0.039 ± 0.003	0.197 ± 0.008	0.039 ± 0.003	0.197 ± 0.009	—	—	-0.002	-0.002	0.1693	0.1693	1.576	96.792 ± 0.008	1.219 ± 0.351	3.208 ± 0.008
	beta34	0.000	-0.006	-0.006 ± 0.012	—	0.079 ± 0.008	0.280 ± 0.014	0.078 ± 0.008	0.280 ± 0.015	—	—	0.002	0.002	0.2322	0.2322	1.520	98.868 ± 0.005	1.670 ± 0.477	1.132 ± 0.005
	beta35	0.000	0.012	0.012 ± 0.011	—	0.066 ± 0.006	0.258 ± 0.011	0.066 ± 0.006	0.258 ± 0.012	—	—	0.002	0.002	0.2141	0.2141	1.994	98.302 ± 0.006	2.015 ± 0.935	1.698 ± 0.006

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S34

Simulation 1 - Condition 32: Linear Model, Distribution = uniform, N = 1000, Reliability = 0.4

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.316	0.000 ± 0.002	0.000 ± 0.007	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.040 ± 0.002	0.200 ± 0.006	0.314	-0.002	0.0604	0.0604	0.972	94.494 ± 0.008	0.241 ± 0.001	—
	beta32	0.316	0.318	0.002 ± 0.002	0.005 ± 0.007	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.040 ± 0.002	0.201 ± 0.006	0.314	-0.002	0.0657	0.0657	0.966	94.494 ± 0.008	0.241 ± 0.001	—
	beta33	0.000	-0.019	-0.019 ± 0.003	—	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	—	—	-0.018	-0.018	0.0776	0.0796	1.058	95.506 ± 0.007	0.337 ± 0.002	4.494 ± 0.007
	beta34	0.000	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	0.000	0.000	0.0613	0.0613	1.047	96.292 ± 0.006	0.241 ± 0.001	3.708 ± 0.006
	beta35	0.000	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.002	-0.002	0.0552	0.0552	1.065	96.404 ± 0.006	0.241 ± 0.001	3.596 ± 0.006
LSAM	beta31	0.316	0.317	0.001 ± 0.002	0.004 ± 0.007	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.049 ± 0.002	0.222 ± 0.006	0.313	-0.003	0.0700	0.0701	1.141	97.978 ± 0.005	0.313 ± 0.002	—
	beta32	0.316	0.320	0.004 ± 0.002	0.013 ± 0.008	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.051 ± 0.002	0.226 ± 0.007	0.318	0.002	0.0725	0.0725	1.095	96.966 ± 0.006	0.306 ± 0.002	—
	beta33	0.000	0.001	0.001 ± 0.003	—	0.009 ± 0.001	0.095 ± 0.003	0.009 ± 0.001	0.095 ± 0.003	—	—	0.004	0.004	0.0880	0.0881	1.119	98.652 ± 0.004	0.416 ± 0.006	1.348 ± 0.004
	beta34	0.000	0.000	-0.000 ± 0.004	—	0.016 ± 0.001	0.127 ± 0.004	0.016 ± 0.001	0.127 ± 0.004	—	—	0.001	0.001	0.1184	0.1184	1.059	98.764 ± 0.004	0.528 ± 0.008	1.236 ± 0.004
	beta35	0.000	0.000	-0.000 ± 0.004	—	0.015 ± 0.001	0.124 ± 0.004	0.015 ± 0.001	0.124 ± 0.004	—	—	-0.002	-0.002	0.1134	0.1134	1.110	99.101 ± 0.003	0.541 ± 0.008	0.899 ± 0.003
QML	beta31	0.316	0.316	0.000 ± 0.002	0.001 ± 0.007	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.040 ± 0.002	0.200 ± 0.006	0.314	-0.002	0.0605	0.0606	0.971	94.382 ± 0.008	0.241 ± 0.001	—
	beta32	0.316	0.318	0.002 ± 0.002	0.006 ± 0.007	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.040 ± 0.002	0.201 ± 0.006	0.314	-0.002	0.0652	0.0652	0.966	94.494 ± 0.008	0.241 ± 0.001	—
	beta33	0.000	-0.018	-0.018 ± 0.003	—	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	—	—	-0.017	-0.017	0.0784	0.0802	1.060	95.618 ± 0.007	0.338 ± 0.002	4.382 ± 0.007
	beta34	0.000	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	0.000	0.000	0.0606	0.0606	1.051	96.292 ± 0.006	0.242 ± 0.001	3.708 ± 0.006
	beta35	0.000	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.001	-0.001	0.0550	0.0550	1.071	96.517 ± 0.006	0.242 ± 0.001	3.483 ± 0.006
UPI	beta31	0.316	0.320	0.004 ± 0.002	0.014 ± 0.008	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.052 ± 0.003	0.229 ± 0.007	0.318	0.002	0.0729	0.0730	1.004	96.067 ± 0.007	0.284 ± 0.004	—
	beta32	0.316	0.321	0.005 ± 0.002	0.014 ± 0.008	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.054 ± 0.003	0.232 ± 0.007	0.319	0.003	0.0748	0.0749	0.990	95.730 ± 0.007	0.284 ± 0.003	—
	beta33	0.000	0.003	0.003 ± 0.003	—	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.004	—	—	0.008	0.008	0.0907	0.0910	1.046	96.742 ± 0.006	0.412 ± 0.008	3.258 ± 0.006
	beta34	0.000	-0.003	-0.003 ± 0.005	—	0.025 ± 0.002	0.159 ± 0.005	0.025 ± 0.002	0.159 ± 0.006	—	—	-0.001	-0.001	0.1300	0.1300	0.925	98.090 ± 0.005	0.576 ± 0.015	1.910 ± 0.005
	beta35	0.000	0.000	0.000 ± 0.005	—	0.023 ± 0.002	0.151 ± 0.005	0.023 ± 0.002	0.151 ± 0.006	—	—	0.000	0.000	0.1273	0.1273	0.965	98.427 ± 0.004	0.573 ± 0.015	1.573 ± 0.004

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S35

Simulation 1 - Condition 33: Linear Model, Distribution = uniform, N = 400, Reliability = 0.6

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.314	-0.002 ± 0.002	-0.006 ± 0.007	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.052 ± 0.002	0.228 ± 0.006	0.311	-0.005	0.0731	0.0733	0.999	95.528 ± 0.007	0.282 ± 0.001	—
	beta32	0.316	0.322	0.006 ± 0.002	0.020 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.054 ± 0.003	0.233 ± 0.007	0.320	0.004	0.0708	0.0709	0.985	93.902 ± 0.008	0.284 ± 0.001	—
	beta33	0.000	-0.007	-0.007 ± 0.003	—	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	—	—	-0.005	-0.005	0.0824	0.0826	1.028	95.935 ± 0.006	0.355 ± 0.002	4.065 ± 0.006
	beta34	0.000	-0.006	-0.006 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	-0.006	-0.006	0.0679	0.0682	1.035	96.545 ± 0.006	0.285 ± 0.001	3.455 ± 0.006
	beta35	0.000	-0.002	-0.002 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	-0.002	-0.002	0.0682	0.0682	1.027	96.138 ± 0.006	0.288 ± 0.001	3.862 ± 0.006
LSAM	beta31	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.008	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.059 ± 0.003	0.243 ± 0.007	0.313	-0.003	0.0805	0.0806	1.062	96.240 ± 0.006	0.319 ± 0.001	—
	beta32	0.316	0.322	0.006 ± 0.002	0.020 ± 0.008	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.061 ± 0.003	0.247 ± 0.007	0.321	0.005	0.0774	0.0776	1.031	95.224 ± 0.007	0.315 ± 0.001	—
	beta33	0.000	0.004	0.004 ± 0.003	—	0.009 ± 0.000	0.093 ± 0.002	0.009 ± 0.000	0.093 ± 0.003	—	—	0.004	0.004	0.0853	0.0853	1.018	95.935 ± 0.006	0.372 ± 0.003	4.065 ± 0.006
	beta34	0.000	-0.008	-0.008 ± 0.004	—	0.013 ± 0.001	0.113 ± 0.003	0.013 ± 0.001	0.113 ± 0.004	—	—	-0.008	-0.008	0.1041	0.1045	1.037	96.748 ± 0.006	0.458 ± 0.004	3.252 ± 0.006
	beta35	0.000	-0.001	-0.001 ± 0.004	—	0.014 ± 0.001	0.116 ± 0.003	0.013 ± 0.001	0.116 ± 0.004	—	—	-0.001	-0.001	0.1094	0.1094	1.017	96.240 ± 0.006	0.463 ± 0.004	3.760 ± 0.006
QML	beta31	0.316	0.314	-0.002 ± 0.002	-0.006 ± 0.007	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.052 ± 0.002	0.228 ± 0.006	0.311	-0.005	0.0727	0.0728	0.998	95.224 ± 0.007	0.282 ± 0.001	—
	beta32	0.316	0.322	0.006 ± 0.002	0.020 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.054 ± 0.003	0.233 ± 0.007	0.320	0.004	0.0705	0.0706	0.985	93.902 ± 0.008	0.284 ± 0.001	—
	beta33	0.000	-0.007	-0.007 ± 0.003	—	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	—	—	-0.005	-0.005	0.0828	0.0829	1.029	95.935 ± 0.006	0.355 ± 0.002	4.065 ± 0.006
	beta34	0.000	-0.006	-0.006 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	-0.006	-0.006	0.0672	0.0674	1.041	96.545 ± 0.006	0.285 ± 0.001	3.455 ± 0.006
	beta35	0.000	-0.002	-0.002 ± 0.002	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	-0.002	-0.002	0.0679	0.0679	1.031	96.341 ± 0.006	0.288 ± 0.001	3.659 ± 0.006
UPI	beta31	0.316	0.316	0.000 ± 0.003	0.001 ± 0.008	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.062 ± 0.003	0.248 ± 0.007	0.317	0.001	0.0807	0.0807	0.981	94.919 ± 0.007	0.302 ± 0.001	—
	beta32	0.316	0.324	0.008 ± 0.003	0.026 ± 0.008	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.065 ± 0.003	0.255 ± 0.007	0.322	0.006	0.0765	0.0768	0.966	94.106 ± 0.008	0.303 ± 0.001	—
	beta33	0.000	0.004	0.004 ± 0.003	—	0.009 ± 0.000	0.096 ± 0.002	0.009 ± 0.000	0.096 ± 0.003	—	—	0.000	0.000	0.0859	0.0859	0.985	95.732 ± 0.006	0.371 ± 0.003	4.268 ± 0.006
	beta34	0.000	-0.006	-0.006 ± 0.004	—	0.017 ± 0.001	0.129 ± 0.004	0.017 ± 0.001	0.129 ± 0.004	—	—	-0.009	-0.009	0.1173	0.1176	0.904	95.630 ± 0.007	0.457 ± 0.005	4.370 ± 0.007
	beta35	0.000	-0.001	-0.001 ± 0.004	—	0.018 ± 0.001	0.135 ± 0.004	0.018 ± 0.001	0.135 ± 0.005	—	—	0.001	0.001	0.1195	0.1195	0.893	95.224 ± 0.007	0.473 ± 0.005	4.776 ± 0.007

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S36

Simulation 1 - Condition 34: Linear Model, Distribution = uniform, N = 1000, Reliability = 0.6

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.316	0.000 ± 0.001	0.000 ± 0.004	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.141 ± 0.004	0.316	0.000	0.0426	0.0426	1.018	95.046 ± 0.007	0.178 ± 0.000	—
	beta32	0.316	0.317	0.001 ± 0.001	0.002 ± 0.004	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.020 ± 0.001	0.140 ± 0.004	0.315	-0.001	0.0419	0.0419	1.022	95.248 ± 0.007	0.177 ± 0.000	—
	beta33	0.000	-0.010	-0.010 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.010	-0.010	0.0553	0.0563	1.031	96.057 ± 0.006	0.220 ± 0.001	3.943 ± 0.006
	beta34	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.001	-0.001	0.0461	0.0462	1.000	95.349 ± 0.007	0.179 ± 0.001	4.651 ± 0.007
	beta35	0.000	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.001	-0.001	0.0453	0.0453	1.010	96.158 ± 0.006	0.178 ± 0.001	3.842 ± 0.006
LSAM	beta31	0.316	0.317	0.001 ± 0.001	0.002 ± 0.005	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.148 ± 0.004	0.316	0.000	0.0450	0.0450	1.058	95.248 ± 0.007	0.194 ± 0.000	—
	beta32	0.316	0.318	0.002 ± 0.001	0.005 ± 0.005	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.315	-0.001	0.0444	0.0444	1.035	96.057 ± 0.006	0.189 ± 0.000	—
	beta33	0.000	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.002	-0.002	0.0570	0.0570	1.028	95.854 ± 0.006	0.223 ± 0.001	4.146 ± 0.006
	beta34	0.000	0.002	0.002 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	0.001	0.001	0.0683	0.0683	0.965	95.248 ± 0.007	0.274 ± 0.001	4.752 ± 0.007
	beta35	0.000	0.001	0.001 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.000	0.000	0.0662	0.0662	1.009	96.360 ± 0.006	0.273 ± 0.001	3.640 ± 0.006
QML	beta31	0.316	0.316	0.000 ± 0.001	0.000 ± 0.004	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.141 ± 0.004	0.316	0.000	0.0427	0.0427	1.018	95.046 ± 0.007	0.178 ± 0.000	—
	beta32	0.316	0.317	0.001 ± 0.001	0.002 ± 0.004	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.020 ± 0.001	0.140 ± 0.004	0.316	0.000	0.0421	0.0421	1.022	95.147 ± 0.007	0.177 ± 0.000	—
	beta33	0.000	-0.011	-0.011 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.011	-0.011	0.0557	0.0567	1.038	96.057 ± 0.006	0.221 ± 0.001	3.943 ± 0.006
	beta34	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.001	-0.001	0.0451	0.0451	1.006	95.450 ± 0.007	0.179 ± 0.001	4.550 ± 0.007
	beta35	0.000	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.001	-0.001	0.0447	0.0447	1.017	96.158 ± 0.006	0.178 ± 0.001	3.842 ± 0.006
UPI	beta31	0.316	0.317	0.001 ± 0.001	0.005 ± 0.005	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.148 ± 0.004	0.316	0.000	0.0452	0.0452	1.007	94.843 ± 0.007	0.185 ± 0.000	—
	beta32	0.316	0.318	0.002 ± 0.001	0.006 ± 0.005	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.148 ± 0.004	0.316	0.000	0.0450	0.0450	1.008	95.248 ± 0.007	0.185 ± 0.000	—
	beta33	0.000	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.002	-0.002	0.0589	0.0589	1.012	95.956 ± 0.006	0.224 ± 0.001	4.044 ± 0.006
	beta34	0.000	0.001	0.001 ± 0.002	—	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	—	—	0.000	0.000	0.0685	0.0685	0.925	94.742 ± 0.007	0.274 ± 0.002	5.258 ± 0.007
	beta35	0.000	0.001	0.001 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	—	—	0.001	0.001	0.0695	0.0695	0.953	94.540 ± 0.007	0.274 ± 0.002	5.460 ± 0.007

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S37

Simulation 1 - Condition 35: Linear Model, Distribution = uniform, N = 400, Reliability = 0.8

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.315	-0.001 ± 0.002	-0.002 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.185 ± 0.005	0.314	-0.002	0.0587	0.0587	1.016	94.965 ± 0.007	0.233 ± 0.000	—
	beta32	0.316	0.318	0.002 ± 0.002	0.006 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.189 ± 0.005	0.317	0.001	0.0580	0.0580	0.996	95.569 ± 0.007	0.233 ± 0.000	—
	beta33	0.000	-0.004	-0.004 ± 0.002	—	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	—	—	-0.004	-0.004	0.0666	0.0667	1.030	95.267 ± 0.007	0.269 ± 0.001	4.733 ± 0.007
	beta34	0.000	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.003	-0.003	0.0608	0.0609	1.036	96.073 ± 0.006	0.245 ± 0.001	3.927 ± 0.006
	beta35	0.000	0.000	-0.000 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	0.000	0.000	0.0577	0.0577	1.033	95.770 ± 0.006	0.245 ± 0.001	4.230 ± 0.006
LSAM	beta31	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.006	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.188 ± 0.005	0.314	-0.002	0.0592	0.0592	1.041	95.770 ± 0.006	0.243 ± 0.001	—
	beta32	0.316	0.318	0.002 ± 0.002	0.008 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.193 ± 0.005	0.318	0.002	0.0587	0.0587	1.001	95.972 ± 0.006	0.240 ± 0.001	—
	beta33	0.000	0.000	-0.000 ± 0.002	—	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	—	—	0.001	0.001	0.0673	0.0673	1.017	94.562 ± 0.007	0.267 ± 0.001	5.438 ± 0.007
	beta34	0.000	-0.001	-0.001 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	—	—	-0.002	-0.002	0.0738	0.0739	1.037	96.677 ± 0.006	0.303 ± 0.001	3.323 ± 0.006
	beta35	0.000	0.000	-0.000 ± 0.002	—	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	—	—	0.001	0.001	0.0667	0.0667	1.026	94.864 ± 0.007	0.304 ± 0.001	5.136 ± 0.007
QML	beta31	0.316	0.315	-0.001 ± 0.002	-0.002 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.185 ± 0.005	0.314	-0.002	0.0587	0.0587	1.015	94.965 ± 0.007	0.233 ± 0.000	—
	beta32	0.316	0.318	0.002 ± 0.002	0.006 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.189 ± 0.005	0.316	0.000	0.0580	0.0580	0.996	95.569 ± 0.007	0.233 ± 0.000	—
	beta33	0.000	-0.005	-0.005 ± 0.002	—	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	—	—	-0.004	-0.004	0.0669	0.0670	1.028	95.166 ± 0.007	0.269 ± 0.001	4.834 ± 0.007
	beta34	0.000	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.003	-0.003	0.0609	0.0610	1.039	96.073 ± 0.006	0.245 ± 0.001	3.927 ± 0.006
	beta35	0.000	0.000	-0.000 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.001	-0.001	0.0577	0.0577	1.036	95.770 ± 0.006	0.245 ± 0.001	4.230 ± 0.006
UPI	beta31	0.316	0.316	-0.000 ± 0.002	-0.000 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.189 ± 0.005	0.316	0.000	0.0595	0.0595	1.011	95.065 ± 0.007	0.237 ± 0.000	—
	beta32	0.316	0.319	0.003 ± 0.002	0.009 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.038 ± 0.002	0.194 ± 0.005	0.318	0.002	0.0598	0.0598	0.988	95.368 ± 0.007	0.237 ± 0.000	—
	beta33	0.000	0.000	-0.000 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	0.000	0.000	0.0677	0.0677	1.005	94.864 ± 0.007	0.268 ± 0.001	5.136 ± 0.007
	beta34	0.000	-0.001	-0.001 ± 0.002	—	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	—	—	-0.004	-0.004	0.0750	0.0751	1.008	95.670 ± 0.006	0.301 ± 0.001	4.330 ± 0.006
	beta35	0.000	0.000	0.000 ± 0.002	—	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	—	—	0.002	0.002	0.0726	0.0727	0.985	95.065 ± 0.007	0.303 ± 0.001	4.935 ± 0.007

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S38

Simulation 1 - Condition 36: Linear Model, Distribution = uniform, N = 1000, Reliability = 0.8

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.315	-0.001 ± 0.001	-0.003 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.314	-0.002	0.0375	0.0375	1.013	95.046 ± 0.007	0.147 ± 0.000	—
	beta32	0.316	0.316	0.000 ± 0.001	0.001 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.315	-0.001	0.0363	0.0363	0.997	94.439 ± 0.007	0.147 ± 0.000	—
	beta33	0.000	-0.004	-0.004 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	-0.003	-0.003	0.0431	0.0432	0.982	93.832 ± 0.008	0.169 ± 0.000	6.168 ± 0.008
	beta34	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.001	-0.001	0.0385	0.0385	1.004	94.540 ± 0.007	0.154 ± 0.000	5.460 ± 0.007
	beta35	0.000	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.001	-0.001	0.0400	0.0400	0.961	93.428 ± 0.008	0.154 ± 0.000	6.572 ± 0.008
LSAM	beta31	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.314	-0.002	0.0383	0.0384	1.040	96.057 ± 0.006	0.153 ± 0.000	—
	beta32	0.316	0.317	0.001 ± 0.001	0.002 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.015 ± 0.001	0.121 ± 0.003	0.315	-0.001	0.0367	0.0367	1.003	94.641 ± 0.007	0.150 ± 0.000	—
	beta33	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.000	0.000	0.0437	0.0437	0.978	93.225 ± 0.008	0.168 ± 0.000	6.775 ± 0.008
	beta34	0.000	0.001	0.001 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.001	-0.001	0.0487	0.0487	0.999	94.742 ± 0.007	0.190 ± 0.000	5.258 ± 0.007
	beta35	0.000	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	—	—	0.000	0.000	0.0489	0.0489	0.961	93.630 ± 0.008	0.190 ± 0.000	6.370 ± 0.008
QML	beta31	0.316	0.315	-0.001 ± 0.001	-0.003 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.314	-0.002	0.0373	0.0374	1.012	95.046 ± 0.007	0.147 ± 0.000	—
	beta32	0.316	0.316	0.000 ± 0.001	0.001 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.315	-0.001	0.0362	0.0362	0.996	94.439 ± 0.007	0.147 ± 0.000	—
	beta33	0.000	-0.004	-0.004 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	-0.004	-0.004	0.0430	0.0431	0.981	93.832 ± 0.008	0.169 ± 0.000	6.168 ± 0.008
	beta34	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.001	-0.001	0.0382	0.0382	1.005	94.641 ± 0.007	0.154 ± 0.000	5.359 ± 0.007
	beta35	0.000	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.001	-0.001	0.0398	0.0398	0.963	93.428 ± 0.008	0.154 ± 0.000	6.572 ± 0.008
UPI	beta31	0.316	0.316	-0.000 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.314	-0.002	0.0384	0.0384	1.012	95.248 ± 0.007	0.149 ± 0.000	—
	beta32	0.316	0.317	0.001 ± 0.001	0.003 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.015 ± 0.001	0.121 ± 0.003	0.316	0.000	0.0367	0.0367	0.996	94.439 ± 0.007	0.149 ± 0.000	—
	beta33	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.000	0.000	0.0431	0.0431	0.978	93.832 ± 0.008	0.167 ± 0.000	6.168 ± 0.008
	beta34	0.000	0.001	0.001 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.001	-0.001	0.0484	0.0484	0.988	94.843 ± 0.007	0.190 ± 0.001	5.157 ± 0.007
	beta35	0.000	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	-0.001	-0.001	0.0497	0.0497	0.953	93.327 ± 0.008	0.189 ± 0.000	6.673 ± 0.008

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S39

Simulation 2 - Condition 1: Full Model, Distribution = normal, N = 400, Reliability = 0.4

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.004	-0.000 ± 0.018	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	0.250 ± 0.015	0.500 ± 0.017	0.202	0.002	0.0998	0.0998	1.066	97.286 ± 0.006	0.419 ± 0.004	48.982 ± 0.018
	beta42	0.20	0.199	-0.001 ± 0.004	-0.006 ± 0.019	0.010 ± 0.001	0.102 ± 0.003	0.010 ± 0.001	0.102 ± 0.003	0.261 ± 0.015	0.510 ± 0.017	0.193	-0.007	0.0974	0.0977	1.041	96.472 ± 0.007	0.417 ± 0.004	50.204 ± 0.018
	beta43	0.20	0.207	0.007 ± 0.003	0.035 ± 0.017	0.008 ± 0.000	0.090 ± 0.002	0.008 ± 0.000	0.090 ± 0.003	0.204 ± 0.011	0.451 ± 0.014	0.208	0.008	0.0910	0.0914	1.060	96.065 ± 0.007	0.374 ± 0.003	62.144 ± 0.018
	beta44	0.11	0.110	0.000 ± 0.005	0.001 ± 0.048	0.021 ± 0.001	0.144 ± 0.005	0.021 ± 0.001	0.144 ± 0.005	1.716 ± 0.114	1.310 ± 0.050	0.106	-0.004	0.1314	0.1314	1.080	97.829 ± 0.005	0.611 ± 0.010	9.769 ± 0.011
	beta45	0.11	0.112	0.002 ± 0.004	0.019 ± 0.040	0.014 ± 0.001	0.118 ± 0.004	0.014 ± 0.001	0.118 ± 0.004	1.156 ± 0.072	1.075 ± 0.039	0.113	0.003	0.1073	0.1074	1.039	97.422 ± 0.006	0.482 ± 0.006	13.704 ± 0.013
	beta46	0.08	0.097	0.017 ± 0.004	0.216 ± 0.050	0.012 ± 0.001	0.108 ± 0.003	0.012 ± 0.001	0.110 ± 0.004	1.877 ± 0.124	1.370 ± 0.052	0.091	0.011	0.1021	0.1026	1.007	97.151 ± 0.006	0.428 ± 0.007	10.855 ± 0.011
	beta47	0.08	0.086	0.006 ± 0.004	0.072 ± 0.048	0.011 ± 0.001	0.104 ± 0.004	0.011 ± 0.001	0.105 ± 0.004	1.707 ± 0.120	1.307 ± 0.052	0.081	0.001	0.0949	0.0949	1.018	97.422 ± 0.006	0.417 ± 0.008	10.991 ± 0.012
	beta51	0.16	0.188	0.028 ± 0.003	0.172 ± 0.022	0.009 ± 0.000	0.094 ± 0.002	0.010 ± 0.001	0.098 ± 0.003	0.373 ± 0.020	0.611 ± 0.019	0.195	0.035	0.0922	0.0988	1.061	96.472 ± 0.007	0.390 ± 0.003	48.711 ± 0.018
	beta52	0.16	0.191	0.031 ± 0.004	0.195 ± 0.023	0.010 ± 0.000	0.099 ± 0.002	0.011 ± 0.001	0.104 ± 0.003	0.419 ± 0.022	0.648 ± 0.020	0.190	0.030	0.1017	0.1060	1.031	96.608 ± 0.007	0.399 ± 0.003	49.525 ± 0.018
	beta53	0.16	0.163	0.003 ± 0.004	0.017 ± 0.022	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.096 ± 0.003	0.357 ± 0.021	0.598 ± 0.021	0.155	-0.005	0.0914	0.0916	0.998	96.336 ± 0.007	0.374 ± 0.003	41.520 ± 0.018
	beta54	0.16	0.155	-0.005 ± 0.004	-0.031 ± 0.025	0.012 ± 0.001	0.109 ± 0.003	0.012 ± 0.001	0.109 ± 0.004	0.466 ± 0.027	0.683 ± 0.023	0.154	-0.006	0.0992	0.0994	0.907	93.351 ± 0.009	0.388 ± 0.003	38.128 ± 0.018
	beta55	0.08	0.088	0.008 ± 0.005	0.105 ± 0.061	0.017 ± 0.001	0.132 ± 0.004	0.017 ± 0.001	0.132 ± 0.005	2.714 ± 0.170	1.647 ± 0.060	0.081	0.001	0.1229	0.1229	1.011	96.608 ± 0.007	0.522 ± 0.009	9.498 ± 0.011
	beta56	0.08	0.085	0.005 ± 0.005	0.057 ± 0.057	0.015 ± 0.001	0.123 ± 0.004	0.015 ± 0.001	0.123 ± 0.004	2.373 ± 0.148	1.540 ± 0.056	0.089	0.009	0.1145	0.1148	0.967	95.522 ± 0.008	0.467 ± 0.007	10.991 ± 0.012
	beta57	0.06	0.070	0.010 ± 0.004	0.175 ± 0.062	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.101 ± 0.004	2.816 ± 0.168	1.678 ± 0.059	0.070	0.010	0.0944	0.0949	0.977	95.929 ± 0.007	0.384 ± 0.007	9.634 ± 0.011
	beta58	0.06	0.059	-0.001 ± 0.003	-0.009 ± 0.048	0.006 ± 0.000	0.079 ± 0.003	0.006 ± 0.000	0.078 ± 0.003	1.711 ± 0.109	1.308 ± 0.048	0.060	0.000	0.0696	0.0696	1.026	97.558 ± 0.006	0.316 ± 0.004	7.870 ± 0.010
QML	beta41	0.20	0.183	-0.017 ± 0.003	-0.085 ± 0.016	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.090 ± 0.003	0.201 ± 0.010	0.448 ± 0.014	0.181	-0.019	0.0863	0.0882	0.943	91.995 ± 0.010	0.326 ± 0.002	62.008 ± 0.018
	beta42	0.20	0.186	-0.014 ± 0.003	-0.070 ± 0.016	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.090 ± 0.003	0.202 ± 0.011	0.450 ± 0.014	0.184	-0.016	0.0853	0.0869	0.964	94.166 ± 0.009	0.336 ± 0.002	59.294 ± 0.018
	beta43	0.20	0.188	-0.012 ± 0.003	-0.062 ± 0.015	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	0.175 ± 0.009	0.418 ± 0.013	0.190	-0.010	0.0825	0.0831	0.989	92.673 ± 0.010	0.321 ± 0.002	66.621 ± 0.017
	beta44	0.11	0.095	-0.015 ± 0.004	-0.134 ± 0.038	0.013 ± 0.001	0.114 ± 0.003	0.013 ± 0.001	0.115 ± 0.004	1.097 ± 0.059	1.047 ± 0.034	0.094	-0.016	0.1192	0.1202	1.014	95.251 ± 0.008	0.454 ± 0.004	12.890 ± 0.012
	beta45	0.11	0.101	-0.009 ± 0.003	-0.079 ± 0.031	0.009 ± 0.000	0.094 ± 0.003	0.009 ± 0.000	0.094 ± 0.003	0.731 ± 0.039	0.855 ± 0.028	0.103	-0.007	0.0934	0.0937	1.023	96.065 ± 0.007	0.376 ± 0.003	18.046 ± 0.014
	beta46	0.08	0.087	0.007 ± 0.003	0.083 ± 0.037	0.007 ± 0.000	0.081 ± 0.003	0.007 ± 0.000	0.081 ± 0.003	1.027 ± 0.066	1.013 ± 0.038	0.083	0.003	0.0781	0.0781	0.967	95.794 ± 0.007	0.307 ± 0.003	17.775 ± 0.014
	beta47	0.08	0.075	-0.005 ± 0.003	-0.056 ± 0.036	0.006 ± 0.000	0.079 ± 0.003	0.006 ± 0.000	0.079 ± 0.003	0.975 ± 0.065	0.988 ± 0.038	0.076	-0.004	0.0727	0.0728	0.958	94.708 ± 0.008	0.297 ± 0.003	15.061 ± 0.013
	beta51	0.16	0.175	0.015 ± 0.004	0.097 ± 0.023	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	0.393 ± 0.022	0.627 ± 0.021	0.179	0.019	0.0953	0.0972	0.980	94.980 ± 0.008	0.381 ± 0.003	46.540 ± 0.018
	beta52	0.16	0.161	0.001 ± 0.003	0.007 ± 0.022	0.009 ± 0.000	0.094 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	0.348 ± 0.018	0.590 ± 0.019	0.159	-0.001	0.0967	0.0967	1.011	96.608 ± 0.007	0.374 ± 0.002	40.027 ± 0.018
	beta53	0.16	0.162	0.002 ± 0.003	0.010 ± 0.020	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	0.290 ± 0.016	0.539 ± 0.018	0.159	-0.001	0.0867	0.0867	0.978	94.708 ± 0.008	0.331 ± 0.002	48.033 ± 0.018
	beta54	0.16	0.178	0.018 ± 0.004	0.113 ± 0.025	0.012 ± 0.001	0.109 ± 0.003	0.012 ± 0.001	0.111 ± 0.004	0.477 ± 0.027	0.691 ± 0.023	0.176	0.016	0.1037	0.1048	0.909	94.301 ± 0.009	0.389 ± 0.003	45.590 ± 0.018
	beta55	0.08	0.092	0.012 ± 0.004	0.146 ± 0.054	0.014 ± 0.001	0.118 ± 0.003	0.014 ± 0.001	0.119 ± 0.004	2.205 ± 0.121	1.485 ± 0.049	0.090	0.010	0.1186	0.1190	0.984	95.251 ± 0.008	0.456 ± 0.004	12.212 ± 0.012
	beta56	0.08	0.083	0.003 ± 0.004	0.038 ± 0.050	0.012 ± 0.001	0.109 ± 0.003	0.012 ± 0.001	0.109 ± 0.004	1.851 ± 0.098	1.360 ± 0.044	0.091	0.011	0.1114	0.1119	0.956	95.115 ± 0.008	0.408 ± 0.003	13.026 ± 0.012
	beta57	0.06	0.050	-0.010 ± 0.003	-0.163 ± 0.048	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.003	1.747 ± 0.096	1.322 ± 0.043	0.048	-0.012	0.0801	0.0810	0.940	93.351 ± 0.009	0.290 ± 0.003	11.805 ± 0.012
	beta58	0.06	0.046	-0.014 ± 0.002	-0.231 ± 0.039	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	1.169 ± 0.069	1.081 ± 0.037	0.048	-0.012	0.0579	0.0592	0.995	95.115 ± 0.008	0.247 ± 0.002	9.769 ± 0.011

Table S39

Simulation 2 - Condition 1: Full Model, Distribution = normal, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta41	0.20	0.206	0.006 ± 0.004	0.031 ± 0.020	0.012 ± 0.001	0.110 ± 0.003	0.012 ± 0.001	0.110 ± 0.004	0.303 ± 0.018	0.550 ± 0.019	0.206	0.006	0.1149	0.1151	0.874	92.130 ± 0.010	0.377 ± 0.003	59.701 ± 0.018
	beta42	0.20	0.199	-0.001 ± 0.004	-0.003 ± 0.021	0.013 ± 0.001	0.112 ± 0.003	0.013 ± 0.001	0.112 ± 0.004	0.314 ± 0.018	0.561 ± 0.019	0.199	-0.001	0.1080	0.1080	0.886	94.030 ± 0.009	0.390 ± 0.003	56.581 ± 0.018
	beta43	0.20	0.208	0.008 ± 0.004	0.040 ± 0.018	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.098 ± 0.003	0.239 ± 0.014	0.489 ± 0.017	0.208	0.008	0.1018	0.1021	0.937	92.944 ± 0.009	0.358 ± 0.002	64.858 ± 0.018
	beta44	0.11	0.131	0.021 ± 0.007	0.193 ± 0.067	0.040 ± 0.003	0.200 ± 0.007	0.040 ± 0.003	0.201 ± 0.008	3.336 ± 0.238	1.826 ± 0.073	0.105	-0.005	0.1776	0.1777	0.739	91.859 ± 0.010	0.579 ± 0.011	18.046 ± 0.014
	beta45	0.11	0.119	0.009 ± 0.006	0.080 ± 0.051	0.023 ± 0.002	0.153 ± 0.006	0.023 ± 0.002	0.153 ± 0.006	1.935 ± 0.143	1.391 ± 0.058	0.110	0.000	0.1332	0.1332	0.787	93.623 ± 0.009	0.471 ± 0.009	19.946 ± 0.015
	beta46	0.08	0.086	0.006 ± 0.005	0.078 ± 0.068	0.022 ± 0.002	0.148 ± 0.006	0.022 ± 0.002	0.148 ± 0.006	3.439 ± 0.258	1.855 ± 0.078	0.085	0.005	0.1227	0.1228	0.687	89.552 ± 0.011	0.399 ± 0.008	19.267 ± 0.015
	beta47	0.08	0.085	0.005 ± 0.005	0.067 ± 0.062	0.018 ± 0.001	0.134 ± 0.005	0.018 ± 0.001	0.134 ± 0.006	2.817 ± 0.215	1.678 ± 0.071	0.076	-0.004	0.1162	0.1163	0.739	91.316 ± 0.010	0.389 ± 0.009	17.503 ± 0.014
	beta51	0.16	0.192	0.032 ± 0.004	0.198 ± 0.022	0.009 ± 0.001	0.096 ± 0.003	0.010 ± 0.001	0.101 ± 0.003	0.400 ± 0.022	0.632 ± 0.020	0.199	0.039	0.0944	0.1021	0.951	94.030 ± 0.009	0.358 ± 0.002	60.109 ± 0.018
	beta52	0.16	0.189	0.029 ± 0.004	0.182 ± 0.023	0.010 ± 0.000	0.098 ± 0.002	0.010 ± 0.001	0.102 ± 0.003	0.407 ± 0.020	0.638 ± 0.019	0.190	0.030	0.1013	0.1058	0.985	95.522 ± 0.008	0.378 ± 0.002	52.510 ± 0.018
	beta53	0.16	0.163	0.003 ± 0.004	0.019 ± 0.022	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.096 ± 0.003	0.363 ± 0.020	0.602 ± 0.020	0.160	0.000	0.0901	0.0901	0.930	94.030 ± 0.009	0.352 ± 0.002	46.269 ± 0.018
	beta54	0.16	0.166	0.006 ± 0.004	0.035 ± 0.025	0.012 ± 0.001	0.110 ± 0.003	0.012 ± 0.001	0.110 ± 0.004	0.470 ± 0.029	0.686 ± 0.025	0.162	0.002	0.1007	0.1008	0.864	93.351 ± 0.009	0.371 ± 0.002	44.505 ± 0.018
	beta55	0.08	0.087	0.007 ± 0.006	0.093 ± 0.069	0.023 ± 0.002	0.150 ± 0.005	0.023 ± 0.002	0.150 ± 0.006	3.534 ± 0.243	1.880 ± 0.073	0.081	0.001	0.1371	0.1371	0.828	93.894 ± 0.009	0.488 ± 0.010	14.383 ± 0.013
	beta56	0.08	0.082	0.002 ± 0.005	0.020 ± 0.064	0.019 ± 0.001	0.139 ± 0.005	0.019 ± 0.001	0.139 ± 0.006	3.037 ± 0.215	1.743 ± 0.070	0.077	-0.003	0.1229	0.1229	0.818	91.588 ± 0.010	0.447 ± 0.009	15.739 ± 0.013
	beta57	0.06	0.071	0.011 ± 0.004	0.184 ± 0.071	0.013 ± 0.001	0.115 ± 0.004	0.013 ± 0.001	0.116 ± 0.005	3.714 ± 0.280	1.927 ± 0.081	0.069	0.009	0.1043	0.1047	0.757	90.502 ± 0.011	0.342 ± 0.006	14.654 ± 0.013
	beta58	0.06	0.063	0.003 ± 0.003	0.049 ± 0.056	0.008 ± 0.001	0.091 ± 0.003	0.008 ± 0.001	0.091 ± 0.004	2.287 ± 0.164	1.512 ± 0.061	0.059	-0.001	0.0837	0.0837	0.856	91.588 ± 0.010	0.305 ± 0.006	10.448 ± 0.011

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S40

Simulation 2 - Condition 2: Full Model, Distribution = normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.200	0.000 ± 0.002	0.002 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.080 ± 0.004	0.283 ± 0.008	0.199	-0.001	0.0592	0.0592	1.049	96.545 ± 0.006	0.233 ± 0.001	93.801 ± 0.008
	beta42	0.20	0.203	0.003 ± 0.002	0.017 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.083 ± 0.004	0.287 ± 0.009	0.201	0.001	0.0548	0.0548	1.040	95.935 ± 0.006	0.234 ± 0.001	94.411 ± 0.007
	beta43	0.20	0.201	0.001 ± 0.002	0.003 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.073 ± 0.003	0.269 ± 0.008	0.198	-0.002	0.0527	0.0527	1.021	95.935 ± 0.006	0.216 ± 0.001	97.053 ± 0.005
	beta44	0.11	0.116	0.006 ± 0.003	0.059 ± 0.024	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.568 ± 0.029	0.753 ± 0.023	0.117	0.007	0.0811	0.0815	0.992	96.240 ± 0.006	0.322 ± 0.003	32.927 ± 0.015
	beta45	0.11	0.113	0.003 ± 0.002	0.029 ± 0.019	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.367 ± 0.017	0.606 ± 0.017	0.112	0.002	0.0619	0.0619	1.009	95.325 ± 0.007	0.263 ± 0.002	38.516 ± 0.016
	beta46	0.08	0.083	0.003 ± 0.002	0.042 ± 0.023	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.502 ± 0.026	0.708 ± 0.022	0.082	0.002	0.0532	0.0532	1.022	95.528 ± 0.007	0.227 ± 0.002	29.268 ± 0.015
	beta47	0.08	0.084	0.004 ± 0.002	0.047 ± 0.022	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.466 ± 0.023	0.682 ± 0.020	0.083	0.003	0.0509	0.0510	1.023	95.833 ± 0.006	0.219 ± 0.002	33.841 ± 0.015
	beta51	0.16	0.186	0.026 ± 0.002	0.163 ± 0.011	0.003 ± 0.000	0.056 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.147 ± 0.007	0.384 ± 0.010	0.186	0.026	0.0539	0.0599	1.020	93.902 ± 0.008	0.222 ± 0.001	91.667 ± 0.009
	beta52	0.16	0.188	0.028 ± 0.002	0.173 ± 0.012	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.163 ± 0.008	0.404 ± 0.011	0.185	0.025	0.0586	0.0639	1.016	93.496 ± 0.008	0.233 ± 0.001	90.650 ± 0.009
	beta53	0.16	0.161	0.001 ± 0.002	0.008 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.116 ± 0.006	0.341 ± 0.010	0.162	0.002	0.0550	0.0550	1.021	95.732 ± 0.006	0.218 ± 0.001	85.061 ± 0.011
	beta54	0.16	0.156	-0.004 ± 0.002	-0.027 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.141 ± 0.007	0.376 ± 0.011	0.154	-0.006	0.0568	0.0571	0.982	94.309 ± 0.007	0.231 ± 0.001	76.321 ± 0.014
	beta55	0.08	0.083	0.003 ± 0.002	0.033 ± 0.030	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.857 ± 0.040	0.926 ± 0.026	0.080	0.000	0.0748	0.0748	0.956	94.614 ± 0.007	0.277 ± 0.002	23.577 ± 0.014
	beta56	0.08	0.081	0.001 ± 0.002	0.013 ± 0.027	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.732 ± 0.034	0.856 ± 0.024	0.080	0.000	0.0669	0.0669	0.973	94.614 ± 0.007	0.261 ± 0.002	23.374 ± 0.013
	beta57	0.06	0.066	0.006 ± 0.002	0.108 ± 0.028	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.777 ± 0.043	0.882 ± 0.028	0.064	0.004	0.0494	0.0496	0.996	96.850 ± 0.006	0.205 ± 0.002	22.764 ± 0.013
	beta58	0.06	0.062	0.002 ± 0.001	0.041 ± 0.024	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.561 ± 0.029	0.749 ± 0.023	0.061	0.001	0.0428	0.0428	1.009	95.630 ± 0.007	0.178 ± 0.001	28.049 ± 0.014
QML	beta41	0.20	0.183	-0.017 ± 0.002	-0.084 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.072 ± 0.003	0.269 ± 0.007	0.183	-0.017	0.0542	0.0567	1.009	93.801 ± 0.008	0.202 ± 0.001	95.630 ± 0.007
	beta42	0.20	0.190	-0.010 ± 0.002	-0.048 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.069 ± 0.003	0.263 ± 0.007	0.188	-0.012	0.0518	0.0531	1.031	95.020 ± 0.007	0.209 ± 0.001	96.443 ± 0.006
	beta43	0.20	0.182	-0.018 ± 0.002	-0.091 ± 0.008	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.071 ± 0.003	0.267 ± 0.007	0.180	-0.020	0.0508	0.0545	1.008	92.988 ± 0.008	0.198 ± 0.001	95.935 ± 0.006
	beta44	0.11	0.104	-0.006 ± 0.002	-0.057 ± 0.021	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.427 ± 0.020	0.654 ± 0.019	0.104	-0.006	0.0727	0.0729	0.998	95.224 ± 0.007	0.280 ± 0.001	32.724 ± 0.015
	beta45	0.11	0.107	-0.003 ± 0.002	-0.031 ± 0.017	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.296 ± 0.014	0.544 ± 0.015	0.107	-0.003	0.0608	0.0608	0.991	94.919 ± 0.007	0.232 ± 0.001	44.512 ± 0.016
	beta46	0.08	0.078	-0.002 ± 0.001	-0.026 ± 0.018	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.328 ± 0.015	0.572 ± 0.016	0.079	-0.001	0.0437	0.0437	1.040	95.325 ± 0.007	0.187 ± 0.001	38.110 ± 0.015
	beta47	0.08	0.076	-0.004 ± 0.001	-0.044 ± 0.018	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.328 ± 0.015	0.572 ± 0.016	0.075	-0.005	0.0436	0.0439	1.013	95.833 ± 0.006	0.181 ± 0.001	36.077 ± 0.015
	beta51	0.16	0.173	0.013 ± 0.002	0.080 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.145 ± 0.007	0.381 ± 0.011	0.175	0.015	0.0571	0.0590	0.979	94.411 ± 0.007	0.229 ± 0.001	83.841 ± 0.012
	beta52	0.16	0.161	0.001 ± 0.002	0.007 ± 0.011	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.124 ± 0.006	0.352 ± 0.010	0.160	0.000	0.0566	0.0566	1.030	94.817 ± 0.007	0.228 ± 0.001	80.894 ± 0.013
	beta53	0.16	0.161	0.001 ± 0.002	0.003 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.106 ± 0.005	0.326 ± 0.009	0.160	0.000	0.0524	0.0524	0.990	94.512 ± 0.007	0.203 ± 0.001	88.821 ± 0.010
	beta54	0.16	0.174	0.014 ± 0.002	0.088 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.152 ± 0.007	0.390 ± 0.011	0.173	0.013	0.0598	0.0612	0.989	94.715 ± 0.007	0.236 ± 0.001	83.333 ± 0.012
	beta55	0.08	0.093	0.013 ± 0.002	0.162 ± 0.029	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.855 ± 0.037	0.925 ± 0.025	0.092	0.012	0.0747	0.0758	0.968	94.411 ± 0.007	0.276 ± 0.001	27.236 ± 0.014
	beta56	0.08	0.077	-0.003 ± 0.002	-0.033 ± 0.026	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.688 ± 0.032	0.830 ± 0.023	0.075	-0.005	0.0649	0.0651	0.955	94.207 ± 0.007	0.248 ± 0.001	23.577 ± 0.014
	beta57	0.06	0.049	-0.011 ± 0.001	-0.178 ± 0.024	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.606 ± 0.028	0.778 ± 0.022	0.049	-0.011	0.0439	0.0453	0.992	94.106 ± 0.008	0.177 ± 0.001	18.394 ± 0.012
	beta58	0.06	0.050	-0.010 ± 0.001	-0.172 ± 0.021	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.455 ± 0.021	0.675 ± 0.019	0.050	-0.010	0.0377	0.0392	0.989	93.699 ± 0.008	0.152 ± 0.001	24.390 ± 0.014

Table S40

Simulation 2 - Condition 2: Full Model, Distribution = normal, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta41	0.20	0.202	0.002 ± 0.002	0.010 ± 0.009	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.083 ± 0.004	0.289 ± 0.008	0.202	0.002	0.0575	0.0575	0.967	94.309 ± 0.007	0.219 ± 0.001	95.732 ± 0.006
	beta42	0.20	0.204	0.004 ± 0.002	0.020 ± 0.009	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.085 ± 0.004	0.292 ± 0.009	0.200	0.000	0.0577	0.0577	0.993	94.411 ± 0.007	0.227 ± 0.001	95.630 ± 0.007
	beta43	0.20	0.201	0.001 ± 0.002	0.005 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.074 ± 0.004	0.273 ± 0.008	0.199	-0.001	0.0515	0.0515	0.986	94.614 ± 0.007	0.211 ± 0.001	96.545 ± 0.006
	beta44	0.11	0.120	0.010 ± 0.003	0.095 ± 0.027	0.009 ± 0.000	0.093 ± 0.003	0.009 ± 0.000	0.093 ± 0.003	0.716 ± 0.041	0.846 ± 0.028	0.120	0.010	0.0811	0.0817	0.855	92.886 ± 0.008	0.310 ± 0.003	39.736 ± 0.016
	beta45	0.11	0.115	0.005 ± 0.002	0.043 ± 0.021	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.450 ± 0.023	0.671 ± 0.020	0.111	0.001	0.0694	0.0694	0.888	93.293 ± 0.008	0.256 ± 0.002	44.715 ± 0.016
	beta46	0.08	0.081	0.001 ± 0.002	0.015 ± 0.026	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.655 ± 0.038	0.809 ± 0.027	0.079	-0.001	0.0606	0.0606	0.856	92.886 ± 0.008	0.217 ± 0.002	33.333 ± 0.015
	beta47	0.08	0.084	0.004 ± 0.002	0.048 ± 0.024	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.059 ± 0.002	0.552 ± 0.030	0.743 ± 0.023	0.081	0.001	0.0549	0.0549	0.900	93.801 ± 0.008	0.209 ± 0.002	36.077 ± 0.015
	beta51	0.16	0.188	0.028 ± 0.002	0.173 ± 0.011	0.003 ± 0.000	0.056 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.152 ± 0.007	0.389 ± 0.010	0.189	0.029	0.0550	0.0624	0.975	93.191 ± 0.008	0.214 ± 0.001	92.886 ± 0.008
	beta52	0.16	0.188	0.028 ± 0.002	0.172 ± 0.012	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.160 ± 0.007	0.400 ± 0.011	0.186	0.026	0.0575	0.0632	1.011	93.394 ± 0.008	0.229 ± 0.001	91.565 ± 0.009
	beta53	0.16	0.162	0.002 ± 0.002	0.010 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.116 ± 0.006	0.341 ± 0.010	0.161	0.001	0.0542	0.0543	0.987	94.817 ± 0.007	0.211 ± 0.001	86.382 ± 0.011
	beta54	0.16	0.158	-0.002 ± 0.002	-0.014 ± 0.012	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.137 ± 0.007	0.370 ± 0.011	0.155	-0.005	0.0563	0.0565	0.976	94.614 ± 0.007	0.227 ± 0.001	79.472 ± 0.013
	beta55	0.08	0.084	0.004 ± 0.002	0.048 ± 0.030	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.902 ± 0.045	0.950 ± 0.028	0.079	-0.001	0.0744	0.0744	0.903	94.817 ± 0.007	0.269 ± 0.002	26.829 ± 0.014
	beta56	0.08	0.080	0.000 ± 0.002	0.004 ± 0.028	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.777 ± 0.040	0.881 ± 0.027	0.076	-0.004	0.0655	0.0656	0.916	93.496 ± 0.008	0.253 ± 0.002	24.797 ± 0.014
	beta57	0.06	0.066	0.006 ± 0.002	0.098 ± 0.030	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	0.872 ± 0.049	0.934 ± 0.030	0.061	0.001	0.0543	0.0543	0.892	93.496 ± 0.008	0.195 ± 0.002	27.846 ± 0.014
	beta58	0.06	0.063	0.003 ± 0.002	0.052 ± 0.026	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.002	0.688 ± 0.041	0.830 ± 0.028	0.060	0.000	0.0471	0.0471	0.887	93.699 ± 0.008	0.173 ± 0.002	30.589 ± 0.015

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S41

Simulation 2 - Condition 3: Full Model, Distribution = normal, N = 400, Reliability = 0.6

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.002	0.003 ± 0.011	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.116 ± 0.005	0.341 ± 0.009	0.197	-0.003	0.0674	0.0675	0.976	94.141 ± 0.007	0.261 ± 0.001	86.263 ± 0.011
	beta42	0.20	0.201	0.001 ± 0.002	0.006 ± 0.011	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.116 ± 0.006	0.341 ± 0.010	0.200	0.000	0.0697	0.0697	0.998	95.859 ± 0.006	0.267 ± 0.001	84.848 ± 0.011
	beta43	0.20	0.200	0.000 ± 0.002	0.002 ± 0.010	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.098 ± 0.005	0.314 ± 0.009	0.201	0.001	0.0636	0.0636	1.014	95.455 ± 0.007	0.249 ± 0.001	90.000 ± 0.010
	beta44	0.11	0.111	0.001 ± 0.003	0.011 ± 0.024	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.565 ± 0.027	0.752 ± 0.022	0.106	-0.004	0.0766	0.0767	0.974	93.636 ± 0.008	0.316 ± 0.002	29.798 ± 0.015
	beta45	0.11	0.113	0.003 ± 0.002	0.031 ± 0.020	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.407 ± 0.018	0.638 ± 0.018	0.116	0.006	0.0696	0.0698	0.986	94.848 ± 0.007	0.271 ± 0.002	39.293 ± 0.016
	beta46	0.08	0.082	0.002 ± 0.002	0.027 ± 0.024	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.553 ± 0.027	0.744 ± 0.022	0.080	0.000	0.0599	0.0599	0.956	93.939 ± 0.008	0.223 ± 0.002	32.727 ± 0.015
	beta47	0.08	0.083	0.003 ± 0.002	0.036 ± 0.023	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.534 ± 0.027	0.731 ± 0.022	0.078	-0.002	0.0560	0.0560	0.960	93.232 ± 0.008	0.220 ± 0.002	32.626 ± 0.015
	beta51	0.16	0.187	0.027 ± 0.002	0.167 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.005 ± 0.000	0.069 ± 0.002	0.188 ± 0.009	0.433 ± 0.012	0.182	0.022	0.0604	0.0643	1.018	93.434 ± 0.008	0.255 ± 0.001	81.818 ± 0.012
	beta52	0.16	0.185	0.025 ± 0.002	0.155 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.210 ± 0.009	0.458 ± 0.012	0.187	0.027	0.0682	0.0735	0.993	94.343 ± 0.007	0.269 ± 0.001	78.384 ± 0.013
	beta53	0.16	0.157	-0.003 ± 0.002	-0.017 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.161 ± 0.007	0.402 ± 0.011	0.157	-0.003	0.0666	0.0666	0.995	95.354 ± 0.007	0.251 ± 0.001	70.303 ± 0.015
	beta54	0.16	0.162	0.002 ± 0.002	0.012 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.185 ± 0.008	0.430 ± 0.012	0.160	0.000	0.0671	0.0671	0.993	94.747 ± 0.007	0.268 ± 0.001	65.758 ± 0.015
	beta55	0.08	0.086	0.006 ± 0.002	0.080 ± 0.030	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.895 ± 0.044	0.946 ± 0.028	0.088	0.008	0.0713	0.0718	0.983	94.747 ± 0.007	0.291 ± 0.002	22.121 ± 0.013
	beta56	0.08	0.082	0.002 ± 0.002	0.030 ± 0.030	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.890 ± 0.048	0.943 ± 0.029	0.079	-0.001	0.0718	0.0718	0.919	94.242 ± 0.007	0.272 ± 0.002	24.242 ± 0.014
	beta57	0.06	0.058	-0.002 ± 0.002	-0.040 ± 0.030	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.869 ± 0.040	0.932 ± 0.026	0.055	-0.005	0.0549	0.0551	0.964	94.343 ± 0.007	0.211 ± 0.002	20.404 ± 0.013
	beta58	0.06	0.062	0.002 ± 0.002	0.032 ± 0.026	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.671 ± 0.033	0.819 ± 0.024	0.061	0.001	0.0467	0.0467	0.970	95.859 ± 0.006	0.187 ± 0.001	26.465 ± 0.014
QML	beta41	0.20	0.191	-0.009 ± 0.002	-0.044 ± 0.010	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.104 ± 0.005	0.322 ± 0.009	0.188	-0.012	0.0639	0.0650	0.987	94.343 ± 0.007	0.247 ± 0.001	86.162 ± 0.011
	beta42	0.20	0.196	-0.004 ± 0.002	-0.022 ± 0.010	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.107 ± 0.005	0.327 ± 0.009	0.194	-0.006	0.0659	0.0662	0.990	95.152 ± 0.007	0.253 ± 0.001	85.758 ± 0.011
	beta43	0.20	0.189	-0.011 ± 0.002	-0.053 ± 0.010	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.095 ± 0.004	0.308 ± 0.009	0.190	-0.010	0.0621	0.0628	1.004	94.949 ± 0.007	0.239 ± 0.001	88.889 ± 0.010
	beta44	0.11	0.102	-0.008 ± 0.002	-0.073 ± 0.022	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.503 ± 0.024	0.709 ± 0.020	0.098	-0.012	0.0723	0.0732	0.991	94.040 ± 0.008	0.302 ± 0.001	25.455 ± 0.014
	beta45	0.11	0.111	0.001 ± 0.002	0.006 ± 0.019	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.361 ± 0.016	0.601 ± 0.017	0.113	0.003	0.0653	0.0654	0.998	95.960 ± 0.006	0.259 ± 0.001	39.798 ± 0.016
	beta46	0.08	0.078	-0.002 ± 0.002	-0.028 ± 0.021	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.435 ± 0.020	0.660 ± 0.018	0.077	-0.003	0.0530	0.0531	0.993	94.848 ± 0.007	0.205 ± 0.001	32.828 ± 0.015
	beta47	0.08	0.079	-0.001 ± 0.002	-0.016 ± 0.021	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.429 ± 0.021	0.655 ± 0.019	0.076	-0.004	0.0502	0.0504	0.978	94.040 ± 0.008	0.201 ± 0.001	33.535 ± 0.015
	beta51	0.16	0.169	0.009 ± 0.002	0.059 ± 0.013	0.004 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.178 ± 0.008	0.422 ± 0.012	0.168	0.008	0.0649	0.0654	1.029	96.162 ± 0.006	0.270 ± 0.001	70.606 ± 0.014
	beta52	0.16	0.162	0.002 ± 0.002	0.010 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.196 ± 0.009	0.443 ± 0.013	0.164	0.004	0.0693	0.0694	0.985	94.444 ± 0.007	0.274 ± 0.001	65.253 ± 0.015
	beta53	0.16	0.157	-0.003 ± 0.002	-0.021 ± 0.013	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.156 ± 0.007	0.395 ± 0.011	0.156	-0.004	0.0643	0.0645	0.988	95.354 ± 0.007	0.244 ± 0.001	70.808 ± 0.014
	beta54	0.16	0.174	0.014 ± 0.002	0.086 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.202 ± 0.009	0.450 ± 0.012	0.172	0.012	0.0686	0.0696	1.002	94.444 ± 0.007	0.278 ± 0.001	69.091 ± 0.015
	beta55	0.08	0.092	0.012 ± 0.002	0.144 ± 0.030	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.889 ± 0.041	0.943 ± 0.026	0.091	0.011	0.0705	0.0713	1.013	94.646 ± 0.007	0.296 ± 0.001	21.111 ± 0.013
	beta56	0.08	0.082	0.002 ± 0.002	0.025 ± 0.029	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.848 ± 0.041	0.921 ± 0.027	0.081	0.001	0.0723	0.0723	0.947	93.939 ± 0.008	0.273 ± 0.001	21.515 ± 0.013
	beta57	0.06	0.049	-0.011 ± 0.002	-0.190 ± 0.027	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.767 ± 0.035	0.876 ± 0.024	0.049	-0.011	0.0501	0.0512	0.983	93.939 ± 0.008	0.198 ± 0.001	16.869 ± 0.012
	beta58	0.06	0.055	-0.005 ± 0.001	-0.090 ± 0.024	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.569 ± 0.028	0.755 ± 0.022	0.054	-0.006	0.0439	0.0443	0.987	95.556 ± 0.007	0.174 ± 0.001	24.343 ± 0.014

Table S41

Simulation 2 - Condition 3: Full Model, Distribution = normal, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta41	0.20	0.201	0.001 ± 0.002	0.006 ± 0.011	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.116 ± 0.005	0.341 ± 0.009	0.198	-0.002	0.0686	0.0686	0.941	93.434 ± 0.008	0.252 ± 0.001	86.869 ± 0.011
	beta42	0.20	0.201	0.001 ± 0.002	0.007 ± 0.011	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.120 ± 0.006	0.346 ± 0.010	0.200	0.000	0.0707	0.0707	0.960	94.545 ± 0.007	0.261 ± 0.001	85.657 ± 0.011
	beta43	0.20	0.201	0.001 ± 0.002	0.003 ± 0.010	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.100 ± 0.005	0.317 ± 0.009	0.201	0.001	0.0664	0.0664	0.985	94.646 ± 0.007	0.245 ± 0.001	90.101 ± 0.009
	beta44	0.11	0.113	0.003 ± 0.003	0.029 ± 0.025	0.007 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.003	0.620 ± 0.031	0.787 ± 0.023	0.109	-0.001	0.0815	0.0815	0.921	92.929 ± 0.008	0.313 ± 0.002	31.313 ± 0.015
	beta45	0.11	0.114	0.004 ± 0.002	0.034 ± 0.021	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.435 ± 0.020	0.660 ± 0.019	0.114	0.004	0.0703	0.0704	0.938	94.141 ± 0.007	0.267 ± 0.002	42.020 ± 0.016
	beta46	0.08	0.082	0.002 ± 0.002	0.021 ± 0.026	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.653 ± 0.032	0.808 ± 0.024	0.078	-0.002	0.0632	0.0632	0.861	91.212 ± 0.009	0.218 ± 0.002	33.737 ± 0.015
	beta47	0.08	0.084	0.004 ± 0.002	0.051 ± 0.025	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.614 ± 0.036	0.784 ± 0.026	0.079	-0.001	0.0564	0.0564	0.881	92.222 ± 0.009	0.216 ± 0.002	33.232 ± 0.015
	beta51	0.16	0.188	0.028 ± 0.002	0.177 ± 0.013	0.004 ± 0.000	0.065 ± 0.001	0.005 ± 0.000	0.071 ± 0.002	0.194 ± 0.010	0.441 ± 0.012	0.184	0.024	0.0600	0.0645	0.993	92.121 ± 0.009	0.251 ± 0.001	84.141 ± 0.012
	beta52	0.16	0.185	0.025 ± 0.002	0.158 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.214 ± 0.010	0.463 ± 0.012	0.187	0.027	0.0689	0.0741	0.977	93.737 ± 0.008	0.267 ± 0.001	79.293 ± 0.013
	beta53	0.16	0.157	-0.003 ± 0.002	-0.016 ± 0.013	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.164 ± 0.007	0.405 ± 0.011	0.157	-0.003	0.0651	0.0652	0.974	94.242 ± 0.007	0.248 ± 0.001	70.505 ± 0.014
	beta54	0.16	0.164	0.004 ± 0.002	0.023 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.185 ± 0.008	0.430 ± 0.012	0.162	0.002	0.0663	0.0664	0.988	94.141 ± 0.007	0.266 ± 0.001	66.970 ± 0.015
	beta55	0.08	0.087	0.007 ± 0.002	0.082 ± 0.031	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.934 ± 0.048	0.966 ± 0.029	0.086	0.006	0.0703	0.0706	0.954	93.535 ± 0.008	0.288 ± 0.002	21.717 ± 0.013
	beta56	0.08	0.081	0.001 ± 0.002	0.018 ± 0.031	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.932 ± 0.051	0.966 ± 0.031	0.080	0.000	0.0740	0.0740	0.891	93.535 ± 0.008	0.270 ± 0.002	23.838 ± 0.014
	beta57	0.06	0.057	-0.003 ± 0.002	-0.042 ± 0.031	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	0.971 ± 0.052	0.985 ± 0.031	0.057	-0.003	0.0548	0.0549	0.891	92.626 ± 0.008	0.206 ± 0.002	21.919 ± 0.013
	beta58	0.06	0.063	0.003 ± 0.002	0.050 ± 0.028	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.755 ± 0.037	0.869 ± 0.025	0.061	0.001	0.0516	0.0516	0.905	94.141 ± 0.007	0.185 ± 0.001	28.485 ± 0.014

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S42

Simulation 2 - Condition 4: Full Model, Distribution = normal, N = 1000, Reliability = 0.6

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.203	0.003 ± 0.001	0.013 ± 0.007	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.043 ± 0.002	0.208 ± 0.006	0.202	0.002	0.0407	0.0407	0.991	94.985 ± 0.007	0.161 ± 0.000	100.000 ± 0.000
	beta42	0.20	0.197	-0.003 ± 0.001	-0.013 ± 0.007	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.044 ± 0.002	0.209 ± 0.006	0.198	-0.002	0.0414	0.0415	1.007	94.885 ± 0.007	0.165 ± 0.000	100.000 ± 0.000
	beta43	0.20	0.202	0.002 ± 0.001	0.008 ± 0.007	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.043 ± 0.002	0.206 ± 0.006	0.202	0.002	0.0393	0.0393	0.966	93.982 ± 0.008	0.156 ± 0.000	99.799 ± 0.001
	beta44	0.11	0.112	0.002 ± 0.002	0.014 ± 0.014	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.202 ± 0.009	0.449 ± 0.012	0.112	0.002	0.0510	0.0510	0.985	95.687 ± 0.006	0.191 ± 0.001	63.190 ± 0.015
	beta45	0.11	0.112	0.002 ± 0.001	0.020 ± 0.013	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.158 ± 0.007	0.397 ± 0.011	0.113	0.003	0.0403	0.0404	0.976	94.584 ± 0.007	0.167 ± 0.001	75.226 ± 0.014
	beta46	0.08	0.081	0.001 ± 0.001	0.014 ± 0.014	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.194 ± 0.009	0.441 ± 0.012	0.080	0.000	0.0354	0.0354	0.973	95.186 ± 0.007	0.135 ± 0.001	66.299 ± 0.015
	beta47	0.08	0.081	0.001 ± 0.001	0.014 ± 0.014	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.183 ± 0.009	0.428 ± 0.012	0.081	0.001	0.0347	0.0347	0.987	94.183 ± 0.007	0.133 ± 0.001	69.007 ± 0.015
	beta51	0.16	0.185	0.025 ± 0.001	0.154 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.089 ± 0.004	0.299 ± 0.008	0.184	0.024	0.0402	0.0466	0.980	90.572 ± 0.009	0.158 ± 0.000	99.900 ± 0.001
	beta52	0.16	0.187	0.027 ± 0.001	0.167 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.096 ± 0.004	0.311 ± 0.008	0.186	0.026	0.0424	0.0496	1.011	91.374 ± 0.009	0.166 ± 0.000	99.498 ± 0.002
	beta53	0.16	0.161	0.001 ± 0.001	0.006 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.056 ± 0.002	0.237 ± 0.006	0.161	0.001	0.0378	0.0378	1.052	96.088 ± 0.006	0.157 ± 0.000	98.596 ± 0.004
	beta54	0.16	0.160	-0.000 ± 0.001	-0.001 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.074 ± 0.003	0.272 ± 0.008	0.159	-0.001	0.0433	0.0433	0.979	94.584 ± 0.007	0.167 ± 0.000	95.787 ± 0.006
	beta55	0.08	0.082	0.002 ± 0.001	0.031 ± 0.018	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.333 ± 0.016	0.577 ± 0.016	0.082	0.002	0.0450	0.0450	0.971	95.186 ± 0.007	0.176 ± 0.001	45.938 ± 0.016
	beta56	0.08	0.079	-0.001 ± 0.001	-0.018 ± 0.017	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.302 ± 0.013	0.550 ± 0.015	0.077	-0.003	0.0454	0.0455	0.973	94.283 ± 0.007	0.168 ± 0.001	44.935 ± 0.016
	beta57	0.06	0.059	-0.001 ± 0.001	-0.013 ± 0.018	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.313 ± 0.015	0.559 ± 0.016	0.059	-0.001	0.0329	0.0329	0.968	94.383 ± 0.007	0.127 ± 0.001	46.239 ± 0.016
	beta58	0.06	0.062	0.002 ± 0.001	0.035 ± 0.016	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.243 ± 0.012	0.493 ± 0.014	0.062	0.002	0.0284	0.0285	0.991	94.784 ± 0.007	0.115 ± 0.000	59.679 ± 0.016
QML	beta41	0.20	0.193	-0.007 ± 0.001	-0.033 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.041 ± 0.002	0.203 ± 0.006	0.192	-0.008	0.0388	0.0395	0.985	94.383 ± 0.007	0.155 ± 0.000	100.000 ± 0.000
	beta42	0.20	0.192	-0.008 ± 0.001	-0.041 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.042 ± 0.002	0.206 ± 0.006	0.191	-0.009	0.0408	0.0417	1.004	94.082 ± 0.007	0.159 ± 0.000	99.900 ± 0.001
	beta43	0.20	0.191	-0.009 ± 0.001	-0.047 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.043 ± 0.002	0.207 ± 0.006	0.190	-0.010	0.0384	0.0397	0.958	92.979 ± 0.008	0.151 ± 0.000	99.699 ± 0.002
	beta44	0.11	0.105	-0.005 ± 0.002	-0.044 ± 0.014	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.197 ± 0.009	0.444 ± 0.012	0.106	-0.004	0.0488	0.0490	0.984	94.684 ± 0.007	0.188 ± 0.001	59.278 ± 0.016
	beta45	0.11	0.110	0.000 ± 0.001	0.004 ± 0.012	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.146 ± 0.007	0.382 ± 0.011	0.110	0.000	0.0399	0.0399	0.988	95.085 ± 0.007	0.163 ± 0.000	76.630 ± 0.013
	beta46	0.08	0.079	-0.001 ± 0.001	-0.012 ± 0.013	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.167 ± 0.007	0.409 ± 0.011	0.078	-0.002	0.0342	0.0342	0.993	95.486 ± 0.007	0.127 ± 0.000	68.205 ± 0.015
	beta47	0.08	0.078	-0.002 ± 0.001	-0.023 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.161 ± 0.007	0.401 ± 0.011	0.080	0.000	0.0314	0.0314	0.995	94.684 ± 0.007	0.125 ± 0.000	69.509 ± 0.015
	beta51	0.16	0.168	0.008 ± 0.001	0.047 ± 0.009	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.075 ± 0.003	0.274 ± 0.008	0.168	0.008	0.0429	0.0436	0.989	94.684 ± 0.007	0.167 ± 0.000	97.593 ± 0.005
	beta52	0.16	0.163	0.003 ± 0.001	0.021 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.070 ± 0.003	0.264 ± 0.007	0.164	0.004	0.0435	0.0436	1.027	95.888 ± 0.006	0.170 ± 0.000	96.389 ± 0.006
	beta53	0.16	0.160	0.000 ± 0.001	0.003 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.055 ± 0.002	0.234 ± 0.006	0.160	0.000	0.0375	0.0375	1.046	95.888 ± 0.006	0.154 ± 0.000	98.596 ± 0.004
	beta54	0.16	0.170	0.010 ± 0.001	0.062 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.081 ± 0.004	0.285 ± 0.008	0.170	0.010	0.0444	0.0456	0.984	94.584 ± 0.007	0.172 ± 0.000	96.690 ± 0.006
	beta55	0.08	0.090	0.010 ± 0.002	0.122 ± 0.019	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.366 ± 0.017	0.605 ± 0.017	0.089	0.009	0.0474	0.0483	0.977	94.483 ± 0.007	0.181 ± 0.001	50.853 ± 0.016
	beta56	0.08	0.078	-0.002 ± 0.001	-0.020 ± 0.018	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.308 ± 0.013	0.555 ± 0.014	0.077	-0.003	0.0476	0.0477	0.974	94.784 ± 0.007	0.169 ± 0.000	44.835 ± 0.016
	beta57	0.06	0.051	-0.009 ± 0.001	-0.156 ± 0.017	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.312 ± 0.014	0.558 ± 0.015	0.050	-0.010	0.0315	0.0330	0.965	93.079 ± 0.008	0.122 ± 0.000	36.008 ± 0.015
	beta58	0.06	0.056	-0.004 ± 0.001	-0.070 ± 0.015	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.220 ± 0.011	0.469 ± 0.014	0.056	-0.004	0.0272	0.0274	0.999	95.286 ± 0.007	0.109 ± 0.000	53.862 ± 0.016

Table S42

Simulation 2 - Condition 4: Full Model, Distribution = normal, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta41	0.20	0.203	0.003 ± 0.001	0.014 ± 0.007	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.044 ± 0.002	0.209 ± 0.006	0.201	0.001	0.0401	0.0402	0.955	93.681 ± 0.008	0.157 ± 0.000	100.000 ± 0.000
	beta42	0.20	0.198	-0.002 ± 0.001	-0.012 ± 0.007	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.044 ± 0.002	0.209 ± 0.006	0.197	-0.003	0.0428	0.0429	0.987	95.085 ± 0.007	0.162 ± 0.000	100.000 ± 0.000
	beta43	0.20	0.202	0.002 ± 0.001	0.008 ± 0.007	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.043 ± 0.002	0.206 ± 0.006	0.201	0.001	0.0403	0.0403	0.952	93.982 ± 0.008	0.154 ± 0.000	99.799 ± 0.001
	beta44	0.11	0.112	0.002 ± 0.002	0.015 ± 0.015	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.215 ± 0.010	0.464 ± 0.013	0.111	0.001	0.0511	0.0511	0.943	94.283 ± 0.007	0.189 ± 0.001	63.892 ± 0.015
	beta45	0.11	0.113	0.003 ± 0.001	0.024 ± 0.013	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.169 ± 0.008	0.411 ± 0.012	0.113	0.003	0.0419	0.0421	0.933	93.581 ± 0.008	0.165 ± 0.001	75.827 ± 0.014
	beta46	0.08	0.081	0.001 ± 0.001	0.008 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.206 ± 0.010	0.454 ± 0.013	0.080	0.000	0.0374	0.0374	0.930	94.283 ± 0.007	0.132 ± 0.001	67.101 ± 0.015
	beta47	0.08	0.082	0.002 ± 0.001	0.019 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.200 ± 0.009	0.448 ± 0.013	0.082	0.002	0.0370	0.0370	0.933	93.581 ± 0.008	0.131 ± 0.001	69.709 ± 0.015
	beta51	0.16	0.185	0.025 ± 0.001	0.157 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.090 ± 0.004	0.300 ± 0.008	0.184	0.024	0.0403	0.0471	0.972	89.569 ± 0.010	0.156 ± 0.000	99.799 ± 0.001
	beta52	0.16	0.187	0.027 ± 0.001	0.167 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.096 ± 0.004	0.310 ± 0.008	0.187	0.027	0.0420	0.0498	1.013	91.474 ± 0.009	0.166 ± 0.000	99.498 ± 0.002
	beta53	0.16	0.161	0.001 ± 0.001	0.007 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.056 ± 0.002	0.238 ± 0.006	0.161	0.001	0.0382	0.0383	1.039	95.888 ± 0.006	0.155 ± 0.000	98.696 ± 0.004
	beta54	0.16	0.161	0.001 ± 0.001	0.004 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.074 ± 0.003	0.273 ± 0.008	0.160	0.000	0.0437	0.0437	0.971	94.684 ± 0.007	0.166 ± 0.000	96.289 ± 0.006
	beta55	0.08	0.082	0.002 ± 0.001	0.026 ± 0.019	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.343 ± 0.017	0.586 ± 0.017	0.080	0.000	0.0441	0.0441	0.950	94.383 ± 0.007	0.174 ± 0.001	46.540 ± 0.016
	beta56	0.08	0.078	-0.002 ± 0.001	-0.026 ± 0.018	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.311 ± 0.014	0.557 ± 0.015	0.077	-0.003	0.0440	0.0440	0.946	93.581 ± 0.008	0.165 ± 0.001	45.336 ± 0.016
	beta57	0.06	0.059	-0.001 ± 0.001	-0.015 ± 0.018	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.327 ± 0.016	0.572 ± 0.017	0.057	-0.003	0.0332	0.0333	0.933	92.979 ± 0.008	0.126 ± 0.001	46.239 ± 0.016
	beta58	0.06	0.062	0.002 ± 0.001	0.038 ± 0.016	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.252 ± 0.013	0.502 ± 0.015	0.062	0.002	0.0296	0.0297	0.965	94.684 ± 0.007	0.114 ± 0.001	60.582 ± 0.015

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S43

Simulation 2 - Condition 5: Full Model, Distribution = normal, N = 400, Reliability = 0.8

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.199	-0.001 ± 0.002	-0.006 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.075 ± 0.003	0.274 ± 0.007	0.198	-0.002	0.0559	0.0559	0.966	94.500 ± 0.007	0.208 ± 0.001	95.800 ± 0.006
	beta42	0.20	0.202	0.002 ± 0.002	0.008 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.079 ± 0.003	0.282 ± 0.008	0.202	0.002	0.0573	0.0574	0.971	94.200 ± 0.007	0.214 ± 0.001	95.100 ± 0.007
	beta43	0.20	0.201	0.001 ± 0.002	0.003 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.072 ± 0.003	0.269 ± 0.007	0.202	0.002	0.0546	0.0546	0.969	94.200 ± 0.007	0.204 ± 0.001	96.600 ± 0.006
	beta44	0.11	0.114	0.004 ± 0.002	0.038 ± 0.017	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.307 ± 0.014	0.554 ± 0.015	0.115	0.005	0.0604	0.0606	0.967	94.200 ± 0.007	0.231 ± 0.001	50.500 ± 0.016
	beta45	0.11	0.110	-0.000 ± 0.002	-0.004 ± 0.016	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.259 ± 0.012	0.509 ± 0.014	0.110	0.000	0.0556	0.0556	0.931	93.400 ± 0.008	0.204 ± 0.001	56.900 ± 0.016
	beta46	0.08	0.080	0.000 ± 0.001	0.005 ± 0.017	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.280 ± 0.013	0.529 ± 0.015	0.080	0.000	0.0423	0.0423	0.972	95.000 ± 0.007	0.161 ± 0.001	53.300 ± 0.016
	beta47	0.08	0.080	0.000 ± 0.001	0.000 ± 0.017	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.304 ± 0.015	0.551 ± 0.016	0.079	-0.001	0.0448	0.0449	0.926	93.900 ± 0.008	0.160 ± 0.001	50.800 ± 0.016
	beta51	0.16	0.185	0.025 ± 0.002	0.156 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.057 ± 0.001	0.127 ± 0.005	0.357 ± 0.009	0.184	0.024	0.0518	0.0571	1.016	93.400 ± 0.008	0.205 ± 0.001	94.100 ± 0.007
	beta52	0.16	0.183	0.023 ± 0.002	0.146 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.140 ± 0.007	0.375 ± 0.010	0.182	0.022	0.0542	0.0584	0.998	92.900 ± 0.008	0.216 ± 0.001	91.700 ± 0.009
	beta53	0.16	0.160	0.000 ± 0.002	0.003 ± 0.010	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.110 ± 0.005	0.331 ± 0.009	0.160	0.000	0.0530	0.0530	0.986	93.700 ± 0.008	0.205 ± 0.001	86.500 ± 0.011
	beta54	0.16	0.160	0.000 ± 0.002	0.003 ± 0.011	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.124 ± 0.006	0.352 ± 0.010	0.159	-0.001	0.0566	0.0566	0.977	94.000 ± 0.008	0.216 ± 0.001	81.900 ± 0.012
	beta55	0.08	0.081	0.001 ± 0.002	0.010 ± 0.023	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.543 ± 0.025	0.737 ± 0.021	0.079	-0.001	0.0578	0.0578	0.925	93.000 ± 0.008	0.214 ± 0.001	33.600 ± 0.015
	beta56	0.08	0.082	0.002 ± 0.002	0.021 ± 0.021	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.458 ± 0.022	0.677 ± 0.019	0.084	0.004	0.0521	0.0523	0.966	94.300 ± 0.007	0.205 ± 0.001	36.300 ± 0.015
	beta57	0.06	0.061	0.001 ± 0.001	0.024 ± 0.022	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.483 ± 0.023	0.695 ± 0.020	0.061	0.001	0.0401	0.0401	0.940	92.800 ± 0.008	0.154 ± 0.001	35.200 ± 0.015
	beta58	0.06	0.060	0.000 ± 0.001	0.008 ± 0.020	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.403 ± 0.018	0.635 ± 0.017	0.061	0.001	0.0375	0.0375	0.946	93.100 ± 0.008	0.141 ± 0.001	40.900 ± 0.016
QML	beta41	0.20	0.195	-0.005 ± 0.002	-0.026 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.073 ± 0.003	0.270 ± 0.007	0.195	-0.005	0.0534	0.0536	0.990	94.500 ± 0.007	0.209 ± 0.000	95.600 ± 0.006
	beta42	0.20	0.199	-0.001 ± 0.002	-0.004 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.001	0.077 ± 0.003	0.278 ± 0.007	0.199	-0.001	0.0565	0.0565	0.976	94.800 ± 0.007	0.212 ± 0.000	95.100 ± 0.007
	beta43	0.20	0.196	-0.004 ± 0.002	-0.019 ± 0.008	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.072 ± 0.003	0.268 ± 0.007	0.197	-0.003	0.0548	0.0548	0.962	93.700 ± 0.008	0.201 ± 0.000	96.300 ± 0.006
	beta44	0.11	0.111	0.001 ± 0.002	0.010 ± 0.017	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.295 ± 0.013	0.543 ± 0.015	0.111	0.001	0.0584	0.0584	0.990	95.400 ± 0.007	0.232 ± 0.001	48.100 ± 0.016
	beta45	0.11	0.109	-0.001 ± 0.002	-0.011 ± 0.016	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.248 ± 0.011	0.498 ± 0.014	0.110	0.000	0.0525	0.0525	0.957	94.200 ± 0.007	0.205 ± 0.001	55.500 ± 0.016
	beta46	0.08	0.079	-0.001 ± 0.001	-0.011 ± 0.016	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.260 ± 0.012	0.510 ± 0.014	0.080	0.000	0.0418	0.0418	1.004	95.600 ± 0.006	0.161 ± 0.001	51.500 ± 0.016
	beta47	0.08	0.079	-0.001 ± 0.001	-0.017 ± 0.017	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.280 ± 0.013	0.529 ± 0.015	0.078	-0.002	0.0433	0.0433	0.954	95.100 ± 0.007	0.158 ± 0.001	50.600 ± 0.016
	beta51	0.16	0.168	0.008 ± 0.002	0.049 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.119 ± 0.005	0.345 ± 0.010	0.167	0.007	0.0530	0.0536	1.025	95.200 ± 0.007	0.220 ± 0.001	85.300 ± 0.011
	beta52	0.16	0.163	0.003 ± 0.002	0.016 ± 0.011	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.128 ± 0.006	0.357 ± 0.010	0.160	0.000	0.0533	0.0533	1.002	94.800 ± 0.007	0.224 ± 0.000	82.100 ± 0.012
	beta53	0.16	0.160	0.000 ± 0.002	0.001 ± 0.010	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.108 ± 0.005	0.328 ± 0.009	0.159	-0.001	0.0529	0.0529	0.993	94.200 ± 0.007	0.205 ± 0.000	86.500 ± 0.011
	beta54	0.16	0.165	0.005 ± 0.002	0.033 ± 0.011	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.128 ± 0.006	0.357 ± 0.010	0.165	0.005	0.0571	0.0573	0.994	94.400 ± 0.007	0.222 ± 0.000	82.400 ± 0.012
	beta55	0.08	0.084	0.004 ± 0.002	0.044 ± 0.024	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.558 ± 0.026	0.747 ± 0.021	0.082	0.002	0.0586	0.0586	0.946	93.900 ± 0.008	0.221 ± 0.001	34.400 ± 0.015
	beta56	0.08	0.082	0.002 ± 0.002	0.025 ± 0.021	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.460 ± 0.022	0.678 ± 0.019	0.084	0.004	0.0514	0.0515	0.991	95.000 ± 0.007	0.211 ± 0.001	35.300 ± 0.015
	beta57	0.06	0.057	-0.003 ± 0.001	-0.042 ± 0.021	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.454 ± 0.021	0.674 ± 0.019	0.057	-0.003	0.0391	0.0392	0.971	93.500 ± 0.008	0.154 ± 0.001	31.600 ± 0.015
	beta58	0.06	0.058	-0.002 ± 0.001	-0.036 ± 0.020	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.387 ± 0.017	0.622 ± 0.017	0.058	-0.002	0.0377	0.0377	0.960	94.400 ± 0.007	0.140 ± 0.001	37.500 ± 0.015

Table S43

Simulation 2 - Condition 5: Full Model, Distribution = normal, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta41	0.20	0.199	-0.001 ± 0.002	-0.004 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.076 ± 0.003	0.275 ± 0.007	0.199	-0.001	0.0559	0.0560	0.946	93.600 ± 0.008	0.204 ± 0.000	96.000 ± 0.006
	beta42	0.20	0.202	0.002 ± 0.002	0.008 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.081 ± 0.003	0.284 ± 0.008	0.201	0.001	0.0569	0.0569	0.945	93.200 ± 0.008	0.210 ± 0.000	95.300 ± 0.007
	beta43	0.20	0.201	0.001 ± 0.002	0.003 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.073 ± 0.003	0.270 ± 0.007	0.202	0.002	0.0544	0.0545	0.953	93.500 ± 0.008	0.202 ± 0.000	96.600 ± 0.006
	beta44	0.11	0.115	0.005 ± 0.002	0.041 ± 0.018	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.322 ± 0.015	0.567 ± 0.016	0.113	0.003	0.0609	0.0610	0.953	94.500 ± 0.007	0.233 ± 0.001	49.100 ± 0.016
	beta45	0.11	0.110	0.000 ± 0.002	0.002 ± 0.016	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.262 ± 0.012	0.512 ± 0.014	0.111	0.001	0.0540	0.0540	0.934	93.600 ± 0.008	0.206 ± 0.001	55.900 ± 0.016
	beta46	0.08	0.080	-0.000 ± 0.001	-0.004 ± 0.017	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.289 ± 0.014	0.538 ± 0.015	0.079	-0.001	0.0422	0.0422	0.962	94.300 ± 0.007	0.162 ± 0.001	50.800 ± 0.016
	beta47	0.08	0.081	0.001 ± 0.001	0.007 ± 0.018	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.317 ± 0.015	0.563 ± 0.016	0.080	0.000	0.0464	0.0464	0.913	94.100 ± 0.007	0.161 ± 0.001	50.500 ± 0.016
	beta51	0.16	0.185	0.025 ± 0.002	0.159 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.057 ± 0.001	0.128 ± 0.005	0.358 ± 0.009	0.184	0.024	0.0525	0.0579	1.015	92.700 ± 0.008	0.205 ± 0.000	94.200 ± 0.007
	beta52	0.16	0.184	0.024 ± 0.002	0.147 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.141 ± 0.007	0.375 ± 0.010	0.182	0.022	0.0550	0.0593	0.994	93.300 ± 0.008	0.215 ± 0.000	91.500 ± 0.009
	beta53	0.16	0.161	0.001 ± 0.002	0.004 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.110 ± 0.005	0.332 ± 0.009	0.160	0.000	0.0537	0.0537	0.977	93.400 ± 0.008	0.204 ± 0.000	86.300 ± 0.011
	beta54	0.16	0.161	0.001 ± 0.002	0.007 ± 0.011	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.124 ± 0.006	0.352 ± 0.010	0.160	0.000	0.0577	0.0577	0.978	94.400 ± 0.007	0.216 ± 0.000	81.900 ± 0.012
	beta55	0.08	0.081	0.001 ± 0.002	0.007 ± 0.023	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.545 ± 0.026	0.738 ± 0.021	0.079	-0.001	0.0560	0.0560	0.933	94.400 ± 0.007	0.216 ± 0.001	32.600 ± 0.015
	beta56	0.08	0.082	0.002 ± 0.002	0.021 ± 0.022	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.471 ± 0.023	0.687 ± 0.020	0.083	0.003	0.0513	0.0514	0.965	94.300 ± 0.007	0.208 ± 0.001	36.200 ± 0.015
	beta57	0.06	0.061	0.001 ± 0.001	0.024 ± 0.022	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.491 ± 0.023	0.701 ± 0.020	0.061	0.001	0.0414	0.0414	0.941	93.400 ± 0.008	0.155 ± 0.001	36.000 ± 0.015
	beta58	0.06	0.060	0.000 ± 0.001	0.007 ± 0.020	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.411 ± 0.019	0.641 ± 0.018	0.059	-0.001	0.0384	0.0384	0.941	93.800 ± 0.008	0.142 ± 0.001	40.100 ± 0.015

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S44

Simulation 2 - Condition 6: Full Model, Distribution = normal, N = 1000, Reliability = 0.8

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.200	0.000 ± 0.001	0.001 ± 0.005	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.029 ± 0.001	0.169 ± 0.005	0.202	0.002	0.0348	0.0348	0.988	95.500 ± 0.007	0.131 ± 0.000	100.000 ± 0.000
	beta42	0.20	0.202	0.002 ± 0.001	0.009 ± 0.006	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.031 ± 0.001	0.175 ± 0.005	0.202	0.002	0.0338	0.0339	0.980	94.400 ± 0.007	0.135 ± 0.000	100.000 ± 0.000
	beta43	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.155 ± 0.004	0.199	-0.001	0.0317	0.0317	1.055	96.500 ± 0.006	0.128 ± 0.000	100.000 ± 0.000
	beta44	0.11	0.113	0.003 ± 0.001	0.023 ± 0.011	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.113 ± 0.005	0.337 ± 0.009	0.113	0.003	0.0365	0.0367	0.990	93.900 ± 0.008	0.143 ± 0.000	86.700 ± 0.011
	beta45	0.11	0.110	-0.000 ± 0.001	-0.003 ± 0.010	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.091 ± 0.004	0.301 ± 0.008	0.108	-0.002	0.0323	0.0324	0.984	94.600 ± 0.007	0.128 ± 0.000	91.300 ± 0.009
	beta46	0.08	0.081	0.001 ± 0.001	0.008 ± 0.011	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	0.113 ± 0.005	0.336 ± 0.009	0.080	0.000	0.0266	0.0266	0.961	94.000 ± 0.008	0.101 ± 0.000	86.600 ± 0.011
	beta47	0.08	0.080	0.000 ± 0.001	0.003 ± 0.011	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	0.112 ± 0.005	0.334 ± 0.009	0.080	0.000	0.0273	0.0273	0.953	93.900 ± 0.008	0.100 ± 0.000	87.500 ± 0.010
	beta51	0.16	0.185	0.025 ± 0.001	0.154 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.063 ± 0.003	0.251 ± 0.006	0.184	0.024	0.0307	0.0391	1.039	88.800 ± 0.010	0.129 ± 0.000	100.000 ± 0.000
	beta52	0.16	0.186	0.026 ± 0.001	0.161 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.071 ± 0.003	0.266 ± 0.006	0.186	0.026	0.0328	0.0417	1.018	89.000 ± 0.010	0.136 ± 0.000	100.000 ± 0.000
	beta53	0.16	0.163	0.003 ± 0.001	0.016 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.204 ± 0.005	0.161	0.001	0.0342	0.0343	1.009	96.100 ± 0.006	0.129 ± 0.000	99.900 ± 0.001
	beta54	0.16	0.159	-0.001 ± 0.001	-0.008 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.048 ± 0.002	0.219 ± 0.006	0.159	-0.001	0.0340	0.0340	0.998	94.200 ± 0.007	0.137 ± 0.000	99.200 ± 0.003
	beta55	0.08	0.081	0.001 ± 0.001	0.009 ± 0.013	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.176 ± 0.008	0.419 ± 0.012	0.080	0.000	0.0324	0.0324	1.020	94.900 ± 0.007	0.134 ± 0.000	66.200 ± 0.015
	beta56	0.08	0.078	-0.002 ± 0.001	-0.019 ± 0.014	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.184 ± 0.008	0.429 ± 0.012	0.080	0.000	0.0335	0.0335	0.958	93.500 ± 0.008	0.129 ± 0.000	66.400 ± 0.015
	beta57	0.06	0.060	0.000 ± 0.001	0.007 ± 0.013	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	0.177 ± 0.008	0.421 ± 0.012	0.062	0.002	0.0262	0.0262	0.980	94.100 ± 0.007	0.097 ± 0.000	68.400 ± 0.015
	beta58	0.06	0.060	0.000 ± 0.001	0.006 ± 0.012	0.001 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.022 ± 0.001	0.140 ± 0.006	0.375 ± 0.010	0.060	0.000	0.0230	0.0230	1.004	94.700 ± 0.007	0.088 ± 0.000	76.100 ± 0.013
QML	beta41	0.20	0.196	-0.004 ± 0.001	-0.019 ± 0.005	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.028 ± 0.001	0.168 ± 0.005	0.197	-0.003	0.0345	0.0346	1.003	95.200 ± 0.007	0.132 ± 0.000	100.000 ± 0.000
	beta42	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.030 ± 0.001	0.173 ± 0.005	0.200	0.000	0.0333	0.0333	0.982	94.500 ± 0.007	0.134 ± 0.000	100.000 ± 0.000
	beta43	0.20	0.195	-0.005 ± 0.001	-0.025 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.156 ± 0.004	0.194	-0.006	0.0316	0.0322	1.049	95.800 ± 0.006	0.127 ± 0.000	100.000 ± 0.000
	beta44	0.11	0.110	0.000 ± 0.001	0.001 ± 0.011	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.112 ± 0.005	0.335 ± 0.009	0.111	0.001	0.0360	0.0360	1.001	94.200 ± 0.007	0.144 ± 0.000	85.500 ± 0.011
	beta45	0.11	0.109	-0.001 ± 0.001	-0.008 ± 0.009	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.090 ± 0.004	0.299 ± 0.008	0.108	-0.002	0.0327	0.0328	0.993	95.000 ± 0.007	0.128 ± 0.000	91.900 ± 0.009
	beta46	0.08	0.080	0.000 ± 0.001	0.002 ± 0.010	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	0.109 ± 0.005	0.331 ± 0.009	0.080	0.000	0.0257	0.0257	0.968	94.400 ± 0.007	0.100 ± 0.000	86.600 ± 0.011
	beta47	0.08	0.079	-0.001 ± 0.001	-0.011 ± 0.010	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	0.105 ± 0.004	0.324 ± 0.009	0.079	-0.001	0.0266	0.0266	0.972	94.600 ± 0.007	0.099 ± 0.000	87.500 ± 0.010
	beta51	0.16	0.168	0.008 ± 0.001	0.047 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.046 ± 0.002	0.215 ± 0.006	0.168	0.008	0.0325	0.0335	1.048	96.000 ± 0.006	0.138 ± 0.000	99.900 ± 0.001
	beta52	0.16	0.166	0.006 ± 0.001	0.035 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.050 ± 0.002	0.224 ± 0.006	0.166	0.006	0.0355	0.0361	1.016	95.700 ± 0.006	0.141 ± 0.000	99.900 ± 0.001
	beta53	0.16	0.162	0.002 ± 0.001	0.015 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.041 ± 0.002	0.203 ± 0.005	0.161	0.001	0.0332	0.0332	1.013	96.300 ± 0.006	0.128 ± 0.000	100.000 ± 0.000
	beta54	0.16	0.163	0.003 ± 0.001	0.018 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.049 ± 0.002	0.222 ± 0.006	0.163	0.003	0.0337	0.0339	1.008	94.400 ± 0.007	0.140 ± 0.000	99.400 ± 0.002
	beta55	0.08	0.084	0.004 ± 0.001	0.047 ± 0.013	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.183 ± 0.008	0.428 ± 0.012	0.084	0.004	0.0328	0.0330	1.036	96.000 ± 0.006	0.138 ± 0.000	67.200 ± 0.015
	beta56	0.08	0.079	-0.001 ± 0.001	-0.012 ± 0.014	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.188 ± 0.009	0.434 ± 0.012	0.080	0.000	0.0335	0.0335	0.968	94.100 ± 0.007	0.132 ± 0.000	65.900 ± 0.015
	beta57	0.06	0.057	-0.003 ± 0.001	-0.053 ± 0.013	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	0.175 ± 0.008	0.418 ± 0.011	0.058	-0.002	0.0255	0.0256	0.986	93.900 ± 0.008	0.096 ± 0.000	65.000 ± 0.015
	beta58	0.06	0.058	-0.002 ± 0.001	-0.031 ± 0.012	0.000 ± 0.000	0.022 ± 0.000	0.000 ± 0.000	0.022 ± 0.001	0.135 ± 0.006	0.367 ± 0.010	0.058	-0.002	0.0220	0.0221	1.018	95.300 ± 0.007	0.088 ± 0.000	75.000 ± 0.014

Table S44

Simulation 2 - Condition 6: Full Model, Distribution = normal, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta41	0.20	0.200	0.000 ± 0.001	0.002 ± 0.005	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.029 ± 0.001	0.169 ± 0.005	0.202	0.002	0.0346	0.0346	0.968	94.600 ± 0.007	0.129 ± 0.000	100.000 ± 0.000
	beta42	0.20	0.202	0.002 ± 0.001	0.009 ± 0.006	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.031 ± 0.001	0.176 ± 0.005	0.202	0.002	0.0345	0.0346	0.961	93.800 ± 0.008	0.132 ± 0.000	100.000 ± 0.000
	beta43	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.155 ± 0.004	0.199	-0.001	0.0315	0.0315	1.040	95.800 ± 0.006	0.127 ± 0.000	100.000 ± 0.000
	beta44	0.11	0.113	0.003 ± 0.001	0.024 ± 0.011	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.115 ± 0.005	0.338 ± 0.010	0.113	0.003	0.0367	0.0369	0.985	93.800 ± 0.008	0.143 ± 0.000	87.400 ± 0.010
	beta45	0.11	0.110	-0.000 ± 0.001	-0.001 ± 0.010	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.093 ± 0.004	0.304 ± 0.008	0.109	-0.001	0.0334	0.0334	0.971	94.800 ± 0.007	0.127 ± 0.000	91.800 ± 0.009
	beta46	0.08	0.080	0.000 ± 0.001	0.005 ± 0.011	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	0.115 ± 0.005	0.339 ± 0.010	0.080	0.000	0.0267	0.0267	0.951	93.300 ± 0.008	0.101 ± 0.000	86.400 ± 0.011
	beta47	0.08	0.080	0.000 ± 0.001	0.004 ± 0.011	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	0.117 ± 0.005	0.341 ± 0.009	0.079	-0.001	0.0279	0.0279	0.931	93.400 ± 0.008	0.100 ± 0.000	87.800 ± 0.010
	beta51	0.16	0.185	0.025 ± 0.001	0.156 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.064 ± 0.003	0.252 ± 0.006	0.185	0.025	0.0309	0.0395	1.036	88.900 ± 0.010	0.129 ± 0.000	100.000 ± 0.000
	beta52	0.16	0.186	0.026 ± 0.001	0.161 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.071 ± 0.003	0.267 ± 0.006	0.186	0.026	0.0325	0.0413	1.015	88.900 ± 0.010	0.136 ± 0.000	100.000 ± 0.000
	beta53	0.16	0.163	0.003 ± 0.001	0.017 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.042 ± 0.002	0.204 ± 0.005	0.162	0.002	0.0341	0.0342	1.005	96.400 ± 0.006	0.128 ± 0.000	100.000 ± 0.000
	beta54	0.16	0.159	-0.001 ± 0.001	-0.007 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.048 ± 0.002	0.218 ± 0.006	0.159	-0.001	0.0336	0.0336	0.999	94.400 ± 0.007	0.137 ± 0.000	99.200 ± 0.003
	beta55	0.08	0.081	0.001 ± 0.001	0.007 ± 0.013	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.176 ± 0.008	0.419 ± 0.012	0.080	0.000	0.0331	0.0331	1.026	95.200 ± 0.007	0.135 ± 0.000	65.800 ± 0.015
	beta56	0.08	0.078	-0.002 ± 0.001	-0.022 ± 0.014	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.183 ± 0.008	0.428 ± 0.012	0.079	-0.001	0.0331	0.0331	0.965	93.500 ± 0.008	0.129 ± 0.000	65.900 ± 0.015
	beta57	0.06	0.060	0.000 ± 0.001	0.006 ± 0.013	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	0.178 ± 0.008	0.422 ± 0.012	0.062	0.002	0.0257	0.0258	0.976	94.100 ± 0.007	0.097 ± 0.000	68.900 ± 0.015
	beta58	0.06	0.060	0.000 ± 0.001	0.005 ± 0.012	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.142 ± 0.006	0.377 ± 0.010	0.060	0.000	0.0227	0.0227	1.000	95.400 ± 0.007	0.089 ± 0.000	76.600 ± 0.013

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S45

Simulation 2 - Condition 7: Full Model, Distribution = right-skewed, N = 400, Reliability = 0.4

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.196	-0.004 ± 0.007	-0.019 ± 0.036	0.035 ± 0.002	0.188 ± 0.006	0.035 ± 0.002	0.188 ± 0.007	0.884 ± 0.056	0.940 ± 0.035	0.195	-0.005	0.1829	0.1830	1.034	98.220 ± 0.005	0.763 ± 0.012	20.030 ± 0.015
	beta42	0.20	0.197	-0.003 ± 0.007	-0.013 ± 0.036	0.036 ± 0.002	0.189 ± 0.006	0.035 ± 0.002	0.188 ± 0.007	0.887 ± 0.056	0.942 ± 0.035	0.200	0.000	0.1861	0.1861	1.065	97.923 ± 0.005	0.787 ± 0.012	21.810 ± 0.016
	beta43	0.20	0.198	-0.002 ± 0.004	-0.008 ± 0.019	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.096 ± 0.004	0.232 ± 0.015	0.482 ± 0.018	0.195	-0.005	0.0958	0.0960	1.071	97.626 ± 0.006	0.405 ± 0.004	52.671 ± 0.019
	beta44	0.11	0.127	0.017 ± 0.006	0.154 ± 0.057	0.026 ± 0.002	0.163 ± 0.006	0.027 ± 0.002	0.164 ± 0.007	2.210 ± 0.159	1.487 ± 0.061	0.111	0.001	0.1456	0.1456	1.079	96.588 ± 0.007	0.688 ± 0.021	11.128 ± 0.012
	beta45	0.11	0.125	0.015 ± 0.006	0.135 ± 0.052	0.022 ± 0.001	0.148 ± 0.005	0.022 ± 0.002	0.149 ± 0.006	1.835 ± 0.125	1.355 ± 0.053	0.117	0.007	0.1317	0.1319	1.048	96.588 ± 0.007	0.610 ± 0.017	14.243 ± 0.013
	beta46	0.08	0.087	0.007 ± 0.003	0.093 ± 0.042	0.008 ± 0.001	0.087 ± 0.003	0.008 ± 0.001	0.087 ± 0.003	1.185 ± 0.083	1.089 ± 0.044	0.089	0.009	0.0782	0.0787	1.015	96.142 ± 0.007	0.346 ± 0.009	22.700 ± 0.016
	beta47	0.08	0.088	0.008 ± 0.003	0.100 ± 0.040	0.007 ± 0.001	0.082 ± 0.003	0.007 ± 0.001	0.082 ± 0.004	1.061 ± 0.083	1.030 ± 0.045	0.083	0.003	0.0743	0.0744	1.033	96.291 ± 0.007	0.332 ± 0.008	23.294 ± 0.016
	beta51	0.16	0.174	0.014 ± 0.007	0.088 ± 0.043	0.032 ± 0.002	0.178 ± 0.006	0.032 ± 0.002	0.178 ± 0.007	1.240 ± 0.078	1.114 ± 0.041	0.174	0.014	0.1749	0.1754	1.069	98.220 ± 0.005	0.745 ± 0.012	15.875 ± 0.014
	beta52	0.16	0.186	0.026 ± 0.005	0.160 ± 0.033	0.019 ± 0.001	0.137 ± 0.004	0.019 ± 0.001	0.139 ± 0.005	0.754 ± 0.049	0.869 ± 0.033	0.184	0.024	0.1344	0.1365	1.008	97.033 ± 0.007	0.540 ± 0.006	27.448 ± 0.017
	beta53	0.16	0.157	-0.003 ± 0.007	-0.022 ± 0.041	0.029 ± 0.002	0.170 ± 0.006	0.029 ± 0.002	0.170 ± 0.007	1.132 ± 0.076	1.064 ± 0.041	0.155	-0.005	0.1599	0.1599	1.035	97.181 ± 0.006	0.691 ± 0.011	13.650 ± 0.013
	beta54	0.16	0.159	-0.001 ± 0.004	-0.006 ± 0.024	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	0.393 ± 0.022	0.627 ± 0.022	0.160	0.000	0.0973	0.0973	0.998	95.252 ± 0.008	0.393 ± 0.003	37.834 ± 0.019
	beta55	0.08	0.077	-0.003 ± 0.005	-0.033 ± 0.059	0.015 ± 0.001	0.122 ± 0.004	0.015 ± 0.001	0.122 ± 0.005	2.311 ± 0.149	1.520 ± 0.057	0.078	-0.002	0.1141	0.1141	1.037	96.142 ± 0.007	0.495 ± 0.012	11.573 ± 0.012
	beta56	0.08	0.087	0.007 ± 0.003	0.091 ± 0.037	0.006 ± 0.000	0.076 ± 0.003	0.006 ± 0.000	0.077 ± 0.003	0.919 ± 0.066	0.959 ± 0.039	0.086	0.006	0.0747	0.0749	0.981	95.697 ± 0.008	0.294 ± 0.007	31.157 ± 0.018
	beta57	0.06	0.072	0.012 ± 0.005	0.196 ± 0.082	0.016 ± 0.001	0.128 ± 0.005	0.016 ± 0.001	0.128 ± 0.005	4.562 ± 0.334	2.136 ± 0.089	0.067	0.007	0.1150	0.1152	1.034	97.923 ± 0.005	0.518 ± 0.013	7.418 ± 0.010
	beta58	0.06	0.063	0.003 ± 0.003	0.057 ± 0.043	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.003	1.265 ± 0.092	1.125 ± 0.046	0.063	0.003	0.0570	0.0570	1.008	96.439 ± 0.007	0.267 ± 0.006	18.101 ± 0.015
QML	beta41	0.20	0.238	0.038 ± 0.004	0.189 ± 0.020	0.011 ± 0.001	0.102 ± 0.003	0.012 ± 0.001	0.109 ± 0.004	0.298 ± 0.017	0.546 ± 0.018	0.236	0.036	0.1042	0.1104	0.928	94.214 ± 0.009	0.373 ± 0.002	70.920 ± 0.017
	beta42	0.20	0.237	0.037 ± 0.004	0.187 ± 0.020	0.011 ± 0.001	0.103 ± 0.003	0.012 ± 0.001	0.109 ± 0.004	0.299 ± 0.018	0.547 ± 0.019	0.238	0.038	0.1031	0.1097	0.960	93.769 ± 0.009	0.388 ± 0.002	67.804 ± 0.018
	beta43	0.20	0.183	-0.017 ± 0.003	-0.086 ± 0.016	0.007 ± 0.000	0.085 ± 0.003	0.008 ± 0.000	0.087 ± 0.003	0.188 ± 0.011	0.433 ± 0.015	0.176	-0.024	0.0864	0.0896	0.978	94.214 ± 0.009	0.326 ± 0.002	60.089 ± 0.019
	beta44	0.11	0.031	-0.079 ± 0.005	-0.714 ± 0.043	0.015 ± 0.001	0.124 ± 0.004	0.021 ± 0.001	0.146 ± 0.005	1.769 ± 0.104	1.330 ± 0.043	0.032	-0.078	0.1327	0.1538	0.950	89.021 ± 0.012	0.460 ± 0.005	7.567 ± 0.010
	beta45	0.11	0.099	-0.011 ± 0.004	-0.102 ± 0.037	0.011 ± 0.001	0.105 ± 0.003	0.011 ± 0.001	0.105 ± 0.003	0.912 ± 0.048	0.955 ± 0.031	0.096	-0.014	0.1074	0.1083	0.935	92.136 ± 0.010	0.383 ± 0.003	18.546 ± 0.015
	beta46	0.08	0.159	0.079 ± 0.003	0.990 ± 0.036	0.006 ± 0.000	0.075 ± 0.002	0.012 ± 0.001	0.109 ± 0.003	1.862 ± 0.099	1.364 ± 0.038	0.157	0.077	0.0737	0.1063	0.959	84.570 ± 0.014	0.283 ± 0.004	66.617 ± 0.018
	beta47	0.08	0.158	0.078 ± 0.003	0.972 ± 0.036	0.006 ± 0.000	0.075 ± 0.003	0.012 ± 0.001	0.108 ± 0.003	1.818 ± 0.106	1.348 ± 0.041	0.152	0.072	0.0692	0.1001	0.946	85.608 ± 0.014	0.277 ± 0.004	67.656 ± 0.018
	beta51	0.16	0.177	0.017 ± 0.005	0.107 ± 0.029	0.014 ± 0.001	0.120 ± 0.004	0.015 ± 0.001	0.121 ± 0.005	0.574 ± 0.037	0.757 ± 0.028	0.177	0.017	0.1238	0.1251	0.932	95.104 ± 0.008	0.438 ± 0.003	38.427 ± 0.019
	beta52	0.16	0.129	-0.031 ± 0.004	-0.191 ± 0.028	0.014 ± 0.001	0.116 ± 0.003	0.014 ± 0.001	0.120 ± 0.004	0.566 ± 0.030	0.752 ± 0.024	0.123	-0.037	0.1191	0.1247	0.953	93.027 ± 0.010	0.435 ± 0.003	23.145 ± 0.016
	beta53	0.16	0.179	0.019 ± 0.004	0.117 ± 0.023	0.009 ± 0.001	0.097 ± 0.003	0.010 ± 0.001	0.099 ± 0.003	0.380 ± 0.022	0.617 ± 0.021	0.180	0.020	0.0985	0.1006	0.983	94.807 ± 0.009	0.373 ± 0.002	49.110 ± 0.019
	beta54	0.16	0.187	0.027 ± 0.004	0.166 ± 0.027	0.012 ± 0.001	0.111 ± 0.003	0.013 ± 0.001	0.114 ± 0.004	0.511 ± 0.028	0.715 ± 0.023	0.180	0.020	0.1084	0.1102	0.930	91.246 ± 0.011	0.406 ± 0.003	44.510 ± 0.019
	beta55	0.08	0.078	-0.002 ± 0.005	-0.027 ± 0.063	0.017 ± 0.001	0.131 ± 0.004	0.017 ± 0.001	0.131 ± 0.005	2.690 ± 0.176	1.640 ± 0.063	0.074	-0.006	0.1332	0.1333	0.818	89.763 ± 0.012	0.421 ± 0.005	18.249 ± 0.015
	beta56	0.08	0.122	0.042 ± 0.003	0.520 ± 0.037	0.006 ± 0.000	0.078 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	1.208 ± 0.064	1.099 ± 0.033	0.120	0.040	0.0815	0.0909	0.926	90.059 ± 0.012	0.281 ± 0.003	44.065 ± 0.019
	beta57	0.06	0.135	0.075 ± 0.005	1.255 ± 0.075	0.014 ± 0.001	0.117 ± 0.004	0.019 ± 0.001	0.139 ± 0.005	5.402 ± 0.319	2.324 ± 0.076	0.136	0.076	0.1203	0.1422	0.844	84.718 ± 0.014	0.389 ± 0.005	31.306 ± 0.018
	beta58	0.06	0.106	0.046 ± 0.002	0.768 ± 0.038	0.004 ± 0.000	0.059 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	1.564 ± 0.091	1.251 ± 0.040	0.104	0.044	0.0548	0.0704	0.967	91.691 ± 0.011	0.225 ± 0.003	52.819 ± 0.019

Table S45

Simulation 2 - Condition 7: Full Model, Distribution = right-skewed, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta41	0.20	0.219	0.019 ± 0.008	0.095 ± 0.040	0.044 ± 0.003	0.210 ± 0.008	0.044 ± 0.003	0.211 ± 0.009	1.109 ± 0.080	1.053 ± 0.043	0.206	0.006	0.2013	0.2014	0.816	91.098 ± 0.011	0.671 ± 0.008	30.267 ± 0.018
	beta42	0.20	0.222	0.022 ± 0.008	0.108 ± 0.042	0.047 ± 0.003	0.217 ± 0.007	0.047 ± 0.003	0.218 ± 0.008	1.183 ± 0.077	1.088 ± 0.041	0.236	0.036	0.2037	0.2068	0.819	92.136 ± 0.010	0.696 ± 0.009	31.306 ± 0.018
	beta43	0.20	0.196	-0.004 ± 0.004	-0.019 ± 0.020	0.011 ± 0.001	0.104 ± 0.003	0.011 ± 0.001	0.104 ± 0.004	0.268 ± 0.016	0.518 ± 0.018	0.194	-0.006	0.1113	0.1114	0.890	92.878 ± 0.010	0.362 ± 0.003	61.276 ± 0.019
	beta44	0.11	0.109	-0.001 ± 0.007	-0.011 ± 0.063	0.033 ± 0.003	0.180 ± 0.007	0.033 ± 0.003	0.180 ± 0.008	2.689 ± 0.210	1.640 ± 0.072	0.100	-0.010	0.1704	0.1707	0.775	91.840 ± 0.011	0.549 ± 0.014	20.326 ± 0.016
	beta45	0.11	0.118	0.008 ± 0.007	0.073 ± 0.068	0.037 ± 0.003	0.194 ± 0.007	0.037 ± 0.003	0.194 ± 0.008	3.099 ± 0.232	1.760 ± 0.074	0.095	-0.015	0.1670	0.1676	0.732	87.685 ± 0.013	0.556 ± 0.016	19.733 ± 0.015
	beta46	0.08	0.079	-0.001 ± 0.004	-0.012 ± 0.045	0.009 ± 0.001	0.094 ± 0.004	0.009 ± 0.001	0.094 ± 0.004	1.370 ± 0.103	1.170 ± 0.050	0.080	0.000	0.0881	0.0881	0.775	89.318 ± 0.012	0.285 ± 0.006	31.157 ± 0.018
	beta47	0.08	0.080	-0.000 ± 0.004	-0.004 ± 0.045	0.009 ± 0.001	0.094 ± 0.004	0.009 ± 0.001	0.094 ± 0.004	1.383 ± 0.105	1.176 ± 0.050	0.070	-0.010	0.0794	0.0800	0.743	90.653 ± 0.011	0.274 ± 0.006	31.306 ± 0.018
	beta51	0.16	0.199	0.039 ± 0.008	0.242 ± 0.047	0.039 ± 0.003	0.197 ± 0.007	0.040 ± 0.003	0.201 ± 0.008	1.574 ± 0.107	1.255 ± 0.049	0.203	0.043	0.2004	0.2051	0.860	93.472 ± 0.010	0.664 ± 0.008	28.338 ± 0.017
	beta52	0.16	0.183	0.023 ± 0.006	0.143 ± 0.035	0.022 ± 0.001	0.147 ± 0.005	0.022 ± 0.001	0.148 ± 0.005	0.860 ± 0.054	0.927 ± 0.034	0.183	0.023	0.1445	0.1464	0.910	93.620 ± 0.009	0.523 ± 0.005	30.564 ± 0.018
	beta53	0.16	0.169	0.009 ± 0.007	0.057 ± 0.045	0.035 ± 0.002	0.188 ± 0.006	0.035 ± 0.002	0.188 ± 0.007	1.376 ± 0.086	1.173 ± 0.043	0.165	0.005	0.1912	0.1913	0.923	95.252 ± 0.008	0.679 ± 0.009	18.694 ± 0.015
	beta54	0.16	0.169	0.009 ± 0.004	0.057 ± 0.025	0.011 ± 0.001	0.103 ± 0.003	0.011 ± 0.001	0.103 ± 0.004	0.414 ± 0.023	0.643 ± 0.022	0.162	0.002	0.1052	0.1052	0.950	94.510 ± 0.009	0.382 ± 0.002	42.285 ± 0.019
	beta55	0.08	0.087	0.007 ± 0.005	0.090 ± 0.063	0.017 ± 0.001	0.131 ± 0.005	0.017 ± 0.001	0.131 ± 0.005	2.680 ± 0.189	1.637 ± 0.066	0.087	0.007	0.1217	0.1219	0.819	92.433 ± 0.010	0.420 ± 0.008	20.475 ± 0.016
	beta56	0.08	0.083	0.003 ± 0.003	0.037 ± 0.039	0.006 ± 0.000	0.080 ± 0.003	0.006 ± 0.000	0.080 ± 0.003	1.011 ± 0.069	1.005 ± 0.040	0.081	0.001	0.0752	0.0752	0.855	93.175 ± 0.010	0.270 ± 0.005	31.899 ± 0.018
	beta57	0.06	0.054	-0.006 ± 0.005	-0.099 ± 0.089	0.019 ± 0.001	0.139 ± 0.005	0.019 ± 0.001	0.139 ± 0.006	5.396 ± 0.412	2.323 ± 0.100	0.047	-0.013	0.1241	0.1247	0.759	91.988 ± 0.010	0.415 ± 0.008	11.573 ± 0.012
	beta58	0.06	0.057	-0.003 ± 0.003	-0.044 ± 0.050	0.006 ± 0.000	0.077 ± 0.003	0.006 ± 0.000	0.077 ± 0.003	1.661 ± 0.134	1.289 ± 0.058	0.058	-0.002	0.0679	0.0679	0.840	93.917 ± 0.009	0.255 ± 0.006	22.107 ± 0.016

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S46

Simulation 2 - Condition 8: Full Model, Distribution = right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.200	0.000 ± 0.004	0.000 ± 0.018	0.012 ± 0.001	0.108 ± 0.003	0.012 ± 0.001	0.108 ± 0.003	0.293 ± 0.015	0.541 ± 0.017	0.196	-0.004	0.1055	0.1056	1.031	96.383 ± 0.006	0.438 ± 0.003	46.064 ± 0.016
	beta42	0.20	0.196	-0.004 ± 0.004	-0.018 ± 0.019	0.014 ± 0.001	0.119 ± 0.003	0.014 ± 0.001	0.119 ± 0.004	0.354 ± 0.018	0.595 ± 0.018	0.204	0.004	0.1152	0.1153	0.984	95.851 ± 0.007	0.459 ± 0.003	44.787 ± 0.016
	beta43	0.20	0.200	0.000 ± 0.002	0.002 ± 0.010	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.085 ± 0.004	0.291 ± 0.009	0.198	-0.002	0.0599	0.0599	1.026	96.277 ± 0.006	0.235 ± 0.001	94.043 ± 0.008
	beta44	0.11	0.118	0.008 ± 0.003	0.074 ± 0.025	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	0.599 ± 0.031	0.774 ± 0.024	0.114	0.004	0.0831	0.0832	0.990	96.170 ± 0.006	0.329 ± 0.004	30.213 ± 0.015
	beta45	0.11	0.117	0.007 ± 0.003	0.062 ± 0.023	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.497 ± 0.026	0.705 ± 0.022	0.115	0.005	0.0710	0.0711	0.997	94.787 ± 0.007	0.302 ± 0.003	34.468 ± 0.016
	beta46	0.08	0.082	0.002 ± 0.001	0.027 ± 0.019	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.329 ± 0.017	0.573 ± 0.018	0.081	0.001	0.0437	0.0437	0.943	94.149 ± 0.008	0.170 ± 0.002	53.511 ± 0.016
	beta47	0.08	0.083	0.003 ± 0.001	0.038 ± 0.019	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.326 ± 0.017	0.571 ± 0.017	0.081	0.001	0.0436	0.0437	0.943	94.787 ± 0.007	0.168 ± 0.002	55.213 ± 0.016
	beta51	0.16	0.190	0.030 ± 0.004	0.185 ± 0.023	0.013 ± 0.001	0.113 ± 0.003	0.014 ± 0.001	0.117 ± 0.003	0.533 ± 0.027	0.730 ± 0.022	0.183	0.023	0.1101	0.1125	1.016	96.383 ± 0.006	0.450 ± 0.004	41.170 ± 0.016
	beta52	0.16	0.187	0.027 ± 0.003	0.169 ± 0.017	0.007 ± 0.000	0.082 ± 0.002	0.008 ± 0.000	0.087 ± 0.003	0.294 ± 0.015	0.542 ± 0.016	0.186	0.026	0.0827	0.0868	0.996	95.532 ± 0.007	0.322 ± 0.002	63.085 ± 0.016
	beta53	0.16	0.160	-0.000 ± 0.003	-0.000 ± 0.022	0.011 ± 0.001	0.106 ± 0.002	0.011 ± 0.001	0.106 ± 0.003	0.440 ± 0.020	0.663 ± 0.019	0.162	0.002	0.1102	0.1102	0.999	96.170 ± 0.006	0.416 ± 0.003	34.894 ± 0.016
	beta54	0.16	0.156	-0.004 ± 0.002	-0.027 ± 0.012	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.138 ± 0.007	0.372 ± 0.011	0.158	-0.002	0.0607	0.0607	1.023	95.638 ± 0.007	0.238 ± 0.001	73.511 ± 0.014
	beta55	0.08	0.083	0.003 ± 0.002	0.035 ± 0.028	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.746 ± 0.040	0.864 ± 0.027	0.083	0.003	0.0639	0.0640	0.965	95.745 ± 0.007	0.261 ± 0.003	30.638 ± 0.015
	beta56	0.08	0.084	0.004 ± 0.001	0.047 ± 0.017	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.287 ± 0.016	0.535 ± 0.018	0.081	0.001	0.0380	0.0380	0.904	93.936 ± 0.008	0.151 ± 0.002	62.340 ± 0.016
	beta57	0.06	0.061	0.001 ± 0.002	0.012 ± 0.039	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	1.465 ± 0.077	1.210 ± 0.037	0.059	-0.001	0.0674	0.0674	0.995	96.383 ± 0.006	0.283 ± 0.004	13.191 ± 0.011
	beta58	0.06	0.063	0.003 ± 0.001	0.049 ± 0.021	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.403 ± 0.021	0.635 ± 0.020	0.061	0.001	0.0372	0.0372	0.970	95.106 ± 0.007	0.145 ± 0.002	46.809 ± 0.016
QML	beta41	0.20	0.235	0.035 ± 0.002	0.173 ± 0.009	0.003 ± 0.000	0.058 ± 0.001	0.005 ± 0.000	0.068 ± 0.002	0.115 ± 0.006	0.339 ± 0.009	0.232	0.032	0.0576	0.0659	1.019	92.447 ± 0.009	0.233 ± 0.001	98.511 ± 0.004
	beta42	0.20	0.231	0.031 ± 0.002	0.155 ± 0.010	0.004 ± 0.000	0.062 ± 0.001	0.005 ± 0.000	0.069 ± 0.002	0.119 ± 0.006	0.345 ± 0.009	0.233	0.033	0.0627	0.0707	0.991	93.404 ± 0.008	0.240 ± 0.001	97.234 ± 0.005
	beta43	0.20	0.184	-0.016 ± 0.002	-0.082 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.073 ± 0.003	0.270 ± 0.007	0.183	-0.017	0.0514	0.0542	1.003	94.362 ± 0.008	0.203 ± 0.001	94.681 ± 0.007
	beta44	0.11	0.030	-0.080 ± 0.002	-0.729 ± 0.022	0.006 ± 0.000	0.075 ± 0.002	0.012 ± 0.000	0.110 ± 0.002	0.999 ± 0.041	1.000 ± 0.022	0.031	-0.079	0.0715	0.1066	0.955	79.894 ± 0.013	0.282 ± 0.002	7.660 ± 0.009
	beta45	0.11	0.099	-0.011 ± 0.002	-0.096 ± 0.019	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.338 ± 0.017	0.581 ± 0.018	0.102	-0.008	0.0577	0.0583	0.963	94.681 ± 0.007	0.238 ± 0.001	38.723 ± 0.016
	beta46	0.08	0.159	0.079 ± 0.001	0.986 ± 0.018	0.002 ± 0.000	0.044 ± 0.001	0.008 ± 0.000	0.090 ± 0.002	1.279 ± 0.042	1.131 ± 0.019	0.155	0.075	0.0428	0.0866	0.974	56.277 ± 0.016	0.169 ± 0.001	97.553 ± 0.005
	beta47	0.08	0.156	0.076 ± 0.001	0.944 ± 0.018	0.002 ± 0.000	0.045 ± 0.001	0.008 ± 0.000	0.088 ± 0.002	1.203 ± 0.040	1.097 ± 0.019	0.151	0.071	0.0433	0.0835	0.939	56.489 ± 0.016	0.165 ± 0.001	96.809 ± 0.006
	beta51	0.16	0.184	0.024 ± 0.002	0.153 ± 0.015	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.237 ± 0.011	0.487 ± 0.013	0.185	0.025	0.0745	0.0785	0.932	92.234 ± 0.009	0.271 ± 0.001	75.426 ± 0.014
	beta52	0.16	0.129	-0.031 ± 0.002	-0.194 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.227 ± 0.010	0.476 ± 0.013	0.126	-0.034	0.0685	0.0763	0.973	91.809 ± 0.009	0.265 ± 0.001	48.723 ± 0.016
	beta53	0.16	0.179	0.019 ± 0.002	0.120 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.150 ± 0.007	0.387 ± 0.010	0.180	0.020	0.0637	0.0668	1.009	95.106 ± 0.007	0.233 ± 0.001	86.489 ± 0.011
	beta54	0.16	0.176	0.016 ± 0.002	0.098 ± 0.013	0.004 ± 0.000	0.066 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.180 ± 0.008	0.424 ± 0.012	0.176	0.016	0.0662	0.0682	0.963	94.043 ± 0.008	0.250 ± 0.001	78.085 ± 0.013
	beta55	0.08	0.074	-0.006 ± 0.003	-0.077 ± 0.035	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	1.125 ± 0.056	1.061 ± 0.032	0.067	-0.013	0.0824	0.0834	0.802	87.234 ± 0.011	0.266 ± 0.002	23.191 ± 0.014
	beta56	0.08	0.124	0.044 ± 0.002	0.552 ± 0.019	0.002 ± 0.000	0.046 ± 0.001	0.004 ± 0.000	0.064 ± 0.001	0.638 ± 0.026	0.799 ± 0.018	0.124	0.044	0.0445	0.0624	0.913	79.894 ± 0.013	0.165 ± 0.001	81.702 ± 0.013
	beta57	0.06	0.142	0.082 ± 0.003	1.371 ± 0.043	0.006 ± 0.000	0.079 ± 0.002	0.013 ± 0.001	0.114 ± 0.002	3.615 ± 0.139	1.901 ± 0.039	0.143	0.083	0.0749	0.1117	0.820	66.915 ± 0.015	0.254 ± 0.002	60.532 ± 0.016
	beta58	0.06	0.112	0.052 ± 0.001	0.864 ± 0.019	0.001 ± 0.000	0.035 ± 0.001	0.004 ± 0.000	0.063 ± 0.001	1.087 ± 0.040	1.042 ± 0.020	0.110	0.050	0.0348	0.0609	0.986	69.681 ± 0.015	0.135 ± 0.001	92.553 ± 0.009

Table S46

Simulation 2 - Condition 8: Full Model, Distribution = right-skewed, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta41	0.20	0.212	0.012 ± 0.004	0.059 ± 0.018	0.012 ± 0.001	0.108 ± 0.003	0.012 ± 0.001	0.109 ± 0.003	0.297 ± 0.015	0.545 ± 0.016	0.209	0.009	0.1068	0.1071	0.916	93.404 ± 0.008	0.389 ± 0.002	57.872 ± 0.016
	beta42	0.20	0.204	0.004 ± 0.004	0.021 ± 0.019	0.014 ± 0.001	0.118 ± 0.003	0.014 ± 0.001	0.118 ± 0.004	0.346 ± 0.018	0.588 ± 0.018	0.205	0.005	0.1072	0.1073	0.883	92.447 ± 0.009	0.407 ± 0.003	53.830 ± 0.016
	beta43	0.20	0.199	-0.001 ± 0.002	-0.003 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.080 ± 0.004	0.283 ± 0.008	0.198	-0.002	0.0559	0.0560	0.943	94.574 ± 0.007	0.209 ± 0.001	95.745 ± 0.007
	beta44	0.11	0.116	0.006 ± 0.003	0.054 ± 0.027	0.008 ± 0.000	0.091 ± 0.003	0.008 ± 0.000	0.091 ± 0.003	0.687 ± 0.040	0.829 ± 0.028	0.109	-0.001	0.0832	0.0832	0.800	89.787 ± 0.010	0.285 ± 0.004	42.021 ± 0.016
	beta45	0.11	0.113	0.003 ± 0.003	0.031 ± 0.026	0.008 ± 0.000	0.087 ± 0.003	0.008 ± 0.000	0.087 ± 0.003	0.631 ± 0.038	0.794 ± 0.027	0.109	-0.001	0.0770	0.0770	0.795	89.043 ± 0.010	0.272 ± 0.004	42.766 ± 0.016
	beta46	0.08	0.078	-0.002 ± 0.002	-0.028 ± 0.019	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.002	0.341 ± 0.019	0.584 ± 0.019	0.075	-0.005	0.0406	0.0410	0.785	88.830 ± 0.010	0.144 ± 0.002	60.532 ± 0.016
	beta47	0.08	0.081	0.001 ± 0.001	0.007 ± 0.018	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.320 ± 0.017	0.566 ± 0.018	0.077	-0.003	0.0404	0.0405	0.810	90.213 ± 0.010	0.144 ± 0.002	61.383 ± 0.016
	beta51	0.16	0.204	0.044 ± 0.004	0.272 ± 0.022	0.012 ± 0.001	0.107 ± 0.003	0.013 ± 0.001	0.116 ± 0.003	0.525 ± 0.028	0.725 ± 0.022	0.203	0.043	0.1013	0.1101	0.951	93.404 ± 0.008	0.401 ± 0.003	53.191 ± 0.016
	beta52	0.16	0.187	0.027 ± 0.003	0.166 ± 0.017	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	0.288 ± 0.015	0.536 ± 0.016	0.188	0.028	0.0823	0.0870	0.953	93.723 ± 0.008	0.305 ± 0.001	67.660 ± 0.015
	beta53	0.16	0.159	-0.001 ± 0.003	-0.005 ± 0.022	0.011 ± 0.001	0.106 ± 0.003	0.011 ± 0.001	0.106 ± 0.003	0.441 ± 0.021	0.664 ± 0.019	0.161	0.001	0.1077	0.1077	0.963	95.745 ± 0.007	0.401 ± 0.003	37.979 ± 0.016
	beta54	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.012	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.130 ± 0.006	0.361 ± 0.010	0.160	0.000	0.0580	0.0580	1.019	96.170 ± 0.006	0.231 ± 0.001	77.660 ± 0.014
	beta55	0.08	0.086	0.006 ± 0.002	0.074 ± 0.026	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.657 ± 0.037	0.811 ± 0.026	0.087	0.007	0.0613	0.0617	0.878	92.660 ± 0.009	0.222 ± 0.002	39.043 ± 0.016
	beta56	0.08	0.082	0.002 ± 0.001	0.028 ± 0.018	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.300 ± 0.017	0.548 ± 0.018	0.079	-0.001	0.0394	0.0394	0.824	91.702 ± 0.009	0.141 ± 0.002	64.255 ± 0.016
	beta57	0.06	0.053	-0.007 ± 0.002	-0.122 ± 0.035	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	1.186 ± 0.060	1.089 ± 0.033	0.051	-0.009	0.0599	0.0606	0.919	93.085 ± 0.008	0.234 ± 0.003	16.596 ± 0.012
	beta58	0.06	0.062	0.002 ± 0.001	0.040 ± 0.021	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.425 ± 0.023	0.652 ± 0.021	0.061	0.001	0.0348	0.0348	0.885	94.255 ± 0.008	0.136 ± 0.002	49.468 ± 0.016

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S47

Simulation 2 - Condition 9: Full Model, Distribution = right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.202	0.002 ± 0.004	0.008 ± 0.019	0.014 ± 0.001	0.117 ± 0.003	0.014 ± 0.001	0.117 ± 0.003	0.340 ± 0.016	0.583 ± 0.017	0.201	0.001	0.1119	0.1119	1.026	95.872 ± 0.006	0.469 ± 0.003	42.002 ± 0.016
	beta42	0.20	0.197	-0.003 ± 0.004	-0.017 ± 0.019	0.014 ± 0.001	0.120 ± 0.003	0.014 ± 0.001	0.120 ± 0.003	0.360 ± 0.017	0.600 ± 0.017	0.200	0.000	0.1161	0.1161	1.031	96.078 ± 0.006	0.485 ± 0.003	38.906 ± 0.016
	beta43	0.20	0.200	-0.000 ± 0.002	-0.000 ± 0.011	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.115 ± 0.006	0.340 ± 0.010	0.201	0.001	0.0636	0.0636	0.998	94.324 ± 0.007	0.266 ± 0.001	84.004 ± 0.012
	beta44	0.11	0.115	0.005 ± 0.003	0.049 ± 0.027	0.008 ± 0.000	0.091 ± 0.002	0.008 ± 0.000	0.092 ± 0.003	0.692 ± 0.036	0.832 ± 0.026	0.112	0.002	0.0856	0.0856	0.953	94.324 ± 0.007	0.341 ± 0.004	28.586 ± 0.015
	beta45	0.11	0.123	0.013 ± 0.003	0.116 ± 0.025	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	0.638 ± 0.034	0.799 ± 0.025	0.122	0.012	0.0861	0.0869	0.923	94.530 ± 0.007	0.315 ± 0.003	37.255 ± 0.016
	beta46	0.08	0.082	0.002 ± 0.002	0.026 ± 0.021	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.427 ± 0.023	0.653 ± 0.020	0.083	0.003	0.0464	0.0465	0.921	94.324 ± 0.007	0.189 ± 0.002	48.297 ± 0.016
	beta47	0.08	0.082	0.002 ± 0.002	0.030 ± 0.020	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	0.398 ± 0.023	0.631 ± 0.021	0.079	-0.001	0.0452	0.0452	0.908	93.292 ± 0.008	0.180 ± 0.002	49.742 ± 0.016
	beta51	0.16	0.191	0.031 ± 0.004	0.197 ± 0.024	0.014 ± 0.001	0.120 ± 0.003	0.015 ± 0.001	0.124 ± 0.003	0.604 ± 0.029	0.777 ± 0.022	0.192	0.032	0.1210	0.1253	0.999	94.530 ± 0.007	0.471 ± 0.003	39.835 ± 0.016
	beta52	0.16	0.183	0.023 ± 0.003	0.143 ± 0.018	0.008 ± 0.000	0.091 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	0.345 ± 0.016	0.588 ± 0.016	0.180	0.020	0.0886	0.0909	0.999	94.014 ± 0.008	0.357 ± 0.002	52.941 ± 0.016
	beta53	0.16	0.158	-0.002 ± 0.004	-0.009 ± 0.024	0.014 ± 0.001	0.118 ± 0.003	0.014 ± 0.001	0.117 ± 0.003	0.539 ± 0.026	0.734 ± 0.021	0.161	0.001	0.1151	0.1151	0.968	94.634 ± 0.007	0.446 ± 0.003	30.650 ± 0.015
	beta54	0.16	0.160	0.000 ± 0.002	0.003 ± 0.014	0.005 ± 0.000	0.070 ± 0.001	0.005 ± 0.000	0.070 ± 0.002	0.189 ± 0.008	0.435 ± 0.012	0.160	0.000	0.0674	0.0674	1.005	95.046 ± 0.007	0.274 ± 0.001	63.261 ± 0.015
	beta55	0.08	0.082	0.002 ± 0.002	0.031 ± 0.031	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.905 ± 0.048	0.951 ± 0.029	0.084	0.004	0.0713	0.0714	0.925	93.705 ± 0.008	0.276 ± 0.003	26.522 ± 0.014
	beta56	0.08	0.080	0.000 ± 0.002	0.004 ± 0.020	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	0.395 ± 0.021	0.628 ± 0.020	0.082	0.002	0.0454	0.0454	0.893	91.228 ± 0.009	0.176 ± 0.002	51.393 ± 0.016
	beta57	0.06	0.060	-0.000 ± 0.003	-0.002 ± 0.042	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.003	1.700 ± 0.096	1.304 ± 0.042	0.060	0.000	0.0691	0.0691	0.932	94.324 ± 0.007	0.286 ± 0.003	15.996 ± 0.012
	beta58	0.06	0.062	0.002 ± 0.001	0.033 ± 0.025	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.002	0.589 ± 0.034	0.767 ± 0.025	0.062	0.002	0.0412	0.0412	0.892	93.602 ± 0.008	0.161 ± 0.002	42.415 ± 0.016
QML	beta41	0.20	0.218	0.018 ± 0.002	0.089 ± 0.012	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.155 ± 0.007	0.394 ± 0.011	0.218	0.018	0.0778	0.0797	1.027	95.253 ± 0.007	0.309 ± 0.001	79.876 ± 0.013
	beta42	0.20	0.211	0.011 ± 0.003	0.055 ± 0.013	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.157 ± 0.007	0.396 ± 0.011	0.212	0.012	0.0781	0.0791	1.026	95.150 ± 0.007	0.316 ± 0.001	75.955 ± 0.014
	beta43	0.20	0.190	-0.010 ± 0.002	-0.051 ± 0.010	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.103 ± 0.005	0.321 ± 0.009	0.192	-0.008	0.0619	0.0624	0.992	93.808 ± 0.008	0.247 ± 0.001	85.759 ± 0.011
	beta44	0.11	0.070	-0.040 ± 0.003	-0.363 ± 0.023	0.006 ± 0.000	0.079 ± 0.002	0.008 ± 0.000	0.089 ± 0.002	0.653 ± 0.030	0.808 ± 0.021	0.071	-0.039	0.0758	0.0850	1.000	92.260 ± 0.009	0.312 ± 0.002	13.829 ± 0.011
	beta45	0.11	0.106	-0.004 ± 0.002	-0.034 ± 0.021	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.430 ± 0.020	0.656 ± 0.018	0.107	-0.003	0.0720	0.0720	0.977	94.840 ± 0.007	0.276 ± 0.001	35.088 ± 0.015
	beta46	0.08	0.115	0.035 ± 0.002	0.433 ± 0.019	0.002 ± 0.000	0.047 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.538 ± 0.026	0.733 ± 0.019	0.113	0.033	0.0437	0.0549	0.961	89.680 ± 0.010	0.178 ± 0.002	74.200 ± 0.014
	beta47	0.08	0.115	0.035 ± 0.001	0.440 ± 0.018	0.002 ± 0.000	0.045 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.503 ± 0.025	0.709 ± 0.019	0.114	0.034	0.0435	0.0550	0.979	89.474 ± 0.010	0.171 ± 0.002	77.193 ± 0.013
	beta51	0.16	0.175	0.015 ± 0.003	0.096 ± 0.016	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.264 ± 0.012	0.514 ± 0.014	0.177	0.017	0.0847	0.0863	1.001	95.769 ± 0.006	0.317 ± 0.001	59.030 ± 0.016
	beta52	0.16	0.147	-0.013 ± 0.003	-0.083 ± 0.017	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	0.278 ± 0.012	0.527 ± 0.014	0.145	-0.015	0.0833	0.0846	0.983	95.046 ± 0.007	0.321 ± 0.001	42.828 ± 0.016
	beta53	0.16	0.167	0.007 ± 0.003	0.043 ± 0.016	0.006 ± 0.000	0.080 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.255 ± 0.012	0.505 ± 0.014	0.170	0.010	0.0818	0.0824	0.966	94.118 ± 0.008	0.305 ± 0.001	58.411 ± 0.016
	beta54	0.16	0.177	0.017 ± 0.002	0.104 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.205 ± 0.009	0.453 ± 0.012	0.177	0.017	0.0689	0.0709	1.009	95.356 ± 0.007	0.279 ± 0.001	69.866 ± 0.015
	beta55	0.08	0.085	0.005 ± 0.002	0.065 ± 0.030	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.889 ± 0.044	0.943 ± 0.028	0.085	0.005	0.0698	0.0700	0.928	93.602 ± 0.008	0.274 ± 0.002	27.554 ± 0.014
	beta56	0.08	0.095	0.015 ± 0.002	0.192 ± 0.020	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.424 ± 0.021	0.651 ± 0.019	0.097	0.017	0.0466	0.0497	0.933	93.602 ± 0.008	0.182 ± 0.002	57.585 ± 0.016
	beta57	0.06	0.088	0.028 ± 0.002	0.467 ± 0.037	0.005 ± 0.000	0.069 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	1.541 ± 0.079	1.241 ± 0.036	0.088	0.028	0.0630	0.0689	0.932	91.744 ± 0.009	0.252 ± 0.002	29.618 ± 0.015
	beta58	0.06	0.083	0.023 ± 0.001	0.385 ± 0.021	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.579 ± 0.027	0.761 ± 0.020	0.082	0.022	0.0369	0.0432	0.953	91.744 ± 0.009	0.147 ± 0.001	63.571 ± 0.015

Table S47

Simulation 2 - Condition 9: Full Model, Distribution = right-skewed, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta41	0.20	0.213	0.013 ± 0.004	0.067 ± 0.019	0.013 ± 0.001	0.116 ± 0.003	0.014 ± 0.001	0.116 ± 0.004	0.338 ± 0.017	0.581 ± 0.018	0.216	0.016	0.1093	0.1106	0.957	94.737 ± 0.007	0.433 ± 0.002	51.393 ± 0.016
	beta42	0.20	0.206	0.006 ± 0.004	0.031 ± 0.019	0.014 ± 0.001	0.119 ± 0.003	0.014 ± 0.001	0.119 ± 0.003	0.357 ± 0.017	0.597 ± 0.017	0.206	0.006	0.1195	0.1197	0.961	95.150 ± 0.007	0.450 ± 0.003	48.194 ± 0.016
	beta43	0.20	0.196	-0.004 ± 0.002	-0.020 ± 0.011	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.117 ± 0.006	0.343 ± 0.010	0.198	-0.002	0.0650	0.0651	0.940	93.498 ± 0.008	0.252 ± 0.001	83.695 ± 0.012
	beta44	0.11	0.113	0.003 ± 0.003	0.023 ± 0.026	0.008 ± 0.000	0.090 ± 0.003	0.008 ± 0.000	0.090 ± 0.003	0.674 ± 0.039	0.821 ± 0.027	0.112	0.002	0.0832	0.0832	0.897	93.086 ± 0.008	0.318 ± 0.003	33.540 ± 0.015
	beta45	0.11	0.120	0.010 ± 0.003	0.089 ± 0.028	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.097 ± 0.003	0.772 ± 0.047	0.879 ± 0.030	0.114	0.004	0.0840	0.0841	0.809	90.299 ± 0.010	0.305 ± 0.004	36.739 ± 0.015
	beta46	0.08	0.078	-0.002 ± 0.002	-0.024 ± 0.021	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.419 ± 0.023	0.648 ± 0.020	0.078	-0.002	0.0457	0.0457	0.859	92.570 ± 0.008	0.174 ± 0.002	48.400 ± 0.016
	beta47	0.08	0.079	-0.001 ± 0.002	-0.014 ± 0.020	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	0.371 ± 0.020	0.609 ± 0.019	0.078	-0.002	0.0443	0.0443	0.882	93.395 ± 0.008	0.169 ± 0.002	53.148 ± 0.016
	beta51	0.16	0.203	0.043 ± 0.004	0.266 ± 0.024	0.015 ± 0.001	0.121 ± 0.003	0.017 ± 0.001	0.128 ± 0.004	0.645 ± 0.031	0.803 ± 0.022	0.203	0.043	0.1184	0.1259	0.920	92.260 ± 0.009	0.438 ± 0.002	47.368 ± 0.016
	beta52	0.16	0.182	0.022 ± 0.003	0.138 ± 0.018	0.008 ± 0.000	0.090 ± 0.002	0.009 ± 0.000	0.093 ± 0.003	0.336 ± 0.016	0.579 ± 0.016	0.182	0.022	0.0867	0.0895	0.980	93.602 ± 0.008	0.346 ± 0.002	55.521 ± 0.016
	beta53	0.16	0.161	0.001 ± 0.004	0.008 ± 0.024	0.014 ± 0.001	0.117 ± 0.003	0.014 ± 0.001	0.117 ± 0.003	0.537 ± 0.026	0.733 ± 0.021	0.165	0.005	0.1163	0.1165	0.940	94.014 ± 0.008	0.432 ± 0.002	34.572 ± 0.015
	beta54	0.16	0.164	0.004 ± 0.002	0.023 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.189 ± 0.008	0.435 ± 0.012	0.161	0.001	0.0696	0.0696	0.988	94.943 ± 0.007	0.269 ± 0.001	66.254 ± 0.015
	beta55	0.08	0.085	0.005 ± 0.002	0.061 ± 0.031	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.078 ± 0.003	0.941 ± 0.053	0.970 ± 0.032	0.084	0.004	0.0699	0.0700	0.866	93.602 ± 0.008	0.263 ± 0.002	29.721 ± 0.015
	beta56	0.08	0.079	-0.001 ± 0.002	-0.010 ± 0.020	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.406 ± 0.023	0.637 ± 0.021	0.079	-0.001	0.0439	0.0439	0.875	92.673 ± 0.008	0.175 ± 0.002	48.504 ± 0.016
	beta57	0.06	0.054	-0.006 ± 0.002	-0.093 ± 0.042	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.003	1.679 ± 0.094	1.296 ± 0.042	0.053	-0.007	0.0717	0.0720	0.856	92.054 ± 0.009	0.260 ± 0.003	16.925 ± 0.012
	beta58	0.06	0.061	0.001 ± 0.001	0.012 ± 0.025	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.590 ± 0.033	0.768 ± 0.025	0.059	-0.001	0.0389	0.0389	0.872	92.570 ± 0.008	0.157 ± 0.002	41.073 ± 0.016

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S48

Simulation 2 - Condition 10: Full Model, Distribution = right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.002	0.006 ± 0.012	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.135 ± 0.006	0.368 ± 0.010	0.199	-0.001	0.0738	0.0738	0.963	94.737 ± 0.007	0.278 ± 0.001	80.162 ± 0.013
	beta42	0.20	0.196	-0.004 ± 0.002	-0.020 ± 0.012	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.135 ± 0.007	0.367 ± 0.011	0.195	-0.005	0.0705	0.0706	1.002	95.850 ± 0.006	0.288 ± 0.001	76.721 ± 0.013
	beta43	0.20	0.201	0.001 ± 0.001	0.005 ± 0.007	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.042 ± 0.002	0.206 ± 0.006	0.201	0.001	0.0414	0.0414	1.012	95.445 ± 0.007	0.163 ± 0.000	99.798 ± 0.001
	beta44	0.11	0.111	0.001 ± 0.002	0.012 ± 0.015	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.211 ± 0.011	0.460 ± 0.014	0.111	0.001	0.0477	0.0477	0.967	94.028 ± 0.008	0.192 ± 0.001	64.069 ± 0.015
	beta45	0.11	0.113	0.003 ± 0.002	0.026 ± 0.014	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.188 ± 0.009	0.433 ± 0.012	0.114	0.004	0.0474	0.0476	0.971	94.534 ± 0.007	0.181 ± 0.001	68.623 ± 0.015
	beta46	0.08	0.081	0.001 ± 0.001	0.019 ± 0.011	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.125 ± 0.006	0.353 ± 0.010	0.081	0.001	0.0264	0.0264	0.941	94.231 ± 0.007	0.104 ± 0.001	85.020 ± 0.011
	beta47	0.08	0.081	0.001 ± 0.001	0.015 ± 0.011	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.120 ± 0.006	0.346 ± 0.011	0.081	0.001	0.0270	0.0270	0.927	93.117 ± 0.008	0.101 ± 0.001	85.931 ± 0.011
	beta51	0.16	0.185	0.025 ± 0.002	0.159 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.224 ± 0.010	0.473 ± 0.013	0.184	0.024	0.0713	0.0754	1.005	93.725 ± 0.008	0.281 ± 0.001	73.684 ± 0.014
	beta52	0.16	0.184	0.024 ± 0.002	0.151 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.136 ± 0.006	0.369 ± 0.010	0.185	0.025	0.0521	0.0577	1.028	94.636 ± 0.007	0.217 ± 0.001	91.802 ± 0.009
	beta53	0.16	0.161	0.001 ± 0.002	0.007 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.199 ± 0.010	0.446 ± 0.013	0.161	0.001	0.0695	0.0695	0.955	94.534 ± 0.007	0.267 ± 0.001	64.777 ± 0.015
	beta54	0.16	0.159	-0.001 ± 0.001	-0.005 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.067 ± 0.003	0.258 ± 0.007	0.159	-0.001	0.0406	0.0406	1.057	96.356 ± 0.006	0.171 ± 0.000	95.445 ± 0.007
	beta55	0.08	0.081	0.001 ± 0.001	0.007 ± 0.016	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.240 ± 0.011	0.490 ± 0.014	0.079	-0.001	0.0384	0.0384	0.994	94.737 ± 0.007	0.153 ± 0.001	55.061 ± 0.016
	beta56	0.08	0.080	-0.000 ± 0.001	-0.002 ± 0.010	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	0.106 ± 0.006	0.325 ± 0.010	0.080	0.000	0.0239	0.0239	0.947	93.219 ± 0.008	0.097 ± 0.001	86.437 ± 0.011
	beta57	0.06	0.061	0.001 ± 0.001	0.020 ± 0.022	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.476 ± 0.023	0.690 ± 0.020	0.060	0.000	0.0391	0.0391	0.994	94.636 ± 0.007	0.161 ± 0.001	34.008 ± 0.015
	beta58	0.06	0.060	-0.000 ± 0.001	-0.004 ± 0.013	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	0.161 ± 0.008	0.401 ± 0.012	0.060	0.000	0.0232	0.0232	0.968	95.445 ± 0.007	0.091 ± 0.001	75.202 ± 0.014
QML	beta41	0.20	0.217	0.017 ± 0.002	0.085 ± 0.008	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.070 ± 0.003	0.265 ± 0.007	0.216	0.016	0.0516	0.0540	0.981	93.623 ± 0.008	0.193 ± 0.000	99.393 ± 0.002
	beta42	0.20	0.210	0.010 ± 0.002	0.049 ± 0.008	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.064 ± 0.003	0.252 ± 0.007	0.209	0.009	0.0481	0.0489	1.015	94.939 ± 0.007	0.197 ± 0.000	98.684 ± 0.004
	beta43	0.20	0.192	-0.008 ± 0.001	-0.040 ± 0.006	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.040 ± 0.002	0.199 ± 0.005	0.192	-0.008	0.0391	0.0399	1.008	93.421 ± 0.008	0.154 ± 0.000	99.798 ± 0.001
	beta44	0.11	0.072	-0.038 ± 0.002	-0.341 ± 0.014	0.002 ± 0.000	0.050 ± 0.001	0.004 ± 0.000	0.062 ± 0.001	0.321 ± 0.014	0.566 ± 0.013	0.074	-0.036	0.0499	0.0617	0.966	86.741 ± 0.011	0.188 ± 0.001	35.020 ± 0.015
	beta45	0.11	0.103	-0.007 ± 0.001	-0.063 ± 0.013	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.161 ± 0.008	0.402 ± 0.011	0.105	-0.005	0.0442	0.0445	0.990	94.332 ± 0.007	0.169 ± 0.001	66.802 ± 0.015
	beta46	0.08	0.115	0.035 ± 0.001	0.442 ± 0.011	0.001 ± 0.000	0.027 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.311 ± 0.012	0.557 ± 0.011	0.113	0.033	0.0253	0.0420	0.991	76.417 ± 0.014	0.106 ± 0.001	98.279 ± 0.004
	beta47	0.08	0.115	0.035 ± 0.001	0.436 ± 0.011	0.001 ± 0.000	0.027 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.304 ± 0.011	0.552 ± 0.011	0.114	0.034	0.0268	0.0430	0.959	73.178 ± 0.014	0.102 ± 0.001	98.988 ± 0.003
	beta51	0.16	0.169	0.009 ± 0.002	0.058 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.101 ± 0.005	0.317 ± 0.009	0.169	0.009	0.0502	0.0511	1.000	94.838 ± 0.007	0.196 ± 0.000	92.915 ± 0.008
	beta52	0.16	0.148	-0.012 ± 0.002	-0.076 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.099 ± 0.004	0.314 ± 0.009	0.148	-0.012	0.0488	0.0503	1.039	94.636 ± 0.007	0.199 ± 0.000	84.312 ± 0.012
	beta53	0.16	0.167	0.007 ± 0.002	0.043 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.106 ± 0.005	0.325 ± 0.009	0.166	0.006	0.0501	0.0505	0.943	93.421 ± 0.008	0.191 ± 0.000	90.992 ± 0.009
	beta54	0.16	0.173	0.013 ± 0.001	0.084 ± 0.009	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.079 ± 0.003	0.281 ± 0.007	0.174	0.014	0.0433	0.0454	1.040	95.243 ± 0.007	0.175 ± 0.000	97.166 ± 0.005
	beta55	0.08	0.081	0.001 ± 0.001	0.009 ± 0.018	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.333 ± 0.015	0.577 ± 0.016	0.079	-0.001	0.0473	0.0473	0.926	94.130 ± 0.007	0.168 ± 0.001	47.976 ± 0.016
	beta56	0.08	0.097	0.017 ± 0.001	0.215 ± 0.011	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.168 ± 0.008	0.410 ± 0.011	0.097	0.017	0.0270	0.0321	0.972	89.372 ± 0.010	0.107 ± 0.001	93.623 ± 0.008
	beta57	0.06	0.095	0.035 ± 0.001	0.585 ± 0.023	0.002 ± 0.000	0.044 ± 0.001	0.003 ± 0.000	0.056 ± 0.001	0.873 ± 0.037	0.934 ± 0.022	0.096	0.036	0.0412	0.0546	0.930	85.020 ± 0.011	0.159 ± 0.001	65.789 ± 0.015
	beta58	0.06	0.083	0.023 ± 0.001	0.384 ± 0.012	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.291 ± 0.012	0.539 ± 0.011	0.083	0.023	0.0221	0.0318	0.996	81.984 ± 0.012	0.089 ± 0.001	94.433 ± 0.007

Table S48

Simulation 2 - Condition 10: Full Model, Distribution = right-skewed, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta41	0.20	0.208	0.008 ± 0.002	0.039 ± 0.012	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.135 ± 0.006	0.367 ± 0.011	0.207	0.007	0.0718	0.0722	0.918	93.421 ± 0.008	0.263 ± 0.001	85.324 ± 0.011
	beta42	0.20	0.199	-0.001 ± 0.002	-0.005 ± 0.011	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.130 ± 0.006	0.361 ± 0.010	0.198	-0.002	0.0713	0.0713	0.961	94.737 ± 0.007	0.272 ± 0.001	81.680 ± 0.012
	beta43	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.007	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.042 ± 0.002	0.205 ± 0.006	0.199	-0.001	0.0407	0.0407	0.964	93.725 ± 0.008	0.155 ± 0.000	99.798 ± 0.001
	beta44	0.11	0.111	0.001 ± 0.002	0.008 ± 0.015	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.228 ± 0.012	0.478 ± 0.014	0.109	-0.001	0.0515	0.0515	0.861	91.599 ± 0.009	0.177 ± 0.001	68.826 ± 0.015
	beta45	0.11	0.112	0.002 ± 0.002	0.020 ± 0.014	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	0.208 ± 0.011	0.456 ± 0.014	0.111	0.001	0.0491	0.0491	0.878	92.510 ± 0.008	0.172 ± 0.001	71.862 ± 0.014
	beta46	0.08	0.079	-0.001 ± 0.001	-0.011 ± 0.012	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.131 ± 0.007	0.362 ± 0.011	0.078	-0.002	0.0259	0.0260	0.841	90.587 ± 0.009	0.096 ± 0.001	86.538 ± 0.011
	beta47	0.08	0.080	0.000 ± 0.001	0.004 ± 0.011	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.119 ± 0.006	0.345 ± 0.011	0.079	-0.001	0.0270	0.0270	0.862	92.004 ± 0.009	0.093 ± 0.001	88.664 ± 0.010
	beta51	0.16	0.191	0.031 ± 0.002	0.193 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.231 ± 0.010	0.481 ± 0.012	0.192	0.032	0.0700	0.0772	0.970	92.308 ± 0.008	0.268 ± 0.001	78.644 ± 0.013
	beta52	0.16	0.184	0.024 ± 0.002	0.149 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.133 ± 0.006	0.365 ± 0.010	0.183	0.023	0.0500	0.0550	1.008	92.611 ± 0.008	0.211 ± 0.001	92.713 ± 0.008
	beta53	0.16	0.162	0.002 ± 0.002	0.011 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.196 ± 0.009	0.443 ± 0.013	0.162	0.002	0.0691	0.0692	0.947	93.725 ± 0.008	0.263 ± 0.001	66.296 ± 0.015
	beta54	0.16	0.161	0.001 ± 0.001	0.005 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.067 ± 0.003	0.259 ± 0.007	0.160	0.000	0.0410	0.0410	1.042	95.445 ± 0.007	0.169 ± 0.000	96.053 ± 0.006
	beta55	0.08	0.082	0.002 ± 0.001	0.021 ± 0.016	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.242 ± 0.012	0.492 ± 0.014	0.080	0.000	0.0377	0.0377	0.942	93.725 ± 0.008	0.145 ± 0.001	60.628 ± 0.016
	beta56	0.08	0.079	-0.001 ± 0.001	-0.009 ± 0.011	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	0.111 ± 0.006	0.333 ± 0.010	0.079	-0.001	0.0240	0.0240	0.923	93.927 ± 0.008	0.096 ± 0.001	86.235 ± 0.011
	beta57	0.06	0.059	-0.001 ± 0.001	-0.022 ± 0.022	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.480 ± 0.024	0.693 ± 0.021	0.057	-0.003	0.0397	0.0399	0.917	93.117 ± 0.008	0.149 ± 0.001	35.020 ± 0.015
	beta58	0.06	0.059	-0.001 ± 0.001	-0.008 ± 0.013	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	0.164 ± 0.008	0.405 ± 0.012	0.060	0.000	0.0232	0.0232	0.943	94.433 ± 0.007	0.090 ± 0.001	75.607 ± 0.014

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S49

Simulation 2 - Condition 11: Full Model, Distribution = right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.203	0.003 ± 0.003	0.015 ± 0.013	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	0.175 ± 0.008	0.418 ± 0.012	0.203	0.003	0.0819	0.0820	1.040	96.061 ± 0.006	0.341 ± 0.001	67.273 ± 0.015
	beta42	0.20	0.201	0.001 ± 0.003	0.003 ± 0.015	0.008 ± 0.000	0.091 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	0.208 ± 0.009	0.456 ± 0.013	0.205	0.005	0.0870	0.0871	0.995	95.354 ± 0.007	0.356 ± 0.002	61.818 ± 0.015
	beta43	0.20	0.198	-0.002 ± 0.002	-0.009 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.073 ± 0.003	0.270 ± 0.007	0.199	-0.001	0.0543	0.0543	0.991	94.646 ± 0.007	0.210 ± 0.001	95.556 ± 0.007
	beta44	0.11	0.115	0.005 ± 0.002	0.042 ± 0.019	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.349 ± 0.017	0.591 ± 0.017	0.115	0.005	0.0607	0.0609	0.905	92.222 ± 0.009	0.230 ± 0.002	51.818 ± 0.016
	beta45	0.11	0.112	0.002 ± 0.002	0.016 ± 0.018	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.325 ± 0.016	0.570 ± 0.017	0.110	0.000	0.0615	0.0615	0.889	91.212 ± 0.009	0.219 ± 0.002	52.222 ± 0.016
	beta46	0.08	0.080	-0.000 ± 0.001	-0.003 ± 0.014	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.192 ± 0.009	0.438 ± 0.013	0.080	0.000	0.0313	0.0313	0.947	94.141 ± 0.007	0.130 ± 0.001	69.293 ± 0.015
	beta47	0.08	0.080	-0.000 ± 0.001	-0.001 ± 0.015	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.213 ± 0.011	0.461 ± 0.014	0.078	-0.002	0.0336	0.0337	0.889	92.424 ± 0.008	0.129 ± 0.001	67.374 ± 0.015
	beta51	0.16	0.188	0.028 ± 0.003	0.172 ± 0.017	0.007 ± 0.000	0.083 ± 0.002	0.008 ± 0.000	0.088 ± 0.002	0.300 ± 0.014	0.548 ± 0.015	0.187	0.027	0.0832	0.0875	1.042	95.455 ± 0.007	0.340 ± 0.001	60.303 ± 0.016
	beta52	0.16	0.183	0.023 ± 0.002	0.147 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.216 ± 0.010	0.464 ± 0.013	0.184	0.024	0.0703	0.0744	0.993	93.131 ± 0.008	0.274 ± 0.001	75.253 ± 0.014
	beta53	0.16	0.158	-0.002 ± 0.003	-0.011 ± 0.016	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.256 ± 0.011	0.506 ± 0.014	0.156	-0.004	0.0776	0.0777	1.038	95.657 ± 0.006	0.329 ± 0.001	47.576 ± 0.016
	beta54	0.16	0.157	-0.003 ± 0.002	-0.020 ± 0.011	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.126 ± 0.006	0.355 ± 0.010	0.156	-0.004	0.0565	0.0567	1.002	95.152 ± 0.007	0.223 ± 0.001	78.586 ± 0.013
	beta55	0.08	0.079	-0.001 ± 0.002	-0.015 ± 0.021	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.436 ± 0.021	0.660 ± 0.019	0.078	-0.002	0.0531	0.0531	0.931	92.828 ± 0.008	0.193 ± 0.001	38.485 ± 0.015
	beta56	0.08	0.083	0.003 ± 0.001	0.037 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.204 ± 0.011	0.452 ± 0.014	0.082	0.002	0.0322	0.0322	0.936	93.232 ± 0.008	0.132 ± 0.001	67.778 ± 0.015
	beta57	0.06	0.061	0.001 ± 0.002	0.019 ± 0.027	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.731 ± 0.036	0.855 ± 0.025	0.061	0.001	0.0463	0.0463	0.953	93.232 ± 0.008	0.192 ± 0.002	26.768 ± 0.014
	beta58	0.06	0.062	0.002 ± 0.001	0.036 ± 0.016	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.266 ± 0.014	0.516 ± 0.016	0.061	0.001	0.0277	0.0277	0.953	94.444 ± 0.007	0.115 ± 0.001	60.202 ± 0.016
QML	beta41	0.20	0.209	0.009 ± 0.002	0.047 ± 0.011	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.121 ± 0.005	0.348 ± 0.009	0.209	0.009	0.0693	0.0698	1.050	95.859 ± 0.006	0.284 ± 0.001	82.727 ± 0.012
	beta42	0.20	0.205	0.005 ± 0.002	0.027 ± 0.012	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.138 ± 0.006	0.372 ± 0.010	0.209	0.009	0.0732	0.0737	0.997	95.051 ± 0.007	0.290 ± 0.001	77.475 ± 0.013
	beta43	0.20	0.194	-0.006 ± 0.002	-0.029 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.070 ± 0.003	0.264 ± 0.007	0.194	-0.006	0.0531	0.0535	1.002	95.051 ± 0.007	0.206 ± 0.001	95.354 ± 0.007
	beta44	0.11	0.099	-0.011 ± 0.002	-0.104 ± 0.018	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.334 ± 0.016	0.578 ± 0.017	0.100	-0.010	0.0607	0.0616	0.965	94.343 ± 0.007	0.237 ± 0.001	38.384 ± 0.015
	beta45	0.11	0.106	-0.004 ± 0.002	-0.037 ± 0.017	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.290 ± 0.013	0.538 ± 0.015	0.106	-0.004	0.0592	0.0594	0.946	94.444 ± 0.007	0.219 ± 0.001	47.475 ± 0.016
	beta46	0.08	0.091	0.011 ± 0.001	0.141 ± 0.013	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.193 ± 0.009	0.440 ± 0.012	0.091	0.011	0.0297	0.0317	1.013	95.152 ± 0.007	0.132 ± 0.001	78.283 ± 0.013
	beta47	0.08	0.092	0.012 ± 0.001	0.147 ± 0.014	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.203 ± 0.011	0.450 ± 0.014	0.089	0.009	0.0309	0.0323	0.964	93.030 ± 0.008	0.129 ± 0.001	78.990 ± 0.013
	beta51	0.16	0.171	0.011 ± 0.002	0.071 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.191 ± 0.009	0.437 ± 0.012	0.170	0.010	0.0709	0.0717	1.038	95.859 ± 0.006	0.281 ± 0.001	67.374 ± 0.015
	beta52	0.16	0.159	-0.001 ± 0.002	-0.009 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.183 ± 0.008	0.428 ± 0.012	0.157	-0.003	0.0682	0.0682	1.006	95.657 ± 0.006	0.270 ± 0.001	63.636 ± 0.015
	beta53	0.16	0.161	0.001 ± 0.002	0.007 ± 0.014	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.181 ± 0.008	0.426 ± 0.012	0.157	-0.003	0.0652	0.0653	1.039	95.556 ± 0.007	0.278 ± 0.001	62.525 ± 0.015
	beta54	0.16	0.164	0.004 ± 0.002	0.024 ± 0.011	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.128 ± 0.006	0.358 ± 0.010	0.164	0.004	0.0570	0.0571	1.000	95.051 ± 0.007	0.225 ± 0.000	82.929 ± 0.012
	beta55	0.08	0.080	-0.000 ± 0.002	-0.000 ± 0.021	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.436 ± 0.021	0.660 ± 0.019	0.079	-0.001	0.0515	0.0515	0.974	94.141 ± 0.007	0.202 ± 0.001	36.465 ± 0.015
	beta56	0.08	0.089	0.009 ± 0.001	0.111 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.213 ± 0.011	0.461 ± 0.014	0.087	0.007	0.0330	0.0338	1.007	95.758 ± 0.006	0.141 ± 0.001	69.192 ± 0.015
	beta57	0.06	0.072	0.012 ± 0.002	0.194 ± 0.025	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.680 ± 0.032	0.825 ± 0.023	0.073	0.013	0.0446	0.0465	0.988	94.444 ± 0.007	0.186 ± 0.001	34.040 ± 0.015
	beta58	0.06	0.070	0.010 ± 0.001	0.174 ± 0.015	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.265 ± 0.012	0.515 ± 0.014	0.069	0.009	0.0272	0.0288	1.004	94.040 ± 0.008	0.114 ± 0.001	70.909 ± 0.014

Table S49

Simulation 2 - Condition 11: Full Model, Distribution = right-skewed, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta41	0.20	0.209	0.009 ± 0.003	0.047 ± 0.013	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	0.175 ± 0.008	0.419 ± 0.012	0.209	0.009	0.0797	0.0802	1.014	95.253 ± 0.007	0.331 ± 0.001	71.919 ± 0.014
	beta42	0.20	0.203	0.003 ± 0.003	0.014 ± 0.014	0.008 ± 0.000	0.091 ± 0.002	0.008 ± 0.000	0.091 ± 0.002	0.207 ± 0.009	0.455 ± 0.012	0.206	0.006	0.0939	0.0941	0.963	94.848 ± 0.007	0.343 ± 0.001	64.242 ± 0.015
	beta43	0.20	0.197	-0.003 ± 0.002	-0.015 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.074 ± 0.003	0.271 ± 0.007	0.198	-0.002	0.0548	0.0548	0.973	94.242 ± 0.007	0.207 ± 0.001	95.152 ± 0.007
	beta44	0.11	0.112	0.002 ± 0.002	0.017 ± 0.019	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.355 ± 0.017	0.596 ± 0.017	0.111	0.001	0.0646	0.0646	0.897	92.929 ± 0.008	0.230 ± 0.002	50.909 ± 0.016
	beta45	0.11	0.112	0.002 ± 0.002	0.016 ± 0.019	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.355 ± 0.018	0.596 ± 0.018	0.109	-0.001	0.0627	0.0627	0.865	92.020 ± 0.009	0.222 ± 0.002	51.212 ± 0.016
	beta46	0.08	0.078	-0.002 ± 0.001	-0.030 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.198 ± 0.010	0.445 ± 0.013	0.078	-0.002	0.0325	0.0326	0.945	93.434 ± 0.008	0.132 ± 0.001	66.364 ± 0.015
	beta47	0.08	0.079	-0.001 ± 0.001	-0.011 ± 0.015	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.218 ± 0.012	0.467 ± 0.015	0.077	-0.003	0.0330	0.0331	0.883	93.333 ± 0.008	0.129 ± 0.001	66.768 ± 0.015
	beta51	0.16	0.192	0.032 ± 0.003	0.199 ± 0.016	0.007 ± 0.000	0.083 ± 0.002	0.008 ± 0.000	0.089 ± 0.002	0.307 ± 0.014	0.554 ± 0.015	0.192	0.032	0.0834	0.0895	1.023	94.444 ± 0.007	0.332 ± 0.001	64.141 ± 0.015
	beta52	0.16	0.183	0.023 ± 0.002	0.145 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.215 ± 0.010	0.464 ± 0.013	0.184	0.024	0.0699	0.0739	0.980	93.333 ± 0.008	0.271 ± 0.001	74.747 ± 0.014
	beta53	0.16	0.160	-0.000 ± 0.003	-0.003 ± 0.016	0.007 ± 0.000	0.081 ± 0.002	0.006 ± 0.000	0.081 ± 0.002	0.254 ± 0.011	0.504 ± 0.014	0.157	-0.003	0.0794	0.0794	1.028	95.455 ± 0.007	0.325 ± 0.001	48.283 ± 0.016
	beta54	0.16	0.158	-0.002 ± 0.002	-0.013 ± 0.011	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.126 ± 0.006	0.355 ± 0.010	0.157	-0.003	0.0560	0.0561	0.990	94.949 ± 0.007	0.221 ± 0.000	80.202 ± 0.013
	beta55	0.08	0.079	-0.001 ± 0.002	-0.009 ± 0.021	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.451 ± 0.021	0.672 ± 0.019	0.078	-0.002	0.0542	0.0542	0.934	93.333 ± 0.008	0.197 ± 0.001	36.869 ± 0.015
	beta56	0.08	0.083	0.003 ± 0.001	0.032 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.208 ± 0.011	0.456 ± 0.014	0.080	0.000	0.0324	0.0324	0.967	94.646 ± 0.007	0.138 ± 0.001	65.758 ± 0.015
	beta57	0.06	0.059	-0.001 ± 0.002	-0.010 ± 0.027	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.731 ± 0.036	0.855 ± 0.025	0.060	0.000	0.0477	0.0477	0.946	93.939 ± 0.008	0.190 ± 0.001	25.657 ± 0.014
	beta58	0.06	0.062	0.002 ± 0.001	0.030 ± 0.017	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.271 ± 0.014	0.520 ± 0.016	0.061	0.001	0.0275	0.0275	0.967	94.747 ± 0.007	0.118 ± 0.001	57.677 ± 0.016

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S50

Simulation 2 - Condition 12: Full Model, Distribution = right-skewed, N = 1000, Reliability = 0.8

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.002	-0.002 ± 0.008	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.069 ± 0.003	0.263 ± 0.008	0.200	0.000	0.0496	0.0496	1.004	95.080 ± 0.007	0.207 ± 0.001	95.582 ± 0.007
	beta42	0.20	0.201	0.001 ± 0.002	0.006 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.079 ± 0.004	0.281 ± 0.008	0.202	0.002	0.0543	0.0543	0.979	94.378 ± 0.007	0.216 ± 0.001	94.378 ± 0.007
	beta43	0.20	0.198	-0.002 ± 0.001	-0.009 ± 0.005	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.030 ± 0.001	0.172 ± 0.005	0.198	-0.002	0.0348	0.0348	0.983	94.679 ± 0.007	0.133 ± 0.000	100.000 ± 0.000
	beta44	0.11	0.112	0.002 ± 0.001	0.015 ± 0.011	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.114 ± 0.005	0.338 ± 0.010	0.111	0.001	0.0376	0.0376	0.940	93.976 ± 0.008	0.137 ± 0.001	86.948 ± 0.011
	beta45	0.11	0.112	0.002 ± 0.001	0.015 ± 0.010	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.103 ± 0.005	0.321 ± 0.009	0.110	0.000	0.0341	0.0341	0.956	93.675 ± 0.008	0.132 ± 0.001	89.458 ± 0.010
	beta46	0.08	0.080	0.000 ± 0.001	0.002 ± 0.008	0.000 ± 0.000	0.020 ± 0.001	0.000 ± 0.000	0.020 ± 0.001	0.063 ± 0.003	0.250 ± 0.007	0.080	0.000	0.0194	0.0194	0.934	93.373 ± 0.008	0.073 ± 0.001	97.189 ± 0.005
	beta47	0.08	0.080	-0.000 ± 0.001	-0.000 ± 0.008	0.000 ± 0.000	0.020 ± 0.001	0.000 ± 0.000	0.020 ± 0.001	0.063 ± 0.003	0.251 ± 0.008	0.080	0.000	0.0185	0.0185	0.925	92.470 ± 0.008	0.073 ± 0.001	96.687 ± 0.006
	beta51	0.16	0.182	0.022 ± 0.002	0.137 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.057 ± 0.001	0.129 ± 0.006	0.359 ± 0.009	0.181	0.021	0.0532	0.0573	1.003	93.474 ± 0.008	0.209 ± 0.000	91.867 ± 0.009
	beta52	0.16	0.187	0.027 ± 0.001	0.168 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.108 ± 0.005	0.328 ± 0.008	0.186	0.026	0.0452	0.0524	0.962	90.161 ± 0.009	0.170 ± 0.000	98.695 ± 0.004
	beta53	0.16	0.160	0.000 ± 0.002	0.002 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.104 ± 0.005	0.323 ± 0.009	0.160	0.000	0.0509	0.0509	1.000	94.578 ± 0.007	0.203 ± 0.001	86.546 ± 0.011
	beta54	0.16	0.161	0.001 ± 0.001	0.005 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.051 ± 0.002	0.227 ± 0.006	0.160	0.000	0.0378	0.0378	0.988	95.482 ± 0.007	0.140 ± 0.000	99.398 ± 0.002
	beta55	0.08	0.079	-0.001 ± 0.001	-0.012 ± 0.012	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.139 ± 0.006	0.373 ± 0.011	0.079	-0.001	0.0278	0.0278	0.939	93.173 ± 0.008	0.110 ± 0.001	78.414 ± 0.013
	beta56	0.08	0.080	0.000 ± 0.001	0.001 ± 0.008	0.000 ± 0.000	0.020 ± 0.001	0.000 ± 0.000	0.020 ± 0.001	0.065 ± 0.003	0.256 ± 0.008	0.079	-0.001	0.0194	0.0194	0.918	92.871 ± 0.008	0.074 ± 0.001	96.285 ± 0.006
	beta57	0.06	0.062	0.002 ± 0.001	0.030 ± 0.016	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.254 ± 0.011	0.504 ± 0.014	0.062	0.002	0.0315	0.0316	0.960	94.378 ± 0.007	0.114 ± 0.001	57.831 ± 0.016
	beta58	0.06	0.061	0.001 ± 0.001	0.013 ± 0.010	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	0.090 ± 0.005	0.301 ± 0.009	0.061	0.001	0.0164	0.0164	0.962	92.871 ± 0.008	0.068 ± 0.001	91.064 ± 0.009
QML	beta41	0.20	0.206	0.006 ± 0.001	0.028 ± 0.007	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.050 ± 0.002	0.223 ± 0.006	0.205	0.005	0.0421	0.0424	1.019	94.880 ± 0.007	0.177 ± 0.000	99.799 ± 0.001
	beta42	0.20	0.205	0.005 ± 0.001	0.027 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.003	0.236 ± 0.007	0.205	0.005	0.0460	0.0463	0.987	94.478 ± 0.007	0.181 ± 0.000	98.996 ± 0.003
	beta43	0.20	0.195	-0.005 ± 0.001	-0.027 ± 0.005	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.029 ± 0.001	0.171 ± 0.005	0.195	-0.005	0.0342	0.0346	0.978	93.574 ± 0.008	0.130 ± 0.000	100.000 ± 0.000
	beta44	0.11	0.098	-0.012 ± 0.001	-0.108 ± 0.011	0.001 ± 0.000	0.037 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.125 ± 0.006	0.353 ± 0.009	0.098	-0.012	0.0373	0.0390	0.976	93.072 ± 0.008	0.142 ± 0.000	76.104 ± 0.014
	beta45	0.11	0.108	-0.002 ± 0.001	-0.015 ± 0.010	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.097 ± 0.005	0.312 ± 0.009	0.109	-0.001	0.0334	0.0334	0.990	95.382 ± 0.007	0.133 ± 0.000	88.855 ± 0.010
	beta46	0.08	0.092	0.012 ± 0.001	0.153 ± 0.008	0.000 ± 0.000	0.020 ± 0.000	0.001 ± 0.000	0.023 ± 0.001	0.083 ± 0.004	0.288 ± 0.008	0.092	0.012	0.0189	0.0224	0.996	91.968 ± 0.009	0.076 ± 0.000	99.398 ± 0.002
	beta47	0.08	0.093	0.013 ± 0.001	0.156 ± 0.008	0.000 ± 0.000	0.019 ± 0.000	0.001 ± 0.000	0.023 ± 0.001	0.083 ± 0.004	0.287 ± 0.007	0.092	0.012	0.0178	0.0218	0.993	90.261 ± 0.009	0.075 ± 0.000	99.197 ± 0.003
	beta51	0.16	0.165	0.005 ± 0.001	0.028 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.076 ± 0.003	0.275 ± 0.007	0.164	0.004	0.0443	0.0445	1.014	95.181 ± 0.007	0.174 ± 0.000	95.482 ± 0.007
	beta52	0.16	0.162	0.002 ± 0.001	0.011 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.076 ± 0.003	0.276 ± 0.008	0.161	0.001	0.0435	0.0435	0.973	93.876 ± 0.008	0.169 ± 0.000	95.783 ± 0.006
	beta53	0.16	0.162	0.002 ± 0.001	0.012 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.077 ± 0.004	0.277 ± 0.008	0.162	0.002	0.0425	0.0425	1.003	94.578 ± 0.007	0.174 ± 0.000	94.880 ± 0.007
	beta54	0.16	0.168	0.008 ± 0.001	0.047 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.054 ± 0.002	0.233 ± 0.006	0.166	0.006	0.0380	0.0385	0.987	94.779 ± 0.007	0.141 ± 0.000	99.398 ± 0.002
	beta55	0.08	0.081	0.001 ± 0.001	0.014 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.149 ± 0.007	0.386 ± 0.011	0.082	0.002	0.0294	0.0294	0.965	95.281 ± 0.007	0.117 ± 0.001	76.205 ± 0.013
	beta56	0.08	0.086	0.006 ± 0.001	0.080 ± 0.008	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	0.074 ± 0.004	0.272 ± 0.008	0.086	0.006	0.0197	0.0206	0.986	94.076 ± 0.007	0.080 ± 0.001	97.390 ± 0.005
	beta57	0.06	0.073	0.013 ± 0.001	0.211 ± 0.016	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.284 ± 0.013	0.533 ± 0.014	0.073	0.013	0.0308	0.0333	0.981	92.470 ± 0.008	0.113 ± 0.000	70.984 ± 0.014
	beta58	0.06	0.070	0.010 ± 0.001	0.162 ± 0.009	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.020 ± 0.001	0.111 ± 0.005	0.333 ± 0.009	0.069	0.009	0.0161	0.0186	1.000	92.369 ± 0.008	0.068 ± 0.000	96.386 ± 0.006

Table S50

Simulation 2 - Condition 12: Full Model, Distribution = right-skewed, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta41	0.20	0.202	0.002 ± 0.002	0.011 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.068 ± 0.003	0.260 ± 0.007	0.202	0.002	0.0480	0.0480	0.991	94.478 ± 0.007	0.202 ± 0.001	96.888 ± 0.006
	beta42	0.20	0.204	0.004 ± 0.002	0.018 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.078 ± 0.004	0.280 ± 0.008	0.203	0.003	0.0558	0.0558	0.962	93.775 ± 0.008	0.211 ± 0.001	95.482 ± 0.007
	beta43	0.20	0.197	-0.003 ± 0.001	-0.013 ± 0.005	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.030 ± 0.001	0.172 ± 0.005	0.197	-0.003	0.0349	0.0350	0.963	93.775 ± 0.008	0.130 ± 0.000	100.000 ± 0.000
	beta44	0.11	0.111	0.001 ± 0.001	0.008 ± 0.011	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.117 ± 0.005	0.342 ± 0.010	0.109	-0.001	0.0378	0.0378	0.909	93.474 ± 0.008	0.134 ± 0.001	88.153 ± 0.010
	beta45	0.11	0.112	0.002 ± 0.001	0.014 ± 0.010	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.105 ± 0.005	0.323 ± 0.009	0.110	0.000	0.0347	0.0347	0.937	94.478 ± 0.007	0.131 ± 0.001	90.663 ± 0.009
	beta46	0.08	0.079	-0.001 ± 0.001	-0.006 ± 0.008	0.000 ± 0.000	0.020 ± 0.000	0.000 ± 0.000	0.020 ± 0.001	0.062 ± 0.003	0.250 ± 0.007	0.079	-0.001	0.0192	0.0192	0.930	93.474 ± 0.008	0.073 ± 0.001	97.892 ± 0.005
	beta47	0.08	0.079	-0.001 ± 0.001	-0.010 ± 0.008	0.000 ± 0.000	0.020 ± 0.001	0.000 ± 0.000	0.020 ± 0.001	0.063 ± 0.003	0.251 ± 0.008	0.079	-0.001	0.0187	0.0187	0.925	93.072 ± 0.008	0.073 ± 0.001	97.289 ± 0.005
	beta51	0.16	0.184	0.024 ± 0.002	0.150 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.058 ± 0.001	0.133 ± 0.006	0.364 ± 0.009	0.184	0.024	0.0537	0.0588	0.986	92.671 ± 0.008	0.206 ± 0.000	93.474 ± 0.008
	beta52	0.16	0.187	0.027 ± 0.001	0.167 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.108 ± 0.005	0.328 ± 0.008	0.187	0.027	0.0469	0.0542	0.954	89.960 ± 0.010	0.169 ± 0.000	98.594 ± 0.004
	beta53	0.16	0.161	0.001 ± 0.002	0.005 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.104 ± 0.005	0.322 ± 0.009	0.161	0.001	0.0495	0.0495	0.998	94.779 ± 0.007	0.202 ± 0.001	87.651 ± 0.010
	beta54	0.16	0.161	0.001 ± 0.001	0.008 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.051 ± 0.002	0.226 ± 0.006	0.160	0.000	0.0378	0.0378	0.981	94.980 ± 0.007	0.139 ± 0.000	99.398 ± 0.002
	beta55	0.08	0.079	-0.001 ± 0.001	-0.010 ± 0.012	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.139 ± 0.006	0.373 ± 0.010	0.080	0.000	0.0284	0.0284	0.950	93.976 ± 0.008	0.111 ± 0.001	78.313 ± 0.013
	beta56	0.08	0.080	-0.000 ± 0.001	-0.003 ± 0.008	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	0.066 ± 0.003	0.257 ± 0.008	0.079	-0.001	0.0196	0.0196	0.951	94.578 ± 0.007	0.077 ± 0.001	95.884 ± 0.006
	beta57	0.06	0.061	0.001 ± 0.001	0.019 ± 0.016	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.256 ± 0.011	0.506 ± 0.014	0.061	0.001	0.0306	0.0306	0.947	94.779 ± 0.007	0.113 ± 0.000	57.430 ± 0.016
	beta58	0.06	0.061	0.001 ± 0.001	0.011 ± 0.010	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	0.092 ± 0.005	0.303 ± 0.009	0.060	0.000	0.0170	0.0170	0.972	93.675 ± 0.008	0.069 ± 0.001	90.763 ± 0.009

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S51

Simulation 2 - Condition 13: Full Model, Distribution = uniform, N = 400, Reliability = 0.4

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.199	-0.001 ± 0.005	-0.004 ± 0.027	0.009 ± 0.001	0.093 ± 0.005	0.009 ± 0.001	0.093 ± 0.006	0.214 ± 0.023	0.463 ± 0.028	0.191	-0.009	0.0966	0.0971	1.351	99.652 ± 0.003	0.491 ± 0.012	39.373 ± 0.029
	beta42	0.20	0.198	-0.002 ± 0.006	-0.008 ± 0.029	0.010 ± 0.001	0.098 ± 0.004	0.010 ± 0.001	0.098 ± 0.005	0.241 ± 0.019	0.491 ± 0.024	0.196	-0.004	0.0948	0.0949	1.236	97.561 ± 0.009	0.477 ± 0.011	41.115 ± 0.029
	beta43	0.20	0.200	0.000 ± 0.006	0.001 ± 0.028	0.009 ± 0.001	0.093 ± 0.005	0.009 ± 0.001	0.093 ± 0.006	0.217 ± 0.023	0.466 ± 0.029	0.196	-0.004	0.0927	0.0927	1.116	97.561 ± 0.009	0.408 ± 0.007	51.916 ± 0.029
	beta44	0.11	0.120	0.010 ± 0.010	0.091 ± 0.086	0.026 ± 0.003	0.161 ± 0.008	0.026 ± 0.003	0.161 ± 0.009	2.143 ± 0.213	1.464 ± 0.085	0.112	0.002	0.1549	0.1550	1.163	97.213 ± 0.010	0.734 ± 0.028	10.105 ± 0.018
	beta45	0.11	0.118	0.008 ± 0.008	0.076 ± 0.072	0.018 ± 0.002	0.133 ± 0.006	0.018 ± 0.002	0.133 ± 0.007	1.470 ± 0.134	1.212 ± 0.066	0.110	0.000	0.1251	0.1251	1.063	97.561 ± 0.009	0.556 ± 0.013	13.589 ± 0.020
	beta46	0.08	0.094	0.014 ± 0.012	0.174 ± 0.150	0.041 ± 0.004	0.203 ± 0.011	0.041 ± 0.004	0.203 ± 0.013	6.448 ± 0.700	2.539 ± 0.158	0.092	0.012	0.1840	0.1844	1.249	99.303 ± 0.005	0.994 ± 0.047	0.697 ± 0.005
	beta47	0.08	0.075	-0.005 ± 0.012	-0.064 ± 0.145	0.039 ± 0.004	0.196 ± 0.010	0.038 ± 0.004	0.196 ± 0.012	6.000 ± 0.616	2.449 ± 0.146	0.079	-0.001	0.1769	0.1769	1.177	98.955 ± 0.006	0.905 ± 0.037	1.394 ± 0.007
	beta51	0.16	0.178	0.018 ± 0.006	0.110 ± 0.037	0.010 ± 0.001	0.101 ± 0.005	0.010 ± 0.001	0.102 ± 0.006	0.408 ± 0.042	0.639 ± 0.038	0.183	0.023	0.1012	0.1037	1.158	97.909 ± 0.008	0.458 ± 0.012	37.979 ± 0.029
	beta52	0.16	0.188	0.028 ± 0.006	0.176 ± 0.037	0.010 ± 0.001	0.102 ± 0.005	0.011 ± 0.001	0.105 ± 0.007	0.432 ± 0.048	0.657 ± 0.041	0.183	0.023	0.0986	0.1012	1.097	96.864 ± 0.010	0.437 ± 0.009	41.115 ± 0.029
	beta53	0.16	0.161	0.001 ± 0.006	0.004 ± 0.036	0.009 ± 0.001	0.096 ± 0.004	0.009 ± 0.001	0.096 ± 0.005	0.361 ± 0.032	0.601 ± 0.032	0.161	0.001	0.0994	0.0994	1.183	96.864 ± 0.010	0.446 ± 0.010	29.965 ± 0.027
	beta54	0.16	0.160	0.000 ± 0.007	0.000 ± 0.041	0.012 ± 0.001	0.111 ± 0.006	0.012 ± 0.001	0.111 ± 0.007	0.479 ± 0.051	0.692 ± 0.042	0.158	-0.002	0.1019	0.1020	0.955	93.728 ± 0.014	0.415 ± 0.006	36.237 ± 0.028
	beta55	0.08	0.075	-0.005 ± 0.009	-0.059 ± 0.111	0.023 ± 0.002	0.150 ± 0.008	0.023 ± 0.002	0.150 ± 0.009	3.527 ± 0.380	1.878 ± 0.116	0.075	-0.005	0.1287	0.1288	1.104	97.213 ± 0.010	0.651 ± 0.023	6.272 ± 0.014
	beta56	0.08	0.082	0.002 ± 0.009	0.020 ± 0.108	0.021 ± 0.002	0.146 ± 0.008	0.021 ± 0.002	0.146 ± 0.009	3.318 ± 0.359	1.822 ± 0.113	0.072	-0.008	0.1283	0.1286	1.013	96.516 ± 0.011	0.579 ± 0.018	7.666 ± 0.016
	beta57	0.06	0.059	-0.001 ± 0.012	-0.024 ± 0.200	0.041 ± 0.005	0.203 ± 0.012	0.041 ± 0.005	0.203 ± 0.014	11.449 ± 1.380	3.384 ± 0.229	0.037	-0.023	0.1811	0.1825	1.053	98.606 ± 0.007	0.839 ± 0.038	1.045 ± 0.006
	beta58	0.06	0.052	-0.008 ± 0.011	-0.126 ± 0.181	0.034 ± 0.004	0.184 ± 0.010	0.034 ± 0.004	0.184 ± 0.012	9.378 ± 1.047	3.062 ± 0.195	0.048	-0.012	0.1541	0.1545	1.033	98.606 ± 0.007	0.745 ± 0.030	2.787 ± 0.010
QML	beta41	0.20	0.189	-0.011 ± 0.004	-0.054 ± 0.022	0.006 ± 0.000	0.075 ± 0.003	0.006 ± 0.000	0.076 ± 0.004	0.143 ± 0.012	0.378 ± 0.019	0.192	-0.008	0.0785	0.0789	1.084	95.122 ± 0.013	0.318 ± 0.003	69.338 ± 0.027
	beta42	0.20	0.188	-0.012 ± 0.005	-0.060 ± 0.024	0.007 ± 0.001	0.082 ± 0.003	0.007 ± 0.001	0.083 ± 0.004	0.171 ± 0.013	0.413 ± 0.020	0.189	-0.011	0.0851	0.0858	1.030	95.819 ± 0.012	0.330 ± 0.003	62.369 ± 0.029
	beta43	0.20	0.186	-0.014 ± 0.005	-0.072 ± 0.023	0.006 ± 0.001	0.080 ± 0.004	0.007 ± 0.001	0.081 ± 0.004	0.163 ± 0.014	0.404 ± 0.021	0.185	-0.015	0.0863	0.0876	1.017	94.774 ± 0.013	0.317 ± 0.003	67.596 ± 0.028
	beta44	0.11	0.121	0.011 ± 0.007	0.097 ± 0.063	0.014 ± 0.001	0.117 ± 0.006	0.014 ± 0.001	0.117 ± 0.007	1.138 ± 0.109	1.067 ± 0.060	0.114	0.004	0.1189	0.1190	0.976	93.728 ± 0.014	0.448 ± 0.006	17.073 ± 0.022
	beta45	0.11	0.107	-0.003 ± 0.006	-0.025 ± 0.055	0.011 ± 0.001	0.103 ± 0.005	0.010 ± 0.001	0.102 ± 0.005	0.866 ± 0.077	0.931 ± 0.050	0.097	-0.013	0.0996	0.1005	0.934	94.077 ± 0.014	0.375 ± 0.004	22.648 ± 0.025
	beta46	0.08	0.026	-0.054 ± 0.005	-0.680 ± 0.057	0.006 ± 0.001	0.077 ± 0.004	0.009 ± 0.001	0.094 ± 0.005	1.377 ± 0.127	1.173 ± 0.059	0.028	-0.052	0.0729	0.0896	1.084	91.289 ± 0.017	0.326 ± 0.005	2.787 ± 0.010
	beta47	0.08	0.027	-0.053 ± 0.004	-0.666 ± 0.056	0.006 ± 0.000	0.076 ± 0.003	0.009 ± 0.001	0.093 ± 0.004	1.338 ± 0.100	1.157 ± 0.049	0.025	-0.055	0.0809	0.0977	1.063	90.941 ± 0.017	0.316 ± 0.004	4.878 ± 0.013
	beta51	0.16	0.172	0.012 ± 0.005	0.073 ± 0.032	0.008 ± 0.001	0.087 ± 0.004	0.008 ± 0.001	0.088 ± 0.005	0.301 ± 0.025	0.548 ± 0.028	0.180	0.020	0.0942	0.0964	1.025	97.213 ± 0.010	0.350 ± 0.004	48.084 ± 0.029
	beta52	0.16	0.174	0.014 ± 0.005	0.089 ± 0.033	0.008 ± 0.001	0.089 ± 0.004	0.008 ± 0.001	0.090 ± 0.005	0.319 ± 0.031	0.565 ± 0.032	0.167	0.007	0.0872	0.0875	1.011	95.122 ± 0.013	0.354 ± 0.003	47.387 ± 0.029
	beta53	0.16	0.158	-0.002 ± 0.005	-0.012 ± 0.031	0.007 ± 0.001	0.083 ± 0.003	0.007 ± 0.001	0.083 ± 0.004	0.267 ± 0.022	0.517 ± 0.027	0.162	0.002	0.0881	0.0881	1.006	95.122 ± 0.013	0.327 ± 0.003	49.826 ± 0.030
	beta54	0.16	0.182	0.022 ± 0.006	0.136 ± 0.039	0.011 ± 0.001	0.104 ± 0.006	0.011 ± 0.001	0.106 ± 0.007	0.442 ± 0.049	0.665 ± 0.042	0.174	0.014	0.0938	0.0949	0.934	94.774 ± 0.013	0.382 ± 0.004	46.690 ± 0.029
	beta55	0.08	0.080	-0.000 ± 0.007	-0.004 ± 0.089	0.015 ± 0.001	0.121 ± 0.005	0.015 ± 0.001	0.121 ± 0.006	2.286 ± 0.194	1.512 ± 0.079	0.085	0.005	0.1232	0.1233	0.978	95.122 ± 0.013	0.464 ± 0.006	11.847 ± 0.019
	beta56	0.08	0.070	-0.010 ± 0.007	-0.121 ± 0.082	0.012 ± 0.001	0.111 ± 0.005	0.012 ± 0.001	0.111 ± 0.006	1.933 ± 0.163	1.390 ± 0.072	0.070	-0.010	0.1105	0.1109	0.956	93.380 ± 0.015	0.416 ± 0.005	9.408 ± 0.017
	beta57	0.06	0.014	-0.046 ± 0.005	-0.762 ± 0.084	0.007 ± 0.001	0.086 ± 0.005	0.009 ± 0.001	0.097 ± 0.005	2.607 ± 0.226	1.615 ± 0.080	0.006	-0.054	0.0843	0.1002	0.937	88.502 ± 0.019	0.314 ± 0.005	4.878 ± 0.013
	beta58	0.06	0.022	-0.038 ± 0.005	-0.641 ± 0.076	0.006 ± 0.000	0.078 ± 0.003	0.008 ± 0.001	0.087 ± 0.004	2.085 ± 0.169	1.444 ± 0.068	0.021	-0.039	0.0790	0.0883	0.917	91.986 ± 0.016	0.280 ± 0.004	7.317 ± 0.015

Table S51

Simulation 2 - Condition 13: Full Model, Distribution = uniform, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta41	0.20	0.194	-0.006 ± 0.008	-0.031 ± 0.039	0.018 ± 0.002	0.134 ± 0.008	0.018 ± 0.002	0.133 ± 0.009	0.445 ± 0.055	0.667 ± 0.046	0.194	-0.006	0.1301	0.1302	0.808	94.425 ± 0.014	0.423 ± 0.012	54.007 ± 0.029
	beta42	0.20	0.208	0.008 ± 0.008	0.039 ± 0.040	0.019 ± 0.002	0.137 ± 0.008	0.019 ± 0.002	0.137 ± 0.009	0.469 ± 0.057	0.685 ± 0.047	0.202	0.002	0.1344	0.1344	0.816	93.380 ± 0.015	0.438 ± 0.010	52.265 ± 0.029
	beta43	0.20	0.199	-0.001 ± 0.007	-0.003 ± 0.034	0.013 ± 0.001	0.116 ± 0.006	0.013 ± 0.001	0.116 ± 0.007	0.334 ± 0.035	0.578 ± 0.035	0.193	-0.007	0.1147	0.1149	0.885	95.819 ± 0.012	0.402 ± 0.009	56.098 ± 0.029
	beta44	0.11	0.156	0.046 ± 0.015	0.420 ± 0.140	0.069 ± 0.009	0.262 ± 0.017	0.070 ± 0.009	0.265 ± 0.019	5.818 ± 0.738	2.412 ± 0.169	0.137	0.027	0.2511	0.2526	0.666	92.334 ± 0.016	0.683 ± 0.026	23.693 ± 0.025
	beta45	0.11	0.119	0.009 ± 0.012	0.081 ± 0.106	0.039 ± 0.005	0.197 ± 0.012	0.039 ± 0.005	0.197 ± 0.013	3.205 ± 0.375	1.790 ± 0.118	0.121	0.011	0.1876	0.1879	0.739	90.244 ± 0.018	0.571 ± 0.021	18.467 ± 0.023
	beta46	0.08	0.094	0.014 ± 0.025	0.174 ± 0.314	0.181 ± 0.024	0.425 ± 0.028	0.181 ± 0.024	0.425 ± 0.031	28.212 ± 3.736	5.312 ± 0.389	0.053	-0.027	0.3418	0.3428	0.650	88.850 ± 0.019	1.085 ± 0.087	6.620 ± 0.015
	beta47	0.08	0.046	-0.034 ± 0.023	-0.428 ± 0.291	0.156 ± 0.019	0.395 ± 0.025	0.156 ± 0.019	0.396 ± 0.027	24.451 ± 3.016	4.945 ± 0.341	0.060	-0.020	0.3353	0.3359	0.692	88.850 ± 0.019	1.071 ± 0.090	6.969 ± 0.015
	beta51	0.16	0.177	0.017 ± 0.006	0.104 ± 0.038	0.011 ± 0.001	0.103 ± 0.004	0.011 ± 0.001	0.104 ± 0.005	0.421 ± 0.035	0.649 ± 0.033	0.181	0.021	0.1168	0.1187	0.940	96.167 ± 0.011	0.378 ± 0.006	48.780 ± 0.030
	beta52	0.16	0.187	0.027 ± 0.007	0.170 ± 0.043	0.014 ± 0.001	0.117 ± 0.006	0.014 ± 0.001	0.120 ± 0.007	0.564 ± 0.056	0.751 ± 0.043	0.174	0.014	0.1216	0.1224	0.881	94.425 ± 0.014	0.405 ± 0.007	47.735 ± 0.029
	beta53	0.16	0.157	-0.003 ± 0.007	-0.019 ± 0.041	0.012 ± 0.001	0.110 ± 0.006	0.012 ± 0.001	0.110 ± 0.007	0.475 ± 0.048	0.689 ± 0.041	0.161	0.001	0.1218	0.1218	0.890	95.470 ± 0.012	0.386 ± 0.008	43.206 ± 0.029
	beta54	0.16	0.170	0.010 ± 0.007	0.061 ± 0.043	0.014 ± 0.002	0.117 ± 0.007	0.014 ± 0.002	0.117 ± 0.008	0.536 ± 0.064	0.732 ± 0.049	0.163	0.003	0.1214	0.1215	0.853	96.167 ± 0.011	0.391 ± 0.005	40.070 ± 0.029
	beta55	0.08	0.092	0.012 ± 0.011	0.148 ± 0.133	0.032 ± 0.004	0.180 ± 0.011	0.032 ± 0.004	0.180 ± 0.012	5.060 ± 0.627	2.250 ± 0.156	0.088	0.008	0.1787	0.1789	0.843	96.516 ± 0.011	0.595 ± 0.020	10.453 ± 0.018
	beta56	0.08	0.076	-0.004 ± 0.011	-0.056 ± 0.133	0.032 ± 0.004	0.180 ± 0.011	0.032 ± 0.004	0.179 ± 0.012	5.025 ± 0.581	2.242 ± 0.147	0.071	-0.009	0.1642	0.1644	0.780	91.638 ± 0.016	0.549 ± 0.023	11.150 ± 0.019
	beta57	0.06	0.036	-0.024 ± 0.017	-0.399 ± 0.276	0.079 ± 0.011	0.281 ± 0.019	0.079 ± 0.011	0.281 ± 0.021	21.972 ± 2.957	4.687 ± 0.348	0.016	-0.044	0.2539	0.2577	0.738	87.108 ± 0.020	0.812 ± 0.056	3.833 ± 0.011
	beta58	0.06	0.020	-0.040 ± 0.015	-0.659 ± 0.254	0.067 ± 0.008	0.258 ± 0.016	0.068 ± 0.008	0.261 ± 0.017	18.899 ± 2.253	4.347 ± 0.290	0.032	-0.028	0.2035	0.2054	0.718	90.244 ± 0.018	0.727 ± 0.043	6.620 ± 0.015

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S52

Simulation 2 - Condition 14: Full Model, Distribution = uniform, N = 1000, Reliability = 0.4

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.202	0.002 ± 0.002	0.010 ± 0.011	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.091 ± 0.005	0.302 ± 0.010	0.201	0.001	0.0598	0.0599	1.049	96.086 ± 0.007	0.248 ± 0.002	91.363 ± 0.010
	beta42	0.20	0.202	0.002 ± 0.002	0.009 ± 0.011	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.091 ± 0.005	0.301 ± 0.010	0.198	-0.002	0.0594	0.0594	1.044	96.356 ± 0.007	0.247 ± 0.002	92.038 ± 0.010
	beta43	0.20	0.203	0.003 ± 0.002	0.014 ± 0.010	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.077 ± 0.004	0.278 ± 0.009	0.200	0.000	0.0531	0.0531	1.036	96.086 ± 0.007	0.226 ± 0.001	95.816 ± 0.007
	beta44	0.11	0.112	0.002 ± 0.003	0.016 ± 0.028	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	0.579 ± 0.032	0.761 ± 0.025	0.112	0.002	0.0818	0.0819	1.058	97.571 ± 0.006	0.348 ± 0.006	25.776 ± 0.016
	beta45	0.11	0.113	0.003 ± 0.003	0.029 ± 0.023	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.397 ± 0.023	0.630 ± 0.022	0.112	0.002	0.0637	0.0638	1.041	96.086 ± 0.007	0.283 ± 0.002	36.707 ± 0.018
	beta46	0.08	0.085	0.005 ± 0.004	0.061 ± 0.055	0.014 ± 0.001	0.119 ± 0.004	0.014 ± 0.001	0.119 ± 0.004	2.221 ± 0.139	1.490 ± 0.054	0.085	0.005	0.1127	0.1129	1.022	97.166 ± 0.006	0.477 ± 0.008	5.533 ± 0.008
	beta47	0.08	0.091	0.011 ± 0.004	0.140 ± 0.053	0.013 ± 0.001	0.115 ± 0.004	0.013 ± 0.001	0.115 ± 0.004	2.082 ± 0.138	1.443 ± 0.055	0.074	-0.006	0.1049	0.1050	1.086	97.841 ± 0.005	0.489 ± 0.013	5.263 ± 0.008
	beta51	0.16	0.184	0.024 ± 0.002	0.148 ± 0.013	0.003 ± 0.000	0.055 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.140 ± 0.008	0.375 ± 0.012	0.183	0.023	0.0554	0.0601	1.099	95.816 ± 0.007	0.237 ± 0.001	88.934 ± 0.012
	beta52	0.16	0.188	0.028 ± 0.002	0.177 ± 0.015	0.004 ± 0.000	0.064 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.193 ± 0.011	0.439 ± 0.015	0.187	0.027	0.0601	0.0657	0.975	92.173 ± 0.010	0.246 ± 0.002	88.259 ± 0.012
	beta53	0.16	0.160	0.000 ± 0.002	0.001 ± 0.013	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.132 ± 0.007	0.364 ± 0.012	0.160	0.000	0.0564	0.0564	1.036	95.816 ± 0.007	0.237 ± 0.002	78.543 ± 0.015
	beta54	0.16	0.157	-0.003 ± 0.002	-0.017 ± 0.014	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.154 ± 0.008	0.393 ± 0.012	0.154	-0.006	0.0622	0.0626	0.988	95.951 ± 0.007	0.243 ± 0.001	71.525 ± 0.017
	beta55	0.08	0.081	0.001 ± 0.003	0.017 ± 0.039	0.007 ± 0.000	0.084 ± 0.003	0.007 ± 0.000	0.084 ± 0.003	1.108 ± 0.068	1.053 ± 0.038	0.079	-0.001	0.0804	0.0804	0.993	97.031 ± 0.006	0.328 ± 0.003	15.924 ± 0.013
	beta56	0.08	0.086	0.006 ± 0.003	0.074 ± 0.035	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.889 ± 0.049	0.943 ± 0.031	0.083	0.003	0.0728	0.0728	1.037	96.356 ± 0.007	0.306 ± 0.003	20.378 ± 0.015
	beta57	0.06	0.070	0.010 ± 0.004	0.166 ± 0.067	0.012 ± 0.001	0.110 ± 0.004	0.012 ± 0.001	0.110 ± 0.004	3.360 ± 0.217	1.833 ± 0.068	0.066	0.006	0.1004	0.1005	1.026	97.841 ± 0.005	0.441 ± 0.007	5.263 ± 0.008
	beta58	0.06	0.067	0.007 ± 0.004	0.111 ± 0.062	0.010 ± 0.001	0.101 ± 0.003	0.010 ± 0.001	0.101 ± 0.004	2.820 ± 0.189	1.679 ± 0.064	0.064	0.004	0.0940	0.0941	1.024	97.976 ± 0.005	0.404 ± 0.008	6.883 ± 0.009
QML	beta41	0.20	0.190	-0.010 ± 0.002	-0.052 ± 0.010	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.072 ± 0.004	0.268 ± 0.008	0.188	-0.012	0.0533	0.0546	0.964	91.633 ± 0.010	0.199 ± 0.001	96.626 ± 0.007
	beta42	0.20	0.195	-0.005 ± 0.002	-0.023 ± 0.010	0.003 ± 0.000	0.053 ± 0.002	0.003 ± 0.000	0.053 ± 0.002	0.070 ± 0.004	0.265 ± 0.009	0.192	-0.008	0.0515	0.0521	0.999	95.816 ± 0.007	0.207 ± 0.001	96.896 ± 0.006
	beta43	0.20	0.188	-0.012 ± 0.002	-0.062 ± 0.009	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.068 ± 0.004	0.261 ± 0.008	0.187	-0.013	0.0514	0.0529	0.989	93.657 ± 0.009	0.197 ± 0.001	96.221 ± 0.007
	beta44	0.11	0.115	0.005 ± 0.003	0.043 ± 0.024	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.430 ± 0.023	0.656 ± 0.021	0.115	0.005	0.0678	0.0680	0.980	95.007 ± 0.008	0.277 ± 0.001	36.707 ± 0.018
	beta45	0.11	0.107	-0.003 ± 0.002	-0.030 ± 0.019	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.281 ± 0.016	0.530 ± 0.018	0.101	-0.009	0.0594	0.0600	1.019	94.602 ± 0.008	0.233 ± 0.001	43.185 ± 0.018
	beta46	0.08	0.026	-0.054 ± 0.002	-0.681 ± 0.023	0.003 ± 0.000	0.051 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.871 ± 0.038	0.933 ± 0.022	0.026	-0.054	0.0523	0.0751	1.006	80.027 ± 0.015	0.202 ± 0.001	6.748 ± 0.009
	beta47	0.08	0.029	-0.051 ± 0.002	-0.631 ± 0.022	0.002 ± 0.000	0.048 ± 0.001	0.005 ± 0.000	0.069 ± 0.002	0.754 ± 0.034	0.868 ± 0.021	0.028	-0.052	0.0481	0.0711	1.055	83.131 ± 0.014	0.197 ± 0.001	7.018 ± 0.009
	beta51	0.16	0.176	0.016 ± 0.002	0.100 ± 0.012	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.117 ± 0.007	0.342 ± 0.011	0.176	0.016	0.0496	0.0520	1.032	94.602 ± 0.008	0.212 ± 0.001	92.308 ± 0.010
	beta52	0.16	0.169	0.009 ± 0.002	0.058 ± 0.013	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.132 ± 0.007	0.363 ± 0.012	0.168	0.008	0.0554	0.0560	0.967	93.792 ± 0.009	0.217 ± 0.001	87.314 ± 0.012
	beta53	0.16	0.158	-0.002 ± 0.002	-0.013 ± 0.012	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.112 ± 0.006	0.334 ± 0.011	0.159	-0.001	0.0561	0.0561	0.963	93.522 ± 0.009	0.202 ± 0.001	87.449 ± 0.012
	beta54	0.16	0.177	0.017 ± 0.002	0.104 ± 0.014	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.155 ± 0.008	0.394 ± 0.012	0.173	0.013	0.0630	0.0642	0.980	95.277 ± 0.008	0.234 ± 0.001	85.290 ± 0.013
	beta55	0.08	0.084	0.004 ± 0.003	0.047 ± 0.033	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.812 ± 0.046	0.901 ± 0.030	0.083	0.003	0.0699	0.0700	1.014	95.951 ± 0.007	0.286 ± 0.001	18.623 ± 0.014
	beta56	0.08	0.077	-0.003 ± 0.002	-0.035 ± 0.030	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.675 ± 0.037	0.822 ± 0.027	0.077	-0.003	0.0641	0.0641	0.998	93.792 ± 0.009	0.257 ± 0.001	20.648 ± 0.015
	beta57	0.06	0.021	-0.039 ± 0.002	-0.649 ± 0.031	0.003 ± 0.000	0.051 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	1.134 ± 0.058	1.065 ± 0.030	0.022	-0.038	0.0518	0.0641	0.997	87.179 ± 0.012	0.198 ± 0.001	6.748 ± 0.009
	beta58	0.06	0.029	-0.031 ± 0.002	-0.512 ± 0.027	0.002 ± 0.000	0.045 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.818 ± 0.037	0.905 ± 0.024	0.027	-0.033	0.0453	0.0562	0.995	87.449 ± 0.012	0.175 ± 0.001	10.796 ± 0.011

Table S52

Simulation 2 - Condition 14: Full Model, Distribution = uniform, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta41	0.20	0.204	0.004 ± 0.002	0.021 ± 0.012	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.100 ± 0.006	0.316 ± 0.011	0.203	0.003	0.0651	0.0652	0.914	92.848 ± 0.009	0.226 ± 0.002	94.197 ± 0.009
	beta42	0.20	0.202	0.002 ± 0.002	0.010 ± 0.012	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.099 ± 0.006	0.314 ± 0.011	0.200	0.000	0.0634	0.0634	0.961	95.277 ± 0.008	0.237 ± 0.002	92.173 ± 0.010
	beta43	0.20	0.202	0.002 ± 0.002	0.010 ± 0.010	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	0.082 ± 0.004	0.286 ± 0.009	0.200	0.000	0.0619	0.0619	0.979	95.007 ± 0.008	0.219 ± 0.001	94.872 ± 0.008
	beta44	0.11	0.121	0.011 ± 0.004	0.102 ± 0.033	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	0.804 ± 0.056	0.897 ± 0.035	0.113	0.003	0.0984	0.0985	0.847	93.657 ± 0.009	0.326 ± 0.004	34.143 ± 0.017
	beta45	0.11	0.113	0.003 ± 0.003	0.027 ± 0.026	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	0.482 ± 0.031	0.694 ± 0.026	0.112	0.002	0.0776	0.0776	0.931	94.467 ± 0.008	0.279 ± 0.003	38.192 ± 0.018
	beta46	0.08	0.089	0.009 ± 0.006	0.113 ± 0.069	0.023 ± 0.002	0.151 ± 0.006	0.023 ± 0.002	0.151 ± 0.006	3.563 ± 0.268	1.887 ± 0.079	0.077	-0.003	0.1307	0.1307	0.780	92.308 ± 0.010	0.461 ± 0.010	8.637 ± 0.010
	beta47	0.08	0.094	0.014 ± 0.005	0.174 ± 0.067	0.021 ± 0.002	0.147 ± 0.005	0.022 ± 0.002	0.147 ± 0.006	3.385 ± 0.249	1.840 ± 0.076	0.081	0.001	0.1222	0.1222	0.830	93.522 ± 0.009	0.477 ± 0.016	8.772 ± 0.010
	beta51	0.16	0.185	0.025 ± 0.002	0.154 ± 0.013	0.003 ± 0.000	0.055 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.142 ± 0.008	0.376 ± 0.012	0.185	0.025	0.0545	0.0599	1.019	93.792 ± 0.009	0.220 ± 0.001	92.308 ± 0.010
	beta52	0.16	0.186	0.026 ± 0.002	0.164 ± 0.015	0.004 ± 0.000	0.065 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.191 ± 0.012	0.437 ± 0.015	0.185	0.025	0.0639	0.0687	0.924	91.903 ± 0.010	0.235 ± 0.001	88.124 ± 0.012
	beta53	0.16	0.160	0.000 ± 0.002	0.001 ± 0.014	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.059 ± 0.002	0.137 ± 0.007	0.370 ± 0.012	0.159	-0.001	0.0602	0.0602	0.945	93.792 ± 0.009	0.219 ± 0.001	82.051 ± 0.014
	beta54	0.16	0.160	-0.000 ± 0.002	-0.002 ± 0.014	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.153 ± 0.008	0.391 ± 0.013	0.156	-0.004	0.0611	0.0613	0.957	94.872 ± 0.008	0.235 ± 0.001	75.709 ± 0.016
	beta55	0.08	0.082	0.002 ± 0.003	0.030 ± 0.040	0.008 ± 0.000	0.087 ± 0.003	0.008 ± 0.000	0.087 ± 0.003	1.188 ± 0.076	1.090 ± 0.040	0.078	-0.002	0.0827	0.0828	0.914	95.682 ± 0.007	0.312 ± 0.004	19.028 ± 0.014
	beta56	0.08	0.085	0.005 ± 0.003	0.068 ± 0.035	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	0.916 ± 0.051	0.957 ± 0.032	0.083	0.003	0.0803	0.0803	0.975	94.602 ± 0.008	0.292 ± 0.003	20.513 ± 0.015
	beta57	0.06	0.069	0.009 ± 0.005	0.153 ± 0.081	0.017 ± 0.001	0.132 ± 0.005	0.017 ± 0.001	0.132 ± 0.005	4.827 ± 0.334	2.197 ± 0.086	0.061	0.001	0.1209	0.1209	0.812	93.927 ± 0.009	0.419 ± 0.009	8.637 ± 0.010
	beta58	0.06	0.063	0.003 ± 0.005	0.054 ± 0.075	0.015 ± 0.001	0.123 ± 0.005	0.015 ± 0.001	0.123 ± 0.005	4.170 ± 0.316	2.042 ± 0.086	0.053	-0.007	0.1030	0.1032	0.844	92.308 ± 0.010	0.405 ± 0.011	5.803 ± 0.009

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S53

Simulation 2 - Condition 15: Full Model, Distribution = uniform, N = 400, Reliability = 0.6

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.200	0.000 ± 0.002	0.002 ± 0.011	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.109 ± 0.005	0.330 ± 0.009	0.201	0.001	0.0667	0.0667	1.017	95.763 ± 0.007	0.263 ± 0.001	85.064 ± 0.012
	beta42	0.20	0.199	-0.001 ± 0.002	-0.004 ± 0.011	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.112 ± 0.006	0.335 ± 0.010	0.197	-0.003	0.0677	0.0678	1.025	95.869 ± 0.006	0.269 ± 0.001	83.475 ± 0.012
	beta43	0.20	0.203	0.003 ± 0.002	0.014 ± 0.010	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.097 ± 0.005	0.312 ± 0.009	0.202	0.002	0.0657	0.0658	1.036	95.975 ± 0.006	0.253 ± 0.001	89.407 ± 0.010
	beta44	0.11	0.112	0.002 ± 0.003	0.016 ± 0.024	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.553 ± 0.026	0.743 ± 0.021	0.110	0.000	0.0839	0.0839	0.993	96.398 ± 0.006	0.318 ± 0.002	30.720 ± 0.015
	beta45	0.11	0.110	0.000 ± 0.002	0.001 ± 0.021	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.413 ± 0.020	0.643 ± 0.019	0.107	-0.003	0.0696	0.0697	1.013	96.292 ± 0.006	0.281 ± 0.001	35.487 ± 0.016
	beta46	0.08	0.082	0.002 ± 0.003	0.027 ± 0.041	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	1.565 ± 0.079	1.251 ± 0.038	0.083	0.003	0.0984	0.0984	1.045	96.822 ± 0.006	0.410 ± 0.003	8.475 ± 0.009
	beta47	0.08	0.078	-0.002 ± 0.003	-0.022 ± 0.042	0.011 ± 0.001	0.104 ± 0.003	0.011 ± 0.001	0.104 ± 0.003	1.689 ± 0.085	1.300 ± 0.039	0.078	-0.002	0.0981	0.0981	0.990	95.445 ± 0.007	0.404 ± 0.003	9.640 ± 0.010
	beta51	0.16	0.188	0.028 ± 0.002	0.177 ± 0.014	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.207 ± 0.011	0.455 ± 0.014	0.186	0.026	0.0624	0.0675	0.992	92.161 ± 0.009	0.261 ± 0.001	82.521 ± 0.012
	beta52	0.16	0.182	0.022 ± 0.002	0.137 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.211 ± 0.010	0.459 ± 0.013	0.184	0.024	0.0730	0.0768	0.977	93.962 ± 0.008	0.269 ± 0.001	75.212 ± 0.014
	beta53	0.16	0.158	-0.002 ± 0.002	-0.014 ± 0.013	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.154 ± 0.007	0.392 ± 0.011	0.158	-0.002	0.0615	0.0615	1.055	96.292 ± 0.006	0.260 ± 0.001	68.644 ± 0.015
	beta54	0.16	0.160	-0.000 ± 0.002	-0.003 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.194 ± 0.009	0.440 ± 0.012	0.160	0.000	0.0703	0.0703	0.986	93.856 ± 0.008	0.272 ± 0.001	63.877 ± 0.016
	beta55	0.08	0.079	-0.001 ± 0.003	-0.007 ± 0.035	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	1.146 ± 0.064	1.070 ± 0.035	0.081	0.001	0.0854	0.0855	0.952	94.597 ± 0.007	0.320 ± 0.002	18.856 ± 0.013
	beta56	0.08	0.081	0.001 ± 0.003	0.010 ± 0.032	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.968 ± 0.044	0.984 ± 0.028	0.080	0.000	0.0795	0.0795	0.976	94.386 ± 0.007	0.301 ± 0.002	20.233 ± 0.013
	beta57	0.06	0.064	0.004 ± 0.003	0.067 ± 0.055	0.010 ± 0.000	0.101 ± 0.002	0.010 ± 0.001	0.101 ± 0.003	2.814 ± 0.139	1.677 ± 0.050	0.061	0.001	0.0937	0.0937	0.997	96.398 ± 0.006	0.393 ± 0.003	7.839 ± 0.009
	beta58	0.06	0.055	-0.005 ± 0.003	-0.091 ± 0.049	0.008 ± 0.000	0.090 ± 0.002	0.008 ± 0.000	0.090 ± 0.003	2.248 ± 0.106	1.499 ± 0.043	0.053	-0.007	0.0867	0.0870	1.027	96.822 ± 0.006	0.362 ± 0.003	7.945 ± 0.009
QML	beta41	0.20	0.195	-0.005 ± 0.002	-0.026 ± 0.010	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.098 ± 0.004	0.313 ± 0.009	0.194	-0.006	0.0637	0.0640	0.995	95.445 ± 0.007	0.244 ± 0.001	88.136 ± 0.011
	beta42	0.20	0.195	-0.005 ± 0.002	-0.027 ± 0.010	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.100 ± 0.005	0.317 ± 0.009	0.191	-0.009	0.0646	0.0653	1.007	96.186 ± 0.006	0.249 ± 0.001	87.394 ± 0.011
	beta43	0.20	0.194	-0.006 ± 0.002	-0.031 ± 0.010	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.091 ± 0.004	0.301 ± 0.009	0.192	-0.008	0.0605	0.0610	1.017	95.445 ± 0.007	0.239 ± 0.001	89.301 ± 0.010
	beta44	0.11	0.116	0.006 ± 0.003	0.050 ± 0.023	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.491 ± 0.023	0.701 ± 0.020	0.114	0.004	0.0811	0.0812	0.992	96.292 ± 0.006	0.299 ± 0.001	33.263 ± 0.015
	beta45	0.11	0.109	-0.001 ± 0.002	-0.011 ± 0.019	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.353 ± 0.016	0.594 ± 0.017	0.108	-0.002	0.0682	0.0682	1.024	96.822 ± 0.006	0.262 ± 0.001	38.665 ± 0.016
	beta46	0.08	0.040	-0.040 ± 0.002	-0.494 ± 0.025	0.004 ± 0.000	0.061 ± 0.001	0.005 ± 0.000	0.072 ± 0.002	0.821 ± 0.039	0.906 ± 0.024	0.043	-0.037	0.0591	0.0697	1.033	91.525 ± 0.009	0.246 ± 0.001	8.686 ± 0.009
	beta47	0.08	0.041	-0.039 ± 0.002	-0.487 ± 0.025	0.004 ± 0.000	0.062 ± 0.001	0.005 ± 0.000	0.074 ± 0.002	0.845 ± 0.038	0.919 ± 0.023	0.043	-0.037	0.0587	0.0696	0.985	90.042 ± 0.010	0.241 ± 0.001	10.699 ± 0.010
	beta51	0.16	0.178	0.018 ± 0.002	0.114 ± 0.014	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.189 ± 0.009	0.435 ± 0.013	0.176	0.016	0.0622	0.0643	0.978	93.008 ± 0.008	0.258 ± 0.001	78.072 ± 0.013
	beta52	0.16	0.164	0.004 ± 0.002	0.028 ± 0.014	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.183 ± 0.008	0.428 ± 0.012	0.164	0.004	0.0713	0.0715	0.975	95.127 ± 0.007	0.262 ± 0.001	68.856 ± 0.015
	beta53	0.16	0.157	-0.003 ± 0.002	-0.019 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.140 ± 0.007	0.375 ± 0.011	0.157	-0.003	0.0575	0.0576	1.041	96.081 ± 0.006	0.245 ± 0.001	71.928 ± 0.015
	beta54	0.16	0.172	0.012 ± 0.002	0.074 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.194 ± 0.009	0.440 ± 0.012	0.172	0.012	0.0683	0.0693	1.012	95.127 ± 0.007	0.276 ± 0.001	69.703 ± 0.015
	beta55	0.08	0.081	0.001 ± 0.003	0.011 ± 0.033	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	1.019 ± 0.053	1.009 ± 0.031	0.084	0.004	0.0797	0.0798	0.978	94.809 ± 0.007	0.310 ± 0.001	17.161 ± 0.012
	beta56	0.08	0.078	-0.002 ± 0.002	-0.027 ± 0.030	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.866 ± 0.040	0.931 ± 0.026	0.078	-0.002	0.0738	0.0738	0.981	95.021 ± 0.007	0.286 ± 0.001	19.492 ± 0.013
	beta57	0.06	0.033	-0.027 ± 0.002	-0.455 ± 0.033	0.004 ± 0.000	0.062 ± 0.001	0.005 ± 0.000	0.067 ± 0.002	1.264 ± 0.057	1.124 ± 0.030	0.033	-0.027	0.0617	0.0673	1.000	92.373 ± 0.009	0.242 ± 0.001	7.945 ± 0.009
	beta58	0.06	0.032	-0.028 ± 0.002	-0.463 ± 0.030	0.003 ± 0.000	0.054 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	1.037 ± 0.045	1.019 ± 0.026	0.033	-0.027	0.0544	0.0608	1.031	91.631 ± 0.009	0.220 ± 0.001	8.581 ± 0.009

Table S53

Simulation 2 - Condition 15: Full Model, Distribution = uniform, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta41	0.20	0.201	0.001 ± 0.002	0.007 ± 0.011	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.113 ± 0.005	0.337 ± 0.009	0.199	-0.001	0.0712	0.0712	0.955	93.962 ± 0.008	0.252 ± 0.001	87.394 ± 0.011
	beta42	0.20	0.200	0.000 ± 0.002	0.000 ± 0.011	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.118 ± 0.006	0.344 ± 0.010	0.195	-0.005	0.0678	0.0680	0.969	95.657 ± 0.007	0.261 ± 0.001	85.805 ± 0.011
	beta43	0.20	0.203	0.003 ± 0.002	0.016 ± 0.010	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.101 ± 0.005	0.317 ± 0.009	0.202	0.002	0.0643	0.0643	0.999	95.551 ± 0.007	0.248 ± 0.001	90.254 ± 0.010
	beta44	0.11	0.116	0.006 ± 0.003	0.051 ± 0.026	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.003	0.631 ± 0.032	0.794 ± 0.024	0.115	0.005	0.0863	0.0864	0.917	93.750 ± 0.008	0.313 ± 0.002	34.322 ± 0.015
	beta45	0.11	0.113	0.003 ± 0.002	0.024 ± 0.023	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.481 ± 0.025	0.694 ± 0.021	0.109	-0.001	0.0740	0.0741	0.927	95.021 ± 0.007	0.277 ± 0.002	38.030 ± 0.016
	beta46	0.08	0.079	-0.001 ± 0.004	-0.019 ± 0.047	0.013 ± 0.001	0.116 ± 0.003	0.013 ± 0.001	0.116 ± 0.004	2.104 ± 0.119	1.451 ± 0.048	0.078	-0.002	0.1082	0.1082	0.855	93.432 ± 0.008	0.389 ± 0.004	14.407 ± 0.011
	beta47	0.08	0.083	0.003 ± 0.004	0.035 ± 0.052	0.017 ± 0.001	0.129 ± 0.004	0.017 ± 0.001	0.129 ± 0.004	2.586 ± 0.152	1.608 ± 0.054	0.075	-0.005	0.1135	0.1136	0.791	91.208 ± 0.009	0.399 ± 0.005	15.572 ± 0.012
	beta51	0.16	0.189	0.029 ± 0.002	0.179 ± 0.014	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.211 ± 0.011	0.459 ± 0.013	0.187	0.027	0.0648	0.0702	0.955	91.314 ± 0.009	0.253 ± 0.001	84.428 ± 0.012
	beta52	0.16	0.181	0.021 ± 0.002	0.130 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.212 ± 0.010	0.461 ± 0.013	0.181	0.021	0.0718	0.0749	0.954	93.326 ± 0.008	0.265 ± 0.001	76.059 ± 0.014
	beta53	0.16	0.158	-0.002 ± 0.002	-0.013 ± 0.013	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.156 ± 0.007	0.395 ± 0.011	0.160	0.000	0.0619	0.0619	1.016	95.445 ± 0.007	0.252 ± 0.001	69.809 ± 0.015
	beta54	0.16	0.163	0.003 ± 0.002	0.016 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.190 ± 0.009	0.436 ± 0.012	0.161	0.001	0.0686	0.0686	0.983	94.174 ± 0.008	0.269 ± 0.001	66.314 ± 0.015
	beta55	0.08	0.081	0.001 ± 0.003	0.008 ± 0.035	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	1.143 ± 0.061	1.069 ± 0.033	0.083	0.003	0.0813	0.0813	0.929	94.068 ± 0.008	0.312 ± 0.002	18.326 ± 0.013
	beta56	0.08	0.083	0.003 ± 0.003	0.040 ± 0.034	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	1.106 ± 0.057	1.052 ± 0.032	0.081	0.001	0.0787	0.0787	0.909	93.644 ± 0.008	0.300 ± 0.002	21.716 ± 0.013
	beta57	0.06	0.059	-0.001 ± 0.004	-0.013 ± 0.060	0.012 ± 0.001	0.111 ± 0.003	0.012 ± 0.001	0.111 ± 0.004	3.445 ± 0.195	1.856 ± 0.061	0.056	-0.004	0.1015	0.1016	0.850	93.644 ± 0.008	0.371 ± 0.004	9.958 ± 0.010
	beta58	0.06	0.053	-0.007 ± 0.003	-0.122 ± 0.056	0.011 ± 0.001	0.104 ± 0.003	0.011 ± 0.001	0.104 ± 0.003	3.023 ± 0.162	1.739 ± 0.055	0.047	-0.013	0.0959	0.0967	0.860	92.267 ± 0.009	0.351 ± 0.004	9.640 ± 0.010

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S54

Simulation 2 - Condition 16: Full Model, Distribution = uniform, N = 1000, Reliability = 0.6

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.198	-0.002 ± 0.001	-0.009 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.040 ± 0.002	0.201 ± 0.006	0.199	-0.001	0.0380	0.0380	1.011	95.080 ± 0.007	0.159 ± 0.000	99.799 ± 0.001
	beta42	0.20	0.202	0.002 ± 0.001	0.010 ± 0.006	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.042 ± 0.002	0.204 ± 0.006	0.202	0.002	0.0388	0.0389	1.019	94.779 ± 0.007	0.163 ± 0.000	100.000 ± 0.000
	beta43	0.20	0.199	-0.001 ± 0.001	-0.004 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.040 ± 0.002	0.201 ± 0.006	0.199	-0.001	0.0393	0.0393	0.987	95.382 ± 0.007	0.155 ± 0.000	99.799 ± 0.001
	beta44	0.11	0.112	0.002 ± 0.002	0.020 ± 0.014	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.193 ± 0.009	0.440 ± 0.013	0.114	0.004	0.0444	0.0445	1.001	94.177 ± 0.007	0.190 ± 0.001	65.964 ± 0.015
	beta45	0.11	0.110	0.000 ± 0.001	0.004 ± 0.013	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.156 ± 0.007	0.395 ± 0.011	0.110	0.000	0.0433	0.0433	0.989	94.076 ± 0.007	0.169 ± 0.001	74.699 ± 0.014
	beta46	0.08	0.081	0.001 ± 0.002	0.013 ± 0.024	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.584 ± 0.027	0.764 ± 0.021	0.081	0.001	0.0597	0.0598	0.997	95.582 ± 0.007	0.239 ± 0.001	25.602 ± 0.014
	beta47	0.08	0.080	-0.000 ± 0.002	-0.003 ± 0.024	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.558 ± 0.026	0.747 ± 0.021	0.081	0.001	0.0580	0.0581	1.018	95.080 ± 0.007	0.239 ± 0.001	25.201 ± 0.014
	beta51	0.16	0.185	0.025 ± 0.001	0.158 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.089 ± 0.004	0.298 ± 0.007	0.186	0.026	0.0414	0.0488	0.998	91.165 ± 0.009	0.158 ± 0.000	99.498 ± 0.002
	beta52	0.16	0.187	0.027 ± 0.001	0.166 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.098 ± 0.004	0.313 ± 0.008	0.185	0.025	0.0419	0.0490	0.998	90.462 ± 0.009	0.166 ± 0.000	99.297 ± 0.003
	beta53	0.16	0.161	0.001 ± 0.001	0.003 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.160	0.000	0.0396	0.0396	0.998	95.382 ± 0.007	0.158 ± 0.000	97.590 ± 0.005
	beta54	0.16	0.158	-0.002 ± 0.001	-0.010 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.080 ± 0.004	0.283 ± 0.008	0.159	-0.001	0.0440	0.0440	0.958	92.671 ± 0.008	0.170 ± 0.000	94.478 ± 0.007
	beta55	0.08	0.084	0.004 ± 0.001	0.044 ± 0.019	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.349 ± 0.016	0.591 ± 0.017	0.083	0.003	0.0466	0.0467	1.042	95.783 ± 0.006	0.193 ± 0.001	40.562 ± 0.016
	beta56	0.08	0.076	-0.004 ± 0.002	-0.055 ± 0.019	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.370 ± 0.017	0.609 ± 0.017	0.074	-0.006	0.0457	0.0461	0.971	94.478 ± 0.007	0.185 ± 0.001	35.040 ± 0.015
	beta57	0.06	0.061	0.001 ± 0.002	0.022 ± 0.032	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.995 ± 0.047	0.998 ± 0.028	0.062	0.002	0.0570	0.0571	0.988	94.980 ± 0.007	0.232 ± 0.001	17.369 ± 0.012
	beta58	0.06	0.061	0.001 ± 0.002	0.014 ± 0.029	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.831 ± 0.037	0.912 ± 0.025	0.059	-0.001	0.0557	0.0557	1.020	96.486 ± 0.006	0.219 ± 0.001	18.976 ± 0.012
QML	beta41	0.20	0.192	-0.008 ± 0.001	-0.038 ± 0.006	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.039 ± 0.002	0.198 ± 0.006	0.194	-0.006	0.0364	0.0368	0.996	94.076 ± 0.007	0.152 ± 0.000	99.900 ± 0.001
	beta42	0.20	0.198	-0.002 ± 0.001	-0.009 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.039 ± 0.002	0.199 ± 0.005	0.199	-0.001	0.0396	0.0396	1.004	94.779 ± 0.007	0.156 ± 0.000	100.000 ± 0.000
	beta43	0.20	0.190	-0.010 ± 0.001	-0.051 ± 0.006	0.001 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.040 ± 0.002	0.200 ± 0.005	0.190	-0.010	0.0374	0.0388	0.986	93.574 ± 0.008	0.150 ± 0.000	99.799 ± 0.001
	beta44	0.11	0.115	0.005 ± 0.001	0.049 ± 0.014	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.185 ± 0.009	0.431 ± 0.012	0.117	0.007	0.0426	0.0431	1.009	94.880 ± 0.007	0.186 ± 0.000	69.980 ± 0.015
	beta45	0.11	0.110	0.000 ± 0.001	0.004 ± 0.012	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.144 ± 0.006	0.379 ± 0.010	0.109	-0.001	0.0415	0.0415	1.000	95.281 ± 0.007	0.163 ± 0.000	76.606 ± 0.013
	beta46	0.08	0.041	-0.039 ± 0.001	-0.484 ± 0.016	0.002 ± 0.000	0.039 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.476 ± 0.019	0.690 ± 0.015	0.041	-0.039	0.0389	0.0553	0.987	81.727 ± 0.012	0.152 ± 0.000	18.675 ± 0.012
	beta47	0.08	0.044	-0.036 ± 0.001	-0.456 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.435 ± 0.017	0.660 ± 0.014	0.044	-0.036	0.0357	0.0509	1.009	84.237 ± 0.012	0.151 ± 0.000	19.177 ± 0.012
	beta51	0.16	0.174	0.014 ± 0.001	0.086 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.070 ± 0.003	0.265 ± 0.007	0.173	0.013	0.0384	0.0407	1.008	93.675 ± 0.008	0.159 ± 0.000	98.996 ± 0.003
	beta52	0.16	0.170	0.010 ± 0.001	0.060 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.071 ± 0.003	0.266 ± 0.007	0.168	0.008	0.0408	0.0416	1.004	94.277 ± 0.007	0.163 ± 0.000	98.695 ± 0.004
	beta53	0.16	0.160	-0.000 ± 0.001	-0.002 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.061 ± 0.003	0.246 ± 0.007	0.159	-0.001	0.0387	0.0387	0.991	95.482 ± 0.007	0.153 ± 0.000	98.092 ± 0.004
	beta54	0.16	0.169	0.009 ± 0.001	0.059 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.083 ± 0.004	0.288 ± 0.008	0.169	0.009	0.0437	0.0447	0.972	93.574 ± 0.008	0.172 ± 0.000	96.486 ± 0.006
	beta55	0.08	0.085	0.005 ± 0.001	0.060 ± 0.018	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.333 ± 0.015	0.577 ± 0.016	0.085	0.005	0.0459	0.0462	1.067	96.185 ± 0.006	0.192 ± 0.000	41.466 ± 0.016
	beta56	0.08	0.073	-0.007 ± 0.001	-0.087 ± 0.018	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.342 ± 0.016	0.585 ± 0.016	0.071	-0.009	0.0444	0.0453	0.987	94.076 ± 0.007	0.179 ± 0.000	33.635 ± 0.015
	beta57	0.06	0.033	-0.027 ± 0.001	-0.455 ± 0.020	0.001 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.621 ± 0.028	0.788 ± 0.020	0.034	-0.026	0.0375	0.0455	0.993	88.253 ± 0.010	0.150 ± 0.000	12.952 ± 0.011
	beta58	0.06	0.037	-0.023 ± 0.001	-0.391 ± 0.018	0.001 ± 0.000	0.035 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.485 ± 0.020	0.697 ± 0.016	0.036	-0.024	0.0353	0.0427	1.019	89.759 ± 0.010	0.138 ± 0.000	17.871 ± 0.012

Table S54

Simulation 2 - Condition 16: Full Model, Distribution = uniform, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta41	0.20	0.199	-0.001 ± 0.001	-0.007 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.041 ± 0.002	0.202 ± 0.006	0.200	0.000	0.0392	0.0392	0.977	94.779 ± 0.007	0.155 ± 0.000	99.699 ± 0.002
	beta42	0.20	0.202	0.002 ± 0.001	0.012 ± 0.007	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.043 ± 0.002	0.206 ± 0.006	0.203	0.003	0.0394	0.0394	0.995	94.679 ± 0.007	0.161 ± 0.000	100.000 ± 0.000
	beta43	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.041 ± 0.002	0.201 ± 0.006	0.200	0.000	0.0399	0.0399	0.973	94.880 ± 0.007	0.154 ± 0.000	99.799 ± 0.001
	beta44	0.11	0.114	0.004 ± 0.002	0.033 ± 0.014	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.209 ± 0.010	0.457 ± 0.013	0.114	0.004	0.0471	0.0473	0.965	93.876 ± 0.008	0.190 ± 0.001	66.566 ± 0.015
	beta45	0.11	0.111	0.001 ± 0.001	0.010 ± 0.013	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.167 ± 0.007	0.409 ± 0.011	0.111	0.001	0.0445	0.0446	0.956	94.578 ± 0.007	0.169 ± 0.001	75.100 ± 0.014
	beta46	0.08	0.082	0.002 ± 0.002	0.019 ± 0.026	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.657 ± 0.032	0.810 ± 0.024	0.080	0.000	0.0616	0.0616	0.925	94.478 ± 0.007	0.235 ± 0.001	27.510 ± 0.014
	beta47	0.08	0.081	0.001 ± 0.002	0.017 ± 0.025	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.639 ± 0.030	0.799 ± 0.023	0.079	-0.001	0.0589	0.0589	0.935	93.173 ± 0.008	0.235 ± 0.001	27.108 ± 0.014
	beta51	0.16	0.186	0.026 ± 0.001	0.161 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.090 ± 0.004	0.300 ± 0.007	0.186	0.026	0.0409	0.0487	0.981	90.361 ± 0.009	0.156 ± 0.000	99.699 ± 0.002
	beta52	0.16	0.186	0.026 ± 0.001	0.165 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.004	0.312 ± 0.008	0.185	0.025	0.0416	0.0488	0.996	90.161 ± 0.009	0.165 ± 0.000	99.398 ± 0.002
	beta53	0.16	0.161	0.001 ± 0.001	0.005 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.160	0.000	0.0407	0.0407	0.983	95.582 ± 0.007	0.156 ± 0.000	97.590 ± 0.005
	beta54	0.16	0.159	-0.001 ± 0.001	-0.007 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.080 ± 0.003	0.282 ± 0.008	0.158	-0.002	0.0454	0.0454	0.958	93.574 ± 0.008	0.170 ± 0.000	94.578 ± 0.007
	beta55	0.08	0.084	0.004 ± 0.002	0.046 ± 0.019	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.359 ± 0.017	0.599 ± 0.017	0.083	0.003	0.0480	0.0481	1.027	95.884 ± 0.006	0.192 ± 0.001	39.056 ± 0.015
	beta56	0.08	0.076	-0.004 ± 0.002	-0.047 ± 0.019	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.378 ± 0.018	0.615 ± 0.017	0.075	-0.005	0.0467	0.0470	0.959	94.177 ± 0.007	0.184 ± 0.001	35.542 ± 0.015
	beta57	0.06	0.061	0.001 ± 0.002	0.020 ± 0.033	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	1.069 ± 0.053	1.034 ± 0.030	0.062	0.002	0.0590	0.0591	0.937	93.775 ± 0.008	0.228 ± 0.001	17.771 ± 0.012
	beta58	0.06	0.060	0.000 ± 0.002	0.001 ± 0.031	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.939 ± 0.044	0.969 ± 0.027	0.058	-0.002	0.0564	0.0564	0.955	94.679 ± 0.007	0.218 ± 0.001	19.578 ± 0.013

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S55

Simulation 2 - Condition 17: Full Model, Distribution = uniform, N = 400, Reliability = 0.8

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.197	-0.003 ± 0.002	-0.016 ± 0.008	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.072 ± 0.003	0.268 ± 0.008	0.197	-0.003	0.0547	0.0548	0.980	94.885 ± 0.007	0.206 ± 0.000	96.690 ± 0.006
	beta42	0.20	0.202	0.002 ± 0.002	0.012 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.080 ± 0.004	0.283 ± 0.008	0.204	0.004	0.0559	0.0560	0.964	94.483 ± 0.007	0.214 ± 0.001	95.587 ± 0.007
	beta43	0.20	0.201	0.001 ± 0.002	0.006 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.068 ± 0.003	0.261 ± 0.007	0.199	-0.001	0.0534	0.0534	0.992	95.286 ± 0.007	0.203 ± 0.000	97.793 ± 0.005
	beta44	0.11	0.111	0.001 ± 0.002	0.009 ± 0.017	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.300 ± 0.014	0.547 ± 0.015	0.109	-0.001	0.0603	0.0603	0.965	94.082 ± 0.007	0.228 ± 0.001	50.150 ± 0.016
	beta45	0.11	0.111	0.001 ± 0.002	0.009 ± 0.016	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.248 ± 0.012	0.498 ± 0.014	0.111	0.001	0.0523	0.0523	0.971	94.183 ± 0.007	0.209 ± 0.001	56.971 ± 0.016
	beta46	0.08	0.078	-0.002 ± 0.002	-0.028 ± 0.028	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.770 ± 0.033	0.877 ± 0.023	0.078	-0.002	0.0702	0.0703	0.956	94.584 ± 0.007	0.263 ± 0.001	21.866 ± 0.013
	beta47	0.08	0.082	0.002 ± 0.002	0.023 ± 0.026	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.695 ± 0.034	0.834 ± 0.025	0.081	0.001	0.0663	0.0663	1.011	95.186 ± 0.007	0.264 ± 0.001	21.464 ± 0.013
	beta51	0.16	0.185	0.025 ± 0.002	0.155 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.138 ± 0.006	0.371 ± 0.010	0.185	0.025	0.0531	0.0585	0.968	92.177 ± 0.009	0.205 ± 0.000	93.079 ± 0.008
	beta52	0.16	0.185	0.025 ± 0.002	0.155 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.004 ± 0.000	0.060 ± 0.001	0.142 ± 0.006	0.377 ± 0.009	0.184	0.024	0.0576	0.0622	0.994	92.076 ± 0.009	0.214 ± 0.001	92.477 ± 0.008
	beta53	0.16	0.160	0.000 ± 0.002	0.003 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.114 ± 0.005	0.337 ± 0.009	0.161	0.001	0.0534	0.0534	0.973	94.483 ± 0.007	0.206 ± 0.000	85.155 ± 0.011
	beta54	0.16	0.161	0.001 ± 0.002	0.005 ± 0.012	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.133 ± 0.006	0.365 ± 0.010	0.160	0.000	0.0577	0.0577	0.956	93.882 ± 0.008	0.219 ± 0.001	81.846 ± 0.012
	beta55	0.08	0.084	0.004 ± 0.002	0.049 ± 0.024	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.579 ± 0.026	0.761 ± 0.021	0.086	0.006	0.0604	0.0607	0.970	93.781 ± 0.008	0.231 ± 0.001	29.689 ± 0.014
	beta56	0.08	0.079	-0.001 ± 0.002	-0.013 ± 0.023	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.549 ± 0.023	0.741 ± 0.020	0.082	0.002	0.0599	0.0599	0.965	94.082 ± 0.007	0.224 ± 0.001	29.789 ± 0.014
	beta57	0.06	0.057	-0.003 ± 0.002	-0.049 ± 0.035	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	1.188 ± 0.054	1.090 ± 0.030	0.057	-0.003	0.0643	0.0644	1.002	95.687 ± 0.006	0.257 ± 0.001	14.042 ± 0.011
	beta58	0.06	0.056	-0.004 ± 0.002	-0.074 ± 0.033	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	1.110 ± 0.051	1.054 ± 0.029	0.054	-0.006	0.0626	0.0629	0.992	94.483 ± 0.007	0.245 ± 0.001	14.945 ± 0.011
QML	beta41	0.20	0.194	-0.006 ± 0.002	-0.029 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.069 ± 0.003	0.263 ± 0.007	0.194	-0.006	0.0533	0.0537	1.002	95.286 ± 0.007	0.205 ± 0.000	96.790 ± 0.006
	beta42	0.20	0.201	0.001 ± 0.002	0.005 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.076 ± 0.003	0.275 ± 0.008	0.202	0.002	0.0549	0.0550	0.968	94.283 ± 0.007	0.209 ± 0.000	96.189 ± 0.006
	beta43	0.20	0.197	-0.003 ± 0.002	-0.017 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.067 ± 0.003	0.258 ± 0.007	0.195	-0.005	0.0528	0.0530	0.988	95.386 ± 0.007	0.200 ± 0.000	97.292 ± 0.005
	beta44	0.11	0.113	0.003 ± 0.002	0.024 ± 0.017	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.296 ± 0.014	0.544 ± 0.015	0.112	0.002	0.0599	0.0600	0.974	94.784 ± 0.007	0.228 ± 0.001	49.147 ± 0.016
	beta45	0.11	0.112	0.002 ± 0.002	0.016 ± 0.016	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.242 ± 0.011	0.492 ± 0.014	0.113	0.003	0.0530	0.0531	0.976	94.885 ± 0.007	0.207 ± 0.000	58.375 ± 0.016
	beta46	0.08	0.058	-0.022 ± 0.002	-0.280 ± 0.022	0.003 ± 0.000	0.057 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.580 ± 0.024	0.762 ± 0.019	0.057	-0.023	0.0579	0.0622	0.951	90.973 ± 0.009	0.211 ± 0.001	20.361 ± 0.013
	beta47	0.08	0.062	-0.018 ± 0.002	-0.222 ± 0.021	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.486 ± 0.022	0.697 ± 0.019	0.061	-0.019	0.0518	0.0552	1.009	94.082 ± 0.007	0.209 ± 0.001	20.662 ± 0.013
	beta51	0.16	0.170	0.010 ± 0.002	0.062 ± 0.011	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.125 ± 0.006	0.353 ± 0.010	0.170	0.010	0.0544	0.0553	0.974	93.982 ± 0.008	0.212 ± 0.000	87.563 ± 0.010
	beta52	0.16	0.167	0.007 ± 0.002	0.044 ± 0.011	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.001	0.124 ± 0.005	0.351 ± 0.009	0.167	0.007	0.0572	0.0576	0.994	94.183 ± 0.007	0.217 ± 0.000	85.858 ± 0.011
	beta53	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.112 ± 0.005	0.334 ± 0.009	0.161	0.001	0.0529	0.0530	0.972	94.383 ± 0.007	0.204 ± 0.000	85.356 ± 0.011
	beta54	0.16	0.166	0.006 ± 0.002	0.040 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.137 ± 0.006	0.370 ± 0.010	0.165	0.005	0.0576	0.0578	0.970	94.383 ± 0.007	0.224 ± 0.000	83.551 ± 0.012
	beta55	0.08	0.085	0.005 ± 0.002	0.065 ± 0.024	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.584 ± 0.026	0.764 ± 0.021	0.088	0.008	0.0612	0.0617	0.984	94.584 ± 0.007	0.235 ± 0.001	28.786 ± 0.014
	beta56	0.08	0.079	-0.001 ± 0.002	-0.019 ± 0.023	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.540 ± 0.023	0.735 ± 0.019	0.080	0.000	0.0595	0.0595	0.973	94.584 ± 0.007	0.224 ± 0.001	30.090 ± 0.015
	beta57	0.06	0.042	-0.018 ± 0.002	-0.306 ± 0.028	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.056 ± 0.001	0.864 ± 0.039	0.929 ± 0.025	0.041	-0.019	0.0533	0.0565	1.007	92.778 ± 0.008	0.208 ± 0.001	12.136 ± 0.010
	beta58	0.06	0.044	-0.016 ± 0.002	-0.274 ± 0.027	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.781 ± 0.035	0.884 ± 0.023	0.041	-0.019	0.0502	0.0535	0.989	92.879 ± 0.008	0.196 ± 0.001	14.343 ± 0.011

Table S55

Simulation 2 - Condition 17: Full Model, Distribution = uniform, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta41	0.20	0.197	-0.003 ± 0.002	-0.013 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.072 ± 0.003	0.269 ± 0.008	0.197	-0.003	0.0543	0.0544	0.958	93.781 ± 0.008	0.202 ± 0.000	96.891 ± 0.005
	beta42	0.20	0.202	0.002 ± 0.002	0.012 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.080 ± 0.004	0.283 ± 0.008	0.203	0.003	0.0571	0.0572	0.939	93.781 ± 0.008	0.208 ± 0.000	95.988 ± 0.006
	beta43	0.20	0.202	0.002 ± 0.002	0.008 ± 0.008	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.069 ± 0.003	0.263 ± 0.007	0.200	0.000	0.0534	0.0534	0.974	94.985 ± 0.007	0.201 ± 0.000	97.994 ± 0.004
	beta44	0.11	0.112	0.002 ± 0.002	0.016 ± 0.018	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.306 ± 0.014	0.554 ± 0.016	0.110	0.000	0.0617	0.0617	0.956	94.383 ± 0.007	0.228 ± 0.001	48.245 ± 0.016
	beta45	0.11	0.112	0.002 ± 0.002	0.016 ± 0.016	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.260 ± 0.013	0.510 ± 0.015	0.112	0.002	0.0548	0.0548	0.947	93.982 ± 0.008	0.208 ± 0.001	58.175 ± 0.016
	beta46	0.08	0.077	-0.003 ± 0.002	-0.040 ± 0.029	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.817 ± 0.037	0.904 ± 0.025	0.076	-0.004	0.0729	0.0730	0.911	91.976 ± 0.009	0.258 ± 0.001	22.969 ± 0.013
	beta47	0.08	0.082	0.002 ± 0.002	0.021 ± 0.027	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.751 ± 0.038	0.867 ± 0.026	0.080	0.000	0.0674	0.0674	0.949	94.082 ± 0.007	0.258 ± 0.001	24.072 ± 0.014
	beta51	0.16	0.185	0.025 ± 0.002	0.158 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.140 ± 0.006	0.374 ± 0.010	0.185	0.025	0.0523	0.0579	0.959	91.775 ± 0.009	0.204 ± 0.000	93.781 ± 0.008
	beta52	0.16	0.185	0.025 ± 0.002	0.154 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.004 ± 0.000	0.060 ± 0.001	0.143 ± 0.006	0.378 ± 0.009	0.185	0.025	0.0568	0.0620	0.985	92.177 ± 0.009	0.213 ± 0.000	92.377 ± 0.008
	beta53	0.16	0.161	0.001 ± 0.002	0.005 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.115 ± 0.005	0.339 ± 0.009	0.161	0.001	0.0533	0.0533	0.958	93.681 ± 0.008	0.204 ± 0.000	85.757 ± 0.011
	beta54	0.16	0.161	0.001 ± 0.002	0.008 ± 0.012	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.132 ± 0.006	0.364 ± 0.010	0.160	0.000	0.0583	0.0583	0.959	93.882 ± 0.008	0.219 ± 0.000	82.247 ± 0.012
	beta55	0.08	0.083	0.003 ± 0.002	0.041 ± 0.024	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.577 ± 0.026	0.760 ± 0.021	0.085	0.005	0.0595	0.0597	0.974	94.483 ± 0.007	0.232 ± 0.001	29.488 ± 0.014
	beta56	0.08	0.078	-0.002 ± 0.002	-0.019 ± 0.024	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.561 ± 0.024	0.749 ± 0.020	0.080	0.000	0.0601	0.0601	0.956	94.082 ± 0.007	0.225 ± 0.001	29.990 ± 0.015
	beta57	0.06	0.056	-0.004 ± 0.002	-0.070 ± 0.035	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	1.225 ± 0.057	1.107 ± 0.031	0.054	-0.006	0.0636	0.0639	0.970	94.885 ± 0.007	0.252 ± 0.001	14.644 ± 0.011
	beta58	0.06	0.056	-0.004 ± 0.002	-0.066 ± 0.034	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	1.184 ± 0.056	1.088 ± 0.031	0.055	-0.005	0.0658	0.0660	0.945	93.982 ± 0.008	0.242 ± 0.001	15.547 ± 0.011

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S56

Simulation 2 - Condition 18: Full Model, Distribution = uniform, N = 1000, Reliability = 0.8

			Mean-based										Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.198	-0.002 ± 0.001	-0.011 ± 0.005	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.028 ± 0.001	0.167 ± 0.005	0.197	-0.003	0.0329	0.0330	0.989	94.995 ± 0.007	0.129 ± 0.000	100.000 ± 0.000
	beta42	0.20	0.202	0.002 ± 0.001	0.011 ± 0.005	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.030 ± 0.001	0.172 ± 0.005	0.203	0.003	0.0330	0.0331	0.991	94.695 ± 0.007	0.133 ± 0.000	100.000 ± 0.000
	beta43	0.20	0.201	0.001 ± 0.001	0.004 ± 0.005	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.027 ± 0.001	0.164 ± 0.004	0.200	0.000	0.0316	0.0316	0.996	95.395 ± 0.007	0.128 ± 0.000	100.000 ± 0.000
	beta44	0.11	0.112	0.002 ± 0.001	0.020 ± 0.010	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.107 ± 0.005	0.328 ± 0.009	0.114	0.004	0.0345	0.0347	1.015	94.595 ± 0.007	0.143 ± 0.000	86.687 ± 0.011
	beta45	0.11	0.111	0.001 ± 0.001	0.005 ± 0.010	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.102 ± 0.004	0.319 ± 0.009	0.110	0.000	0.0343	0.0343	0.952	92.793 ± 0.008	0.131 ± 0.000	89.289 ± 0.010
	beta46	0.08	0.082	0.002 ± 0.001	0.029 ± 0.017	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.287 ± 0.013	0.536 ± 0.015	0.082	0.002	0.0433	0.0434	0.978	94.795 ± 0.007	0.164 ± 0.000	49.049 ± 0.016
	beta47	0.08	0.079	-0.001 ± 0.001	-0.016 ± 0.017	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.282 ± 0.012	0.531 ± 0.014	0.078	-0.002	0.0422	0.0422	0.988	94.394 ± 0.007	0.164 ± 0.000	46.446 ± 0.016
	beta51	0.16	0.184	0.024 ± 0.001	0.153 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.067 ± 0.003	0.259 ± 0.006	0.184	0.024	0.0346	0.0420	0.986	88.188 ± 0.010	0.129 ± 0.000	100.000 ± 0.000
	beta52	0.16	0.187	0.027 ± 0.001	0.167 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.072 ± 0.003	0.269 ± 0.006	0.187	0.027	0.0335	0.0430	1.021	88.589 ± 0.010	0.135 ± 0.000	99.900 ± 0.001
	beta53	0.16	0.160	-0.000 ± 0.001	-0.000 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.039 ± 0.002	0.198 ± 0.005	0.160	0.000	0.0320	0.0320	1.043	95.896 ± 0.006	0.129 ± 0.000	100.000 ± 0.000
	beta54	0.16	0.159	-0.001 ± 0.001	-0.007 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.048 ± 0.002	0.218 ± 0.006	0.157	-0.003	0.0335	0.0337	1.017	95.596 ± 0.006	0.139 ± 0.000	99.600 ± 0.002
	beta55	0.08	0.080	0.000 ± 0.001	0.002 ± 0.015	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.217 ± 0.010	0.466 ± 0.013	0.080	0.000	0.0367	0.0367	0.999	94.895 ± 0.007	0.146 ± 0.000	56.356 ± 0.016
	beta56	0.08	0.081	0.001 ± 0.001	0.010 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.224 ± 0.010	0.474 ± 0.013	0.081	0.001	0.0382	0.0382	0.950	92.492 ± 0.008	0.141 ± 0.000	60.661 ± 0.015
	beta57	0.06	0.061	0.001 ± 0.001	0.025 ± 0.021	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.436 ± 0.021	0.660 ± 0.019	0.062	0.002	0.0389	0.0389	1.038	95.896 ± 0.006	0.161 ± 0.000	31.131 ± 0.015
	beta58	0.06	0.060	0.000 ± 0.001	0.003 ± 0.020	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.412 ± 0.019	0.642 ± 0.018	0.059	-0.001	0.0380	0.0380	1.019	95.395 ± 0.007	0.154 ± 0.000	31.231 ± 0.015
QML	beta41	0.20	0.195	-0.005 ± 0.001	-0.024 ± 0.005	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.027 ± 0.001	0.165 ± 0.004	0.195	-0.005	0.0319	0.0323	1.009	95.195 ± 0.007	0.129 ± 0.000	100.000 ± 0.000
	beta42	0.20	0.200	0.000 ± 0.001	0.002 ± 0.005	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.028 ± 0.001	0.169 ± 0.005	0.202	0.002	0.0327	0.0327	0.990	94.494 ± 0.007	0.131 ± 0.000	100.000 ± 0.000
	beta43	0.20	0.196	-0.004 ± 0.001	-0.020 ± 0.005	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.027 ± 0.001	0.165 ± 0.004	0.196	-0.004	0.0316	0.0318	0.981	94.394 ± 0.007	0.126 ± 0.000	100.000 ± 0.000
	beta44	0.11	0.114	0.004 ± 0.001	0.035 ± 0.010	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.108 ± 0.005	0.328 ± 0.009	0.115	0.005	0.0349	0.0353	1.019	94.795 ± 0.007	0.143 ± 0.000	88.188 ± 0.010
	beta45	0.11	0.111	0.001 ± 0.001	0.010 ± 0.010	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.098 ± 0.004	0.314 ± 0.009	0.111	0.001	0.0335	0.0335	0.961	93.393 ± 0.008	0.130 ± 0.000	89.890 ± 0.010
	beta46	0.08	0.062	-0.018 ± 0.001	-0.229 ± 0.014	0.001 ± 0.000	0.035 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.239 ± 0.010	0.488 ± 0.012	0.062	-0.018	0.0348	0.0393	0.978	90.490 ± 0.009	0.133 ± 0.000	43.744 ± 0.016
	beta47	0.08	0.060	-0.020 ± 0.001	-0.245 ± 0.013	0.001 ± 0.000	0.034 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.242 ± 0.010	0.492 ± 0.012	0.060	-0.020	0.0342	0.0398	0.983	89.990 ± 0.009	0.131 ± 0.000	43.243 ± 0.016
	beta51	0.16	0.170	0.010 ± 0.001	0.062 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.049 ± 0.002	0.220 ± 0.006	0.169	0.009	0.0349	0.0360	1.002	94.494 ± 0.007	0.133 ± 0.000	99.800 ± 0.001
	beta52	0.16	0.168	0.008 ± 0.001	0.051 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.049 ± 0.002	0.221 ± 0.006	0.169	0.009	0.0344	0.0357	1.012	94.795 ± 0.007	0.137 ± 0.000	99.900 ± 0.001
	beta53	0.16	0.160	-0.000 ± 0.001	-0.003 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.038 ± 0.002	0.196 ± 0.005	0.159	-0.001	0.0314	0.0314	1.046	96.096 ± 0.006	0.128 ± 0.000	100.000 ± 0.000
	beta54	0.16	0.164	0.004 ± 0.001	0.025 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.049 ± 0.002	0.220 ± 0.006	0.162	0.002	0.0331	0.0332	1.028	95.596 ± 0.006	0.141 ± 0.000	99.700 ± 0.002
	beta55	0.08	0.081	0.001 ± 0.001	0.012 ± 0.015	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.220 ± 0.010	0.469 ± 0.013	0.081	0.001	0.0372	0.0373	1.006	95.095 ± 0.007	0.148 ± 0.000	56.957 ± 0.016
	beta56	0.08	0.080	0.000 ± 0.001	0.005 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.222 ± 0.009	0.471 ± 0.013	0.081	0.001	0.0371	0.0372	0.955	93.594 ± 0.008	0.141 ± 0.000	59.960 ± 0.016
	beta57	0.06	0.046	-0.014 ± 0.001	-0.230 ± 0.017	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.338 ± 0.015	0.582 ± 0.015	0.046	-0.014	0.0306	0.0337	1.045	94.294 ± 0.007	0.131 ± 0.000	26.827 ± 0.014
	beta58	0.06	0.048	-0.012 ± 0.001	-0.206 ± 0.016	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.307 ± 0.014	0.554 ± 0.015	0.047	-0.013	0.0299	0.0327	1.020	94.294 ± 0.007	0.124 ± 0.000	30.831 ± 0.015

Table S56

Simulation 2 - Condition 18: Full Model, Distribution = uniform, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta41	0.20	0.198	-0.002 ± 0.001	-0.010 ± 0.005	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.028 ± 0.001	0.167 ± 0.005	0.198	-0.002	0.0332	0.0333	0.971	94.995 ± 0.007	0.127 ± 0.000	100.000 ± 0.000
	beta42	0.20	0.202	0.002 ± 0.001	0.011 ± 0.005	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.030 ± 0.001	0.172 ± 0.005	0.203	0.003	0.0325	0.0326	0.971	93.794 ± 0.008	0.131 ± 0.000	100.000 ± 0.000
	beta43	0.20	0.201	0.001 ± 0.001	0.004 ± 0.005	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.027 ± 0.001	0.164 ± 0.004	0.201	0.001	0.0316	0.0316	0.981	94.895 ± 0.007	0.126 ± 0.000	100.000 ± 0.000
	beta44	0.11	0.112	0.002 ± 0.001	0.023 ± 0.010	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.109 ± 0.005	0.330 ± 0.009	0.115	0.005	0.0351	0.0354	1.008	94.795 ± 0.007	0.143 ± 0.000	86.987 ± 0.011
	beta45	0.11	0.111	0.001 ± 0.001	0.005 ± 0.010	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.102 ± 0.004	0.319 ± 0.009	0.109	-0.001	0.0340	0.0340	0.947	92.993 ± 0.008	0.130 ± 0.000	89.489 ± 0.010
	beta46	0.08	0.082	0.002 ± 0.001	0.026 ± 0.017	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.297 ± 0.013	0.545 ± 0.015	0.081	0.001	0.0435	0.0435	0.954	94.494 ± 0.007	0.163 ± 0.000	51.051 ± 0.016
	beta47	0.08	0.079	-0.001 ± 0.001	-0.013 ± 0.017	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.290 ± 0.012	0.539 ± 0.014	0.077	-0.003	0.0429	0.0430	0.961	93.994 ± 0.008	0.162 ± 0.000	46.847 ± 0.016
	beta51	0.16	0.184	0.024 ± 0.001	0.153 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.067 ± 0.003	0.259 ± 0.006	0.184	0.024	0.0345	0.0419	0.980	88.088 ± 0.010	0.128 ± 0.000	100.000 ± 0.000
	beta52	0.16	0.187	0.027 ± 0.001	0.167 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.072 ± 0.003	0.269 ± 0.006	0.188	0.028	0.0336	0.0434	1.017	88.689 ± 0.010	0.135 ± 0.000	99.900 ± 0.001
	beta53	0.16	0.160	0.000 ± 0.001	0.001 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.039 ± 0.002	0.197 ± 0.005	0.160	0.000	0.0314	0.0314	1.039	96.296 ± 0.006	0.129 ± 0.000	100.000 ± 0.000
	beta54	0.16	0.159	-0.001 ± 0.001	-0.007 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.047 ± 0.002	0.217 ± 0.006	0.157	-0.003	0.0331	0.0332	1.019	95.696 ± 0.006	0.139 ± 0.000	99.700 ± 0.002
	beta55	0.08	0.080	0.000 ± 0.001	0.001 ± 0.015	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.218 ± 0.010	0.467 ± 0.013	0.080	0.000	0.0371	0.0371	0.999	94.695 ± 0.007	0.146 ± 0.000	57.658 ± 0.016
	beta56	0.08	0.081	0.001 ± 0.001	0.010 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.224 ± 0.009	0.474 ± 0.012	0.081	0.001	0.0379	0.0379	0.951	93.093 ± 0.008	0.141 ± 0.000	60.861 ± 0.015
	beta57	0.06	0.061	0.001 ± 0.001	0.024 ± 0.021	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.439 ± 0.021	0.663 ± 0.019	0.061	0.001	0.0380	0.0380	1.026	95.395 ± 0.007	0.160 ± 0.000	30.731 ± 0.015
	beta58	0.06	0.060	0.000 ± 0.001	0.003 ± 0.021	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.425 ± 0.019	0.652 ± 0.018	0.059	-0.001	0.0387	0.0387	0.997	94.895 ± 0.007	0.153 ± 0.000	31.932 ± 0.015

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S57

Simulation 2 - Condition 19: Linear Model, Distribution = normal, N = 400, Reliability = 0.4

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	0.000 ± 0.003	0.001 ± 0.017	0.009 ± 0.000	0.092 ± 0.003	0.009 ± 0.000	0.092 ± 0.003	0.213 ± 0.012	0.461 ± 0.016	0.201	0.001	0.0969	0.0969	1.062	97.000 ± 0.006	0.384 ± 0.004	—
	beta42	0.20	0.207	0.007 ± 0.003	0.037 ± 0.017	0.008 ± 0.000	0.090 ± 0.003	0.008 ± 0.000	0.091 ± 0.003	0.205 ± 0.012	0.453 ± 0.015	0.206	0.006	0.0913	0.0915	1.099	96.857 ± 0.007	0.389 ± 0.004	—
	beta43	0.20	0.195	-0.005 ± 0.003	-0.023 ± 0.016	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	0.183 ± 0.010	0.428 ± 0.014	0.188	-0.012	0.0796	0.0804	1.045	95.000 ± 0.008	0.350 ± 0.003	—
	beta44	0.00	0.004	0.004 ± 0.006	—	0.023 ± 0.002	0.152 ± 0.005	0.023 ± 0.002	0.152 ± 0.006	—	—	-0.004	-0.004	0.1400	0.1400	0.969	96.714 ± 0.007	0.576 ± 0.011	3.286 ± 0.007
	beta45	0.00	-0.003	-0.003 ± 0.004	—	0.011 ± 0.001	0.104 ± 0.003	0.011 ± 0.001	0.104 ± 0.004	—	—	-0.001	-0.001	0.1009	0.1009	1.114	97.286 ± 0.006	0.455 ± 0.007	2.714 ± 0.006
	beta46	0.00	0.000	0.000 ± 0.004	—	0.011 ± 0.001	0.104 ± 0.004	0.011 ± 0.001	0.104 ± 0.004	—	—	0.007	0.007	0.0904	0.0906	0.992	98.286 ± 0.005	0.405 ± 0.008	1.714 ± 0.005
	beta47	0.00	-0.003	-0.003 ± 0.004	—	0.010 ± 0.001	0.101 ± 0.004	0.010 ± 0.001	0.101 ± 0.004	—	—	-0.003	-0.003	0.0930	0.0930	0.976	98.714 ± 0.004	0.386 ± 0.007	1.286 ± 0.004
	beta51	0.16	0.162	0.002 ± 0.003	0.010 ± 0.022	0.008 ± 0.000	0.092 ± 0.002	0.008 ± 0.000	0.092 ± 0.003	0.328 ± 0.017	0.573 ± 0.019	0.155	-0.005	0.0901	0.0902	1.043	96.857 ± 0.007	0.375 ± 0.003	—
	beta52	0.16	0.166	0.006 ± 0.004	0.039 ± 0.023	0.009 ± 0.001	0.097 ± 0.003	0.009 ± 0.001	0.097 ± 0.004	0.368 ± 0.023	0.607 ± 0.022	0.161	0.001	0.0914	0.0914	0.994	95.571 ± 0.008	0.378 ± 0.003	—
	beta53	0.16	0.161	0.001 ± 0.003	0.004 ± 0.022	0.008 ± 0.000	0.092 ± 0.003	0.008 ± 0.000	0.092 ± 0.003	0.331 ± 0.019	0.575 ± 0.019	0.161	0.001	0.0927	0.0927	1.009	95.857 ± 0.008	0.364 ± 0.003	—
	beta54	0.16	0.167	0.007 ± 0.004	0.044 ± 0.025	0.011 ± 0.001	0.105 ± 0.003	0.011 ± 0.001	0.105 ± 0.004	0.429 ± 0.025	0.655 ± 0.023	0.161	0.001	0.1051	0.1051	0.944	95.429 ± 0.008	0.387 ± 0.003	—
	beta55	0.00	0.008	0.008 ± 0.006	—	0.023 ± 0.002	0.151 ± 0.005	0.023 ± 0.002	0.151 ± 0.006	—	—	0.001	0.001	0.1301	0.1301	1.009	97.143 ± 0.006	0.599 ± 0.011	2.857 ± 0.006
	beta56	0.00	-0.008	-0.008 ± 0.005	—	0.017 ± 0.001	0.130 ± 0.004	0.017 ± 0.001	0.130 ± 0.005	—	—	0.000	0.000	0.1152	0.1152	0.975	96.714 ± 0.007	0.496 ± 0.008	3.286 ± 0.007
	beta57	0.00	0.001	0.001 ± 0.004	—	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	—	—	-0.001	-0.001	0.0927	0.0927	1.024	97.714 ± 0.006	0.396 ± 0.008	2.286 ± 0.006
	beta58	0.00	0.005	0.005 ± 0.003	—	0.006 ± 0.000	0.078 ± 0.003	0.006 ± 0.000	0.078 ± 0.003	—	—	0.005	0.005	0.0742	0.0744	1.023	98.143 ± 0.005	0.313 ± 0.005	1.857 ± 0.005
QML	beta41	0.20	0.198	-0.002 ± 0.003	-0.010 ± 0.016	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	0.169 ± 0.009	0.411 ± 0.013	0.196	-0.004	0.0852	0.0853	0.987	95.571 ± 0.008	0.318 ± 0.002	—
	beta42	0.20	0.205	0.005 ± 0.003	0.026 ± 0.015	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	0.167 ± 0.010	0.409 ± 0.014	0.200	0.000	0.0793	0.0793	1.031	96.286 ± 0.007	0.330 ± 0.002	—
	beta43	0.20	0.194	-0.006 ± 0.003	-0.028 ± 0.015	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.003	0.159 ± 0.009	0.399 ± 0.013	0.192	-0.008	0.0772	0.0776	1.007	94.143 ± 0.009	0.314 ± 0.002	—
	beta44	0.00	-0.016	-0.016 ± 0.005	—	0.014 ± 0.001	0.120 ± 0.004	0.015 ± 0.001	0.121 ± 0.004	—	—	-0.022	-0.022	0.1226	0.1246	0.935	94.429 ± 0.009	0.441 ± 0.004	5.571 ± 0.009
	beta45	0.00	-0.015	-0.015 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.002	0.008 ± 0.000	0.087 ± 0.003	—	—	-0.015	-0.015	0.0854	0.0867	1.076	96.286 ± 0.007	0.361 ± 0.003	3.714 ± 0.007
	beta46	0.00	0.003	0.003 ± 0.003	—	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	—	—	0.004	0.004	0.0715	0.0716	0.980	96.429 ± 0.007	0.294 ± 0.003	3.571 ± 0.007
	beta47	0.00	-0.006	-0.006 ± 0.003	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.003	—	—	-0.002	-0.002	0.0731	0.0731	0.969	95.714 ± 0.008	0.284 ± 0.003	4.286 ± 0.008
	beta51	0.16	0.159	-0.001 ± 0.003	-0.006 ± 0.021	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	0.304 ± 0.015	0.551 ± 0.018	0.158	-0.002	0.0876	0.0876	0.997	95.714 ± 0.008	0.345 ± 0.002	—
	beta52	0.16	0.165	0.005 ± 0.003	0.031 ± 0.022	0.009 ± 0.000	0.092 ± 0.003	0.009 ± 0.000	0.092 ± 0.003	0.333 ± 0.020	0.577 ± 0.020	0.157	-0.003	0.0894	0.0894	0.968	95.000 ± 0.008	0.350 ± 0.002	—
	beta53	0.16	0.160	-0.000 ± 0.003	-0.001 ± 0.021	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.003	0.296 ± 0.016	0.544 ± 0.018	0.159	-0.001	0.0888	0.0888	0.958	93.714 ± 0.009	0.327 ± 0.002	—
	beta54	0.16	0.171	0.011 ± 0.004	0.070 ± 0.025	0.011 ± 0.001	0.104 ± 0.003	0.011 ± 0.001	0.104 ± 0.004	0.425 ± 0.026	0.652 ± 0.023	0.162	0.002	0.1024	0.1024	0.960	95.571 ± 0.008	0.390 ± 0.003	—
	beta55	0.00	0.003	0.003 ± 0.005	—	0.015 ± 0.001	0.121 ± 0.003	0.015 ± 0.001	0.121 ± 0.004	—	—	0.001	0.001	0.1157	0.1157	1.019	95.857 ± 0.008	0.482 ± 0.004	4.143 ± 0.008
	beta56	0.00	-0.015	-0.015 ± 0.004	—	0.011 ± 0.001	0.105 ± 0.003	0.011 ± 0.001	0.106 ± 0.004	—	—	-0.011	-0.011	0.1008	0.1015	0.974	96.286 ± 0.007	0.402 ± 0.003	3.714 ± 0.007
	beta57	0.00	-0.005	-0.005 ± 0.003	—	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.003	—	—	-0.002	-0.002	0.0745	0.0745	0.984	95.857 ± 0.008	0.300 ± 0.003	4.143 ± 0.008
	beta58	0.00	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.002	-0.002	0.0606	0.0606	1.005	96.429 ± 0.007	0.246 ± 0.002	3.571 ± 0.007

Table S57

Simulation 2 - Condition 19: Linear Model, Distribution = normal, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta41	0.20	0.203	0.003 ± 0.004	0.017 ± 0.019	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.004	0.251 ± 0.016	0.501 ± 0.019	0.202	0.002	0.1096	0.1096	0.913	94.857 ± 0.008	0.359 ± 0.003	—
	beta42	0.20	0.211	0.011 ± 0.004	0.056 ± 0.019	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.101 ± 0.004	0.253 ± 0.017	0.503 ± 0.019	0.207	0.007	0.1013	0.1016	0.946	95.000 ± 0.008	0.371 ± 0.003	—
	beta43	0.20	0.199	-0.001 ± 0.004	-0.007 ± 0.018	0.009 ± 0.000	0.093 ± 0.003	0.009 ± 0.000	0.093 ± 0.003	0.216 ± 0.012	0.464 ± 0.016	0.193	-0.007	0.0952	0.0954	0.939	93.714 ± 0.009	0.342 ± 0.002	—
	beta44	0.00	-0.005	-0.005 ± 0.007	—	0.032 ± 0.002	0.179 ± 0.007	0.032 ± 0.002	0.179 ± 0.007	—	—	0.000	0.000	0.1674	0.1674	0.759	92.000 ± 0.010	0.532 ± 0.010	8.000 ± 0.010
	beta45	0.00	-0.003	-0.003 ± 0.005	—	0.018 ± 0.001	0.132 ± 0.004	0.018 ± 0.001	0.132 ± 0.005	—	—	0.001	0.001	0.1238	0.1238	0.840	94.714 ± 0.008	0.436 ± 0.008	5.286 ± 0.008
	beta46	0.00	0.007	0.007 ± 0.005	—	0.016 ± 0.001	0.125 ± 0.005	0.016 ± 0.001	0.126 ± 0.005	—	—	0.007	0.007	0.1089	0.1092	0.748	92.429 ± 0.010	0.368 ± 0.008	7.571 ± 0.010
	beta47	0.00	-0.002	-0.002 ± 0.005	—	0.016 ± 0.001	0.126 ± 0.005	0.016 ± 0.001	0.126 ± 0.005	—	—	-0.003	-0.003	0.1092	0.1093	0.756	92.714 ± 0.010	0.373 ± 0.010	7.286 ± 0.010
	beta51	0.16	0.163	0.003 ± 0.004	0.021 ± 0.023	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.096 ± 0.003	0.359 ± 0.020	0.599 ± 0.020	0.164	0.004	0.0981	0.0982	0.936	94.857 ± 0.008	0.352 ± 0.002	—
	beta52	0.16	0.169	0.009 ± 0.004	0.057 ± 0.024	0.010 ± 0.001	0.101 ± 0.003	0.010 ± 0.001	0.102 ± 0.004	0.403 ± 0.026	0.635 ± 0.024	0.163	0.003	0.0994	0.0994	0.923	94.429 ± 0.009	0.366 ± 0.003	—
	beta53	0.16	0.163	0.003 ± 0.004	0.022 ± 0.023	0.009 ± 0.000	0.095 ± 0.003	0.009 ± 0.000	0.095 ± 0.003	0.354 ± 0.019	0.595 ± 0.020	0.165	0.005	0.0928	0.0929	0.933	94.286 ± 0.009	0.348 ± 0.002	—
	beta54	0.16	0.167	0.007 ± 0.004	0.044 ± 0.024	0.011 ± 0.001	0.103 ± 0.003	0.011 ± 0.001	0.103 ± 0.004	0.417 ± 0.024	0.646 ± 0.022	0.158	-0.002	0.1065	0.1065	0.939	94.714 ± 0.008	0.380 ± 0.003	—
	beta55	0.00	0.004	0.004 ± 0.006	—	0.028 ± 0.002	0.167 ± 0.007	0.028 ± 0.002	0.167 ± 0.007	—	—	-0.004	-0.004	0.1395	0.1396	0.841	93.286 ± 0.009	0.550 ± 0.011	6.714 ± 0.009
	beta56	0.00	-0.008	-0.008 ± 0.006	—	0.022 ± 0.002	0.147 ± 0.005	0.022 ± 0.002	0.147 ± 0.006	—	—	-0.004	-0.004	0.1161	0.1162	0.809	94.571 ± 0.009	0.467 ± 0.009	5.429 ± 0.009
	beta57	0.00	0.001	0.001 ± 0.004	—	0.013 ± 0.001	0.113 ± 0.004	0.013 ± 0.001	0.113 ± 0.005	—	—	-0.001	-0.001	0.0948	0.0948	0.802	94.286 ± 0.009	0.355 ± 0.007	5.714 ± 0.009
	beta58	0.00	0.010	0.010 ± 0.004	—	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.094 ± 0.004	—	—	0.005	0.005	0.0776	0.0777	0.815	95.571 ± 0.008	0.299 ± 0.008	4.429 ± 0.008

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S58

Simulation 2 - Condition 20: Linear Model, Distribution = normal, N = 1000, Reliability = 0.4

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.202	0.002 ± 0.002	0.009 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.078 ± 0.004	0.279 ± 0.008	0.199	-0.001	0.0552	0.0552	0.987	94.262 ± 0.007	0.216 ± 0.001	—
	beta42	0.20	0.203	0.003 ± 0.002	0.017 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.078 ± 0.004	0.279 ± 0.008	0.203	0.003	0.0531	0.0532	1.012	95.287 ± 0.007	0.221 ± 0.001	—
	beta43	0.20	0.199	-0.001 ± 0.002	-0.004 ± 0.009	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.071 ± 0.003	0.267 ± 0.007	0.197	-0.003	0.0540	0.0541	0.985	94.467 ± 0.007	0.206 ± 0.001	—
	beta44	0.00	0.000	-0.000 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	—	—	0.001	0.001	0.0723	0.0723	1.012	96.619 ± 0.006	0.296 ± 0.002	3.381 ± 0.006
	beta45	0.00	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	0.003	0.003	0.0589	0.0590	0.987	95.902 ± 0.006	0.245 ± 0.002	4.098 ± 0.006
	beta46	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	0.000	0.000	0.0492	0.0492	0.980	95.799 ± 0.006	0.205 ± 0.002	4.201 ± 0.006
	beta47	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	0.002	0.002	0.0470	0.0470	0.990	96.107 ± 0.006	0.202 ± 0.002	3.893 ± 0.006
	beta51	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.110 ± 0.005	0.332 ± 0.009	0.161	0.001	0.0531	0.0531	1.021	95.492 ± 0.007	0.212 ± 0.001	—
	beta52	0.16	0.160	-0.000 ± 0.002	-0.002 ± 0.011	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.129 ± 0.006	0.359 ± 0.010	0.158	-0.002	0.0582	0.0583	0.978	95.287 ± 0.007	0.220 ± 0.001	—
	beta53	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.114 ± 0.005	0.338 ± 0.009	0.161	0.001	0.0526	0.0526	0.989	94.877 ± 0.007	0.210 ± 0.001	—
	beta54	0.16	0.162	0.002 ± 0.002	0.013 ± 0.012	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.150 ± 0.006	0.387 ± 0.010	0.162	0.002	0.0630	0.0631	0.950	94.262 ± 0.007	0.230 ± 0.001	—
	beta55	0.00	0.002	0.002 ± 0.002	—	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	—	—	0.004	0.004	0.0729	0.0730	1.018	96.824 ± 0.006	0.306 ± 0.002	3.176 ± 0.006
	beta56	0.00	0.001	0.001 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	-0.001	-0.001	0.0656	0.0656	0.990	95.902 ± 0.006	0.266 ± 0.002	4.098 ± 0.006
	beta57	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	-0.002	-0.002	0.0482	0.0483	1.001	96.516 ± 0.006	0.201 ± 0.002	3.484 ± 0.006
	beta58	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	-0.001	-0.001	0.0380	0.0380	0.986	95.902 ± 0.006	0.170 ± 0.001	4.098 ± 0.006
QML	beta41	0.20	0.199	-0.001 ± 0.002	-0.006 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.066 ± 0.003	0.257 ± 0.007	0.197	-0.003	0.0528	0.0529	0.987	94.057 ± 0.008	0.199 ± 0.001	—
	beta42	0.20	0.202	0.002 ± 0.002	0.010 ± 0.009	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.071 ± 0.004	0.266 ± 0.008	0.200	0.000	0.0545	0.0545	0.988	95.287 ± 0.007	0.206 ± 0.001	—
	beta43	0.20	0.197	-0.003 ± 0.002	-0.014 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.065 ± 0.003	0.255 ± 0.007	0.197	-0.003	0.0532	0.0533	0.989	94.365 ± 0.007	0.197 ± 0.001	—
	beta44	0.00	-0.016	-0.016 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	-0.017	-0.017	0.0725	0.0744	1.007	95.799 ± 0.006	0.273 ± 0.001	4.201 ± 0.006
	beta45	0.00	-0.011	-0.011 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.010	-0.010	0.0545	0.0554	0.994	95.184 ± 0.007	0.225 ± 0.001	4.816 ± 0.007
	beta46	0.00	0.001	0.001 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.001	0.001	0.0451	0.0451	0.979	95.594 ± 0.007	0.180 ± 0.001	4.406 ± 0.007
	beta47	0.00	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.004	-0.004	0.0420	0.0421	0.996	95.799 ± 0.006	0.174 ± 0.001	4.201 ± 0.006
	beta51	0.16	0.158	-0.002 ± 0.002	-0.014 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.105 ± 0.005	0.324 ± 0.009	0.160	0.000	0.0514	0.0514	1.015	95.594 ± 0.007	0.206 ± 0.001	—
	beta52	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.011	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.121 ± 0.005	0.348 ± 0.009	0.156	-0.004	0.0543	0.0544	0.972	94.980 ± 0.007	0.212 ± 0.001	—
	beta53	0.16	0.158	-0.002 ± 0.002	-0.010 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.108 ± 0.005	0.329 ± 0.009	0.158	-0.002	0.0504	0.0504	0.978	94.570 ± 0.007	0.202 ± 0.001	—
	beta54	0.16	0.163	0.003 ± 0.002	0.022 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.145 ± 0.006	0.381 ± 0.010	0.163	0.003	0.0634	0.0635	0.980	94.877 ± 0.007	0.234 ± 0.001	—
	beta55	0.00	0.000	0.000 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	0.002	0.002	0.0734	0.0734	1.022	96.311 ± 0.006	0.289 ± 0.002	3.689 ± 0.006
	beta56	0.00	-0.010	-0.010 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.011	-0.011	0.0623	0.0633	0.986	95.389 ± 0.007	0.243 ± 0.001	4.611 ± 0.007
	beta57	0.00	-0.006	-0.006 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.008	-0.008	0.0451	0.0458	1.014	96.414 ± 0.006	0.181 ± 0.001	3.586 ± 0.006
	beta58	0.00	-0.006	-0.006 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.007	-0.007	0.0349	0.0357	0.995	94.672 ± 0.007	0.150 ± 0.001	5.328 ± 0.007

Table S58

Simulation 2 - Condition 20: Linear Model, Distribution = normal, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta41	0.20	0.203	0.003 ± 0.002	0.015 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.078 ± 0.004	0.279 ± 0.008	0.201	0.001	0.0556	0.0557	0.971	94.160 ± 0.008	0.212 ± 0.001	—
	beta42	0.20	0.205	0.005 ± 0.002	0.025 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.083 ± 0.004	0.288 ± 0.008	0.205	0.005	0.0567	0.0569	0.982	95.184 ± 0.007	0.221 ± 0.001	—
	beta43	0.20	0.200	0.000 ± 0.002	0.000 ± 0.009	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.071 ± 0.003	0.267 ± 0.008	0.198	-0.002	0.0543	0.0544	0.981	94.365 ± 0.007	0.206 ± 0.001	—
	beta44	0.00	0.001	0.001 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	—	—	0.002	0.002	0.0773	0.0773	0.899	94.365 ± 0.007	0.291 ± 0.003	5.635 ± 0.007
	beta45	0.00	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	0.002	0.002	0.0634	0.0635	0.931	94.672 ± 0.007	0.242 ± 0.002	5.328 ± 0.007
	beta46	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	0.000	0.000	0.0510	0.0510	0.906	93.545 ± 0.008	0.200 ± 0.002	6.455 ± 0.008
	beta47	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	—	—	0.001	0.001	0.0546	0.0546	0.875	95.184 ± 0.007	0.200 ± 0.002	4.816 ± 0.007
	beta51	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.112 ± 0.005	0.335 ± 0.010	0.161	0.001	0.0545	0.0545	0.994	94.775 ± 0.007	0.209 ± 0.001	—
	beta52	0.16	0.160	0.000 ± 0.002	0.003 ± 0.012	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.130 ± 0.006	0.361 ± 0.010	0.159	-0.001	0.0592	0.0592	0.963	94.980 ± 0.007	0.218 ± 0.001	—
	beta53	0.16	0.160	0.000 ± 0.002	0.002 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.114 ± 0.005	0.337 ± 0.010	0.161	0.001	0.0527	0.0527	0.980	94.467 ± 0.007	0.207 ± 0.001	—
	beta54	0.16	0.162	0.002 ± 0.002	0.012 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.143 ± 0.006	0.378 ± 0.010	0.161	0.001	0.0619	0.0619	0.975	95.082 ± 0.007	0.231 ± 0.001	—
	beta55	0.00	0.001	0.001 ± 0.003	—	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	—	—	0.003	0.003	0.0777	0.0778	0.941	95.287 ± 0.007	0.299 ± 0.003	4.713 ± 0.007
	beta56	0.00	0.001	0.001 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	0.000	0.000	0.0668	0.0668	0.941	95.492 ± 0.007	0.257 ± 0.002	4.508 ± 0.007
	beta57	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.002	-0.002	0.0503	0.0504	0.911	94.775 ± 0.007	0.195 ± 0.002	5.225 ± 0.007
	beta58	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.000	0.000	0.0400	0.0400	0.910	94.570 ± 0.007	0.164 ± 0.002	5.430 ± 0.007

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S59

Simulation 2 - Condition 21: Linear Model, Distribution = normal, N = 400, Reliability = 0.6

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.002	-0.001 ± 0.010	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.104 ± 0.005	0.323 ± 0.009	0.199	-0.001	0.0655	0.0655	0.986	95.408 ± 0.007	0.250 ± 0.001	—
	beta42	0.20	0.203	0.003 ± 0.002	0.014 ± 0.011	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.110 ± 0.005	0.332 ± 0.010	0.202	0.002	0.0683	0.0684	0.987	95.000 ± 0.007	0.257 ± 0.001	—
	beta43	0.20	0.197	-0.003 ± 0.002	-0.013 ± 0.010	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.095 ± 0.004	0.308 ± 0.009	0.196	-0.004	0.0577	0.0578	1.014	95.510 ± 0.007	0.245 ± 0.001	—
	beta44	0.00	-0.001	-0.001 ± 0.003	—	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	—	—	-0.001	-0.001	0.0759	0.0759	0.975	94.796 ± 0.007	0.303 ± 0.002	5.204 ± 0.007
	beta45	0.00	0.000	0.000 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	—	—	-0.003	-0.003	0.0731	0.0732	0.929	93.673 ± 0.008	0.265 ± 0.002	6.327 ± 0.008
	beta46	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.002	-0.002	0.0560	0.0561	0.944	93.571 ± 0.008	0.215 ± 0.002	6.429 ± 0.008
	beta47	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	0.001	0.001	0.0545	0.0545	0.945	95.510 ± 0.007	0.210 ± 0.002	4.490 ± 0.007
	beta51	0.16	0.159	-0.001 ± 0.002	-0.003 ± 0.012	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.148 ± 0.007	0.385 ± 0.011	0.161	0.001	0.0619	0.0619	1.025	95.714 ± 0.006	0.248 ± 0.001	—
	beta52	0.16	0.162	0.002 ± 0.002	0.011 ± 0.013	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.177 ± 0.008	0.421 ± 0.012	0.161	0.001	0.0690	0.0690	0.972	95.000 ± 0.007	0.257 ± 0.001	—
	beta53	0.16	0.162	0.002 ± 0.002	0.012 ± 0.013	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.157 ± 0.007	0.396 ± 0.011	0.159	-0.001	0.0645	0.0645	0.996	95.306 ± 0.007	0.248 ± 0.001	—
	beta54	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.194 ± 0.009	0.441 ± 0.013	0.160	0.000	0.0724	0.0724	0.971	95.204 ± 0.007	0.269 ± 0.001	—
	beta55	0.00	-0.003	-0.003 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	—	—	-0.003	-0.003	0.0833	0.0834	0.970	94.388 ± 0.007	0.316 ± 0.002	5.612 ± 0.007
	beta56	0.00	0.000	-0.000 ± 0.002	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	0.001	0.001	0.0711	0.0711	1.003	94.490 ± 0.007	0.279 ± 0.002	5.510 ± 0.007
	beta57	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	—	—	-0.002	-0.002	0.0536	0.0537	0.953	95.408 ± 0.007	0.211 ± 0.002	4.592 ± 0.007
	beta58	0.00	0.002	0.002 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.001	0.001	0.0458	0.0458	0.980	94.286 ± 0.007	0.185 ± 0.001	5.714 ± 0.007
QML	beta41	0.20	0.198	-0.002 ± 0.002	-0.010 ± 0.010	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.097 ± 0.004	0.311 ± 0.009	0.198	-0.002	0.0594	0.0594	0.984	94.898 ± 0.007	0.240 ± 0.001	—
	beta42	0.20	0.202	0.002 ± 0.002	0.009 ± 0.010	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.102 ± 0.005	0.319 ± 0.009	0.200	0.000	0.0649	0.0649	0.987	95.306 ± 0.007	0.247 ± 0.001	—
	beta43	0.20	0.197	-0.003 ± 0.002	-0.016 ± 0.010	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.091 ± 0.004	0.302 ± 0.008	0.196	-0.004	0.0572	0.0573	1.004	95.408 ± 0.007	0.238 ± 0.001	—
	beta44	0.00	-0.010	-0.010 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	—	—	-0.010	-0.010	0.0763	0.0770	0.996	95.612 ± 0.007	0.294 ± 0.001	4.388 ± 0.007
	beta45	0.00	-0.007	-0.007 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	-0.011	-0.011	0.0705	0.0713	0.945	94.286 ± 0.007	0.255 ± 0.001	5.714 ± 0.007
	beta46	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	0.000	0.000	0.0506	0.0506	0.976	95.408 ± 0.007	0.201 ± 0.001	4.592 ± 0.007
	beta47	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	-0.001	-0.001	0.0488	0.0488	0.953	95.204 ± 0.007	0.195 ± 0.001	4.796 ± 0.007
	beta51	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.012	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.149 ± 0.007	0.385 ± 0.011	0.160	0.000	0.0636	0.0636	1.023	96.531 ± 0.006	0.247 ± 0.001	—
	beta52	0.16	0.161	0.001 ± 0.002	0.005 ± 0.013	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.171 ± 0.008	0.413 ± 0.012	0.158	-0.002	0.0680	0.0680	0.977	95.306 ± 0.007	0.253 ± 0.001	—
	beta53	0.16	0.161	0.001 ± 0.002	0.009 ± 0.012	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.152 ± 0.007	0.390 ± 0.011	0.159	-0.001	0.0627	0.0627	0.991	95.102 ± 0.007	0.243 ± 0.001	—
	beta54	0.16	0.161	0.001 ± 0.002	0.003 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.191 ± 0.009	0.437 ± 0.012	0.161	0.001	0.0724	0.0724	1.003	96.429 ± 0.006	0.275 ± 0.001	—
	beta55	0.00	-0.005	-0.005 ± 0.003	—	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	—	—	-0.006	-0.006	0.0810	0.0812	1.000	95.510 ± 0.007	0.309 ± 0.001	4.490 ± 0.007
	beta56	0.00	-0.005	-0.005 ± 0.002	—	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	—	—	-0.004	-0.004	0.0669	0.0670	1.023	95.816 ± 0.006	0.270 ± 0.001	4.184 ± 0.006
	beta57	0.00	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	-0.005	-0.005	0.0496	0.0498	0.973	95.510 ± 0.007	0.200 ± 0.001	4.490 ± 0.007
	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	-0.002	-0.002	0.0422	0.0423	1.016	94.898 ± 0.007	0.175 ± 0.001	5.102 ± 0.007

Table S59

Simulation 2 - Condition 21: Linear Model, Distribution = normal, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta41	0.20	0.201	0.001 ± 0.002	0.004 ± 0.010	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.105 ± 0.005	0.324 ± 0.009	0.199	-0.001	0.0639	0.0639	0.976	95.000 ± 0.007	0.248 ± 0.001	—
	beta42	0.20	0.204	0.004 ± 0.002	0.020 ± 0.011	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.112 ± 0.005	0.335 ± 0.010	0.203	0.003	0.0689	0.0690	0.976	94.388 ± 0.007	0.256 ± 0.001	—
	beta43	0.20	0.199	-0.001 ± 0.002	-0.006 ± 0.010	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.095 ± 0.004	0.309 ± 0.009	0.198	-0.002	0.0582	0.0582	1.005	95.102 ± 0.007	0.243 ± 0.001	—
	beta44	0.00	-0.001	-0.001 ± 0.003	—	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	—	—	-0.004	-0.004	0.0800	0.0801	0.938	95.306 ± 0.007	0.302 ± 0.002	4.694 ± 0.007
	beta45	0.00	0.000	-0.000 ± 0.002	—	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	—	—	-0.001	-0.001	0.0745	0.0745	0.885	93.367 ± 0.008	0.264 ± 0.002	6.633 ± 0.008
	beta46	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.001	-0.001	0.0570	0.0571	0.869	93.265 ± 0.008	0.214 ± 0.002	6.735 ± 0.008
	beta47	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	—	—	0.001	0.001	0.0560	0.0560	0.906	94.592 ± 0.007	0.208 ± 0.002	5.408 ± 0.007
	beta51	0.16	0.160	0.000 ± 0.002	0.002 ± 0.012	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.152 ± 0.007	0.390 ± 0.011	0.163	0.003	0.0634	0.0634	1.008	96.020 ± 0.006	0.247 ± 0.001	—
	beta52	0.16	0.162	0.002 ± 0.002	0.013 ± 0.014	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.179 ± 0.009	0.423 ± 0.012	0.164	0.004	0.0691	0.0692	0.963	94.592 ± 0.007	0.256 ± 0.001	—
	beta53	0.16	0.163	0.003 ± 0.002	0.018 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.158 ± 0.007	0.398 ± 0.011	0.160	0.000	0.0639	0.0639	0.987	94.898 ± 0.007	0.246 ± 0.001	—
	beta54	0.16	0.160	-0.000 ± 0.002	-0.003 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.190 ± 0.009	0.436 ± 0.012	0.159	-0.001	0.0705	0.0705	0.989	95.816 ± 0.006	0.270 ± 0.001	—
	beta55	0.00	-0.002	-0.002 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.002	—	—	0.000	0.000	0.0838	0.0838	0.940	95.204 ± 0.007	0.315 ± 0.002	4.796 ± 0.007
	beta56	0.00	0.000	0.000 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	0.001	0.001	0.0714	0.0714	0.971	95.102 ± 0.007	0.276 ± 0.002	4.898 ± 0.007
	beta57	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.003	-0.003	0.0539	0.0539	0.904	93.878 ± 0.008	0.209 ± 0.002	6.122 ± 0.008
	beta58	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	—	—	0.001	0.001	0.0455	0.0455	0.944	94.388 ± 0.007	0.185 ± 0.001	5.612 ± 0.007

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S60

Simulation 2 - Condition 22: Linear Model, Distribution = normal, N = 1000, Reliability = 0.6

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.006	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.042 ± 0.002	0.204 ± 0.006	0.198	-0.002	0.0419	0.0419	0.961	94.551 ± 0.007	0.154 ± 0.000	—
	beta42	0.20	0.201	0.001 ± 0.001	0.003 ± 0.007	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.044 ± 0.002	0.209 ± 0.006	0.199	-0.001	0.0413	0.0413	0.966	93.946 ± 0.008	0.158 ± 0.000	—
	beta43	0.20	0.200	0.000 ± 0.001	0.001 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.039 ± 0.002	0.198 ± 0.006	0.199	-0.001	0.0391	0.0391	0.977	94.450 ± 0.007	0.152 ± 0.000	—
	beta44	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.001	-0.001	0.0478	0.0478	0.977	94.854 ± 0.007	0.182 ± 0.001	5.146 ± 0.007
	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	-0.001	-0.001	0.0438	0.0438	0.929	93.340 ± 0.008	0.158 ± 0.001	6.660 ± 0.008
	beta46	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.001	-0.001	0.0333	0.0333	0.943	93.340 ± 0.008	0.128 ± 0.001	6.660 ± 0.008
	beta47	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.003	0.003	0.0295	0.0296	0.987	95.055 ± 0.007	0.126 ± 0.001	4.945 ± 0.007
	beta51	0.16	0.160	0.000 ± 0.001	0.000 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.060 ± 0.003	0.245 ± 0.007	0.160	0.000	0.0396	0.0396	0.997	94.450 ± 0.007	0.153 ± 0.000	—
	beta52	0.16	0.160	0.000 ± 0.001	0.003 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.061 ± 0.003	0.248 ± 0.007	0.160	0.000	0.0396	0.0396	1.029	96.165 ± 0.006	0.160 ± 0.000	—
	beta53	0.16	0.160	0.000 ± 0.001	0.002 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.061 ± 0.003	0.246 ± 0.007	0.161	0.001	0.0402	0.0402	0.995	94.753 ± 0.007	0.154 ± 0.000	—
	beta54	0.16	0.162	0.002 ± 0.001	0.015 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.077 ± 0.004	0.277 ± 0.008	0.160	0.000	0.0428	0.0428	0.974	94.753 ± 0.007	0.169 ± 0.000	—
	beta55	0.00	0.000	-0.000 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.001	0.001	0.0472	0.0472	1.019	95.156 ± 0.007	0.191 ± 0.001	4.844 ± 0.007
	beta56	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.002	0.002	0.0442	0.0443	0.973	94.652 ± 0.007	0.171 ± 0.001	5.348 ± 0.007
	beta57	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.000	0.000	0.0316	0.0316	1.006	94.955 ± 0.007	0.127 ± 0.001	5.045 ± 0.007
	beta58	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.001	0.001	0.0292	0.0292	1.008	96.266 ± 0.006	0.112 ± 0.001	3.734 ± 0.006
QML	beta41	0.20	0.198	-0.002 ± 0.001	-0.010 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.039 ± 0.002	0.198 ± 0.005	0.197	-0.003	0.0416	0.0417	0.963	94.450 ± 0.007	0.150 ± 0.000	—
	beta42	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.041 ± 0.002	0.201 ± 0.006	0.199	-0.001	0.0401	0.0401	0.979	94.955 ± 0.007	0.155 ± 0.000	—
	beta43	0.20	0.199	-0.001 ± 0.001	-0.005 ± 0.006	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.037 ± 0.002	0.193 ± 0.005	0.198	-0.002	0.0368	0.0369	0.983	94.652 ± 0.007	0.149 ± 0.000	—
	beta44	0.00	-0.009	-0.009 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.010	-0.010	0.0476	0.0485	1.002	94.955 ± 0.007	0.182 ± 0.001	5.045 ± 0.007
	beta45	0.00	-0.008	-0.008 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	-0.007	-0.007	0.0425	0.0431	0.940	94.046 ± 0.008	0.157 ± 0.000	5.954 ± 0.008
	beta46	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.000	0.000	0.0317	0.0317	0.969	94.248 ± 0.007	0.123 ± 0.000	5.752 ± 0.007
	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.000	0.000	0.0290	0.0290	1.017	95.358 ± 0.007	0.121 ± 0.000	4.642 ± 0.007
	beta51	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.059 ± 0.003	0.243 ± 0.007	0.159	-0.001	0.0394	0.0394	1.002	94.753 ± 0.007	0.153 ± 0.000	—
	beta52	0.16	0.160	-0.000 ± 0.001	-0.003 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.060 ± 0.003	0.245 ± 0.007	0.159	-0.001	0.0390	0.0390	1.027	95.358 ± 0.007	0.158 ± 0.000	—
	beta53	0.16	0.160	-0.000 ± 0.001	-0.002 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.060 ± 0.003	0.244 ± 0.007	0.160	0.000	0.0399	0.0399	0.993	94.551 ± 0.007	0.152 ± 0.000	—
	beta54	0.16	0.163	0.003 ± 0.001	0.020 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.078 ± 0.004	0.278 ± 0.008	0.161	0.001	0.0436	0.0436	0.984	94.854 ± 0.007	0.172 ± 0.000	—
	beta55	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.002	-0.002	0.0466	0.0466	1.028	95.964 ± 0.006	0.190 ± 0.001	4.036 ± 0.006
	beta56	0.00	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	-0.002	-0.002	0.0439	0.0440	0.979	94.955 ± 0.007	0.168 ± 0.000	5.045 ± 0.007
	beta57	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.003	-0.003	0.0291	0.0292	1.020	94.955 ± 0.007	0.123 ± 0.000	5.045 ± 0.007
	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	-0.002	-0.002	0.0277	0.0278	1.024	96.266 ± 0.006	0.108 ± 0.000	3.734 ± 0.006

Table S60

Simulation 2 - Condition 22: Linear Model, Distribution = normal, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta41	0.20	0.200	0.000 ± 0.001	0.001 ± 0.007	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.043 ± 0.002	0.206 ± 0.006	0.199	-0.001	0.0431	0.0431	0.951	94.147 ± 0.007	0.154 ± 0.000	—
	beta42	0.20	0.201	0.001 ± 0.001	0.005 ± 0.007	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.044 ± 0.002	0.210 ± 0.006	0.200	0.000	0.0422	0.0422	0.966	94.046 ± 0.008	0.159 ± 0.000	—
	beta43	0.20	0.201	0.001 ± 0.001	0.004 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.039 ± 0.002	0.198 ± 0.006	0.199	-0.001	0.0385	0.0386	0.974	94.147 ± 0.007	0.152 ± 0.000	—
	beta44	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.001	-0.001	0.0470	0.0471	0.970	94.955 ± 0.007	0.183 ± 0.001	5.045 ± 0.007
	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	-0.001	-0.001	0.0440	0.0440	0.931	94.147 ± 0.007	0.159 ± 0.001	5.853 ± 0.007
	beta46	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.001	-0.001	0.0345	0.0345	0.933	92.634 ± 0.008	0.128 ± 0.001	7.366 ± 0.008
	beta47	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.001	0.001	0.0300	0.0300	0.979	94.349 ± 0.007	0.126 ± 0.001	5.651 ± 0.007
	beta51	0.16	0.160	0.000 ± 0.001	0.002 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.060 ± 0.003	0.245 ± 0.007	0.161	0.001	0.0399	0.0399	1.000	94.753 ± 0.007	0.154 ± 0.000	—
	beta52	0.16	0.161	0.001 ± 0.001	0.005 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.062 ± 0.003	0.248 ± 0.007	0.160	0.000	0.0394	0.0394	1.027	95.762 ± 0.006	0.160 ± 0.000	—
	beta53	0.16	0.161	0.001 ± 0.001	0.003 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.061 ± 0.003	0.247 ± 0.007	0.161	0.001	0.0400	0.0400	0.992	94.854 ± 0.007	0.154 ± 0.000	—
	beta54	0.16	0.162	0.002 ± 0.001	0.016 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.077 ± 0.004	0.277 ± 0.008	0.161	0.001	0.0427	0.0427	0.981	94.955 ± 0.007	0.170 ± 0.000	—
	beta55	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	0.001	0.001	0.0472	0.0472	1.000	95.560 ± 0.007	0.190 ± 0.001	4.440 ± 0.007
	beta56	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.002	0.002	0.0445	0.0446	0.963	94.854 ± 0.007	0.170 ± 0.001	5.146 ± 0.007
	beta57	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.001	-0.001	0.0309	0.0309	0.980	94.955 ± 0.007	0.126 ± 0.001	5.045 ± 0.007
	beta58	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.001	0.001	0.0296	0.0296	0.997	95.055 ± 0.007	0.112 ± 0.001	4.945 ± 0.007

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S61

Simulation 2 - Condition 23: Linear Model, Distribution = normal, N = 400, Reliability = 0.8

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.002	-0.000 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.066 ± 0.003	0.257 ± 0.007	0.199	-0.001	0.0518	0.0518	1.008	95.779 ± 0.006	0.203 ± 0.001	—
	beta42	0.20	0.199	-0.001 ± 0.002	-0.007 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.074 ± 0.003	0.273 ± 0.008	0.202	0.002	0.0528	0.0528	0.978	94.472 ± 0.007	0.209 ± 0.001	—
	beta43	0.20	0.201	0.001 ± 0.002	0.003 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.064 ± 0.003	0.254 ± 0.007	0.200	0.000	0.0485	0.0485	1.017	95.075 ± 0.007	0.202 ± 0.001	—
	beta44	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	0.001	0.001	0.0527	0.0527	0.973	94.171 ± 0.007	0.224 ± 0.001	5.829 ± 0.007
	beta45	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	0.001	0.001	0.0490	0.0490	0.995	93.869 ± 0.008	0.200 ± 0.001	6.131 ± 0.008
	beta46	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.002	0.002	0.0411	0.0412	0.912	92.462 ± 0.008	0.157 ± 0.001	7.538 ± 0.008
	beta47	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.002	-0.002	0.0398	0.0399	0.988	95.276 ± 0.007	0.157 ± 0.001	4.724 ± 0.007
	beta51	0.16	0.163	0.003 ± 0.002	0.019 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.113 ± 0.005	0.337 ± 0.010	0.162	0.002	0.0547	0.0548	0.964	93.769 ± 0.008	0.203 ± 0.001	—
	beta52	0.16	0.161	0.001 ± 0.002	0.005 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.117 ± 0.006	0.341 ± 0.010	0.159	-0.001	0.0531	0.0531	0.985	94.472 ± 0.007	0.211 ± 0.001	—
	beta53	0.16	0.157	-0.003 ± 0.002	-0.018 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.104 ± 0.005	0.323 ± 0.009	0.158	-0.002	0.0506	0.0507	1.006	95.377 ± 0.007	0.204 ± 0.001	—
	beta54	0.16	0.161	0.001 ± 0.002	0.006 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.135 ± 0.006	0.367 ± 0.010	0.161	0.001	0.0560	0.0560	0.964	93.166 ± 0.008	0.222 ± 0.001	—
	beta55	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	0.002	0.002	0.0629	0.0629	0.958	94.573 ± 0.007	0.235 ± 0.001	5.427 ± 0.007
	beta56	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	0.002	0.002	0.0544	0.0545	0.952	94.271 ± 0.007	0.214 ± 0.001	5.729 ± 0.007
	beta57	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.000	0.000	0.0393	0.0393	0.974	93.769 ± 0.008	0.156 ± 0.001	6.231 ± 0.008
	beta58	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	0.0378	0.0378	0.963	94.271 ± 0.007	0.141 ± 0.001	5.729 ± 0.007
QML	beta41	0.20	0.199	-0.001 ± 0.002	-0.005 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.064 ± 0.003	0.254 ± 0.007	0.199	-0.001	0.0515	0.0515	1.010	95.980 ± 0.006	0.201 ± 0.000	—
	beta42	0.20	0.198	-0.002 ± 0.002	-0.009 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.072 ± 0.003	0.269 ± 0.008	0.200	0.000	0.0517	0.0517	0.980	94.472 ± 0.007	0.207 ± 0.000	—
	beta43	0.20	0.200	0.000 ± 0.002	0.001 ± 0.008	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.063 ± 0.003	0.251 ± 0.007	0.200	0.000	0.0487	0.0487	1.007	95.176 ± 0.007	0.198 ± 0.000	—
	beta44	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.002	-0.002	0.0530	0.0531	1.001	94.774 ± 0.007	0.228 ± 0.001	5.226 ± 0.007
	beta45	0.00	-0.004	-0.004 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	-0.002	-0.002	0.0484	0.0485	1.009	94.573 ± 0.007	0.202 ± 0.001	5.427 ± 0.007
	beta46	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.001	0.001	0.0401	0.0401	0.943	93.467 ± 0.008	0.157 ± 0.001	6.533 ± 0.008
	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.002	-0.002	0.0384	0.0385	1.021	96.181 ± 0.006	0.156 ± 0.001	3.819 ± 0.006
	beta51	0.16	0.163	0.003 ± 0.002	0.016 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.113 ± 0.005	0.336 ± 0.010	0.161	0.001	0.0543	0.0543	0.975	94.070 ± 0.007	0.205 ± 0.000	—
	beta52	0.16	0.161	0.001 ± 0.002	0.003 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.115 ± 0.006	0.339 ± 0.010	0.159	-0.001	0.0529	0.0529	0.989	94.472 ± 0.007	0.211 ± 0.000	—
	beta53	0.16	0.157	-0.003 ± 0.002	-0.018 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.104 ± 0.004	0.322 ± 0.009	0.158	-0.002	0.0504	0.0504	1.006	95.075 ± 0.007	0.203 ± 0.000	—
	beta54	0.16	0.161	0.001 ± 0.002	0.009 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.135 ± 0.006	0.368 ± 0.010	0.161	0.001	0.0563	0.0563	0.978	93.568 ± 0.008	0.225 ± 0.000	—
	beta55	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	0.001	0.001	0.0637	0.0637	0.978	94.975 ± 0.007	0.238 ± 0.001	5.025 ± 0.007
	beta56	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	-0.001	-0.001	0.0552	0.0552	0.970	95.477 ± 0.007	0.215 ± 0.001	4.523 ± 0.007
	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.001	-0.001	0.0381	0.0381	1.003	95.377 ± 0.007	0.156 ± 0.001	4.623 ± 0.007
	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.002	-0.002	0.0369	0.0370	0.989	95.477 ± 0.007	0.141 ± 0.001	4.523 ± 0.007

Table S61

Simulation 2 - Condition 23: Linear Model, Distribution = normal, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta41	0.20	0.200	0.000 ± 0.002	0.001 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.066 ± 0.003	0.258 ± 0.007	0.199	-0.001	0.0523	0.0523	1.003	95.980 ± 0.006	0.203 ± 0.000	—
	beta42	0.20	0.199	-0.001 ± 0.002	-0.004 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.074 ± 0.003	0.273 ± 0.008	0.202	0.002	0.0534	0.0535	0.977	94.573 ± 0.007	0.209 ± 0.000	—
	beta43	0.20	0.201	0.001 ± 0.002	0.004 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.065 ± 0.003	0.255 ± 0.007	0.200	0.000	0.0498	0.0499	0.999	94.573 ± 0.007	0.200 ± 0.000	—
	beta44	0.00	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	—	—	0.000	0.000	0.0570	0.0570	0.981	94.171 ± 0.007	0.228 ± 0.001	5.829 ± 0.007
	beta45	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	0.001	0.001	0.0488	0.0488	0.991	93.769 ± 0.008	0.203 ± 0.001	6.231 ± 0.008
	beta46	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.001	0.001	0.0411	0.0411	0.920	93.266 ± 0.008	0.161 ± 0.001	6.734 ± 0.008
	beta47	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.001	-0.001	0.0396	0.0397	0.992	95.176 ± 0.007	0.159 ± 0.001	4.824 ± 0.007
	beta51	0.16	0.163	0.003 ± 0.002	0.020 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.114 ± 0.005	0.337 ± 0.010	0.162	0.002	0.0544	0.0544	0.965	93.769 ± 0.008	0.204 ± 0.000	—
	beta52	0.16	0.161	0.001 ± 0.002	0.006 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.118 ± 0.006	0.343 ± 0.010	0.160	0.000	0.0541	0.0541	0.979	94.171 ± 0.007	0.211 ± 0.000	—
	beta53	0.16	0.157	-0.003 ± 0.002	-0.016 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.105 ± 0.005	0.324 ± 0.009	0.158	-0.002	0.0510	0.0511	1.004	95.276 ± 0.007	0.204 ± 0.000	—
	beta54	0.16	0.161	0.001 ± 0.002	0.006 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.135 ± 0.006	0.367 ± 0.010	0.160	0.000	0.0565	0.0565	0.968	93.065 ± 0.008	0.223 ± 0.000	—
	beta55	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	—	—	0.002	0.002	0.0650	0.0650	0.952	94.774 ± 0.007	0.238 ± 0.001	5.226 ± 0.007
	beta56	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	0.002	0.002	0.0557	0.0557	0.949	94.573 ± 0.007	0.216 ± 0.001	5.427 ± 0.007
	beta57	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.000	0.000	0.0388	0.0388	0.979	95.075 ± 0.007	0.158 ± 0.001	4.925 ± 0.007
	beta58	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.001	-0.001	0.0362	0.0362	0.967	95.075 ± 0.007	0.143 ± 0.001	4.925 ± 0.007

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S62

Simulation 2 - Condition 24: Linear Model, Distribution = normal, N = 1000, Reliability = 0.8

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.026 ± 0.001	0.160 ± 0.004	0.200	0.000	0.0317	0.0317	1.019	95.560 ± 0.007	0.128 ± 0.000	—
	beta42	0.20	0.200	-0.000 ± 0.001	-0.000 ± 0.005	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.028 ± 0.001	0.168 ± 0.005	0.201	0.001	0.0324	0.0324	0.995	94.753 ± 0.007	0.131 ± 0.000	—
	beta43	0.20	0.201	0.001 ± 0.001	0.005 ± 0.005	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.026 ± 0.001	0.162 ± 0.005	0.201	0.001	0.0315	0.0316	0.997	94.551 ± 0.007	0.127 ± 0.000	—
	beta44	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.002	-0.002	0.0387	0.0387	0.953	94.147 ± 0.007	0.141 ± 0.000	5.853 ± 0.007
	beta45	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.002	0.002	0.0322	0.0323	0.981	94.551 ± 0.007	0.125 ± 0.000	5.449 ± 0.007
	beta46	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	0.001	0.001	0.0262	0.0262	0.979	94.147 ± 0.007	0.098 ± 0.000	5.853 ± 0.007
	beta47	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	-0.001	-0.001	0.0268	0.0268	0.973	94.955 ± 0.007	0.097 ± 0.000	5.045 ± 0.007
	beta51	0.16	0.161	0.001 ± 0.001	0.008 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.049 ± 0.002	0.221 ± 0.006	0.162	0.002	0.0350	0.0350	0.926	92.936 ± 0.008	0.128 ± 0.000	—
	beta52	0.16	0.158	-0.002 ± 0.001	-0.011 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.046 ± 0.002	0.214 ± 0.006	0.158	-0.002	0.0329	0.0329	0.993	94.147 ± 0.007	0.133 ± 0.000	—
	beta53	0.16	0.160	0.000 ± 0.001	0.001 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.038 ± 0.002	0.196 ± 0.005	0.159	-0.001	0.0322	0.0322	1.044	96.065 ± 0.006	0.128 ± 0.000	—
	beta54	0.16	0.162	0.002 ± 0.001	0.010 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.054 ± 0.002	0.232 ± 0.006	0.162	0.002	0.0370	0.0370	0.969	94.349 ± 0.007	0.141 ± 0.000	—
	beta55	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.003	0.003	0.0393	0.0394	0.942	93.744 ± 0.008	0.147 ± 0.000	6.256 ± 0.008
	beta56	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.003	0.003	0.0353	0.0354	0.978	94.349 ± 0.007	0.133 ± 0.000	5.651 ± 0.007
	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	-0.001	-0.001	0.0259	0.0259	0.959	94.349 ± 0.007	0.098 ± 0.000	5.651 ± 0.007
	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.022 ± 0.001	—	—	-0.003	-0.003	0.0218	0.0220	1.010	95.055 ± 0.007	0.088 ± 0.000	4.945 ± 0.007
QML	beta41	0.20	0.199	-0.001 ± 0.001	-0.005 ± 0.005	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.025 ± 0.001	0.158 ± 0.004	0.199	-0.001	0.0318	0.0318	1.019	95.661 ± 0.006	0.126 ± 0.000	—
	beta42	0.20	0.199	-0.001 ± 0.001	-0.003 ± 0.005	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.028 ± 0.001	0.166 ± 0.005	0.200	0.000	0.0320	0.0320	0.997	94.854 ± 0.007	0.130 ± 0.000	—
	beta43	0.20	0.200	0.000 ± 0.001	0.002 ± 0.005	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.026 ± 0.001	0.161 ± 0.005	0.200	0.000	0.0312	0.0312	0.990	94.955 ± 0.007	0.125 ± 0.000	—
	beta44	0.00	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.005	-0.005	0.0383	0.0386	0.965	94.955 ± 0.007	0.142 ± 0.000	5.045 ± 0.007
	beta45	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.000	0.000	0.0319	0.0319	0.993	94.753 ± 0.007	0.126 ± 0.000	5.247 ± 0.007
	beta46	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	—	—	0.001	0.001	0.0255	0.0255	0.998	95.358 ± 0.007	0.098 ± 0.000	4.642 ± 0.007
	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	—	—	-0.001	-0.001	0.0263	0.0263	0.991	95.762 ± 0.006	0.097 ± 0.000	4.238 ± 0.006
	beta51	0.16	0.161	0.001 ± 0.001	0.007 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.048 ± 0.002	0.219 ± 0.006	0.162	0.002	0.0348	0.0349	0.934	93.340 ± 0.008	0.129 ± 0.000	—
	beta52	0.16	0.158	-0.002 ± 0.001	-0.013 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.045 ± 0.002	0.213 ± 0.006	0.158	-0.002	0.0333	0.0334	0.999	94.551 ± 0.007	0.133 ± 0.001	—
	beta53	0.16	0.160	0.000 ± 0.001	0.000 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.038 ± 0.002	0.195 ± 0.005	0.159	-0.001	0.0320	0.0320	1.043	96.266 ± 0.006	0.128 ± 0.000	—
	beta54	0.16	0.162	0.002 ± 0.001	0.011 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.054 ± 0.002	0.232 ± 0.006	0.162	0.002	0.0370	0.0371	0.976	94.753 ± 0.007	0.142 ± 0.000	—
	beta55	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.003	0.003	0.0394	0.0394	0.957	94.248 ± 0.007	0.150 ± 0.001	5.752 ± 0.007
	beta56	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.001	0.001	0.0357	0.0357	1.007	94.652 ± 0.007	0.136 ± 0.003	5.348 ± 0.007
	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	-0.002	-0.002	0.0256	0.0257	0.974	94.753 ± 0.007	0.098 ± 0.000	5.247 ± 0.007
	beta58	0.00	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	-0.003	-0.003	0.0221	0.0223	1.023	95.459 ± 0.007	0.088 ± 0.000	4.541 ± 0.007

Table S62

Simulation 2 - Condition 24: Linear Model, Distribution = normal, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta41	0.20	0.200	-0.000 ± 0.001	-0.000 ± 0.005	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.026 ± 0.001	0.160 ± 0.004	0.200	0.000	0.0321	0.0321	1.014	95.661 ± 0.006	0.127 ± 0.000	—
	beta42	0.20	0.200	0.000 ± 0.001	0.000 ± 0.005	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.028 ± 0.001	0.168 ± 0.005	0.201	0.001	0.0321	0.0321	0.994	94.753 ± 0.007	0.131 ± 0.000	—
	beta43	0.20	0.201	0.001 ± 0.001	0.005 ± 0.005	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.026 ± 0.001	0.162 ± 0.005	0.201	0.001	0.0314	0.0315	0.988	94.652 ± 0.007	0.126 ± 0.000	—
	beta44	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.002	-0.002	0.0387	0.0387	0.956	95.055 ± 0.007	0.142 ± 0.000	4.945 ± 0.007
	beta45	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.002	0.002	0.0326	0.0327	0.990	95.055 ± 0.007	0.126 ± 0.000	4.945 ± 0.007
	beta46	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	0.001	0.001	0.0268	0.0268	0.980	94.652 ± 0.007	0.100 ± 0.000	5.348 ± 0.007
	beta47	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	0.000	0.000	0.0268	0.0268	0.975	95.358 ± 0.007	0.098 ± 0.000	4.642 ± 0.007
	beta51	0.16	0.161	0.001 ± 0.001	0.008 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.049 ± 0.002	0.221 ± 0.006	0.162	0.002	0.0353	0.0353	0.926	92.936 ± 0.008	0.128 ± 0.000	—
	beta52	0.16	0.158	-0.002 ± 0.001	-0.010 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.046 ± 0.002	0.214 ± 0.006	0.159	-0.001	0.0330	0.0330	0.993	94.551 ± 0.007	0.133 ± 0.000	—
	beta53	0.16	0.160	0.000 ± 0.001	0.002 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.038 ± 0.002	0.196 ± 0.005	0.159	-0.001	0.0322	0.0322	1.041	96.266 ± 0.006	0.128 ± 0.000	—
	beta54	0.16	0.162	0.002 ± 0.001	0.010 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.054 ± 0.002	0.232 ± 0.006	0.162	0.002	0.0370	0.0370	0.973	94.450 ± 0.007	0.141 ± 0.000	—
	beta55	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.003	0.003	0.0391	0.0392	0.947	94.349 ± 0.007	0.148 ± 0.000	5.651 ± 0.007
	beta56	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.003	0.003	0.0350	0.0351	0.978	94.046 ± 0.008	0.134 ± 0.000	5.954 ± 0.008
	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	-0.001	-0.001	0.0258	0.0258	0.960	94.753 ± 0.007	0.099 ± 0.000	5.247 ± 0.007
	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.022 ± 0.001	—	—	-0.003	-0.003	0.0218	0.0219	1.015	95.156 ± 0.007	0.089 ± 0.000	4.844 ± 0.007

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S63

Simulation 2 - Condition 25: Linear Model, Distribution = right-skewed, N = 400, Reliability = 0.4

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.186	-0.014 ± 0.008	-0.069 ± 0.041	0.037 ± 0.003	0.191 ± 0.008	0.037 ± 0.003	0.192 ± 0.009	0.918 ± 0.073	0.958 ± 0.043	0.189	-0.011	0.1771	0.1774	1.021	96.673 ± 0.008	0.766 ± 0.014	—
	beta42	0.20	0.191	-0.009 ± 0.008	-0.045 ± 0.041	0.036 ± 0.003	0.189 ± 0.007	0.036 ± 0.003	0.189 ± 0.008	0.894 ± 0.069	0.946 ± 0.042	0.184	-0.016	0.1840	0.1848	1.045	97.043 ± 0.007	0.775 ± 0.014	—
	beta43	0.20	0.203	0.003 ± 0.004	0.015 ± 0.021	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.096 ± 0.004	0.229 ± 0.015	0.478 ± 0.018	0.203	0.003	0.0965	0.0966	1.028	95.379 ± 0.009	0.386 ± 0.005	—
	beta44	0.00	0.001	0.001 ± 0.006	—	0.022 ± 0.002	0.149 ± 0.006	0.022 ± 0.002	0.149 ± 0.007	—	—	0.007	0.007	0.1344	0.1346	1.039	97.412 ± 0.007	0.608 ± 0.019	2.588 ± 0.007
	beta45	0.00	0.004	0.004 ± 0.006	—	0.017 ± 0.001	0.129 ± 0.005	0.017 ± 0.001	0.129 ± 0.006	—	—	-0.002	-0.002	0.1269	0.1269	1.074	96.673 ± 0.008	0.543 ± 0.015	3.327 ± 0.008
	beta46	0.00	0.004	0.004 ± 0.003	—	0.006 ± 0.001	0.080 ± 0.003	0.006 ± 0.001	0.080 ± 0.004	—	—	-0.001	-0.001	0.0721	0.0721	1.046	97.412 ± 0.007	0.327 ± 0.008	2.588 ± 0.007
	beta47	0.00	0.003	0.003 ± 0.003	—	0.005 ± 0.000	0.072 ± 0.003	0.005 ± 0.000	0.072 ± 0.003	—	—	0.001	0.001	0.0663	0.0663	1.066	97.782 ± 0.006	0.303 ± 0.008	2.218 ± 0.006
	beta51	0.16	0.164	0.004 ± 0.008	0.024 ± 0.051	0.035 ± 0.003	0.188 ± 0.007	0.035 ± 0.003	0.188 ± 0.008	1.380 ± 0.101	1.175 ± 0.050	0.170	0.010	0.1783	0.1786	0.977	97.043 ± 0.007	0.721 ± 0.012	—
	beta52	0.16	0.158	-0.002 ± 0.005	-0.013 ± 0.033	0.015 ± 0.001	0.122 ± 0.004	0.015 ± 0.001	0.121 ± 0.005	0.576 ± 0.037	0.759 ± 0.029	0.156	-0.004	0.1201	0.1202	1.068	96.858 ± 0.008	0.509 ± 0.009	—
	beta53	0.16	0.163	0.003 ± 0.007	0.016 ± 0.046	0.029 ± 0.002	0.171 ± 0.006	0.029 ± 0.002	0.171 ± 0.007	1.137 ± 0.079	1.066 ± 0.044	0.159	-0.001	0.1671	0.1671	1.026	96.858 ± 0.008	0.687 ± 0.011	—
	beta54	0.16	0.167	0.007 ± 0.004	0.042 ± 0.028	0.011 ± 0.001	0.103 ± 0.004	0.011 ± 0.001	0.104 ± 0.004	0.418 ± 0.029	0.647 ± 0.026	0.157	-0.003	0.0983	0.0983	0.954	94.455 ± 0.010	0.387 ± 0.003	—
	beta55	0.00	-0.004	-0.004 ± 0.007	—	0.027 ± 0.002	0.164 ± 0.006	0.027 ± 0.002	0.164 ± 0.007	—	—	0.000	0.000	0.1577	0.1577	0.994	97.782 ± 0.006	0.639 ± 0.016	2.218 ± 0.006
	beta56	0.00	0.002	0.002 ± 0.005	—	0.016 ± 0.001	0.128 ± 0.005	0.016 ± 0.001	0.127 ± 0.006	—	—	0.003	0.003	0.1149	0.1150	1.054	97.043 ± 0.007	0.527 ± 0.013	2.957 ± 0.007
	beta57	0.00	0.000	0.000 ± 0.004	—	0.008 ± 0.001	0.089 ± 0.004	0.008 ± 0.001	0.088 ± 0.004	—	—	-0.001	-0.001	0.0760	0.0760	0.958	97.967 ± 0.006	0.333 ± 0.009	2.033 ± 0.006
	beta58	0.00	-0.002	-0.002 ± 0.003	—	0.004 ± 0.000	0.064 ± 0.003	0.004 ± 0.000	0.064 ± 0.003	—	—	0.002	0.002	0.0548	0.0548	1.015	97.043 ± 0.007	0.253 ± 0.006	2.957 ± 0.007
QML	beta41	0.20	0.168	-0.032 ± 0.004	-0.158 ± 0.021	0.009 ± 0.001	0.097 ± 0.003	0.010 ± 0.001	0.102 ± 0.004	0.260 ± 0.017	0.510 ± 0.020	0.168	-0.032	0.0941	0.0994	0.939	92.606 ± 0.011	0.357 ± 0.003	—
	beta42	0.20	0.168	-0.032 ± 0.004	-0.162 ± 0.021	0.010 ± 0.001	0.098 ± 0.003	0.011 ± 0.001	0.103 ± 0.004	0.267 ± 0.016	0.517 ± 0.018	0.166	-0.034	0.0984	0.1042	0.958	92.421 ± 0.011	0.369 ± 0.003	—
	beta43	0.20	0.207	0.007 ± 0.004	0.036 ± 0.019	0.007 ± 0.000	0.086 ± 0.003	0.008 ± 0.000	0.087 ± 0.003	0.188 ± 0.012	0.433 ± 0.017	0.209	0.009	0.0807	0.0813	0.948	93.161 ± 0.011	0.321 ± 0.002	—
	beta44	0.00	-0.027	-0.027 ± 0.005	—	0.012 ± 0.001	0.108 ± 0.004	0.012 ± 0.001	0.111 ± 0.004	—	—	-0.027	-0.027	0.1131	0.1161	1.008	96.303 ± 0.008	0.427 ± 0.005	3.697 ± 0.008
	beta45	0.00	0.006	0.006 ± 0.004	—	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.094 ± 0.004	—	—	0.001	0.001	0.0938	0.0938	0.983	95.749 ± 0.009	0.361 ± 0.004	4.251 ± 0.009
	beta46	0.00	0.042	0.042 ± 0.003	—	0.004 ± 0.000	0.064 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	—	—	0.039	0.039	0.0601	0.0714	0.991	91.497 ± 0.012	0.249 ± 0.004	8.503 ± 0.012
	beta47	0.00	0.044	0.044 ± 0.003	—	0.003 ± 0.000	0.058 ± 0.002	0.005 ± 0.000	0.073 ± 0.003	—	—	0.040	0.040	0.0574	0.0698	1.024	94.455 ± 0.010	0.235 ± 0.004	5.545 ± 0.010
	beta51	0.16	0.145	-0.015 ± 0.004	-0.094 ± 0.027	0.010 ± 0.001	0.101 ± 0.003	0.010 ± 0.001	0.102 ± 0.004	0.406 ± 0.026	0.637 ± 0.024	0.150	-0.010	0.1050	0.1054	0.952	94.824 ± 0.010	0.376 ± 0.003	—
	beta52	0.16	0.155	-0.005 ± 0.004	-0.028 ± 0.026	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.096 ± 0.004	0.357 ± 0.022	0.598 ± 0.022	0.159	-0.001	0.0955	0.0955	0.992	95.564 ± 0.009	0.372 ± 0.003	—
	beta53	0.16	0.138	-0.022 ± 0.004	-0.138 ± 0.026	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.100 ± 0.004	0.391 ± 0.024	0.625 ± 0.023	0.137	-0.023	0.0994	0.1019	0.955	93.346 ± 0.011	0.366 ± 0.003	—
	beta54	0.16	0.177	0.017 ± 0.004	0.105 ± 0.027	0.010 ± 0.001	0.101 ± 0.003	0.010 ± 0.001	0.102 ± 0.004	0.407 ± 0.027	0.638 ± 0.025	0.171	0.011	0.1004	0.1010	0.975	95.379 ± 0.009	0.385 ± 0.003	—
	beta55	0.00	-0.020	-0.020 ± 0.005	—	0.015 ± 0.001	0.121 ± 0.004	0.015 ± 0.001	0.123 ± 0.005	—	—	-0.017	-0.017	0.1301	0.1313	0.974	96.673 ± 0.008	0.463 ± 0.005	3.327 ± 0.008
	beta56	0.00	0.024	0.024 ± 0.004	—	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.097 ± 0.004	—	—	0.023	0.023	0.0992	0.1019	1.003	95.749 ± 0.009	0.371 ± 0.004	4.251 ± 0.009
	beta57	0.00	0.032	0.032 ± 0.003	—	0.004 ± 0.000	0.066 ± 0.002	0.005 ± 0.000	0.073 ± 0.003	—	—	0.028	0.028	0.0682	0.0735	0.997	95.009 ± 0.009	0.259 ± 0.004	4.991 ± 0.009
	beta58	0.00	0.028	0.028 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	—	—	0.029	0.029	0.0496	0.0575	0.995	94.640 ± 0.010	0.200 ± 0.003	5.360 ± 0.010

Table S63

Simulation 2 - Condition 25: Linear Model, Distribution = right-skewed, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta41	0.20	0.211	0.011 ± 0.009	0.057 ± 0.047	0.049 ± 0.004	0.221 ± 0.009	0.049 ± 0.004	0.221 ± 0.010	1.218 ± 0.098	1.103 ± 0.051	0.212	0.012	0.2267	0.2270	0.790	92.421 ± 0.011	0.683 ± 0.010	—
	beta42	0.20	0.192	-0.008 ± 0.010	-0.042 ± 0.049	0.051 ± 0.005	0.226 ± 0.010	0.051 ± 0.005	0.226 ± 0.011	1.276 ± 0.114	1.129 ± 0.056	0.185	-0.015	0.2408	0.2413	0.779	92.791 ± 0.011	0.690 ± 0.010	—
	beta43	0.20	0.207	0.007 ± 0.004	0.034 ± 0.021	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.098 ± 0.004	0.241 ± 0.017	0.491 ± 0.020	0.209	0.009	0.1075	0.1079	0.916	93.161 ± 0.011	0.352 ± 0.003	—
	beta44	0.00	0.009	0.009 ± 0.008	—	0.033 ± 0.003	0.181 ± 0.009	0.033 ± 0.003	0.181 ± 0.010	—	—	0.003	0.003	0.1595	0.1596	0.778	92.237 ± 0.012	0.552 ± 0.019	7.763 ± 0.012
	beta45	0.00	0.006	0.006 ± 0.007	—	0.027 ± 0.002	0.163 ± 0.007	0.027 ± 0.002	0.163 ± 0.008	—	—	-0.007	-0.007	0.1429	0.1430	0.801	93.161 ± 0.011	0.512 ± 0.027	6.839 ± 0.011
	beta46	0.00	-0.006	-0.006 ± 0.004	—	0.009 ± 0.001	0.095 ± 0.004	0.009 ± 0.001	0.095 ± 0.005	—	—	-0.004	-0.004	0.0904	0.0905	0.774	92.791 ± 0.011	0.288 ± 0.007	7.209 ± 0.011
	beta47	0.00	0.003	0.003 ± 0.004	—	0.007 ± 0.001	0.085 ± 0.004	0.007 ± 0.001	0.085 ± 0.004	—	—	0.000	0.000	0.0826	0.0826	0.785	92.791 ± 0.011	0.260 ± 0.006	7.209 ± 0.011
	beta51	0.16	0.169	0.009 ± 0.008	0.056 ± 0.051	0.036 ± 0.003	0.189 ± 0.007	0.036 ± 0.003	0.189 ± 0.008	1.394 ± 0.101	1.181 ± 0.050	0.177	0.017	0.1808	0.1816	0.863	94.085 ± 0.010	0.639 ± 0.008	—
	beta52	0.16	0.165	0.005 ± 0.005	0.030 ± 0.033	0.015 ± 0.001	0.123 ± 0.004	0.015 ± 0.001	0.123 ± 0.005	0.590 ± 0.038	0.768 ± 0.030	0.155	-0.005	0.1319	0.1320	0.947	95.194 ± 0.009	0.456 ± 0.006	—
	beta53	0.16	0.167	0.007 ± 0.008	0.044 ± 0.048	0.031 ± 0.002	0.177 ± 0.006	0.031 ± 0.002	0.177 ± 0.007	1.228 ± 0.078	1.108 ± 0.043	0.167	0.007	0.1721	0.1723	0.915	94.270 ± 0.010	0.636 ± 0.008	—
	beta54	0.16	0.168	0.008 ± 0.004	0.049 ± 0.027	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.004	0.392 ± 0.026	0.626 ± 0.024	0.158	-0.002	0.1026	0.1026	0.969	95.194 ± 0.009	0.380 ± 0.003	—
	beta55	0.00	0.002	0.002 ± 0.007	—	0.030 ± 0.003	0.173 ± 0.007	0.030 ± 0.003	0.173 ± 0.008	—	—	-0.006	-0.006	0.1531	0.1532	0.840	93.900 ± 0.010	0.569 ± 0.017	6.100 ± 0.010
	beta56	0.00	-0.010	-0.010 ± 0.006	—	0.018 ± 0.001	0.133 ± 0.005	0.018 ± 0.001	0.134 ± 0.006	—	—	-0.001	-0.001	0.1234	0.1234	0.881	94.455 ± 0.010	0.461 ± 0.012	5.545 ± 0.010
	beta57	0.00	-0.003	-0.003 ± 0.004	—	0.008 ± 0.001	0.088 ± 0.004	0.008 ± 0.001	0.088 ± 0.004	—	—	-0.005	-0.005	0.0771	0.0773	0.831	94.270 ± 0.010	0.288 ± 0.006	5.730 ± 0.010
	beta58	0.00	-0.002	-0.002 ± 0.003	—	0.004 ± 0.000	0.065 ± 0.003	0.004 ± 0.000	0.065 ± 0.003	—	—	0.001	0.001	0.0578	0.0578	0.898	94.085 ± 0.010	0.230 ± 0.005	5.915 ± 0.010

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S64

Simulation 2 - Condition 26: Linear Model, Distribution = right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.198	-0.002 ± 0.003	-0.008 ± 0.017	0.010 ± 0.001	0.102 ± 0.003	0.010 ± 0.001	0.102 ± 0.003	0.260 ± 0.014	0.510 ± 0.016	0.196	-0.004	0.1015	0.1016	1.051	96.117 ± 0.006	0.420 ± 0.003	—
	beta42	0.20	0.197	-0.003 ± 0.004	-0.016 ± 0.018	0.012 ± 0.001	0.108 ± 0.003	0.012 ± 0.001	0.108 ± 0.003	0.294 ± 0.015	0.542 ± 0.016	0.195	-0.005	0.0993	0.0994	1.044	95.901 ± 0.007	0.444 ± 0.004	—
	beta43	0.20	0.200	-0.000 ± 0.002	-0.002 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.077 ± 0.004	0.278 ± 0.009	0.201	0.001	0.0520	0.0520	1.018	94.283 ± 0.008	0.222 ± 0.001	—
	beta44	0.00	0.004	0.004 ± 0.002	—	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.003	—	—	0.003	0.003	0.0658	0.0659	0.985	95.901 ± 0.007	0.285 ± 0.003	4.099 ± 0.007
	beta45	0.00	0.001	0.001 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.002	0.002	0.0654	0.0654	0.978	96.009 ± 0.006	0.266 ± 0.003	3.991 ± 0.006
	beta46	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.001	0.001	0.0344	0.0344	1.019	96.009 ± 0.006	0.152 ± 0.002	3.991 ± 0.006
	beta47	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.000	0.000	0.0333	0.0333	1.032	96.872 ± 0.006	0.153 ± 0.002	3.128 ± 0.006
	beta51	0.16	0.162	0.002 ± 0.003	0.010 ± 0.021	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	0.391 ± 0.020	0.625 ± 0.019	0.158	-0.002	0.0938	0.0938	1.028	96.548 ± 0.006	0.403 ± 0.003	—
	beta52	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.015	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.219 ± 0.010	0.468 ± 0.013	0.159	-0.001	0.0781	0.0781	0.991	96.332 ± 0.006	0.291 ± 0.002	—
	beta53	0.16	0.161	0.001 ± 0.003	0.009 ± 0.022	0.011 ± 0.001	0.106 ± 0.003	0.011 ± 0.001	0.106 ± 0.003	0.436 ± 0.022	0.660 ± 0.020	0.158	-0.002	0.0994	0.0994	0.960	94.606 ± 0.007	0.397 ± 0.003	—
	beta54	0.16	0.160	0.000 ± 0.002	0.001 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.134 ± 0.006	0.365 ± 0.010	0.157	-0.003	0.0592	0.0593	1.004	95.361 ± 0.007	0.230 ± 0.001	—
	beta55	0.00	-0.007	-0.007 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.002	—	—	-0.004	-0.004	0.0831	0.0832	0.988	96.764 ± 0.006	0.331 ± 0.004	3.236 ± 0.006
	beta56	0.00	0.005	0.005 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	0.006	0.006	0.0654	0.0657	1.005	96.548 ± 0.006	0.275 ± 0.003	3.452 ± 0.006
	beta57	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.001	0.001	0.0373	0.0374	0.984	96.440 ± 0.006	0.165 ± 0.002	3.560 ± 0.006
	beta58	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.001	0.001	0.0322	0.0322	0.973	95.901 ± 0.007	0.134 ± 0.002	4.099 ± 0.007
QML	beta41	0.20	0.174	-0.026 ± 0.002	-0.131 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.097 ± 0.004	0.312 ± 0.008	0.171	-0.029	0.0547	0.0617	1.008	92.772 ± 0.009	0.224 ± 0.001	—
	beta42	0.20	0.169	-0.031 ± 0.002	-0.154 ± 0.010	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.067 ± 0.002	0.111 ± 0.005	0.333 ± 0.008	0.169	-0.031	0.0577	0.0655	1.001	90.831 ± 0.009	0.232 ± 0.001	—
	beta43	0.20	0.205	0.005 ± 0.002	0.026 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.065 ± 0.003	0.254 ± 0.008	0.207	0.007	0.0493	0.0497	1.014	95.146 ± 0.007	0.201 ± 0.001	—
	beta44	0.00	-0.021	-0.021 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	-0.022	-0.022	0.0680	0.0714	0.980	93.528 ± 0.008	0.265 ± 0.001	6.472 ± 0.008
	beta45	0.00	0.006	0.006 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	0.006	0.006	0.0579	0.0582	0.965	95.577 ± 0.007	0.226 ± 0.001	4.423 ± 0.007
	beta46	0.00	0.040	0.040 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	—	—	0.039	0.039	0.0385	0.0548	0.978	81.877 ± 0.013	0.146 ± 0.001	18.123 ± 0.013
	beta47	0.00	0.040	0.040 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	—	—	0.037	0.037	0.0362	0.0519	0.981	82.956 ± 0.012	0.143 ± 0.001	17.044 ± 0.012
	beta51	0.16	0.141	-0.019 ± 0.002	-0.117 ± 0.012	0.003 ± 0.000	0.057 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.142 ± 0.007	0.376 ± 0.011	0.139	-0.021	0.0566	0.0602	1.025	94.175 ± 0.008	0.230 ± 0.001	—
	beta52	0.16	0.157	-0.003 ± 0.002	-0.019 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.140 ± 0.006	0.375 ± 0.010	0.157	-0.003	0.0636	0.0637	0.980	95.254 ± 0.007	0.230 ± 0.001	—
	beta53	0.16	0.137	-0.023 ± 0.002	-0.141 ± 0.013	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.166 ± 0.008	0.408 ± 0.011	0.137	-0.023	0.0591	0.0634	0.952	89.968 ± 0.010	0.229 ± 0.001	—
	beta54	0.16	0.166	0.006 ± 0.002	0.035 ± 0.012	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.132 ± 0.006	0.364 ± 0.010	0.162	0.002	0.0599	0.0600	1.025	96.117 ± 0.006	0.233 ± 0.001	—
	beta55	0.00	-0.025	-0.025 ± 0.003	—	0.006 ± 0.000	0.077 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	—	—	-0.020	-0.020	0.0766	0.0792	0.962	94.714 ± 0.007	0.290 ± 0.002	5.286 ± 0.007
	beta56	0.00	0.026	0.026 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	—	—	0.026	0.026	0.0577	0.0634	1.006	92.988 ± 0.008	0.228 ± 0.001	7.012 ± 0.008
	beta57	0.00	0.034	0.034 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	—	—	0.032	0.032	0.0399	0.0515	0.972	87.810 ± 0.011	0.160 ± 0.001	12.190 ± 0.011
	beta58	0.00	0.028	0.028 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.028	0.028	0.0291	0.0404	1.008	87.487 ± 0.011	0.123 ± 0.001	12.513 ± 0.011

Table S64

Simulation 2 - Condition 26: Linear Model, Distribution = right-skewed, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta41	0.20	0.203	0.003 ± 0.004	0.014 ± 0.018	0.012 ± 0.001	0.109 ± 0.003	0.012 ± 0.001	0.109 ± 0.004	0.300 ± 0.017	0.547 ± 0.018	0.203	0.003	0.1059	0.1060	0.938	94.606 ± 0.007	0.403 ± 0.003	—
	beta42	0.20	0.200	-0.000 ± 0.004	-0.002 ± 0.019	0.013 ± 0.001	0.114 ± 0.003	0.013 ± 0.001	0.114 ± 0.003	0.324 ± 0.017	0.569 ± 0.017	0.197	-0.003	0.1062	0.1062	0.941	94.498 ± 0.007	0.420 ± 0.003	—
	beta43	0.20	0.199	-0.001 ± 0.002	-0.006 ± 0.009	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.071 ± 0.004	0.266 ± 0.008	0.199	-0.001	0.0496	0.0496	0.986	95.038 ± 0.007	0.206 ± 0.001	—
	beta44	0.00	0.002	0.002 ± 0.002	—	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	—	—	0.005	0.005	0.0679	0.0681	0.933	94.391 ± 0.008	0.276 ± 0.004	5.609 ± 0.008
	beta45	0.00	0.001	0.001 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.003	—	—	0.000	0.000	0.0662	0.0662	0.886	95.901 ± 0.007	0.262 ± 0.005	4.099 ± 0.007
	beta46	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.000	0.000	0.0376	0.0376	0.911	95.146 ± 0.007	0.147 ± 0.002	4.854 ± 0.007
	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.001	-0.001	0.0332	0.0332	0.940	94.391 ± 0.008	0.146 ± 0.002	5.609 ± 0.008
	beta51	0.16	0.164	0.004 ± 0.003	0.024 ± 0.020	0.010 ± 0.000	0.100 ± 0.002	0.010 ± 0.000	0.100 ± 0.003	0.389 ± 0.019	0.624 ± 0.018	0.159	-0.001	0.0971	0.0971	0.980	95.793 ± 0.007	0.383 ± 0.002	—
	beta52	0.16	0.160	-0.000 ± 0.002	-0.000 ± 0.015	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.205 ± 0.010	0.452 ± 0.013	0.159	-0.001	0.0717	0.0717	0.964	95.361 ± 0.007	0.274 ± 0.002	—
	beta53	0.16	0.163	0.003 ± 0.003	0.019 ± 0.021	0.011 ± 0.001	0.104 ± 0.003	0.011 ± 0.001	0.104 ± 0.003	0.426 ± 0.021	0.653 ± 0.019	0.159	-0.001	0.0982	0.0982	0.936	94.175 ± 0.008	0.383 ± 0.002	—
	beta54	0.16	0.161	0.001 ± 0.002	0.004 ± 0.012	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.127 ± 0.006	0.357 ± 0.010	0.156	-0.004	0.0595	0.0597	1.030	95.685 ± 0.007	0.231 ± 0.001	—
	beta55	0.00	-0.007	-0.007 ± 0.003	—	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.003	—	—	-0.007	-0.007	0.0815	0.0819	0.937	95.361 ± 0.007	0.320 ± 0.004	4.639 ± 0.007
	beta56	0.00	0.004	0.004 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	—	—	0.006	0.006	0.0660	0.0663	0.932	95.469 ± 0.007	0.265 ± 0.003	4.531 ± 0.007
	beta57	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.000	0.000	0.0389	0.0389	0.920	94.175 ± 0.008	0.156 ± 0.002	5.825 ± 0.008
	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.000	0.000	0.0311	0.0311	0.946	95.361 ± 0.007	0.128 ± 0.001	4.639 ± 0.007

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S65

Simulation 2 - Condition 27: Linear Model, Distribution = right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.198	-0.002 ± 0.004	-0.009 ± 0.019	0.014 ± 0.001	0.119 ± 0.003	0.014 ± 0.001	0.119 ± 0.003	0.351 ± 0.017	0.593 ± 0.017	0.196	-0.004	0.1101	0.1101	0.992	94.963 ± 0.007	0.461 ± 0.003	—
	beta42	0.20	0.205	0.005 ± 0.004	0.025 ± 0.020	0.016 ± 0.001	0.125 ± 0.003	0.016 ± 0.001	0.125 ± 0.004	0.389 ± 0.019	0.624 ± 0.018	0.201	0.001	0.1263	0.1263	0.991	95.593 ± 0.007	0.484 ± 0.003	—
	beta43	0.20	0.204	0.004 ± 0.002	0.019 ± 0.011	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.113 ± 0.006	0.337 ± 0.010	0.206	0.006	0.0672	0.0675	0.983	94.753 ± 0.007	0.259 ± 0.001	—
	beta44	0.00	0.003	0.003 ± 0.003	—	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	—	—	0.004	0.004	0.0784	0.0785	0.988	95.173 ± 0.007	0.317 ± 0.003	4.827 ± 0.007
	beta45	0.00	0.000	0.000 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.003	—	—	0.001	0.001	0.0767	0.0767	0.941	94.963 ± 0.007	0.295 ± 0.003	5.037 ± 0.007
	beta46	0.00	0.001	0.001 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	—	—	0.001	0.001	0.0401	0.0401	0.933	94.544 ± 0.007	0.178 ± 0.002	5.456 ± 0.007
	beta47	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.002	—	—	-0.002	-0.002	0.0424	0.0424	0.929	94.858 ± 0.007	0.175 ± 0.002	5.142 ± 0.007
	beta51	0.16	0.160	0.000 ± 0.004	0.003 ± 0.024	0.014 ± 0.001	0.118 ± 0.003	0.014 ± 0.001	0.118 ± 0.003	0.540 ± 0.027	0.735 ± 0.022	0.164	0.004	0.1209	0.1210	0.958	94.229 ± 0.008	0.442 ± 0.003	—
	beta52	0.16	0.162	0.002 ± 0.003	0.013 ± 0.017	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	0.288 ± 0.015	0.537 ± 0.016	0.159	-0.001	0.0837	0.0837	0.968	94.648 ± 0.007	0.326 ± 0.002	—
	beta53	0.16	0.158	-0.002 ± 0.004	-0.013 ± 0.023	0.012 ± 0.001	0.111 ± 0.003	0.012 ± 0.001	0.111 ± 0.003	0.483 ± 0.024	0.695 ± 0.021	0.159	-0.001	0.1074	0.1074	1.007	95.698 ± 0.007	0.439 ± 0.003	—
	beta54	0.16	0.161	0.001 ± 0.002	0.009 ± 0.015	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.202 ± 0.010	0.450 ± 0.013	0.159	-0.001	0.0710	0.0710	0.962	93.704 ± 0.008	0.272 ± 0.001	—
	beta55	0.00	0.003	0.003 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	—	—	0.000	0.000	0.0822	0.0822	0.997	94.229 ± 0.008	0.336 ± 0.003	5.771 ± 0.008
	beta56	0.00	-0.001	-0.001 ± 0.003	—	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	—	—	0.000	0.000	0.0762	0.0762	0.962	94.648 ± 0.007	0.293 ± 0.003	5.352 ± 0.007
	beta57	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	—	—	0.002	0.002	0.0449	0.0449	0.965	93.179 ± 0.008	0.184 ± 0.002	6.821 ± 0.008
	beta58	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.000	0.000	0.0382	0.0382	0.928	94.648 ± 0.007	0.155 ± 0.002	5.352 ± 0.007
QML	beta41	0.20	0.180	-0.020 ± 0.003	-0.099 ± 0.013	0.006 ± 0.000	0.079 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.165 ± 0.008	0.406 ± 0.011	0.179	-0.021	0.0767	0.0797	0.972	92.970 ± 0.008	0.301 ± 0.001	—
	beta42	0.20	0.185	-0.015 ± 0.003	-0.076 ± 0.013	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.171 ± 0.008	0.414 ± 0.012	0.184	-0.016	0.0814	0.0831	0.972	93.704 ± 0.008	0.310 ± 0.001	—
	beta43	0.20	0.208	0.008 ± 0.002	0.041 ± 0.010	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.101 ± 0.005	0.318 ± 0.009	0.209	0.009	0.0643	0.0650	0.985	95.278 ± 0.007	0.244 ± 0.001	—
	beta44	0.00	-0.007	-0.007 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	—	—	-0.006	-0.006	0.0711	0.0714	1.031	95.698 ± 0.007	0.295 ± 0.002	4.302 ± 0.007
	beta45	0.00	-0.001	-0.001 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	-0.001	-0.001	0.0684	0.0684	0.994	96.222 ± 0.006	0.263 ± 0.002	3.778 ± 0.006
	beta46	0.00	0.020	0.020 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.018	0.018	0.0385	0.0427	0.976	92.655 ± 0.008	0.161 ± 0.002	7.345 ± 0.008
	beta47	0.00	0.019	0.019 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.016	0.016	0.0389	0.0422	0.970	93.494 ± 0.008	0.158 ± 0.002	6.506 ± 0.008
	beta51	0.16	0.145	-0.015 ± 0.003	-0.091 ± 0.017	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.270 ± 0.013	0.520 ± 0.015	0.147	-0.013	0.0842	0.0853	0.947	93.284 ± 0.008	0.304 ± 0.001	—
	beta52	0.16	0.162	0.002 ± 0.002	0.013 ± 0.015	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.215 ± 0.010	0.464 ± 0.013	0.160	0.000	0.0733	0.0733	0.965	94.124 ± 0.008	0.281 ± 0.001	—
	beta53	0.16	0.142	-0.018 ± 0.002	-0.113 ± 0.016	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.243 ± 0.012	0.493 ± 0.014	0.143	-0.017	0.0741	0.0760	1.004	94.439 ± 0.007	0.302 ± 0.001	—
	beta54	0.16	0.164	0.004 ± 0.002	0.026 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.200 ± 0.010	0.447 ± 0.013	0.162	0.002	0.0691	0.0691	0.981	94.963 ± 0.007	0.275 ± 0.001	—
	beta55	0.00	-0.005	-0.005 ± 0.002	—	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	—	—	-0.006	-0.006	0.0729	0.0731	1.053	95.803 ± 0.006	0.313 ± 0.002	4.197 ± 0.006
	beta56	0.00	0.010	0.010 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	0.011	0.011	0.0666	0.0676	0.993	95.488 ± 0.007	0.263 ± 0.002	4.512 ± 0.007
	beta57	0.00	0.015	0.015 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.016	0.016	0.0422	0.0452	1.004	95.173 ± 0.007	0.171 ± 0.002	4.827 ± 0.007
	beta58	0.00	0.015	0.015 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.013	0.013	0.0333	0.0356	0.985	93.704 ± 0.008	0.140 ± 0.001	6.296 ± 0.008

Table S65

Simulation 2 - Condition 27: Linear Model, Distribution = right-skewed, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta41	0.20	0.200	0.000 ± 0.004	0.001 ± 0.019	0.014 ± 0.001	0.120 ± 0.003	0.014 ± 0.001	0.120 ± 0.004	0.361 ± 0.018	0.601 ± 0.018	0.198	-0.002	0.1128	0.1128	0.933	94.648 ± 0.007	0.440 ± 0.002	—
	beta42	0.20	0.207	0.007 ± 0.004	0.036 ± 0.021	0.016 ± 0.001	0.127 ± 0.003	0.016 ± 0.001	0.127 ± 0.004	0.404 ± 0.019	0.636 ± 0.018	0.203	0.003	0.1224	0.1225	0.919	94.439 ± 0.007	0.457 ± 0.003	—
	beta43	0.20	0.204	0.004 ± 0.002	0.021 ± 0.011	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.112 ± 0.006	0.335 ± 0.010	0.204	0.004	0.0677	0.0679	0.958	94.544 ± 0.007	0.251 ± 0.001	—
	beta44	0.00	0.001	0.001 ± 0.003	—	0.008 ± 0.000	0.090 ± 0.003	0.008 ± 0.000	0.090 ± 0.003	—	—	0.004	0.004	0.0834	0.0835	0.916	94.439 ± 0.007	0.324 ± 0.004	5.561 ± 0.007
	beta45	0.00	-0.001	-0.001 ± 0.003	—	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.003	—	—	-0.002	-0.002	0.0750	0.0750	0.940	94.963 ± 0.007	0.300 ± 0.005	5.037 ± 0.007
	beta46	0.00	0.000	0.000 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	—	—	0.000	0.000	0.0408	0.0408	0.904	94.334 ± 0.007	0.175 ± 0.002	5.666 ± 0.007
	beta47	0.00	-0.002	-0.002 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	—	—	-0.001	-0.001	0.0423	0.0423	0.889	94.439 ± 0.007	0.171 ± 0.002	5.561 ± 0.007
	beta51	0.16	0.161	0.001 ± 0.004	0.006 ± 0.023	0.013 ± 0.001	0.114 ± 0.003	0.013 ± 0.001	0.114 ± 0.003	0.503 ± 0.024	0.709 ± 0.021	0.158	-0.002	0.1121	0.1121	0.954	93.389 ± 0.008	0.425 ± 0.002	—
	beta52	0.16	0.163	0.003 ± 0.003	0.019 ± 0.017	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	0.275 ± 0.014	0.525 ± 0.016	0.159	-0.001	0.0822	0.0822	0.956	94.858 ± 0.007	0.315 ± 0.002	—
	beta53	0.16	0.160	-0.000 ± 0.004	-0.002 ± 0.023	0.012 ± 0.001	0.111 ± 0.003	0.012 ± 0.001	0.111 ± 0.003	0.484 ± 0.024	0.695 ± 0.020	0.159	-0.001	0.1071	0.1071	0.976	95.383 ± 0.007	0.426 ± 0.002	—
	beta54	0.16	0.162	0.002 ± 0.002	0.011 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.199 ± 0.010	0.446 ± 0.013	0.158	-0.002	0.0679	0.0679	0.969	94.334 ± 0.007	0.271 ± 0.001	—
	beta55	0.00	0.003	0.003 ± 0.003	—	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	—	—	0.003	0.003	0.0822	0.0823	0.966	94.334 ± 0.007	0.339 ± 0.004	5.666 ± 0.007
	beta56	0.00	-0.001	-0.001 ± 0.003	—	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	—	—	0.000	0.000	0.0719	0.0719	0.953	94.439 ± 0.007	0.291 ± 0.003	5.561 ± 0.007
	beta57	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	—	—	0.002	0.002	0.0430	0.0431	0.946	94.544 ± 0.007	0.181 ± 0.002	5.456 ± 0.007
	beta58	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	-0.001	-0.001	0.0368	0.0368	0.919	95.173 ± 0.007	0.154 ± 0.002	4.827 ± 0.007

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S66

Simulation 2 - Condition 28: Linear Model, Distribution = right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.204	0.004 ± 0.002	0.020 ± 0.011	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.116 ± 0.005	0.341 ± 0.010	0.201	0.001	0.0654	0.0654	1.014	96.061 ± 0.006	0.270 ± 0.001	—
	beta42	0.20	0.198	-0.002 ± 0.002	-0.010 ± 0.011	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.126 ± 0.006	0.354 ± 0.010	0.197	-0.003	0.0694	0.0695	1.023	95.758 ± 0.006	0.284 ± 0.001	—
	beta43	0.20	0.199	-0.001 ± 0.001	-0.005 ± 0.006	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.041 ± 0.002	0.203 ± 0.006	0.199	-0.001	0.0381	0.0381	0.990	93.535 ± 0.008	0.158 ± 0.000	—
	beta44	0.00	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.003	-0.003	0.0442	0.0442	0.963	93.333 ± 0.008	0.170 ± 0.001	6.667 ± 0.008
	beta45	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.004	0.004	0.0419	0.0420	0.944	93.838 ± 0.008	0.162 ± 0.001	6.162 ± 0.008
	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	—	—	-0.001	-0.001	0.0246	0.0247	0.965	94.848 ± 0.007	0.094 ± 0.001	5.152 ± 0.007
	beta47	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	0.000	0.000	0.0222	0.0222	0.981	94.343 ± 0.007	0.094 ± 0.001	5.657 ± 0.007
	beta51	0.16	0.162	0.002 ± 0.002	0.016 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.188 ± 0.009	0.433 ± 0.012	0.161	0.001	0.0666	0.0666	0.966	94.444 ± 0.007	0.262 ± 0.001	—
	beta52	0.16	0.156	-0.004 ± 0.002	-0.025 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.103 ± 0.005	0.322 ± 0.009	0.156	-0.004	0.0503	0.0505	0.980	94.040 ± 0.008	0.197 ± 0.001	—
	beta53	0.16	0.161	0.001 ± 0.002	0.005 ± 0.014	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.182 ± 0.008	0.427 ± 0.012	0.164	0.004	0.0678	0.0679	0.976	94.949 ± 0.007	0.261 ± 0.001	—
	beta54	0.16	0.159	-0.001 ± 0.001	-0.007 ± 0.009	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.072 ± 0.003	0.269 ± 0.007	0.161	0.001	0.0422	0.0422	1.001	94.646 ± 0.007	0.169 ± 0.000	—
	beta55	0.00	-0.005	-0.005 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	-0.002	-0.002	0.0484	0.0484	0.989	93.939 ± 0.008	0.195 ± 0.001	6.061 ± 0.008
	beta56	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.003	0.003	0.0431	0.0432	0.925	92.828 ± 0.008	0.170 ± 0.001	7.172 ± 0.008
	beta57	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.001	0.001	0.0270	0.0270	0.950	93.434 ± 0.008	0.101 ± 0.001	6.566 ± 0.008
	beta58	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	-0.001	-0.001	0.0228	0.0228	0.937	94.646 ± 0.007	0.087 ± 0.001	5.354 ± 0.007
QML	beta41	0.20	0.184	-0.016 ± 0.002	-0.081 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.063 ± 0.003	0.251 ± 0.007	0.183	-0.017	0.0464	0.0493	1.003	93.131 ± 0.008	0.187 ± 0.000	—
	beta42	0.20	0.179	-0.021 ± 0.002	-0.103 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.069 ± 0.003	0.263 ± 0.007	0.179	-0.021	0.0479	0.0521	1.018	93.232 ± 0.008	0.193 ± 0.000	—
	beta43	0.20	0.204	0.004 ± 0.001	0.019 ± 0.006	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.038 ± 0.002	0.195 ± 0.006	0.204	0.004	0.0362	0.0364	0.999	94.040 ± 0.008	0.152 ± 0.000	—
	beta44	0.00	-0.011	-0.011 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.012	-0.012	0.0452	0.0467	0.998	94.545 ± 0.007	0.176 ± 0.001	5.455 ± 0.007
	beta45	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.004	0.004	0.0405	0.0407	0.988	95.354 ± 0.007	0.159 ± 0.001	4.646 ± 0.007
	beta46	0.00	0.018	0.018 ± 0.001	—	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.018	0.018	0.0235	0.0293	0.997	88.687 ± 0.010	0.093 ± 0.001	11.313 ± 0.010
	beta47	0.00	0.018	0.018 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.018	0.018	0.0219	0.0282	1.011	89.394 ± 0.010	0.091 ± 0.001	10.606 ± 0.010
	beta51	0.16	0.146	-0.014 ± 0.002	-0.086 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.104 ± 0.005	0.323 ± 0.009	0.145	-0.015	0.0505	0.0527	0.967	93.030 ± 0.008	0.189 ± 0.000	—
	beta52	0.16	0.157	-0.003 ± 0.001	-0.018 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.079 ± 0.004	0.281 ± 0.008	0.158	-0.002	0.0435	0.0435	0.987	93.030 ± 0.008	0.174 ± 0.000	—
	beta53	0.16	0.144	-0.016 ± 0.002	-0.099 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.104 ± 0.005	0.323 ± 0.009	0.147	-0.013	0.0508	0.0525	0.975	92.929 ± 0.008	0.188 ± 0.000	—
	beta54	0.16	0.161	0.001 ± 0.001	0.005 ± 0.009	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.072 ± 0.003	0.268 ± 0.007	0.163	0.003	0.0426	0.0427	1.016	94.848 ± 0.007	0.171 ± 0.000	—
	beta55	0.00	-0.012	-0.012 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	—	—	-0.010	-0.010	0.0491	0.0501	1.013	94.545 ± 0.007	0.193 ± 0.001	5.455 ± 0.007
	beta56	0.00	0.013	0.013 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.013	0.013	0.0399	0.0419	0.947	92.525 ± 0.008	0.159 ± 0.001	7.475 ± 0.008
	beta57	0.00	0.017	0.017 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	0.016	0.016	0.0260	0.0304	0.978	90.101 ± 0.009	0.102 ± 0.001	9.899 ± 0.009
	beta58	0.00	0.014	0.014 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	0.012	0.012	0.0217	0.0250	0.972	90.303 ± 0.009	0.083 ± 0.001	9.697 ± 0.009

Table S66

Simulation 2 - Condition 28: Linear Model, Distribution = right-skewed, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta41	0.20	0.206	0.006 ± 0.002	0.029 ± 0.011	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.117 ± 0.005	0.342 ± 0.009	0.204	0.004	0.0665	0.0667	0.999	95.556 ± 0.007	0.267 ± 0.001	—
	beta42	0.20	0.198	-0.002 ± 0.002	-0.010 ± 0.011	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.123 ± 0.006	0.350 ± 0.010	0.197	-0.003	0.0677	0.0678	1.012	95.455 ± 0.007	0.278 ± 0.001	—
	beta43	0.20	0.199	-0.001 ± 0.001	-0.003 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.039 ± 0.002	0.198 ± 0.006	0.201	0.001	0.0362	0.0362	0.995	94.444 ± 0.007	0.155 ± 0.000	—
	beta44	0.00	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.003	-0.003	0.0427	0.0428	0.960	94.444 ± 0.007	0.174 ± 0.001	5.556 ± 0.007
	beta45	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.002	0.002	0.0418	0.0418	0.947	94.747 ± 0.007	0.164 ± 0.001	5.253 ± 0.007
	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	—	—	-0.001	-0.001	0.0236	0.0236	0.984	95.859 ± 0.006	0.096 ± 0.001	4.141 ± 0.006
	beta47	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	0.000	0.000	0.0208	0.0208	0.990	95.051 ± 0.007	0.093 ± 0.001	4.949 ± 0.007
	beta51	0.16	0.163	0.003 ± 0.002	0.018 ± 0.014	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.181 ± 0.008	0.425 ± 0.012	0.162	0.002	0.0672	0.0673	0.969	94.444 ± 0.007	0.258 ± 0.001	—
	beta52	0.16	0.156	-0.004 ± 0.002	-0.024 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.100 ± 0.005	0.317 ± 0.009	0.156	-0.004	0.0508	0.0509	0.975	93.737 ± 0.008	0.193 ± 0.001	—
	beta53	0.16	0.162	0.002 ± 0.002	0.011 ± 0.013	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.178 ± 0.008	0.422 ± 0.012	0.164	0.004	0.0675	0.0676	0.976	94.848 ± 0.007	0.258 ± 0.001	—
	beta54	0.16	0.159	-0.001 ± 0.001	-0.007 ± 0.009	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.072 ± 0.003	0.268 ± 0.007	0.161	0.001	0.0431	0.0431	1.011	94.646 ± 0.007	0.170 ± 0.000	—
	beta55	0.00	-0.005	-0.005 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	-0.004	-0.004	0.0489	0.0491	0.990	94.848 ± 0.007	0.198 ± 0.001	5.152 ± 0.007
	beta56	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.002	0.002	0.0422	0.0423	0.933	93.434 ± 0.008	0.170 ± 0.001	6.566 ± 0.008
	beta57	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.001	0.001	0.0271	0.0272	0.964	95.253 ± 0.007	0.102 ± 0.001	4.747 ± 0.007
	beta58	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	-0.002	-0.002	0.0228	0.0228	0.956	95.758 ± 0.006	0.088 ± 0.001	4.242 ± 0.006

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S67

Simulation 2 - Condition 29: Linear Model, Distribution = right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.193	-0.007 ± 0.003	-0.036 ± 0.014	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.086 ± 0.002	0.184 ± 0.008	0.428 ± 0.012	0.190	-0.010	0.0859	0.0864	0.992	95.354 ± 0.007	0.332 ± 0.001	—
	beta42	0.20	0.202	0.002 ± 0.003	0.009 ± 0.015	0.009 ± 0.000	0.093 ± 0.002	0.009 ± 0.000	0.093 ± 0.003	0.216 ± 0.010	0.464 ± 0.013	0.202	0.002	0.0877	0.0877	0.959	94.949 ± 0.007	0.349 ± 0.002	—
	beta43	0.20	0.201	0.001 ± 0.002	0.006 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.069 ± 0.003	0.262 ± 0.007	0.201	0.001	0.0507	0.0508	1.011	94.646 ± 0.007	0.208 ± 0.001	—
	beta44	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	0.000	0.000	0.0593	0.0593	0.921	94.040 ± 0.008	0.216 ± 0.002	5.960 ± 0.008
	beta45	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	0.000	0.000	0.0562	0.0562	0.891	92.121 ± 0.009	0.209 ± 0.002	7.879 ± 0.009
	beta46	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.002	0.002	0.0328	0.0329	0.897	92.222 ± 0.009	0.122 ± 0.001	7.778 ± 0.009
	beta47	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.002	-0.002	0.0309	0.0309	0.890	92.121 ± 0.009	0.120 ± 0.001	7.879 ± 0.009
	beta51	0.16	0.158	-0.002 ± 0.003	-0.013 ± 0.017	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.281 ± 0.013	0.530 ± 0.015	0.160	0.000	0.0867	0.0867	0.979	95.556 ± 0.007	0.326 ± 0.001	—
	beta52	0.16	0.160	0.000 ± 0.002	0.003 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.163 ± 0.008	0.404 ± 0.012	0.162	0.002	0.0582	0.0582	0.981	93.434 ± 0.008	0.249 ± 0.001	—
	beta53	0.16	0.159	-0.001 ± 0.003	-0.006 ± 0.017	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	0.277 ± 0.012	0.526 ± 0.014	0.158	-0.002	0.0797	0.0798	0.992	94.444 ± 0.007	0.327 ± 0.001	—
	beta54	0.16	0.161	0.001 ± 0.002	0.004 ± 0.011	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.130 ± 0.006	0.360 ± 0.010	0.159	-0.001	0.0573	0.0573	0.983	94.545 ± 0.007	0.222 ± 0.001	—
	beta55	0.00	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.003	-0.003	0.0636	0.0637	0.942	93.232 ± 0.008	0.243 ± 0.002	6.768 ± 0.008
	beta56	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	—	—	0.000	0.000	0.0541	0.0541	0.947	93.434 ± 0.008	0.212 ± 0.002	6.566 ± 0.008
	beta57	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.002	0.002	0.0337	0.0338	0.897	92.020 ± 0.009	0.129 ± 0.001	7.980 ± 0.009
	beta58	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.000	0.000	0.0297	0.0297	0.893	91.616 ± 0.009	0.114 ± 0.001	8.384 ± 0.009
QML	beta41	0.20	0.184	-0.016 ± 0.002	-0.080 ± 0.011	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.129 ± 0.006	0.359 ± 0.010	0.183	-0.017	0.0704	0.0724	1.003	94.747 ± 0.007	0.275 ± 0.001	—
	beta42	0.20	0.192	-0.008 ± 0.002	-0.042 ± 0.012	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.144 ± 0.007	0.380 ± 0.011	0.191	-0.009	0.0734	0.0739	0.957	93.838 ± 0.008	0.284 ± 0.001	—
	beta43	0.20	0.204	0.004 ± 0.002	0.019 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.066 ± 0.003	0.257 ± 0.007	0.203	0.003	0.0500	0.0501	1.014	95.354 ± 0.007	0.204 ± 0.001	—
	beta44	0.00	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	-0.003	-0.003	0.0586	0.0586	1.003	95.859 ± 0.006	0.224 ± 0.001	4.141 ± 0.006
	beta45	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	0.000	0.000	0.0542	0.0542	0.955	93.636 ± 0.008	0.211 ± 0.001	6.364 ± 0.008
	beta46	0.00	0.009	0.009 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.008	0.008	0.0311	0.0322	0.977	94.141 ± 0.007	0.123 ± 0.001	5.859 ± 0.007
	beta47	0.00	0.007	0.007 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.005	0.005	0.0296	0.0301	0.963	95.556 ± 0.007	0.119 ± 0.001	4.444 ± 0.007
	beta51	0.16	0.150	-0.010 ± 0.002	-0.061 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.202 ± 0.010	0.450 ± 0.013	0.152	-0.008	0.0723	0.0728	0.986	95.253 ± 0.007	0.276 ± 0.001	—
	beta52	0.16	0.161	0.001 ± 0.002	0.009 ± 0.012	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.143 ± 0.007	0.378 ± 0.011	0.162	0.002	0.0555	0.0556	0.994	93.737 ± 0.008	0.235 ± 0.001	—
	beta53	0.16	0.151	-0.009 ± 0.002	-0.059 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.199 ± 0.009	0.446 ± 0.012	0.152	-0.008	0.0696	0.0701	0.993	94.646 ± 0.007	0.275 ± 0.001	—
	beta54	0.16	0.161	0.001 ± 0.002	0.008 ± 0.011	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.130 ± 0.006	0.360 ± 0.010	0.160	0.000	0.0576	0.0576	0.997	94.747 ± 0.007	0.225 ± 0.000	—
	beta55	0.00	-0.005	-0.005 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.005	-0.005	0.0604	0.0606	0.993	94.848 ± 0.007	0.244 ± 0.001	5.152 ± 0.007
	beta56	0.00	0.006	0.006 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.004	0.004	0.0516	0.0517	0.992	94.343 ± 0.007	0.211 ± 0.001	5.657 ± 0.007
	beta57	0.00	0.008	0.008 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.008	0.008	0.0324	0.0334	0.965	94.444 ± 0.007	0.130 ± 0.001	5.556 ± 0.007
	beta58	0.00	0.006	0.006 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	0.005	0.005	0.0270	0.0275	0.955	94.747 ± 0.007	0.113 ± 0.001	5.253 ± 0.007

Table S67

Simulation 2 - Condition 29: Linear Model, Distribution = right-skewed, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta41	0.20	0.194	-0.006 ± 0.003	-0.031 ± 0.013	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.180 ± 0.008	0.424 ± 0.012	0.193	-0.007	0.0867	0.0869	0.991	94.949 ± 0.007	0.328 ± 0.001	—
	beta42	0.20	0.202	0.002 ± 0.003	0.010 ± 0.015	0.008 ± 0.000	0.092 ± 0.002	0.008 ± 0.000	0.092 ± 0.003	0.211 ± 0.010	0.459 ± 0.013	0.202	0.002	0.0888	0.0888	0.952	93.636 ± 0.008	0.343 ± 0.001	—
	beta43	0.20	0.202	0.002 ± 0.002	0.008 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.068 ± 0.003	0.262 ± 0.007	0.201	0.001	0.0521	0.0522	1.009	94.949 ± 0.007	0.207 ± 0.001	—
	beta44	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	0.001	0.001	0.0612	0.0612	0.968	95.253 ± 0.007	0.230 ± 0.002	4.747 ± 0.007
	beta45	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.002	-0.002	0.0566	0.0566	0.935	93.838 ± 0.008	0.219 ± 0.002	6.162 ± 0.008
	beta46	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	0.001	0.001	0.0323	0.0323	0.956	94.343 ± 0.007	0.128 ± 0.001	5.657 ± 0.007
	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.002	-0.002	0.0308	0.0309	0.945	95.152 ± 0.007	0.125 ± 0.001	4.848 ± 0.007
	beta51	0.16	0.158	-0.002 ± 0.003	-0.011 ± 0.017	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	0.279 ± 0.013	0.528 ± 0.015	0.159	-0.001	0.0845	0.0845	0.973	95.556 ± 0.007	0.322 ± 0.001	—
	beta52	0.16	0.161	0.001 ± 0.002	0.007 ± 0.013	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.162 ± 0.008	0.402 ± 0.012	0.162	0.002	0.0593	0.0594	0.983	93.535 ± 0.008	0.248 ± 0.001	—
	beta53	0.16	0.159	-0.001 ± 0.003	-0.006 ± 0.017	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.002	0.275 ± 0.012	0.525 ± 0.014	0.158	-0.002	0.0803	0.0804	0.981	94.242 ± 0.007	0.323 ± 0.001	—
	beta54	0.16	0.161	0.001 ± 0.002	0.005 ± 0.011	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.129 ± 0.006	0.360 ± 0.010	0.159	-0.001	0.0571	0.0571	0.989	94.545 ± 0.007	0.223 ± 0.000	—
	beta55	0.00	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.002	-0.002	0.0649	0.0650	0.967	93.838 ± 0.008	0.252 ± 0.001	6.162 ± 0.008
	beta56	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	—	—	0.000	0.000	0.0542	0.0542	0.975	94.545 ± 0.007	0.220 ± 0.001	5.455 ± 0.007
	beta57	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.002	0.002	0.0331	0.0332	0.936	94.343 ± 0.007	0.135 ± 0.001	5.657 ± 0.007
	beta58	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.001	-0.001	0.0301	0.0301	0.926	93.838 ± 0.008	0.119 ± 0.001	6.162 ± 0.008

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S68

Simulation 2 - Condition 30: Linear Model, Distribution = right-skewed, N = 1000, Reliability = 0.8

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.199	-0.001 ± 0.002	-0.005 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.200	0.000	0.0501	0.0501	1.020	95.243 ± 0.007	0.203 ± 0.001	—
	beta42	0.20	0.201	0.001 ± 0.002	0.006 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.074 ± 0.003	0.272 ± 0.008	0.200	0.000	0.0562	0.0562	0.997	94.130 ± 0.007	0.213 ± 0.001	—
	beta43	0.20	0.200	0.000 ± 0.001	0.001 ± 0.005	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.028 ± 0.001	0.167 ± 0.004	0.201	0.001	0.0354	0.0354	0.998	95.040 ± 0.007	0.131 ± 0.000	—
	beta44	0.00	0.004	0.004 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.003	0.003	0.0344	0.0346	0.954	93.522 ± 0.008	0.129 ± 0.001	6.478 ± 0.008
	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.002	-0.002	0.0326	0.0327	0.968	93.421 ± 0.008	0.126 ± 0.001	6.579 ± 0.008
	beta46	0.00	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.019 ± 0.000	0.000 ± 0.000	0.019 ± 0.001	—	—	0.000	0.000	0.0181	0.0181	0.946	93.219 ± 0.008	0.069 ± 0.001	6.781 ± 0.008
	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.000 ± 0.000	0.019 ± 0.000	0.000 ± 0.000	0.019 ± 0.001	—	—	0.000	0.000	0.0179	0.0179	0.958	93.826 ± 0.008	0.070 ± 0.001	6.174 ± 0.008
	beta51	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.102 ± 0.005	0.319 ± 0.009	0.159	-0.001	0.0509	0.0509	0.999	95.547 ± 0.007	0.200 ± 0.001	—
	beta52	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.066 ± 0.003	0.258 ± 0.007	0.159	-0.001	0.0403	0.0403	0.963	94.231 ± 0.007	0.156 ± 0.000	—
	beta53	0.16	0.159	-0.001 ± 0.002	-0.008 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.105 ± 0.005	0.323 ± 0.009	0.157	-0.003	0.0523	0.0524	0.990	95.445 ± 0.007	0.201 ± 0.001	—
	beta54	0.16	0.162	0.002 ± 0.001	0.010 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.051 ± 0.002	0.225 ± 0.006	0.163	0.003	0.0364	0.0365	0.998	95.142 ± 0.007	0.141 ± 0.000	—
	beta55	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	-0.002	-0.002	0.0382	0.0382	0.974	94.636 ± 0.007	0.148 ± 0.001	5.364 ± 0.007
	beta56	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.003	0.003	0.0349	0.0350	0.946	92.814 ± 0.008	0.130 ± 0.001	7.186 ± 0.008
	beta57	0.00	0.001	0.001 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.001	0.000 ± 0.000	0.020 ± 0.001	—	—	0.002	0.002	0.0172	0.0173	0.969	93.421 ± 0.008	0.074 ± 0.001	6.579 ± 0.008
	beta58	0.00	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	0.000	0.000	0.0172	0.0172	0.941	94.332 ± 0.007	0.066 ± 0.001	5.668 ± 0.007
QML	beta41	0.20	0.190	-0.010 ± 0.001	-0.052 ± 0.007	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.048 ± 0.002	0.219 ± 0.006	0.190	-0.010	0.0427	0.0440	1.027	94.433 ± 0.007	0.171 ± 0.000	—
	beta42	0.20	0.192	-0.008 ± 0.001	-0.042 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.002	0.231 ± 0.006	0.191	-0.009	0.0451	0.0459	0.995	94.231 ± 0.007	0.177 ± 0.000	—
	beta43	0.20	0.203	0.003 ± 0.001	0.014 ± 0.005	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.027 ± 0.001	0.164 ± 0.004	0.203	0.003	0.0341	0.0343	0.993	95.243 ± 0.007	0.128 ± 0.000	—
	beta44	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.001	0.001	0.0346	0.0346	0.998	95.142 ± 0.007	0.135 ± 0.000	4.858 ± 0.007
	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.001	-0.001	0.0312	0.0312	0.999	95.142 ± 0.007	0.128 ± 0.000	4.858 ± 0.007
	beta46	0.00	0.007	0.007 ± 0.001	—	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.019 ± 0.001	—	—	0.006	0.006	0.0178	0.0189	0.998	94.534 ± 0.007	0.071 ± 0.000	5.466 ± 0.007
	beta47	0.00	0.006	0.006 ± 0.001	—	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.019 ± 0.000	—	—	0.006	0.006	0.0177	0.0188	1.012	95.142 ± 0.007	0.071 ± 0.000	4.858 ± 0.007
	beta51	0.16	0.150	-0.010 ± 0.001	-0.060 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.079 ± 0.004	0.280 ± 0.008	0.150	-0.010	0.0439	0.0449	1.002	94.838 ± 0.007	0.172 ± 0.000	—
	beta52	0.16	0.160	0.000 ± 0.001	0.002 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.059 ± 0.003	0.242 ± 0.007	0.161	0.001	0.0375	0.0375	0.972	94.231 ± 0.007	0.148 ± 0.000	—
	beta53	0.16	0.151	-0.009 ± 0.001	-0.058 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.081 ± 0.004	0.284 ± 0.008	0.149	-0.011	0.0453	0.0466	0.989	94.332 ± 0.007	0.172 ± 0.000	—
	beta54	0.16	0.162	0.002 ± 0.001	0.012 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.051 ± 0.002	0.225 ± 0.006	0.163	0.003	0.0367	0.0368	1.005	94.939 ± 0.007	0.142 ± 0.000	—
	beta55	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.004	-0.004	0.0377	0.0378	0.998	96.255 ± 0.006	0.149 ± 0.000	3.745 ± 0.006
	beta56	0.00	0.005	0.005 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	0.006	0.006	0.0342	0.0348	0.978	94.737 ± 0.007	0.129 ± 0.000	5.263 ± 0.007
	beta57	0.00	0.007	0.007 ± 0.001	—	0.000 ± 0.000	0.019 ± 0.000	0.000 ± 0.000	0.020 ± 0.001	—	—	0.007	0.007	0.0172	0.0186	1.021	94.939 ± 0.007	0.077 ± 0.000	5.061 ± 0.007
	beta58	0.00	0.006	0.006 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	0.005	0.005	0.0166	0.0174	0.993	94.534 ± 0.007	0.067 ± 0.000	5.466 ± 0.007

Table S68

Simulation 2 - Condition 30: Linear Model, Distribution = right-skewed, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta41	0.20	0.199	-0.001 ± 0.002	-0.003 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.200	0.000	0.0509	0.0509	1.018	94.838 ± 0.007	0.202 ± 0.001	—
	beta42	0.20	0.201	0.001 ± 0.002	0.007 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.074 ± 0.003	0.272 ± 0.008	0.201	0.001	0.0550	0.0551	0.990	93.927 ± 0.008	0.212 ± 0.001	—
	beta43	0.20	0.200	0.000 ± 0.001	0.002 ± 0.005	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.028 ± 0.001	0.167 ± 0.004	0.200	0.000	0.0348	0.0348	0.985	95.243 ± 0.007	0.129 ± 0.000	—
	beta44	0.00	0.004	0.004 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.002	0.002	0.0342	0.0343	0.983	95.040 ± 0.007	0.133 ± 0.001	4.960 ± 0.007
	beta45	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.003	-0.003	0.0315	0.0316	0.988	94.534 ± 0.007	0.130 ± 0.001	5.466 ± 0.007
	beta46	0.00	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.019 ± 0.000	0.000 ± 0.000	0.019 ± 0.001	—	—	0.000	0.000	0.0180	0.0180	0.990	95.142 ± 0.007	0.073 ± 0.000	4.858 ± 0.007
	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.000 ± 0.000	0.019 ± 0.000	0.000 ± 0.000	0.019 ± 0.001	—	—	0.000	0.000	0.0181	0.0181	0.999	95.850 ± 0.006	0.073 ± 0.001	4.150 ± 0.006
	beta51	0.16	0.159	-0.001 ± 0.002	-0.009 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.100 ± 0.005	0.317 ± 0.009	0.159	-0.001	0.0499	0.0499	1.004	95.445 ± 0.007	0.200 ± 0.000	—
	beta52	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.065 ± 0.003	0.255 ± 0.007	0.159	-0.001	0.0397	0.0397	0.971	94.231 ± 0.007	0.156 ± 0.000	—
	beta53	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.104 ± 0.005	0.322 ± 0.009	0.157	-0.003	0.0506	0.0507	0.993	95.344 ± 0.007	0.200 ± 0.000	—
	beta54	0.16	0.162	0.002 ± 0.001	0.010 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.050 ± 0.002	0.225 ± 0.006	0.162	0.002	0.0366	0.0366	1.002	94.838 ± 0.007	0.141 ± 0.000	—
	beta55	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.002	-0.002	0.0377	0.0378	0.984	95.547 ± 0.007	0.151 ± 0.001	4.453 ± 0.007
	beta56	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.003	0.003	0.0354	0.0356	0.969	94.332 ± 0.007	0.133 ± 0.001	5.668 ± 0.007
	beta57	0.00	0.001	0.001 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.001	0.000 ± 0.000	0.020 ± 0.001	—	—	0.001	0.001	0.0177	0.0177	1.013	94.939 ± 0.007	0.077 ± 0.000	5.061 ± 0.007
	beta58	0.00	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	0.000	0.000	0.0171	0.0171	0.986	96.154 ± 0.006	0.069 ± 0.000	3.846 ± 0.006

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S69

Simulation 2 - Condition 31: Linear Model, Distribution = uniform, N = 400, Reliability = 0.4

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.198	-0.002 ± 0.006	-0.009 ± 0.028	0.009 ± 0.001	0.097 ± 0.005	0.009 ± 0.001	0.097 ± 0.006	0.234 ± 0.025	0.483 ± 0.030	0.202	0.002	0.0934	0.0934	1.251	98.644 ± 0.007	0.475 ± 0.013	—
	beta42	0.20	0.211	0.011 ± 0.006	0.056 ± 0.030	0.010 ± 0.001	0.102 ± 0.005	0.010 ± 0.001	0.102 ± 0.006	0.260 ± 0.024	0.510 ± 0.028	0.202	0.002	0.1041	0.1041	1.150	98.305 ± 0.008	0.458 ± 0.011	—
	beta43	0.20	0.194	-0.006 ± 0.005	-0.029 ± 0.026	0.008 ± 0.001	0.091 ± 0.004	0.008 ± 0.001	0.091 ± 0.005	0.207 ± 0.017	0.455 ± 0.023	0.195	-0.005	0.0870	0.0872	1.114	97.288 ± 0.009	0.397 ± 0.008	—
	beta44	0.00	-0.006	-0.006 ± 0.008	—	0.018 ± 0.002	0.136 ± 0.007	0.018 ± 0.002	0.136 ± 0.008	—	—	-0.013	-0.013	0.1254	0.1261	1.240	99.661 ± 0.003	0.660 ± 0.019	0.339 ± 0.003
	beta45	0.00	0.007	0.007 ± 0.007	—	0.015 ± 0.002	0.123 ± 0.006	0.015 ± 0.002	0.123 ± 0.007	—	—	0.007	0.007	0.1146	0.1148	1.101	97.288 ± 0.009	0.532 ± 0.015	2.712 ± 0.009
	beta46	0.00	-0.008	-0.008 ± 0.012	—	0.045 ± 0.005	0.212 ± 0.012	0.045 ± 0.005	0.211 ± 0.014	—	—	-0.008	-0.008	0.1796	0.1798	1.120	99.322 ± 0.005	0.929 ± 0.039	0.678 ± 0.005
	beta47	0.00	-0.002	-0.002 ± 0.011	—	0.036 ± 0.004	0.189 ± 0.010	0.036 ± 0.004	0.188 ± 0.011	—	—	0.000	0.000	0.1563	0.1563	1.214	98.983 ± 0.006	0.898 ± 0.043	1.017 ± 0.006
	beta51	0.16	0.161	0.001 ± 0.005	0.006 ± 0.032	0.008 ± 0.001	0.087 ± 0.003	0.008 ± 0.001	0.087 ± 0.004	0.297 ± 0.023	0.545 ± 0.027	0.158	-0.002	0.0910	0.0911	1.300	98.644 ± 0.007	0.445 ± 0.011	—
	beta52	0.16	0.164	0.004 ± 0.006	0.026 ± 0.037	0.010 ± 0.001	0.101 ± 0.004	0.010 ± 0.001	0.100 ± 0.005	0.394 ± 0.032	0.628 ± 0.032	0.158	-0.002	0.0996	0.0996	1.069	97.627 ± 0.009	0.421 ± 0.007	—
	beta53	0.16	0.164	0.004 ± 0.006	0.024 ± 0.037	0.011 ± 0.001	0.103 ± 0.005	0.011 ± 0.001	0.103 ± 0.006	0.410 ± 0.040	0.641 ± 0.037	0.156	-0.004	0.0998	0.0999	1.087	97.288 ± 0.009	0.437 ± 0.011	—
	beta54	0.16	0.157	-0.003 ± 0.007	-0.021 ± 0.042	0.013 ± 0.001	0.115 ± 0.005	0.013 ± 0.001	0.115 ± 0.006	0.514 ± 0.047	0.717 ± 0.039	0.161	0.001	0.1184	0.1184	0.908	93.898 ± 0.014	0.409 ± 0.005	—
	beta55	0.00	-0.011	-0.011 ± 0.008	—	0.021 ± 0.002	0.144 ± 0.009	0.021 ± 0.002	0.144 ± 0.010	—	—	-0.004	-0.004	0.1441	0.1441	1.135	97.966 ± 0.008	0.642 ± 0.020	2.034 ± 0.008
	beta56	0.00	0.001	0.001 ± 0.008	—	0.019 ± 0.002	0.137 ± 0.007	0.019 ± 0.002	0.137 ± 0.008	—	—	-0.006	-0.006	0.1329	0.1331	1.030	98.305 ± 0.008	0.554 ± 0.014	1.695 ± 0.008
	beta57	0.00	0.013	0.013 ± 0.012	—	0.039 ± 0.004	0.198 ± 0.010	0.039 ± 0.004	0.198 ± 0.011	—	—	-0.001	-0.001	0.1677	0.1677	1.106	98.983 ± 0.006	0.859 ± 0.036	1.017 ± 0.006
	beta58	0.00	0.002	0.002 ± 0.010	—	0.032 ± 0.003	0.179 ± 0.010	0.032 ± 0.003	0.179 ± 0.011	—	—	0.003	0.003	0.1444	0.1444	1.040	99.322 ± 0.005	0.729 ± 0.030	0.678 ± 0.005
QML	beta41	0.20	0.199	-0.001 ± 0.005	-0.007 ± 0.023	0.006 ± 0.001	0.079 ± 0.003	0.006 ± 0.001	0.079 ± 0.004	0.157 ± 0.013	0.396 ± 0.020	0.198	-0.002	0.0809	0.0809	1.017	95.932 ± 0.012	0.316 ± 0.003	—
	beta42	0.20	0.207	0.007 ± 0.005	0.036 ± 0.025	0.007 ± 0.001	0.086 ± 0.004	0.007 ± 0.001	0.087 ± 0.004	0.187 ± 0.015	0.433 ± 0.022	0.207	0.007	0.0967	0.0970	0.982	95.254 ± 0.012	0.332 ± 0.003	—
	beta43	0.20	0.197	-0.003 ± 0.005	-0.017 ± 0.023	0.006 ± 0.001	0.081 ± 0.004	0.006 ± 0.001	0.080 ± 0.004	0.162 ± 0.014	0.402 ± 0.021	0.198	-0.002	0.0770	0.0770	0.983	94.576 ± 0.013	0.310 ± 0.003	—
	beta44	0.00	-0.011	-0.011 ± 0.007	—	0.013 ± 0.001	0.115 ± 0.005	0.013 ± 0.001	0.115 ± 0.006	—	—	-0.013	-0.013	0.1105	0.1113	0.993	97.966 ± 0.008	0.448 ± 0.006	2.034 ± 0.008
	beta45	0.00	-0.006	-0.006 ± 0.005	—	0.009 ± 0.001	0.094 ± 0.004	0.009 ± 0.001	0.094 ± 0.005	—	—	-0.011	-0.011	0.0985	0.0992	0.982	95.932 ± 0.012	0.362 ± 0.004	4.068 ± 0.012
	beta46	0.00	-0.005	-0.005 ± 0.005	—	0.008 ± 0.001	0.089 ± 0.004	0.008 ± 0.001	0.089 ± 0.005	—	—	-0.004	-0.004	0.0865	0.0866	0.937	94.237 ± 0.014	0.327 ± 0.005	5.763 ± 0.014
	beta47	0.00	-0.005	-0.005 ± 0.005	—	0.007 ± 0.001	0.082 ± 0.003	0.007 ± 0.001	0.082 ± 0.004	—	—	-0.003	-0.003	0.0818	0.0819	1.001	96.271 ± 0.011	0.323 ± 0.005	3.729 ± 0.011
	beta51	0.16	0.161	0.001 ± 0.005	0.003 ± 0.029	0.006 ± 0.001	0.080 ± 0.003	0.006 ± 0.001	0.080 ± 0.004	0.250 ± 0.020	0.500 ± 0.025	0.156	-0.004	0.0816	0.0817	1.109	96.271 ± 0.011	0.348 ± 0.004	—
	beta52	0.16	0.165	0.005 ± 0.005	0.034 ± 0.033	0.008 ± 0.001	0.090 ± 0.003	0.008 ± 0.001	0.090 ± 0.004	0.319 ± 0.024	0.565 ± 0.027	0.165	0.005	0.0912	0.0913	1.002	96.271 ± 0.011	0.355 ± 0.003	—
	beta53	0.16	0.162	0.002 ± 0.005	0.015 ± 0.032	0.008 ± 0.001	0.087 ± 0.004	0.008 ± 0.001	0.087 ± 0.005	0.295 ± 0.027	0.543 ± 0.030	0.158	-0.002	0.0845	0.0846	0.950	95.254 ± 0.012	0.324 ± 0.003	—
	beta54	0.16	0.158	-0.002 ± 0.006	-0.012 ± 0.039	0.012 ± 0.001	0.108 ± 0.005	0.012 ± 0.001	0.108 ± 0.006	0.457 ± 0.039	0.676 ± 0.035	0.170	0.010	0.1108	0.1112	0.929	94.576 ± 0.013	0.394 ± 0.004	—
	beta55	0.00	-0.007	-0.007 ± 0.007	—	0.015 ± 0.001	0.123 ± 0.006	0.015 ± 0.001	0.123 ± 0.007	—	—	-0.008	-0.008	0.1294	0.1296	1.013	96.949 ± 0.010	0.486 ± 0.007	3.051 ± 0.010
	beta56	0.00	-0.007	-0.007 ± 0.006	—	0.012 ± 0.001	0.107 ± 0.005	0.012 ± 0.001	0.107 ± 0.006	—	—	-0.010	-0.010	0.1047	0.1052	0.994	96.271 ± 0.011	0.418 ± 0.005	3.729 ± 0.011
	beta57	0.00	-0.001	-0.001 ± 0.005	—	0.008 ± 0.001	0.088 ± 0.005	0.008 ± 0.001	0.088 ± 0.006	—	—	-0.007	-0.007	0.0796	0.0799	0.954	94.915 ± 0.013	0.329 ± 0.006	5.085 ± 0.013
	beta58	0.00	-0.002	-0.002 ± 0.004	—	0.005 ± 0.000	0.072 ± 0.003	0.005 ± 0.000	0.072 ± 0.004	—	—	-0.005	-0.005	0.0717	0.0718	0.965	95.593 ± 0.012	0.273 ± 0.004	4.407 ± 0.012

Table S69

Simulation 2 - Condition 31: Linear Model, Distribution = uniform, N = 400, Reliability = 0.4 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta41	0.20	0.197	-0.003 ± 0.007	-0.016 ± 0.035	0.014 ± 0.002	0.119 ± 0.007	0.014 ± 0.002	0.119 ± 0.008	0.356 ± 0.041	0.596 ± 0.039	0.207	0.007	0.1263	0.1265	0.882	95.593 ± 0.012	0.413 ± 0.011	—
	beta42	0.20	0.225	0.025 ± 0.007	0.123 ± 0.035	0.015 ± 0.002	0.121 ± 0.007	0.015 ± 0.002	0.123 ± 0.008	0.380 ± 0.044	0.616 ± 0.040	0.211	0.011	0.1170	0.1175	0.909	94.237 ± 0.014	0.431 ± 0.011	—
	beta43	0.20	0.196	-0.004 ± 0.007	-0.019 ± 0.033	0.013 ± 0.001	0.113 ± 0.006	0.013 ± 0.001	0.113 ± 0.007	0.318 ± 0.034	0.564 ± 0.035	0.195	-0.005	0.1180	0.1181	0.906	93.898 ± 0.014	0.401 ± 0.013	—
	beta44	0.00	0.014	0.014 ± 0.014	—	0.055 ± 0.006	0.234 ± 0.014	0.055 ± 0.006	0.234 ± 0.016	—	—	-0.001	-0.001	0.2156	0.2156	0.721	94.576 ± 0.013	0.660 ± 0.030	5.424 ± 0.013
	beta45	0.00	0.002	0.002 ± 0.011	—	0.034 ± 0.004	0.185 ± 0.010	0.034 ± 0.004	0.185 ± 0.011	—	—	0.004	0.004	0.1677	0.1677	0.746	92.881 ± 0.015	0.542 ± 0.021	7.119 ± 0.015
	beta46	0.00	0.016	0.016 ± 0.024	—	0.176 ± 0.020	0.420 ± 0.025	0.176 ± 0.020	0.420 ± 0.027	—	—	0.004	0.004	0.3511	0.3511	0.637	93.559 ± 0.014	1.049 ± 0.070	6.441 ± 0.014
	beta47	0.00	-0.054	-0.054 ± 0.021	—	0.129 ± 0.016	0.360 ± 0.023	0.132 ± 0.017	0.363 ± 0.026	—	—	-0.011	-0.011	0.2937	0.2939	0.689	94.915 ± 0.013	0.970 ± 0.067	5.085 ± 0.013
	beta51	0.16	0.167	0.007 ± 0.005	0.046 ± 0.033	0.008 ± 0.001	0.091 ± 0.004	0.008 ± 0.001	0.091 ± 0.005	0.321 ± 0.027	0.567 ± 0.029	0.159	-0.001	0.1018	0.1018	1.046	96.610 ± 0.011	0.371 ± 0.005	—
	beta52	0.16	0.167	0.007 ± 0.006	0.044 ± 0.038	0.011 ± 0.001	0.104 ± 0.005	0.011 ± 0.001	0.104 ± 0.006	0.426 ± 0.042	0.653 ± 0.038	0.161	0.001	0.1046	0.1046	0.957	94.915 ± 0.013	0.391 ± 0.007	—
	beta53	0.16	0.164	0.004 ± 0.006	0.026 ± 0.037	0.010 ± 0.001	0.102 ± 0.005	0.010 ± 0.001	0.102 ± 0.006	0.408 ± 0.042	0.639 ± 0.038	0.162	0.002	0.1159	0.1159	0.923	95.254 ± 0.012	0.370 ± 0.007	—
	beta54	0.16	0.157	-0.003 ± 0.007	-0.019 ± 0.041	0.013 ± 0.001	0.114 ± 0.005	0.013 ± 0.001	0.113 ± 0.006	0.503 ± 0.041	0.709 ± 0.036	0.161	0.001	0.1284	0.1284	0.896	94.915 ± 0.013	0.399 ± 0.004	—
	beta55	0.00	0.001	0.001 ± 0.011	—	0.034 ± 0.005	0.184 ± 0.013	0.034 ± 0.005	0.184 ± 0.014	—	—	0.009	0.009	0.1670	0.1673	0.792	96.949 ± 0.010	0.572 ± 0.018	3.051 ± 0.010
	beta56	0.00	0.013	0.013 ± 0.010	—	0.029 ± 0.003	0.170 ± 0.010	0.029 ± 0.003	0.170 ± 0.011	—	—	-0.008	-0.008	0.1627	0.1629	0.810	96.610 ± 0.011	0.539 ± 0.017	3.390 ± 0.011
	beta57	0.00	0.010	0.010 ± 0.015	—	0.066 ± 0.009	0.256 ± 0.017	0.066 ± 0.009	0.256 ± 0.019	—	—	0.003	0.003	0.2328	0.2328	0.742	96.610 ± 0.011	0.747 ± 0.040	3.390 ± 0.011
	beta58	0.00	-0.001	-0.001 ± 0.013	—	0.047 ± 0.005	0.216 ± 0.012	0.047 ± 0.005	0.216 ± 0.014	—	—	0.002	0.002	0.1838	0.1838	0.913	95.593 ± 0.012	0.773 ± 0.075	4.407 ± 0.012

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S70

Simulation 2 - Condition 32: Linear Model, Distribution = uniform, N = 1000, Reliability = 0.4

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.002	-0.000 ± 0.010	0.003 ± 0.000	0.053 ± 0.002	0.003 ± 0.000	0.053 ± 0.002	0.071 ± 0.004	0.266 ± 0.009	0.200	0.000	0.0528	0.0528	1.105	97.089 ± 0.006	0.230 ± 0.002	—
	beta42	0.20	0.202	0.002 ± 0.002	0.010 ± 0.011	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.085 ± 0.005	0.292 ± 0.010	0.203	0.003	0.0612	0.0613	1.021	95.633 ± 0.008	0.233 ± 0.002	—
	beta43	0.20	0.202	0.002 ± 0.002	0.008 ± 0.010	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	0.075 ± 0.005	0.274 ± 0.010	0.201	0.001	0.0547	0.0547	0.998	95.197 ± 0.008	0.215 ± 0.001	—
	beta44	0.00	0.000	0.000 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.003	0.006 ± 0.000	0.080 ± 0.003	—	—	0.003	0.003	0.0738	0.0738	1.030	95.924 ± 0.008	0.323 ± 0.006	4.076 ± 0.008
	beta45	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	0.002	0.002	0.0663	0.0664	1.037	95.924 ± 0.008	0.264 ± 0.003	4.076 ± 0.008
	beta46	0.00	0.000	0.000 ± 0.004	—	0.011 ± 0.001	0.105 ± 0.003	0.011 ± 0.001	0.105 ± 0.004	—	—	-0.005	-0.005	0.0976	0.0977	1.043	97.380 ± 0.006	0.431 ± 0.007	2.620 ± 0.006
	beta47	0.00	0.000	-0.000 ± 0.004	—	0.012 ± 0.001	0.108 ± 0.003	0.012 ± 0.001	0.108 ± 0.004	—	—	0.001	0.001	0.1018	0.1018	1.035	98.253 ± 0.005	0.440 ± 0.009	1.747 ± 0.005
	beta51	0.16	0.162	0.002 ± 0.002	0.010 ± 0.014	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.131 ± 0.007	0.362 ± 0.012	0.162	0.002	0.0578	0.0579	0.988	95.197 ± 0.008	0.224 ± 0.001	—
	beta52	0.16	0.161	0.001 ± 0.002	0.006 ± 0.013	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	0.125 ± 0.008	0.353 ± 0.013	0.163	0.003	0.0580	0.0580	1.026	96.943 ± 0.007	0.227 ± 0.001	—
	beta53	0.16	0.161	0.001 ± 0.002	0.006 ± 0.013	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.111 ± 0.006	0.333 ± 0.011	0.159	-0.001	0.0498	0.0498	1.078	96.361 ± 0.007	0.225 ± 0.001	—
	beta54	0.16	0.159	-0.001 ± 0.002	-0.008 ± 0.015	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.154 ± 0.009	0.393 ± 0.013	0.156	-0.004	0.0662	0.0664	0.952	94.760 ± 0.009	0.235 ± 0.001	—
	beta55	0.00	0.001	0.001 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.003	0.007 ± 0.000	0.083 ± 0.003	—	—	-0.002	-0.002	0.0737	0.0737	0.985	95.779 ± 0.008	0.321 ± 0.004	4.221 ± 0.008
	beta56	0.00	0.000	0.000 ± 0.003	—	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	—	—	-0.002	-0.002	0.0729	0.0729	0.983	95.633 ± 0.008	0.293 ± 0.003	4.367 ± 0.008
	beta57	0.00	0.002	0.002 ± 0.004	—	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.098 ± 0.004	—	—	0.005	0.005	0.0908	0.0910	1.063	98.690 ± 0.004	0.407 ± 0.006	1.310 ± 0.004
	beta58	0.00	0.007	0.007 ± 0.004	—	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.098 ± 0.003	—	—	0.008	0.008	0.0953	0.0957	1.006	96.943 ± 0.007	0.386 ± 0.006	3.057 ± 0.007
QML	beta41	0.20	0.198	-0.002 ± 0.002	-0.008 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.002	0.062 ± 0.003	0.249 ± 0.008	0.197	-0.003	0.0509	0.0510	1.007	94.760 ± 0.009	0.197 ± 0.001	—
	beta42	0.20	0.202	0.002 ± 0.002	0.008 ± 0.010	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.071 ± 0.004	0.267 ± 0.009	0.204	0.004	0.0558	0.0559	0.978	95.342 ± 0.008	0.204 ± 0.001	—
	beta43	0.20	0.202	0.002 ± 0.002	0.008 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.068 ± 0.004	0.260 ± 0.009	0.200	0.000	0.0539	0.0539	0.962	94.032 ± 0.009	0.196 ± 0.001	—
	beta44	0.00	-0.008	-0.008 ± 0.003	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	-0.006	-0.006	0.0733	0.0736	0.984	95.051 ± 0.008	0.272 ± 0.001	4.949 ± 0.008
	beta45	0.00	-0.009	-0.009 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.006	-0.006	0.0546	0.0549	1.011	95.051 ± 0.008	0.226 ± 0.001	4.949 ± 0.008
	beta46	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	-0.004	-0.004	0.0504	0.0505	0.986	95.342 ± 0.008	0.199 ± 0.001	4.658 ± 0.008
	beta47	0.00	-0.005	-0.005 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.050 ± 0.002	—	—	-0.006	-0.006	0.0509	0.0513	1.007	95.924 ± 0.008	0.195 ± 0.001	4.076 ± 0.008
	beta51	0.16	0.161	0.001 ± 0.002	0.008 ± 0.013	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	0.122 ± 0.007	0.349 ± 0.012	0.161	0.001	0.0539	0.0539	0.942	93.450 ± 0.009	0.206 ± 0.001	—
	beta52	0.16	0.161	0.001 ± 0.002	0.004 ± 0.013	0.003 ± 0.000	0.053 ± 0.002	0.003 ± 0.000	0.053 ± 0.002	0.108 ± 0.006	0.329 ± 0.011	0.162	0.002	0.0541	0.0542	1.023	96.943 ± 0.007	0.211 ± 0.001	—
	beta53	0.16	0.161	0.001 ± 0.002	0.007 ± 0.012	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	0.100 ± 0.006	0.315 ± 0.011	0.158	-0.002	0.0498	0.0499	1.015	95.488 ± 0.008	0.201 ± 0.001	—
	beta54	0.16	0.160	0.000 ± 0.002	0.001 ± 0.015	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.148 ± 0.008	0.385 ± 0.013	0.159	-0.001	0.0640	0.0640	0.969	94.760 ± 0.009	0.234 ± 0.001	—
	beta55	0.00	0.000	-0.000 ± 0.003	—	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.075 ± 0.003	—	—	-0.004	-0.004	0.0676	0.0677	0.972	94.032 ± 0.009	0.288 ± 0.002	5.968 ± 0.009
	beta56	0.00	-0.006	-0.006 ± 0.003	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.008	-0.008	0.0657	0.0662	0.967	95.342 ± 0.008	0.251 ± 0.001	4.658 ± 0.008
	beta57	0.00	-0.003	-0.003 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	—	—	-0.003	-0.003	0.0489	0.0490	1.043	96.652 ± 0.007	0.198 ± 0.001	3.348 ± 0.007
	beta58	0.00	-0.002	-0.002 ± 0.002	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.002	-0.002	0.0460	0.0461	0.990	96.215 ± 0.007	0.173 ± 0.001	3.785 ± 0.007

Table S70

Simulation 2 - Condition 32: Linear Model, Distribution = uniform, N = 1000, Reliability = 0.4 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta41	0.20	0.199	-0.001 ± 0.002	-0.003 ± 0.011	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	0.081 ± 0.005	0.284 ± 0.010	0.201	0.001	0.0598	0.0598	0.970	95.197 ± 0.008	0.216 ± 0.001	—
	beta42	0.20	0.205	0.005 ± 0.002	0.025 ± 0.012	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.097 ± 0.006	0.311 ± 0.012	0.203	0.003	0.0644	0.0645	0.928	94.178 ± 0.009	0.226 ± 0.001	—
	beta43	0.20	0.202	0.002 ± 0.002	0.012 ± 0.011	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	0.081 ± 0.005	0.284 ± 0.010	0.202	0.002	0.0600	0.0600	0.948	93.886 ± 0.009	0.211 ± 0.001	—
	beta44	0.00	-0.001	-0.001 ± 0.003	—	0.008 ± 0.001	0.087 ± 0.003	0.008 ± 0.001	0.087 ± 0.003	—	—	-0.001	-0.001	0.0863	0.0863	0.877	94.760 ± 0.009	0.300 ± 0.004	5.240 ± 0.009
	beta45	0.00	0.001	0.001 ± 0.003	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.003	—	—	0.001	0.001	0.0716	0.0716	0.910	94.760 ± 0.009	0.256 ± 0.002	5.240 ± 0.009
	beta46	0.00	0.003	0.003 ± 0.005	—	0.019 ± 0.001	0.139 ± 0.005	0.019 ± 0.001	0.139 ± 0.006	—	—	-0.005	-0.005	0.1280	0.1281	0.765	95.488 ± 0.008	0.417 ± 0.009	4.512 ± 0.008
	beta47	0.00	-0.005	-0.005 ± 0.006	—	0.021 ± 0.002	0.144 ± 0.005	0.021 ± 0.002	0.144 ± 0.006	—	—	-0.002	-0.002	0.1253	0.1253	0.766	95.924 ± 0.008	0.433 ± 0.010	4.076 ± 0.008
	beta51	0.16	0.163	0.003 ± 0.002	0.020 ± 0.014	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	0.135 ± 0.007	0.367 ± 0.012	0.161	0.001	0.0637	0.0637	0.925	94.178 ± 0.009	0.213 ± 0.001	—
	beta52	0.16	0.162	0.002 ± 0.002	0.010 ± 0.013	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	0.124 ± 0.007	0.353 ± 0.012	0.166	0.006	0.0600	0.0603	1.002	95.779 ± 0.008	0.222 ± 0.001	—
	beta53	0.16	0.163	0.003 ± 0.002	0.016 ± 0.013	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	0.118 ± 0.007	0.343 ± 0.012	0.162	0.002	0.0548	0.0549	0.992	95.342 ± 0.008	0.213 ± 0.001	—
	beta54	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.015	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.151 ± 0.008	0.388 ± 0.013	0.156	-0.004	0.0624	0.0626	0.957	94.469 ± 0.009	0.233 ± 0.001	—
	beta55	0.00	0.000	0.000 ± 0.003	—	0.008 ± 0.001	0.087 ± 0.003	0.008 ± 0.001	0.087 ± 0.003	—	—	-0.002	-0.002	0.0797	0.0798	0.901	94.323 ± 0.009	0.309 ± 0.004	5.677 ± 0.009
	beta56	0.00	0.001	0.001 ± 0.003	—	0.007 ± 0.000	0.082 ± 0.003	0.007 ± 0.000	0.082 ± 0.003	—	—	0.003	0.003	0.0767	0.0767	0.897	94.469 ± 0.009	0.288 ± 0.003	5.531 ± 0.009
	beta57	0.00	0.001	0.001 ± 0.004	—	0.014 ± 0.001	0.117 ± 0.004	0.014 ± 0.001	0.116 ± 0.005	—	—	0.001	0.001	0.0985	0.0985	0.841	96.507 ± 0.007	0.384 ± 0.008	3.493 ± 0.007
	beta58	0.00	0.008	0.008 ± 0.005	—	0.015 ± 0.001	0.124 ± 0.005	0.015 ± 0.001	0.124 ± 0.005	—	—	0.006	0.006	0.0998	0.1000	0.790	96.070 ± 0.007	0.382 ± 0.011	3.930 ± 0.007

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S71

Simulation 2 - Condition 33: Linear Model, Distribution = uniform, N = 400, Reliability = 0.6

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.002	-0.001 ± 0.011	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.106 ± 0.005	0.326 ± 0.009	0.196	-0.004	0.0615	0.0616	0.995	94.322 ± 0.008	0.254 ± 0.001	—
	beta42	0.20	0.201	0.001 ± 0.002	0.005 ± 0.011	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.115 ± 0.006	0.340 ± 0.010	0.202	0.002	0.0695	0.0695	0.985	95.373 ± 0.007	0.262 ± 0.001	—
	beta43	0.20	0.197	-0.003 ± 0.002	-0.014 ± 0.010	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.097 ± 0.004	0.312 ± 0.009	0.194	-0.006	0.0616	0.0619	1.004	95.373 ± 0.007	0.245 ± 0.001	—
	beta44	0.00	0.001	0.001 ± 0.003	—	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	—	—	0.003	0.003	0.0812	0.0813	1.001	95.373 ± 0.007	0.309 ± 0.002	4.627 ± 0.007
	beta45	0.00	0.001	0.001 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	0.006	0.006	0.0674	0.0677	1.006	94.848 ± 0.007	0.269 ± 0.001	5.152 ± 0.007
	beta46	0.00	-0.002	-0.002 ± 0.003	—	0.011 ± 0.001	0.103 ± 0.002	0.011 ± 0.001	0.102 ± 0.003	—	—	-0.003	-0.003	0.0980	0.0981	0.975	95.584 ± 0.007	0.392 ± 0.003	4.416 ± 0.007
	beta47	0.00	0.000	-0.000 ± 0.003	—	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.099 ± 0.003	—	—	-0.004	-0.004	0.0940	0.0940	0.994	96.635 ± 0.006	0.386 ± 0.003	3.365 ± 0.006
	beta51	0.16	0.158	-0.002 ± 0.002	-0.010 ± 0.013	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.161 ± 0.008	0.401 ± 0.012	0.157	-0.003	0.0629	0.0630	1.004	95.899 ± 0.006	0.253 ± 0.001	—
	beta52	0.16	0.162	0.002 ± 0.002	0.015 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.161 ± 0.007	0.402 ± 0.011	0.162	0.002	0.0642	0.0643	1.031	96.215 ± 0.006	0.260 ± 0.001	—
	beta53	0.16	0.162	0.002 ± 0.002	0.014 ± 0.013	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.160 ± 0.008	0.400 ± 0.012	0.159	-0.001	0.0618	0.0619	1.008	95.163 ± 0.007	0.253 ± 0.001	—
	beta54	0.16	0.158	-0.002 ± 0.002	-0.015 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.197 ± 0.009	0.444 ± 0.012	0.162	0.002	0.0761	0.0761	0.970	94.217 ± 0.008	0.270 ± 0.001	—
	beta55	0.00	0.001	0.001 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	—	—	-0.001	-0.001	0.0793	0.0793	0.977	95.163 ± 0.007	0.318 ± 0.002	4.837 ± 0.007
	beta56	0.00	-0.002	-0.002 ± 0.003	—	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	—	—	-0.001	-0.001	0.0743	0.0743	0.985	94.953 ± 0.007	0.299 ± 0.002	5.047 ± 0.007
	beta57	0.00	-0.002	-0.002 ± 0.003	—	0.010 ± 0.001	0.102 ± 0.003	0.010 ± 0.001	0.102 ± 0.003	—	—	-0.002	-0.002	0.0998	0.0998	0.950	95.899 ± 0.006	0.379 ± 0.003	4.101 ± 0.006
	beta58	0.00	0.004	0.004 ± 0.003	—	0.008 ± 0.000	0.092 ± 0.003	0.008 ± 0.000	0.092 ± 0.003	—	—	0.008	0.008	0.0877	0.0881	0.988	96.740 ± 0.006	0.356 ± 0.003	3.260 ± 0.006
QML	beta41	0.20	0.200	-0.000 ± 0.002	-0.002 ± 0.010	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.098 ± 0.005	0.313 ± 0.009	0.197	-0.003	0.0601	0.0602	0.970	93.796 ± 0.008	0.238 ± 0.001	—
	beta42	0.20	0.201	0.001 ± 0.002	0.003 ± 0.010	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.103 ± 0.005	0.321 ± 0.009	0.201	0.001	0.0657	0.0657	0.979	95.268 ± 0.007	0.246 ± 0.001	—
	beta43	0.20	0.197	-0.003 ± 0.002	-0.013 ± 0.010	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.092 ± 0.004	0.303 ± 0.008	0.194	-0.006	0.0603	0.0606	0.994	95.163 ± 0.007	0.236 ± 0.001	—
	beta44	0.00	-0.003	-0.003 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	—	—	-0.001	-0.001	0.0770	0.0770	0.997	96.215 ± 0.006	0.293 ± 0.001	3.785 ± 0.006
	beta45	0.00	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	0.001	0.001	0.0637	0.0637	1.008	95.373 ± 0.007	0.256 ± 0.001	4.627 ± 0.007
	beta46	0.00	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.003	-0.003	0.0632	0.0633	0.982	95.268 ± 0.007	0.242 ± 0.001	4.732 ± 0.007
	beta47	0.00	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.003	-0.003	0.0593	0.0594	0.992	95.268 ± 0.007	0.238 ± 0.001	4.732 ± 0.007
	beta51	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.013	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.161 ± 0.008	0.401 ± 0.012	0.157	-0.003	0.0649	0.0650	0.982	95.478 ± 0.007	0.247 ± 0.001	—
	beta52	0.16	0.162	0.002 ± 0.002	0.011 ± 0.013	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.152 ± 0.007	0.390 ± 0.011	0.161	0.001	0.0654	0.0654	1.036	96.109 ± 0.006	0.254 ± 0.001	—
	beta53	0.16	0.162	0.002 ± 0.002	0.012 ± 0.013	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.154 ± 0.008	0.393 ± 0.012	0.160	0.000	0.0622	0.0622	0.980	94.111 ± 0.008	0.242 ± 0.001	—
	beta54	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.193 ± 0.008	0.439 ± 0.012	0.165	0.005	0.0745	0.0747	1.000	95.163 ± 0.007	0.276 ± 0.001	—
	beta55	0.00	0.000	-0.000 ± 0.003	—	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	—	—	-0.003	-0.003	0.0752	0.0752	1.003	96.004 ± 0.006	0.308 ± 0.001	3.996 ± 0.006
	beta56	0.00	-0.004	-0.004 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	-0.003	-0.003	0.0737	0.0738	1.003	95.163 ± 0.007	0.283 ± 0.002	4.837 ± 0.007
	beta57	0.00	-0.004	-0.004 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.005	-0.005	0.0641	0.0643	0.969	95.058 ± 0.007	0.238 ± 0.001	4.942 ± 0.007
	beta58	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	0.000	0.000	0.0563	0.0563	1.018	96.004 ± 0.006	0.217 ± 0.001	3.996 ± 0.006

Table S71

Simulation 2 - Condition 33: Linear Model, Distribution = uniform, N = 400, Reliability = 0.6 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta41	0.20	0.201	0.001 ± 0.002	0.006 ± 0.011	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.113 ± 0.005	0.336 ± 0.010	0.199	-0.001	0.0637	0.0637	0.942	92.955 ± 0.008	0.248 ± 0.001	—
	beta42	0.20	0.201	0.001 ± 0.002	0.006 ± 0.011	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.121 ± 0.006	0.348 ± 0.010	0.203	0.003	0.0709	0.0710	0.946	94.742 ± 0.007	0.258 ± 0.001	—
	beta43	0.20	0.199	-0.001 ± 0.002	-0.007 ± 0.010	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.099 ± 0.004	0.315 ± 0.009	0.196	-0.004	0.0652	0.0653	0.986	95.058 ± 0.007	0.244 ± 0.001	—
	beta44	0.00	0.001	0.001 ± 0.003	—	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	—	—	0.001	0.001	0.0835	0.0835	0.925	94.848 ± 0.007	0.305 ± 0.002	5.152 ± 0.007
	beta45	0.00	0.001	0.001 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	0.005	0.005	0.0727	0.0729	0.949	94.006 ± 0.008	0.269 ± 0.001	5.994 ± 0.008
	beta46	0.00	-0.002	-0.002 ± 0.004	—	0.015 ± 0.001	0.121 ± 0.003	0.015 ± 0.001	0.121 ± 0.004	—	—	-0.001	-0.001	0.1065	0.1066	0.801	93.165 ± 0.008	0.379 ± 0.004	6.835 ± 0.008
	beta47	0.00	0.002	0.002 ± 0.004	—	0.014 ± 0.001	0.118 ± 0.003	0.014 ± 0.001	0.118 ± 0.004	—	—	-0.001	-0.001	0.1013	0.1013	0.812	92.324 ± 0.009	0.374 ± 0.004	7.676 ± 0.009
	beta51	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.163 ± 0.008	0.404 ± 0.012	0.158	-0.002	0.0639	0.0639	0.979	95.794 ± 0.007	0.248 ± 0.001	—
	beta52	0.16	0.163	0.003 ± 0.002	0.020 ± 0.013	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.163 ± 0.008	0.404 ± 0.011	0.162	0.002	0.0656	0.0656	1.019	95.584 ± 0.007	0.258 ± 0.001	—
	beta53	0.16	0.163	0.003 ± 0.002	0.020 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.165 ± 0.008	0.406 ± 0.012	0.160	0.000	0.0641	0.0641	0.974	94.217 ± 0.008	0.248 ± 0.001	—
	beta54	0.16	0.158	-0.002 ± 0.002	-0.012 ± 0.014	0.005 ± 0.000	0.070 ± 0.001	0.005 ± 0.000	0.070 ± 0.002	0.191 ± 0.008	0.437 ± 0.012	0.163	0.003	0.0742	0.0743	0.990	94.427 ± 0.007	0.271 ± 0.001	—
	beta55	0.00	0.000	0.000 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	—	—	-0.003	-0.003	0.0810	0.0811	0.933	94.848 ± 0.007	0.313 ± 0.002	5.152 ± 0.007
	beta56	0.00	-0.001	-0.001 ± 0.003	—	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	—	—	0.001	0.001	0.0784	0.0784	0.957	94.427 ± 0.007	0.296 ± 0.002	5.573 ± 0.007
	beta57	0.00	0.000	0.000 ± 0.004	—	0.013 ± 0.001	0.116 ± 0.003	0.013 ± 0.001	0.116 ± 0.004	—	—	0.000	0.000	0.1087	0.1087	0.803	93.586 ± 0.008	0.365 ± 0.004	6.414 ± 0.008
	beta58	0.00	0.003	0.003 ± 0.003	—	0.011 ± 0.001	0.106 ± 0.003	0.011 ± 0.001	0.106 ± 0.004	—	—	0.003	0.003	0.0907	0.0908	0.855	94.742 ± 0.007	0.354 ± 0.005	5.258 ± 0.007

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S72

Simulation 2 - Condition 34: Linear Model, Distribution = uniform, N = 1000, Reliability = 0.6

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.202	0.002 ± 0.001	0.008 ± 0.006	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.038 ± 0.002	0.195 ± 0.006	0.203	0.003	0.0385	0.0386	1.014	95.335 ± 0.007	0.155 ± 0.000	—
	beta42	0.20	0.201	0.001 ± 0.001	0.004 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.040 ± 0.002	0.201 ± 0.006	0.200	0.000	0.0408	0.0408	1.011	95.335 ± 0.007	0.159 ± 0.000	—
	beta43	0.20	0.203	0.003 ± 0.001	0.013 ± 0.006	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.037 ± 0.002	0.193 ± 0.005	0.203	0.003	0.0395	0.0395	1.011	95.538 ± 0.007	0.153 ± 0.000	—
	beta44	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.000	0.000	0.0470	0.0470	0.996	94.118 ± 0.007	0.186 ± 0.001	5.882 ± 0.007
	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	-0.002	-0.002	0.0412	0.0412	0.977	94.726 ± 0.007	0.165 ± 0.001	5.274 ± 0.007
	beta46	0.00	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	0.002	0.002	0.0610	0.0610	0.999	95.842 ± 0.006	0.234 ± 0.001	4.158 ± 0.006
	beta47	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	0.001	0.001	0.0568	0.0568	1.029	95.943 ± 0.006	0.232 ± 0.001	4.057 ± 0.006
	beta51	0.16	0.162	0.002 ± 0.001	0.014 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.065 ± 0.003	0.255 ± 0.007	0.162	0.002	0.0402	0.0403	0.971	94.523 ± 0.007	0.155 ± 0.000	—
	beta52	0.16	0.159	-0.001 ± 0.001	-0.008 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.068 ± 0.003	0.260 ± 0.007	0.158	-0.002	0.0410	0.0410	0.986	94.320 ± 0.007	0.161 ± 0.000	—
	beta53	0.16	0.160	0.000 ± 0.001	0.000 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.065 ± 0.003	0.254 ± 0.007	0.160	0.000	0.0412	0.0412	0.972	94.726 ± 0.007	0.155 ± 0.000	—
	beta54	0.16	0.159	-0.001 ± 0.001	-0.007 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.076 ± 0.003	0.277 ± 0.008	0.158	-0.002	0.0431	0.0431	0.976	94.422 ± 0.007	0.169 ± 0.000	—
	beta55	0.00	0.002	0.002 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	0.000	0.000	0.0497	0.0497	1.006	94.929 ± 0.007	0.192 ± 0.001	5.071 ± 0.007
	beta56	0.00	0.000	-0.000 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.001	0.001	0.0476	0.0476	0.986	94.422 ± 0.007	0.183 ± 0.001	5.578 ± 0.007
	beta57	0.00	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.001	-0.001	0.0535	0.0536	0.991	94.828 ± 0.007	0.227 ± 0.001	5.172 ± 0.007
	beta58	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	-0.004	-0.004	0.0575	0.0577	0.955	95.030 ± 0.007	0.214 ± 0.001	4.970 ± 0.007
QML	beta41	0.20	0.201	0.001 ± 0.001	0.005 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.036 ± 0.002	0.189 ± 0.005	0.202	0.002	0.0384	0.0384	1.010	95.436 ± 0.007	0.149 ± 0.000	—
	beta42	0.20	0.200	0.000 ± 0.001	0.001 ± 0.006	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.038 ± 0.002	0.195 ± 0.006	0.199	-0.001	0.0390	0.0390	1.007	95.639 ± 0.007	0.154 ± 0.000	—
	beta43	0.20	0.202	0.002 ± 0.001	0.008 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.036 ± 0.002	0.189 ± 0.005	0.202	0.002	0.0391	0.0391	1.001	95.639 ± 0.007	0.149 ± 0.000	—
	beta44	0.00	-0.005	-0.005 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.004	-0.004	0.0461	0.0463	0.996	94.016 ± 0.008	0.184 ± 0.001	5.984 ± 0.008
	beta45	0.00	-0.005	-0.005 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.006	-0.006	0.0421	0.0424	0.979	94.422 ± 0.007	0.161 ± 0.000	5.578 ± 0.007
	beta46	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.001	0.001	0.0404	0.0404	1.011	96.247 ± 0.006	0.152 ± 0.000	3.753 ± 0.006
	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.003	-0.003	0.0360	0.0362	1.032	95.639 ± 0.007	0.149 ± 0.000	4.361 ± 0.007
	beta51	0.16	0.162	0.002 ± 0.001	0.015 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.163	0.003	0.0405	0.0406	0.967	94.625 ± 0.007	0.154 ± 0.000	—
	beta52	0.16	0.159	-0.001 ± 0.001	-0.009 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.066 ± 0.003	0.258 ± 0.007	0.157	-0.003	0.0400	0.0401	0.977	94.016 ± 0.008	0.158 ± 0.000	—
	beta53	0.16	0.160	-0.000 ± 0.001	-0.001 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.063 ± 0.003	0.250 ± 0.007	0.160	0.000	0.0397	0.0397	0.967	94.219 ± 0.007	0.152 ± 0.000	—
	beta54	0.16	0.160	-0.000 ± 0.001	-0.002 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.077 ± 0.003	0.277 ± 0.008	0.159	-0.001	0.0436	0.0436	0.989	95.233 ± 0.007	0.172 ± 0.000	—
	beta55	0.00	0.001	0.001 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.001	0.001	0.0485	0.0485	1.011	95.639 ± 0.007	0.191 ± 0.000	4.361 ± 0.007
	beta56	0.00	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.002	-0.002	0.0447	0.0448	0.986	94.422 ± 0.007	0.176 ± 0.000	5.578 ± 0.007
	beta57	0.00	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	-0.004	-0.004	0.0350	0.0353	0.996	94.523 ± 0.007	0.150 ± 0.000	5.477 ± 0.007
	beta58	0.00	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.005	-0.005	0.0364	0.0367	0.960	94.726 ± 0.007	0.137 ± 0.000	5.274 ± 0.007

Table S72

Simulation 2 - Condition 34: Linear Model, Distribution = uniform, N = 1000, Reliability = 0.6 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta41	0.20	0.202	0.002 ± 0.001	0.008 ± 0.006	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.038 ± 0.002	0.195 ± 0.006	0.202	0.002	0.0380	0.0381	1.004	94.625 ± 0.007	0.154 ± 0.000	—
	beta42	0.20	0.201	0.001 ± 0.001	0.005 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.041 ± 0.002	0.202 ± 0.006	0.200	0.000	0.0402	0.0402	1.002	95.233 ± 0.007	0.159 ± 0.000	—
	beta43	0.20	0.203	0.003 ± 0.001	0.014 ± 0.006	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.037 ± 0.002	0.193 ± 0.005	0.203	0.003	0.0395	0.0396	1.003	95.436 ± 0.007	0.152 ± 0.000	—
	beta44	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.000	0.000	0.0457	0.0457	0.980	93.611 ± 0.008	0.184 ± 0.001	6.389 ± 0.008
	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	-0.002	-0.002	0.0425	0.0426	0.956	94.320 ± 0.007	0.164 ± 0.001	5.680 ± 0.007
	beta46	0.00	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	0.003	0.003	0.0622	0.0623	0.932	94.422 ± 0.007	0.230 ± 0.001	5.578 ± 0.007
	beta47	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	0.000	0.000	0.0572	0.0572	0.959	95.233 ± 0.007	0.227 ± 0.001	4.767 ± 0.007
	beta51	0.16	0.163	0.003 ± 0.001	0.016 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.065 ± 0.003	0.255 ± 0.007	0.162	0.002	0.0402	0.0402	0.966	94.726 ± 0.007	0.154 ± 0.000	—
	beta52	0.16	0.159	-0.001 ± 0.001	-0.007 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.069 ± 0.003	0.263 ± 0.007	0.158	-0.002	0.0418	0.0419	0.975	94.422 ± 0.007	0.161 ± 0.000	—
	beta53	0.16	0.160	0.000 ± 0.001	0.001 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.065 ± 0.003	0.254 ± 0.007	0.160	0.000	0.0402	0.0402	0.964	94.219 ± 0.007	0.154 ± 0.000	—
	beta54	0.16	0.159	-0.001 ± 0.001	-0.005 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.076 ± 0.003	0.276 ± 0.008	0.158	-0.002	0.0425	0.0425	0.986	95.132 ± 0.007	0.171 ± 0.000	—
	beta55	0.00	0.002	0.002 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	—	—	0.001	0.001	0.0492	0.0493	0.985	95.538 ± 0.007	0.192 ± 0.001	4.462 ± 0.007
	beta56	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.002	0.002	0.0478	0.0478	0.968	94.726 ± 0.007	0.183 ± 0.001	5.274 ± 0.007
	beta57	0.00	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.002	-0.002	0.0533	0.0533	0.936	94.219 ± 0.007	0.223 ± 0.001	5.781 ± 0.007
	beta58	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.059 ± 0.002	—	—	0.000	0.000	0.0589	0.0589	0.907	94.929 ± 0.007	0.211 ± 0.001	5.071 ± 0.007

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S73

Simulation 2 - Condition 35: Linear Model, Distribution = uniform, N = 400, Reliability = 0.8

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.199	-0.001 ± 0.002	-0.007 ± 0.008	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.069 ± 0.003	0.263 ± 0.007	0.199	-0.001	0.0523	0.0523	0.979	94.165 ± 0.007	0.202 ± 0.000	—
	beta42	0.20	0.201	0.001 ± 0.002	0.004 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.072 ± 0.003	0.269 ± 0.007	0.201	0.001	0.0520	0.0520	0.992	95.473 ± 0.007	0.209 ± 0.001	—
	beta43	0.20	0.201	0.001 ± 0.002	0.006 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.067 ± 0.003	0.259 ± 0.007	0.203	0.003	0.0515	0.0516	0.991	94.869 ± 0.007	0.201 ± 0.000	—
	beta44	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	0.001	0.001	0.0631	0.0631	0.911	92.254 ± 0.008	0.226 ± 0.001	7.746 ± 0.008
	beta45	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.003	-0.003	0.0529	0.0530	0.952	93.260 ± 0.008	0.205 ± 0.001	6.740 ± 0.008
	beta46	0.00	-0.005	-0.005 ± 0.002	—	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	-0.007	-0.007	0.0666	0.0670	0.976	94.266 ± 0.007	0.258 ± 0.001	5.734 ± 0.007
	beta47	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	—	—	0.000	0.000	0.0685	0.0685	1.015	95.171 ± 0.007	0.260 ± 0.001	4.829 ± 0.007
	beta51	0.16	0.162	0.002 ± 0.002	0.013 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.112 ± 0.005	0.335 ± 0.009	0.163	0.003	0.0539	0.0540	0.969	94.567 ± 0.007	0.203 ± 0.000	—
	beta52	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.116 ± 0.005	0.341 ± 0.009	0.160	0.000	0.0536	0.0536	0.984	95.171 ± 0.007	0.211 ± 0.001	—
	beta53	0.16	0.161	0.001 ± 0.002	0.009 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.107 ± 0.005	0.327 ± 0.009	0.163	0.003	0.0536	0.0537	0.996	95.372 ± 0.007	0.205 ± 0.000	—
	beta54	0.16	0.161	0.001 ± 0.002	0.007 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.135 ± 0.006	0.368 ± 0.010	0.160	0.000	0.0602	0.0602	0.964	94.467 ± 0.007	0.222 ± 0.001	—
	beta55	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.003	-0.003	0.0628	0.0628	0.964	94.165 ± 0.007	0.235 ± 0.001	5.835 ± 0.007
	beta56	0.00	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.003	-0.003	0.0582	0.0583	0.929	93.360 ± 0.008	0.225 ± 0.001	6.640 ± 0.008
	beta57	0.00	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.002	-0.002	0.0665	0.0665	0.980	94.668 ± 0.007	0.254 ± 0.001	5.332 ± 0.007
	beta58	0.00	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	0.003	0.003	0.0589	0.0590	1.013	95.573 ± 0.007	0.244 ± 0.001	4.427 ± 0.007
QML	beta41	0.20	0.198	-0.002 ± 0.002	-0.009 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.067 ± 0.003	0.258 ± 0.007	0.199	-0.001	0.0517	0.0518	0.979	94.567 ± 0.007	0.198 ± 0.000	—
	beta42	0.20	0.201	0.001 ± 0.002	0.003 ± 0.008	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.070 ± 0.003	0.264 ± 0.007	0.201	0.001	0.0513	0.0514	0.987	95.272 ± 0.007	0.205 ± 0.000	—
	beta43	0.20	0.201	0.001 ± 0.002	0.007 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.066 ± 0.003	0.256 ± 0.007	0.203	0.003	0.0514	0.0515	0.981	94.668 ± 0.007	0.197 ± 0.000	—
	beta44	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.001	-0.001	0.0625	0.0626	0.922	93.159 ± 0.008	0.227 ± 0.001	6.841 ± 0.008
	beta45	0.00	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	-0.004	-0.004	0.0528	0.0530	0.959	93.461 ± 0.008	0.204 ± 0.000	6.539 ± 0.008
	beta46	0.00	-0.004	-0.004 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.006	-0.006	0.0540	0.0543	0.980	94.366 ± 0.007	0.209 ± 0.001	5.634 ± 0.007
	beta47	0.00	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	-0.001	-0.001	0.0550	0.0550	1.017	95.573 ± 0.007	0.208 ± 0.001	4.427 ± 0.007
	beta51	0.16	0.162	0.002 ± 0.002	0.014 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.112 ± 0.005	0.334 ± 0.009	0.163	0.003	0.0550	0.0551	0.976	95.171 ± 0.007	0.205 ± 0.000	—
	beta52	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.115 ± 0.005	0.339 ± 0.009	0.160	0.000	0.0550	0.0550	0.988	95.272 ± 0.007	0.210 ± 0.000	—
	beta53	0.16	0.161	0.001 ± 0.002	0.009 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.106 ± 0.005	0.325 ± 0.009	0.163	0.003	0.0533	0.0534	0.993	95.573 ± 0.007	0.202 ± 0.000	—
	beta54	0.16	0.162	0.002 ± 0.002	0.011 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.135 ± 0.006	0.368 ± 0.010	0.161	0.001	0.0597	0.0597	0.979	94.970 ± 0.007	0.226 ± 0.001	—
	beta55	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.002	-0.002	0.0632	0.0632	0.980	95.171 ± 0.007	0.237 ± 0.001	4.829 ± 0.007
	beta56	0.00	-0.004	-0.004 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.005	-0.005	0.0585	0.0587	0.943	94.064 ± 0.007	0.223 ± 0.001	5.936 ± 0.007
	beta57	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	—	—	-0.003	-0.003	0.0542	0.0543	0.980	95.171 ± 0.007	0.206 ± 0.001	4.829 ± 0.007
	beta58	0.00	0.000	0.000 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	0.001	0.001	0.0487	0.0487	1.007	95.171 ± 0.007	0.195 ± 0.001	4.829 ± 0.007

Table S73

Simulation 2 - Condition 35: Linear Model, Distribution = uniform, N = 400, Reliability = 0.8 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta41	0.20	0.199	-0.001 ± 0.002	-0.006 ± 0.008	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.069 ± 0.003	0.263 ± 0.007	0.199	-0.001	0.0525	0.0525	0.967	94.165 ± 0.007	0.200 ± 0.000	—
	beta42	0.20	0.201	0.001 ± 0.002	0.005 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.073 ± 0.003	0.269 ± 0.007	0.201	0.001	0.0516	0.0516	0.978	95.372 ± 0.007	0.207 ± 0.000	—
	beta43	0.20	0.202	0.002 ± 0.002	0.008 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.068 ± 0.003	0.261 ± 0.007	0.203	0.003	0.0520	0.0521	0.973	94.266 ± 0.007	0.199 ± 0.000	—
	beta44	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	—	—	0.002	0.002	0.0644	0.0644	0.903	92.857 ± 0.008	0.226 ± 0.001	7.143 ± 0.008
	beta45	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.002	-0.002	0.0549	0.0549	0.944	93.763 ± 0.008	0.205 ± 0.001	6.237 ± 0.008
	beta46	0.00	-0.005	-0.005 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	-0.007	-0.007	0.0686	0.0689	0.941	93.058 ± 0.008	0.254 ± 0.001	6.942 ± 0.008
	beta47	0.00	-0.002	-0.002 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.001	0.005 ± 0.000	0.068 ± 0.002	—	—	-0.001	-0.001	0.0688	0.0688	0.964	94.064 ± 0.007	0.255 ± 0.001	5.936 ± 0.007
	beta51	0.16	0.162	0.002 ± 0.002	0.015 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.112 ± 0.005	0.335 ± 0.009	0.164	0.004	0.0539	0.0540	0.964	94.869 ± 0.007	0.202 ± 0.000	—
	beta52	0.16	0.160	-0.000 ± 0.002	-0.002 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.117 ± 0.005	0.342 ± 0.009	0.160	0.000	0.0540	0.0540	0.977	94.668 ± 0.007	0.210 ± 0.000	—
	beta53	0.16	0.162	0.002 ± 0.002	0.010 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.108 ± 0.005	0.328 ± 0.009	0.164	0.004	0.0536	0.0537	0.987	94.869 ± 0.007	0.203 ± 0.000	—
	beta54	0.16	0.161	0.001 ± 0.002	0.008 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.134 ± 0.006	0.366 ± 0.010	0.161	0.001	0.0598	0.0599	0.971	94.668 ± 0.007	0.223 ± 0.000	—
	beta55	0.00	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.003	-0.003	0.0660	0.0660	0.955	94.668 ± 0.007	0.236 ± 0.001	5.332 ± 0.007
	beta56	0.00	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.004	-0.004	0.0584	0.0586	0.928	93.360 ± 0.008	0.225 ± 0.001	6.640 ± 0.008
	beta57	0.00	0.002	0.002 ± 0.002	—	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	—	—	-0.002	-0.002	0.0670	0.0670	0.943	94.668 ± 0.007	0.249 ± 0.001	5.332 ± 0.007
	beta58	0.00	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	0.001	0.001	0.0603	0.0603	0.981	94.567 ± 0.007	0.240 ± 0.001	5.433 ± 0.007

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S74

Simulation 2 - Condition 36: Linear Model, Distribution = uniform, N = 1000, Reliability = 0.8

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.199	-0.001 ± 0.001	-0.007 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.155 ± 0.004	0.199	-0.001	0.0308	0.0308	1.043	95.753 ± 0.006	0.127 ± 0.000	—
	beta42	0.20	0.199	-0.001 ± 0.001	-0.003 ± 0.005	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.029 ± 0.001	0.171 ± 0.005	0.200	0.000	0.0335	0.0335	0.975	94.237 ± 0.007	0.130 ± 0.000	—
	beta43	0.20	0.201	0.001 ± 0.001	0.004 ± 0.005	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.027 ± 0.001	0.163 ± 0.005	0.201	0.001	0.0318	0.0318	0.988	95.652 ± 0.006	0.126 ± 0.000	—
	beta44	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.000	0.000	0.0348	0.0348	1.013	94.641 ± 0.007	0.141 ± 0.000	5.359 ± 0.007
	beta45	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.001	0.001	0.0336	0.0336	0.975	93.832 ± 0.008	0.128 ± 0.000	6.168 ± 0.008
	beta46	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.001	0.001	0.0418	0.0418	1.018	95.147 ± 0.007	0.162 ± 0.000	4.853 ± 0.007
	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.001	-0.001	0.0433	0.0433	0.991	94.540 ± 0.007	0.162 ± 0.000	5.460 ± 0.007
	beta51	0.16	0.161	0.001 ± 0.001	0.009 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.041 ± 0.002	0.202 ± 0.005	0.162	0.002	0.0333	0.0333	1.010	95.349 ± 0.007	0.128 ± 0.000	—
	beta52	0.16	0.159	-0.001 ± 0.001	-0.006 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.043 ± 0.002	0.208 ± 0.006	0.159	-0.001	0.0334	0.0334	1.016	96.158 ± 0.006	0.132 ± 0.000	—
	beta53	0.16	0.159	-0.001 ± 0.001	-0.006 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.043 ± 0.002	0.207 ± 0.006	0.159	-0.001	0.0329	0.0330	0.988	94.944 ± 0.007	0.128 ± 0.000	—
	beta54	0.16	0.160	0.000 ± 0.001	0.000 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.050 ± 0.002	0.225 ± 0.006	0.160	0.000	0.0350	0.0350	1.002	95.450 ± 0.007	0.141 ± 0.000	—
	beta55	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.001	0.001	0.0373	0.0374	1.041	95.652 ± 0.006	0.148 ± 0.000	4.348 ± 0.006
	beta56	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.000	0.000	0.0360	0.0360	1.001	95.551 ± 0.007	0.142 ± 0.000	4.449 ± 0.007
	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.003	-0.003	0.0388	0.0389	1.032	94.742 ± 0.007	0.160 ± 0.000	5.258 ± 0.007
	beta58	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.000	0.000	0.0381	0.0381	0.983	94.439 ± 0.007	0.152 ± 0.000	5.561 ± 0.007
QML	beta41	0.20	0.198	-0.002 ± 0.001	-0.008 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.154 ± 0.004	0.199	-0.001	0.0304	0.0304	1.036	96.057 ± 0.006	0.125 ± 0.000	—
	beta42	0.20	0.199	-0.001 ± 0.001	-0.004 ± 0.005	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.029 ± 0.001	0.169 ± 0.005	0.200	0.000	0.0339	0.0339	0.968	94.338 ± 0.007	0.128 ± 0.000	—
	beta43	0.20	0.201	0.001 ± 0.001	0.003 ± 0.005	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.026 ± 0.001	0.161 ± 0.005	0.200	0.000	0.0320	0.0320	0.981	95.551 ± 0.007	0.124 ± 0.000	—
	beta44	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.002	-0.002	0.0351	0.0351	1.016	95.450 ± 0.007	0.142 ± 0.000	4.550 ± 0.007
	beta45	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.001	-0.001	0.0333	0.0333	0.976	93.529 ± 0.008	0.128 ± 0.000	6.471 ± 0.008
	beta46	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.001	-0.001	0.0342	0.0342	1.025	95.248 ± 0.007	0.132 ± 0.001	4.752 ± 0.007
	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.002	-0.002	0.0350	0.0351	0.989	94.237 ± 0.007	0.130 ± 0.000	5.763 ± 0.007
	beta51	0.16	0.161	0.001 ± 0.001	0.009 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.041 ± 0.002	0.202 ± 0.005	0.162	0.002	0.0328	0.0329	1.011	95.349 ± 0.007	0.128 ± 0.000	—
	beta52	0.16	0.159	-0.001 ± 0.001	-0.007 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.043 ± 0.002	0.207 ± 0.006	0.159	-0.001	0.0332	0.0332	1.011	96.158 ± 0.006	0.132 ± 0.000	—
	beta53	0.16	0.159	-0.001 ± 0.001	-0.007 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.043 ± 0.002	0.206 ± 0.006	0.158	-0.002	0.0324	0.0324	0.983	95.147 ± 0.007	0.127 ± 0.000	—
	beta54	0.16	0.160	0.000 ± 0.001	0.002 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.050 ± 0.002	0.225 ± 0.006	0.160	0.000	0.0353	0.0353	1.008	95.248 ± 0.007	0.142 ± 0.000	—
	beta55	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.001	0.001	0.0371	0.0371	1.048	96.158 ± 0.006	0.148 ± 0.000	3.842 ± 0.006
	beta56	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.002	-0.002	0.0351	0.0351	1.002	95.652 ± 0.006	0.140 ± 0.000	4.348 ± 0.006
	beta57	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.003	-0.003	0.0312	0.0314	1.039	95.349 ± 0.007	0.130 ± 0.000	4.651 ± 0.007
	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.001	-0.001	0.0313	0.0313	0.988	94.641 ± 0.007	0.122 ± 0.000	5.359 ± 0.007

Table S74

Simulation 2 - Condition 36: Linear Model, Distribution = uniform, N = 1000, Reliability = 0.8 (continued)

			Mean-based									Median-based				Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias (Med)	MAD	RMSE (Med)	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta41	0.20	0.199	-0.001 ± 0.001	-0.006 ± 0.005	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.024 ± 0.001	0.155 ± 0.004	0.199	-0.001	0.0311	0.0311	1.033	95.551 ± 0.007	0.126 ± 0.000	—
	beta42	0.20	0.199	-0.001 ± 0.001	-0.003 ± 0.005	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.029 ± 0.001	0.171 ± 0.005	0.200	0.000	0.0335	0.0335	0.965	93.933 ± 0.008	0.129 ± 0.000	—
	beta43	0.20	0.201	0.001 ± 0.001	0.005 ± 0.005	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.027 ± 0.001	0.163 ± 0.005	0.201	0.001	0.0324	0.0325	0.977	95.450 ± 0.007	0.125 ± 0.000	—
	beta44	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.001	-0.001	0.0357	0.0357	1.004	94.944 ± 0.007	0.141 ± 0.000	5.056 ± 0.007
	beta45	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	0.001	0.001	0.0338	0.0338	0.974	94.237 ± 0.007	0.128 ± 0.000	5.763 ± 0.007
	beta46	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.001	0.001	0.0413	0.0413	1.000	94.944 ± 0.007	0.161 ± 0.000	5.056 ± 0.007
	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.000	0.000	0.0428	0.0428	0.971	94.034 ± 0.008	0.160 ± 0.000	5.966 ± 0.008
	beta51	0.16	0.162	0.002 ± 0.001	0.010 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.041 ± 0.002	0.202 ± 0.005	0.162	0.002	0.0332	0.0333	1.009	95.551 ± 0.007	0.128 ± 0.000	—
	beta52	0.16	0.159	-0.001 ± 0.001	-0.005 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.043 ± 0.002	0.208 ± 0.006	0.159	-0.001	0.0332	0.0332	1.013	95.956 ± 0.006	0.132 ± 0.000	—
	beta53	0.16	0.159	-0.001 ± 0.001	-0.005 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.043 ± 0.002	0.207 ± 0.006	0.159	-0.001	0.0329	0.0329	0.981	94.944 ± 0.007	0.127 ± 0.000	—
	beta54	0.16	0.160	0.000 ± 0.001	0.001 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.050 ± 0.002	0.224 ± 0.006	0.160	0.000	0.0352	0.0352	1.006	95.248 ± 0.007	0.141 ± 0.000	—
	beta55	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.001	0.001	0.0374	0.0375	1.042	95.854 ± 0.006	0.148 ± 0.000	4.146 ± 0.006
	beta56	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.000	0.000	0.0365	0.0365	0.995	95.551 ± 0.007	0.142 ± 0.000	4.449 ± 0.007
	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.003	-0.003	0.0385	0.0386	1.012	94.641 ± 0.007	0.158 ± 0.000	5.359 ± 0.007
	beta58	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.000	0.000	0.0387	0.0387	0.966	95.046 ± 0.007	0.151 ± 0.000	4.954 ± 0.007

Note. MCSE = Monte Carlo Standard Error shown as ± value. Coverage, Type I Error, and Power expressed as percentages. — indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

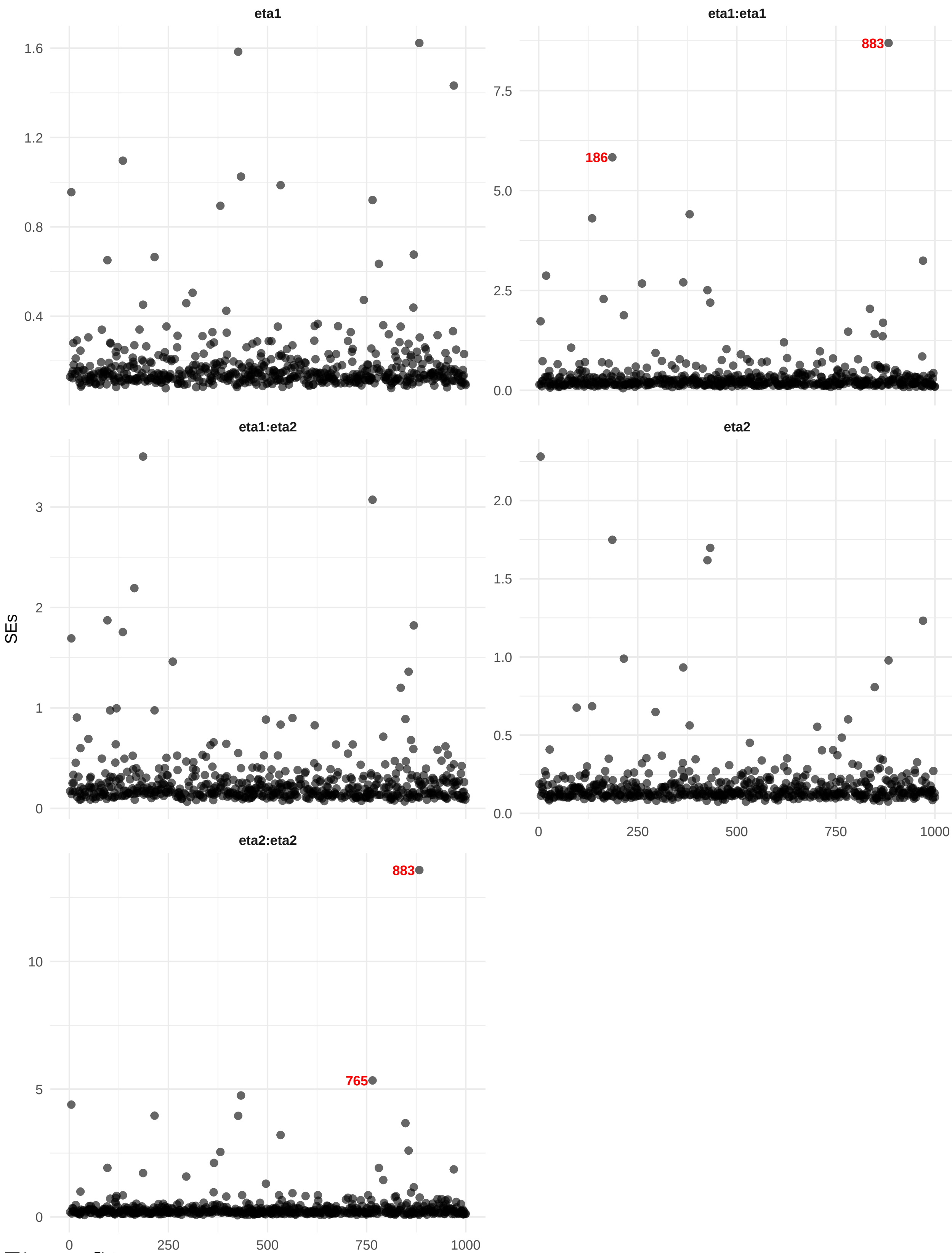

Figure S1

Outliers for the SEs for LSAM in condition 13

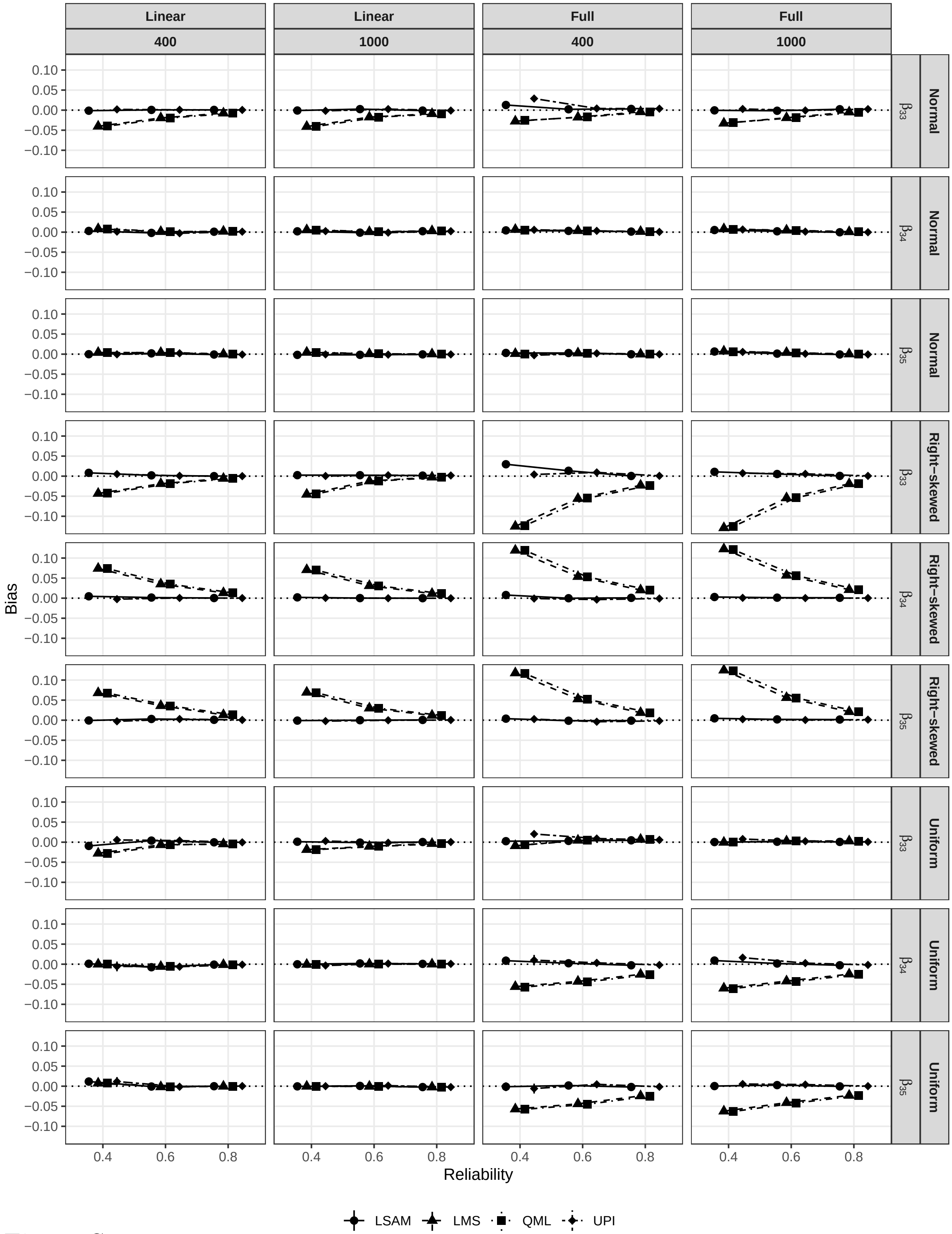

Figure S2

Absolute bias for Simulation 1

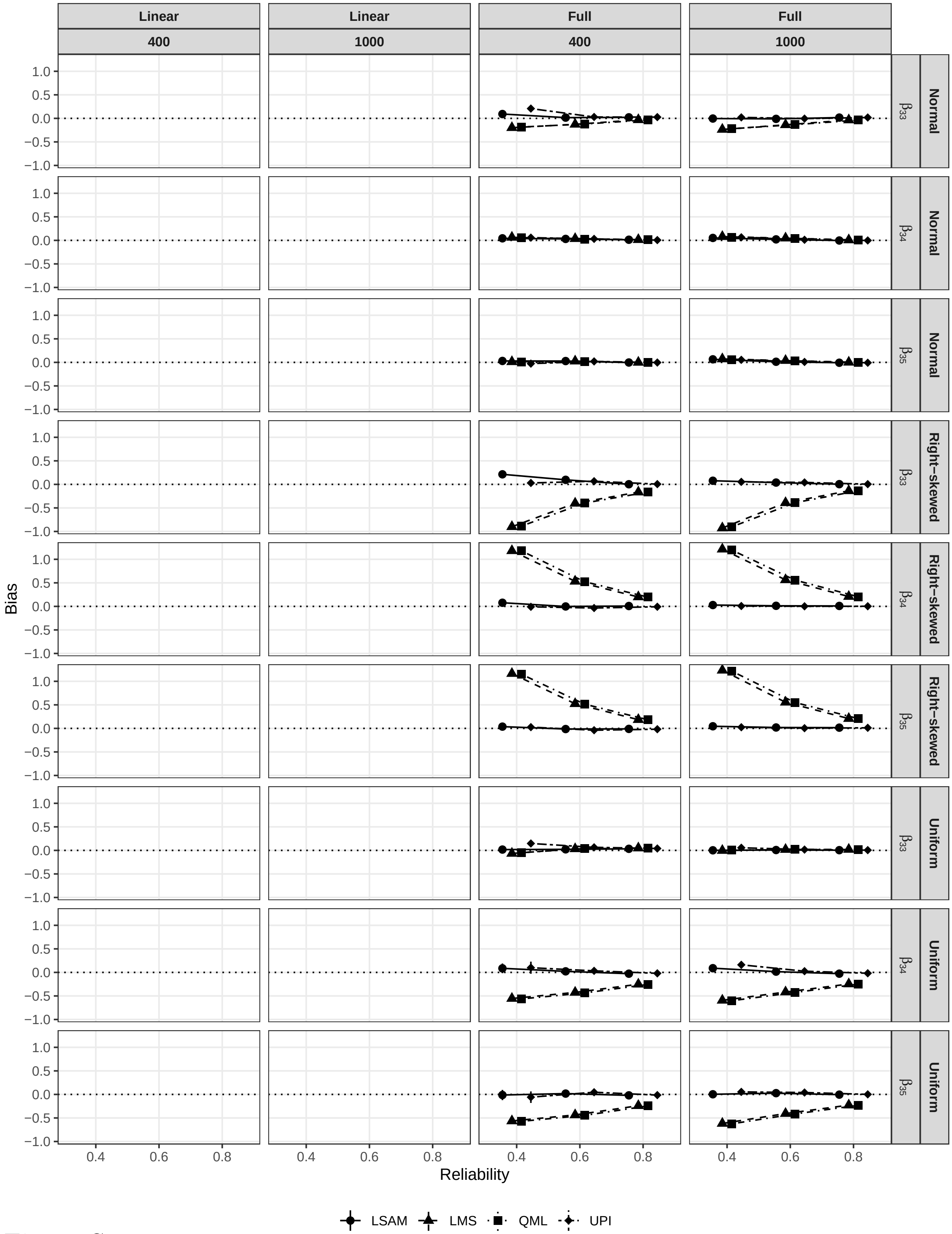

Figure S3

Relative bias for Simulation 1

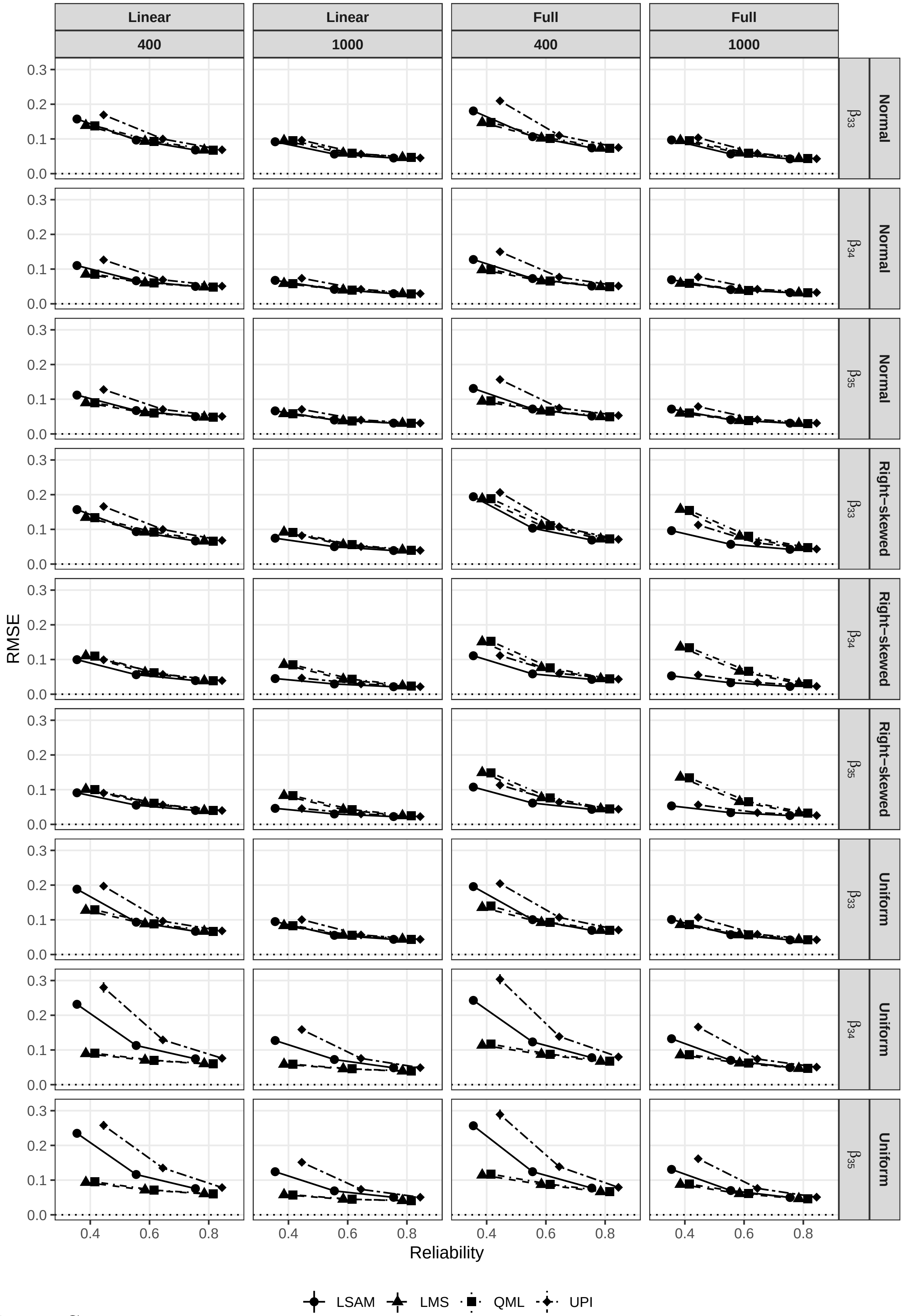

Figure S4

RMSE for Simulation 1

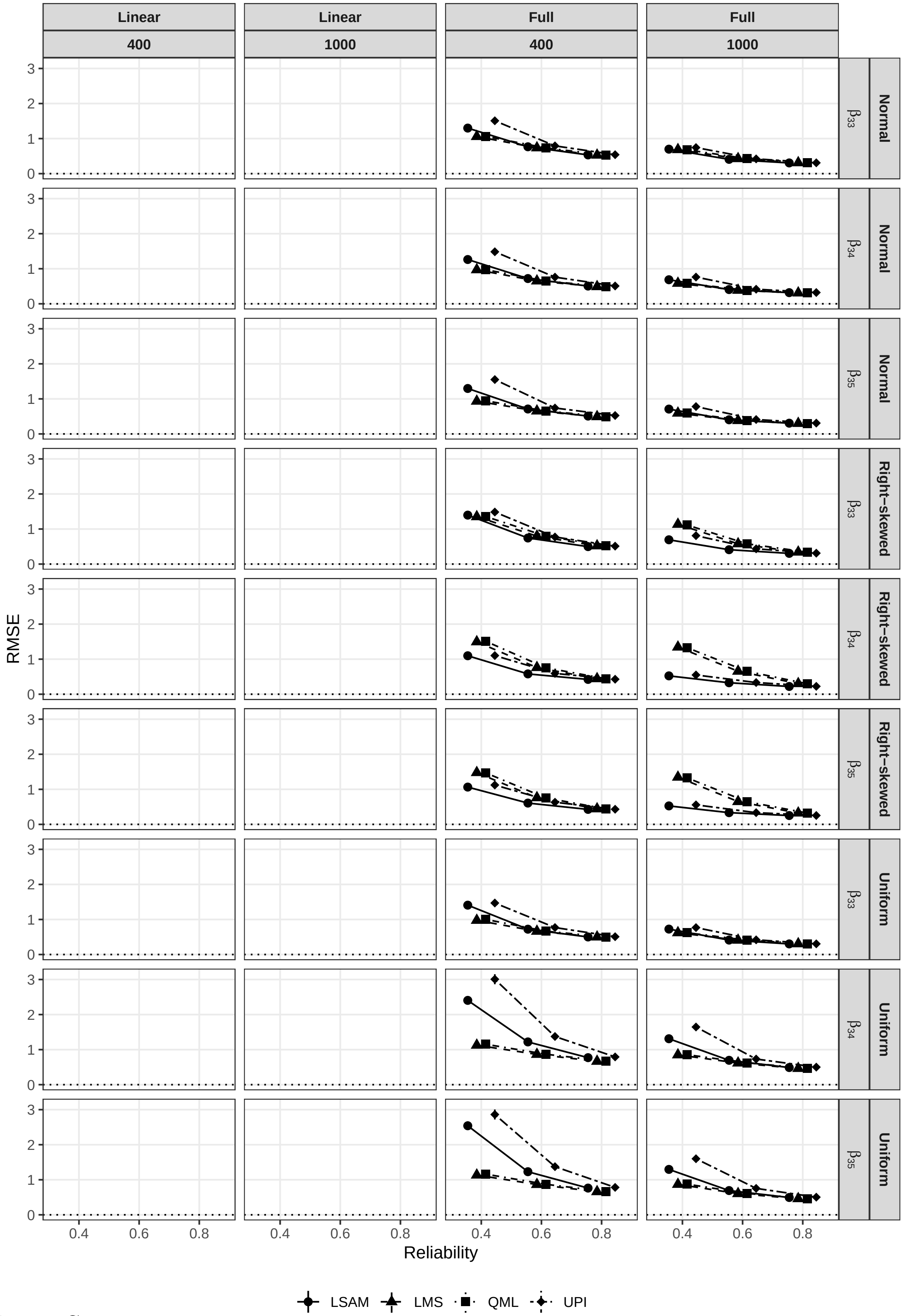

Figure S5

Relative RMSE for Simulation 1

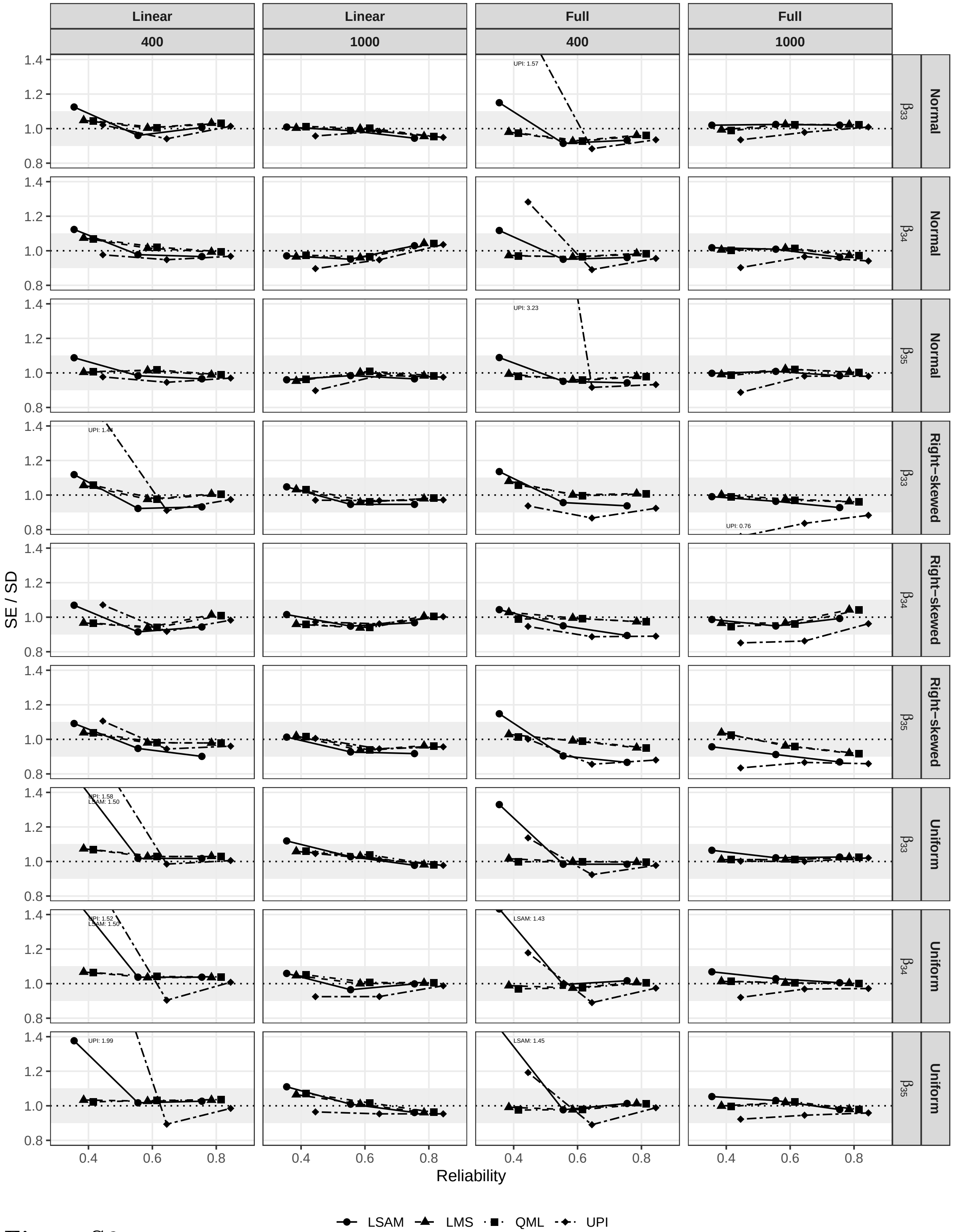

Figure S6

SE/SD ratio for Simulation 1

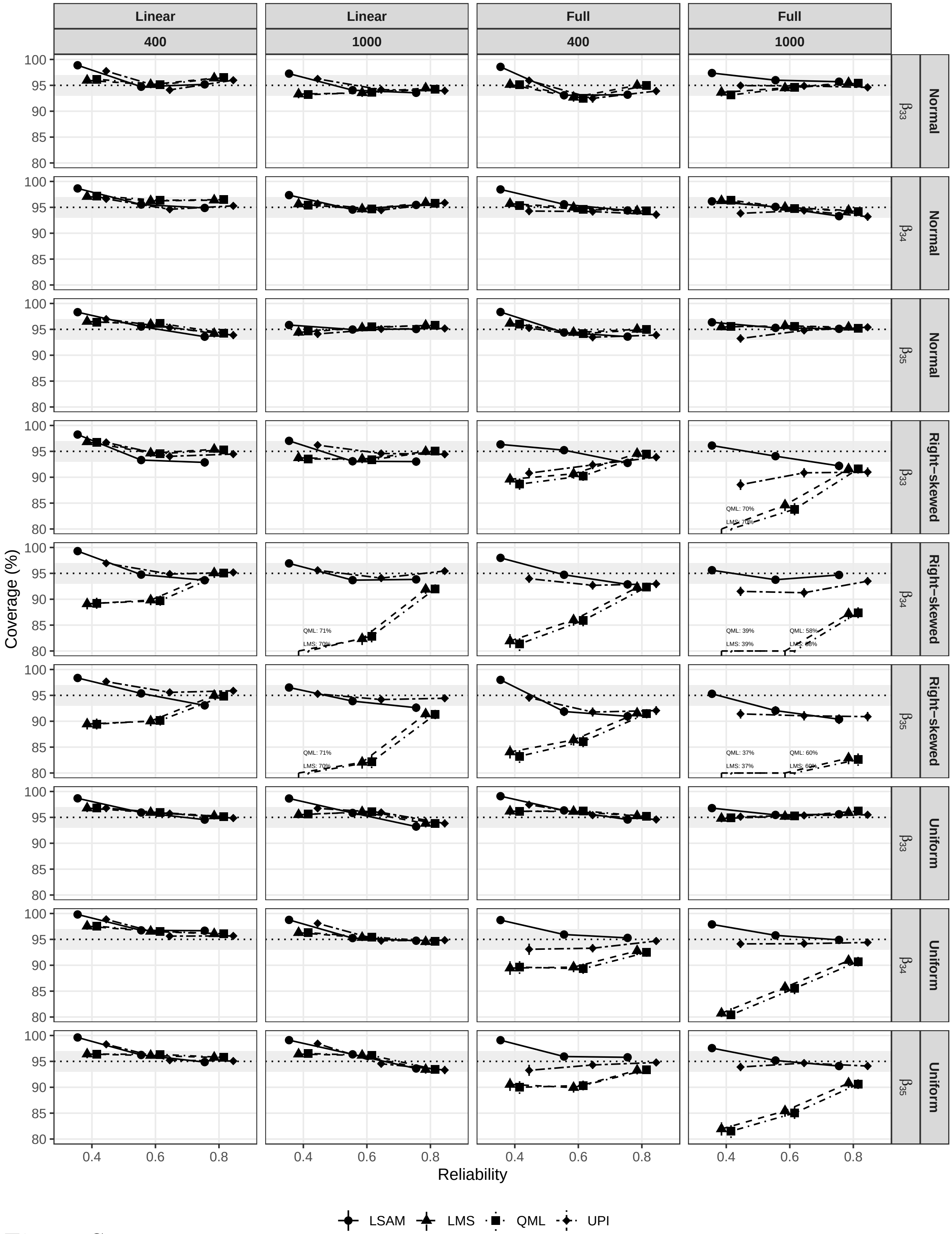

Figure S7

Coverage for Simulation 1

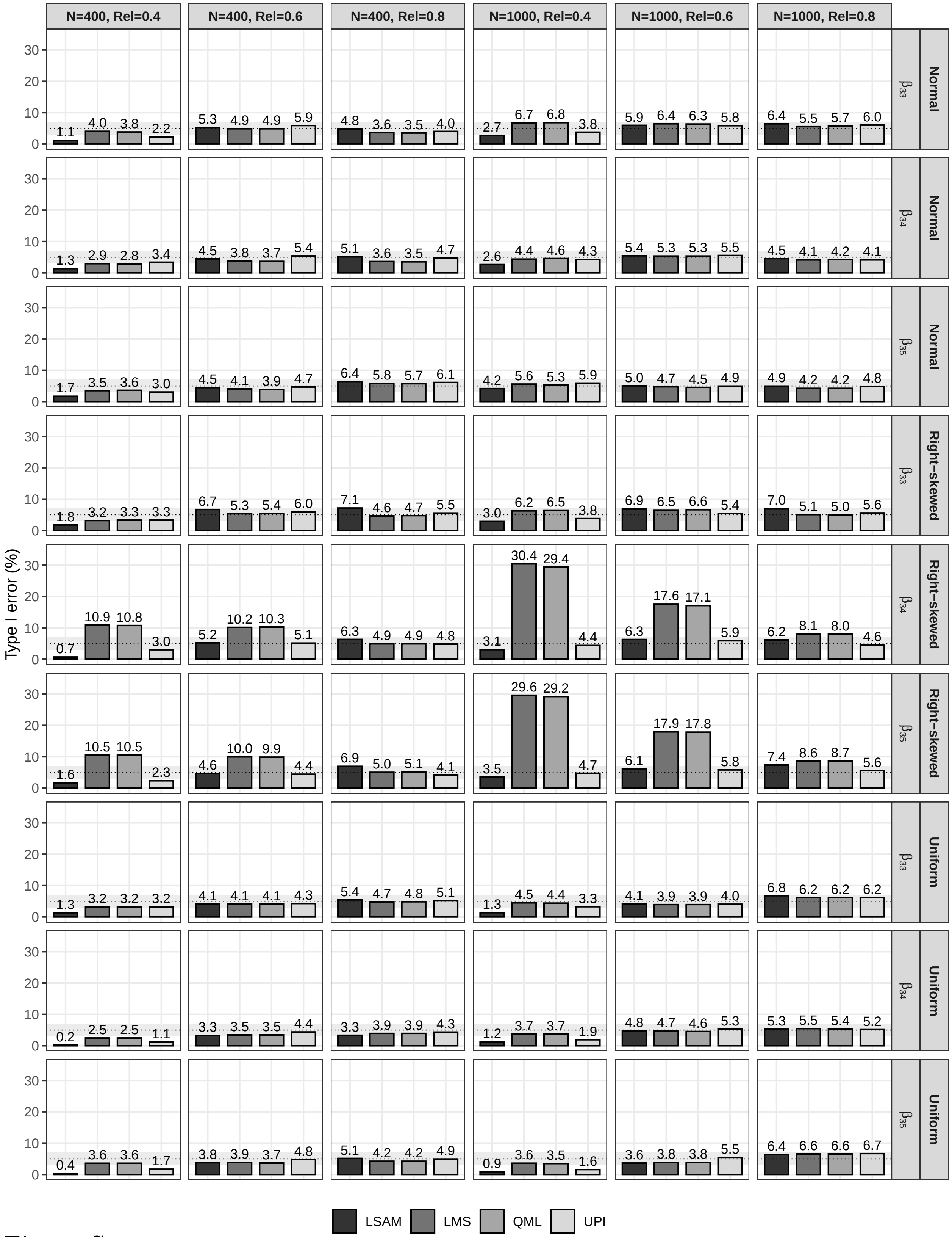

Figure S8

Type I error rate for Simulation 1

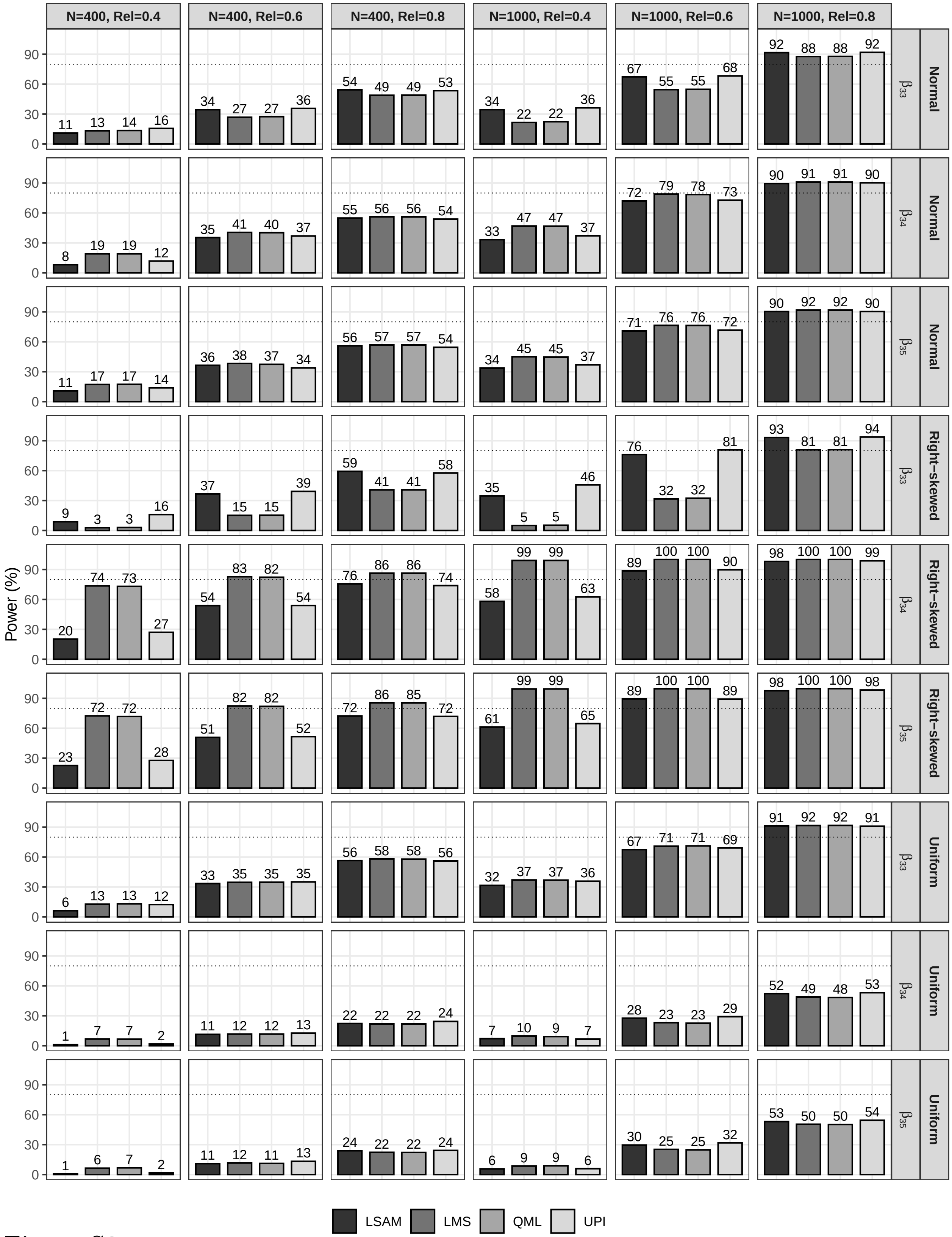

Figure S9

Empirical power for Simulation 1

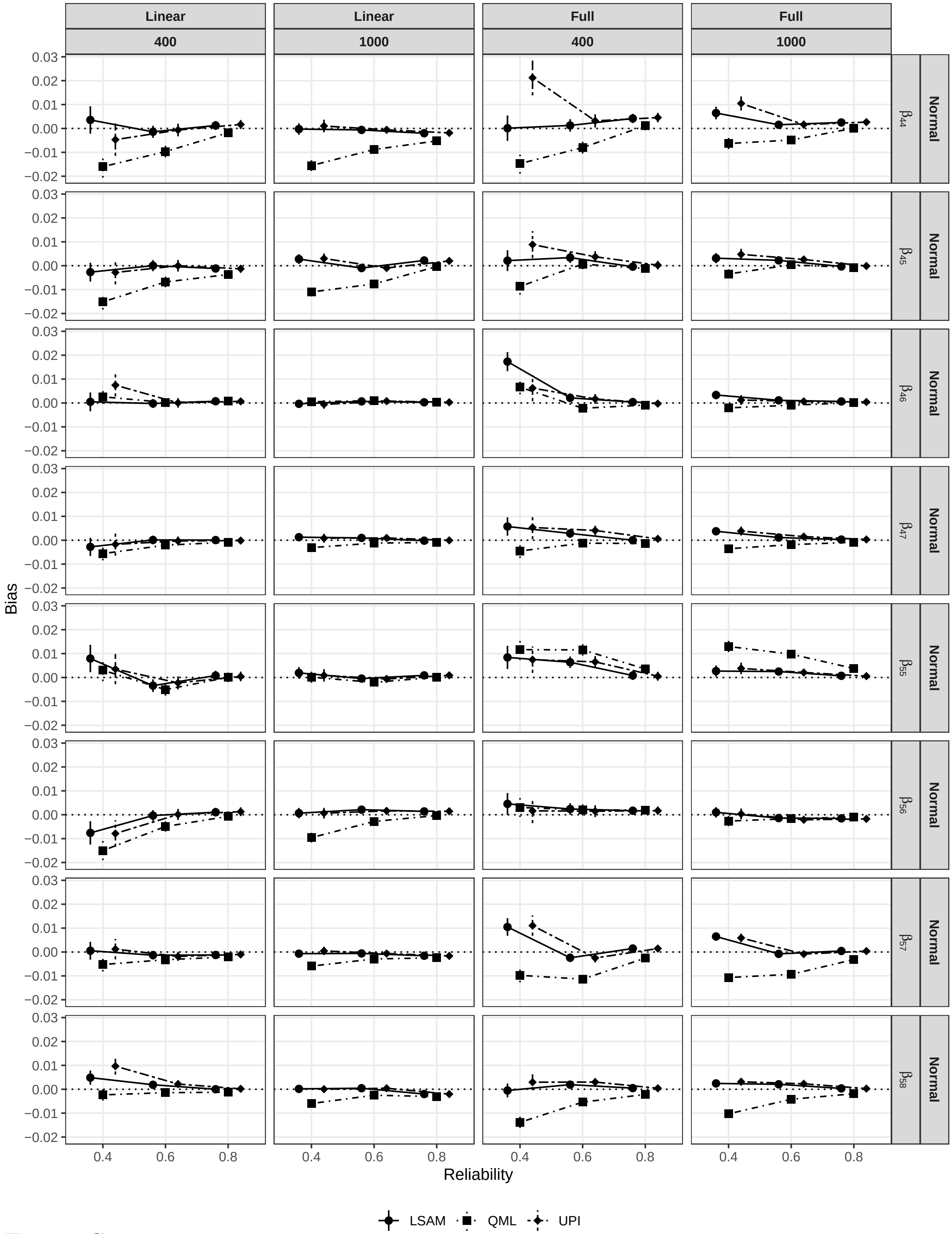

Figure S10

Absolute bias for Simulation 2 (Normal)

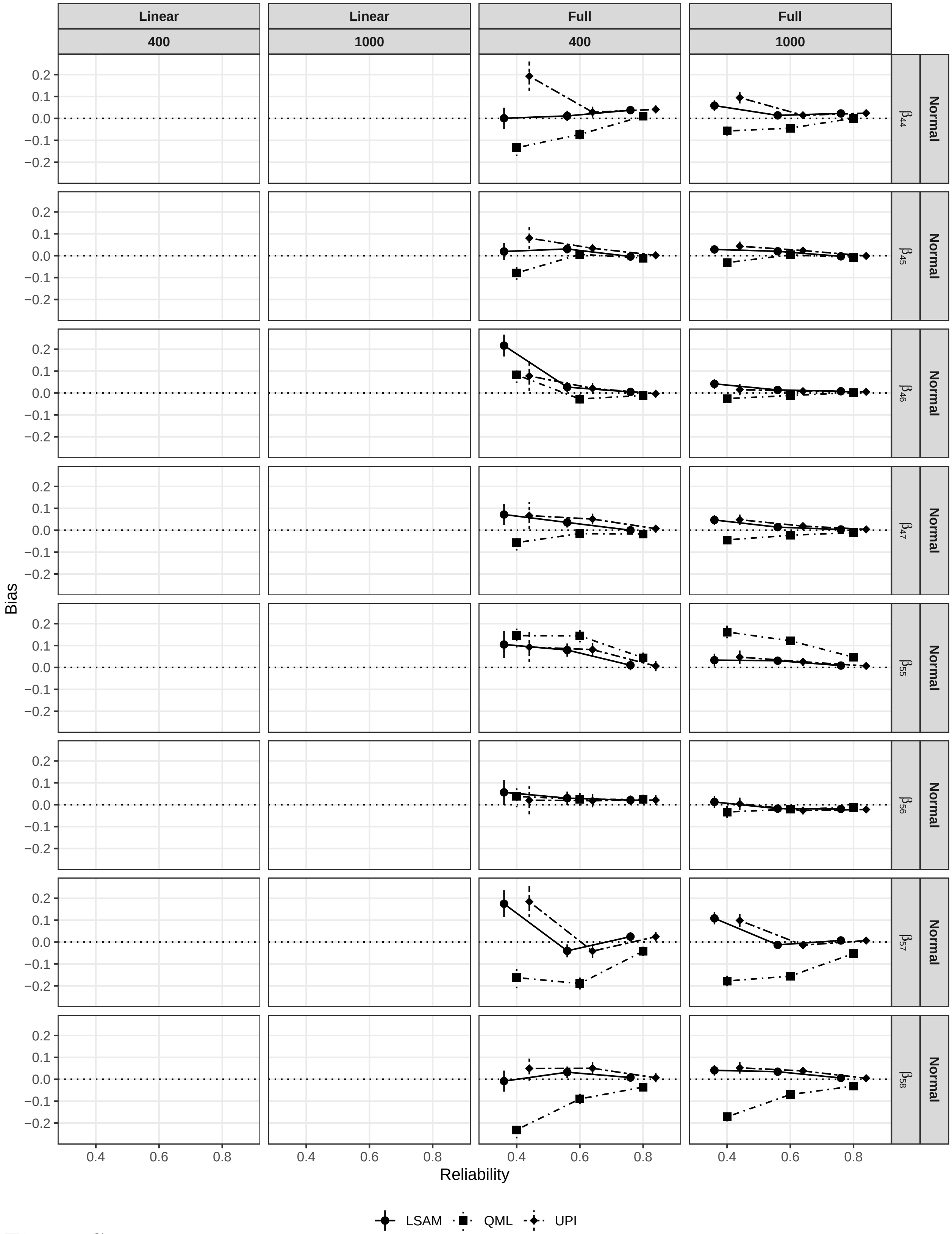

Figure S11

Relative bias for Simulation 2 (Normal)

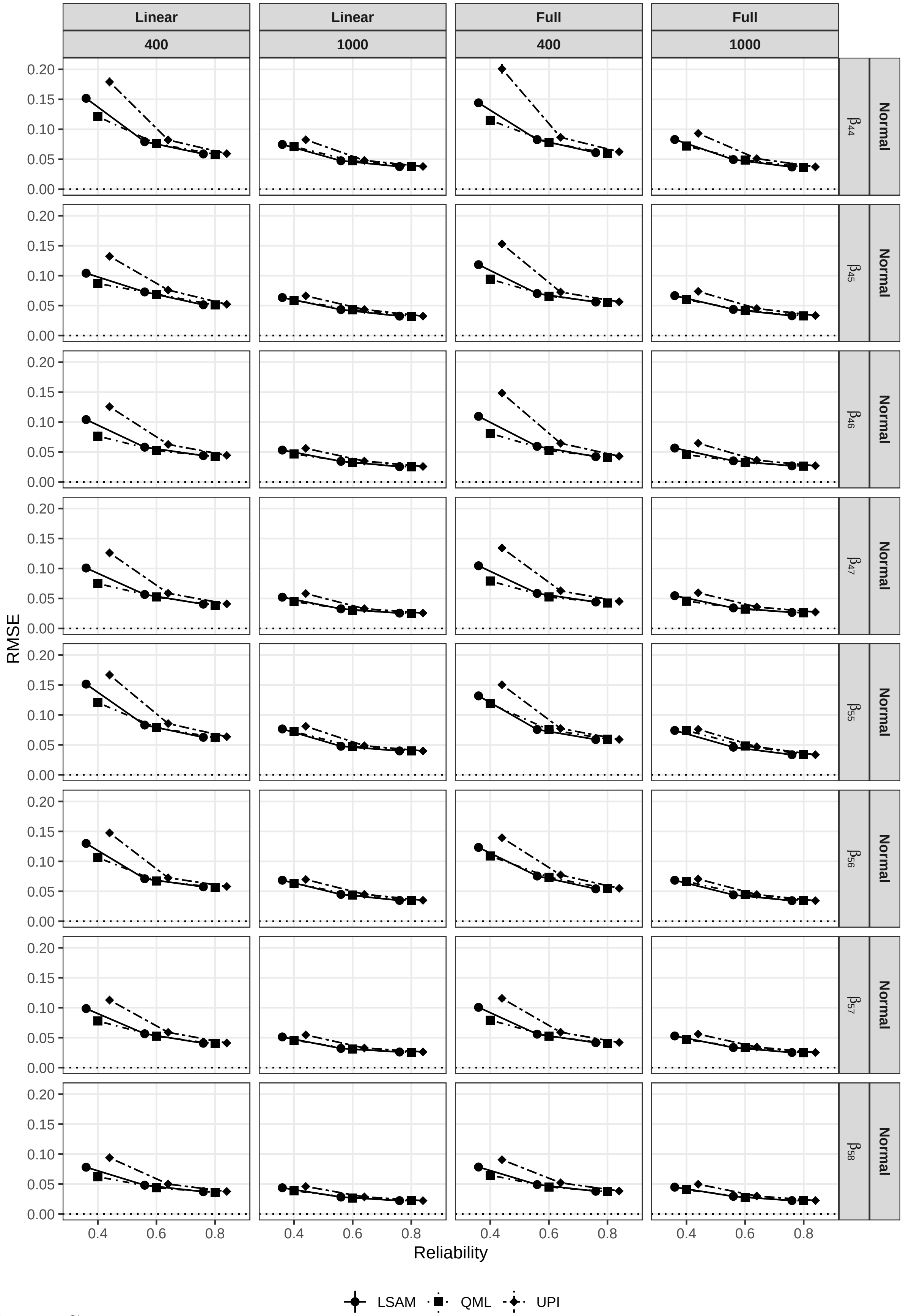

Figure S12

RMSE for Simulation 2 (Normal)

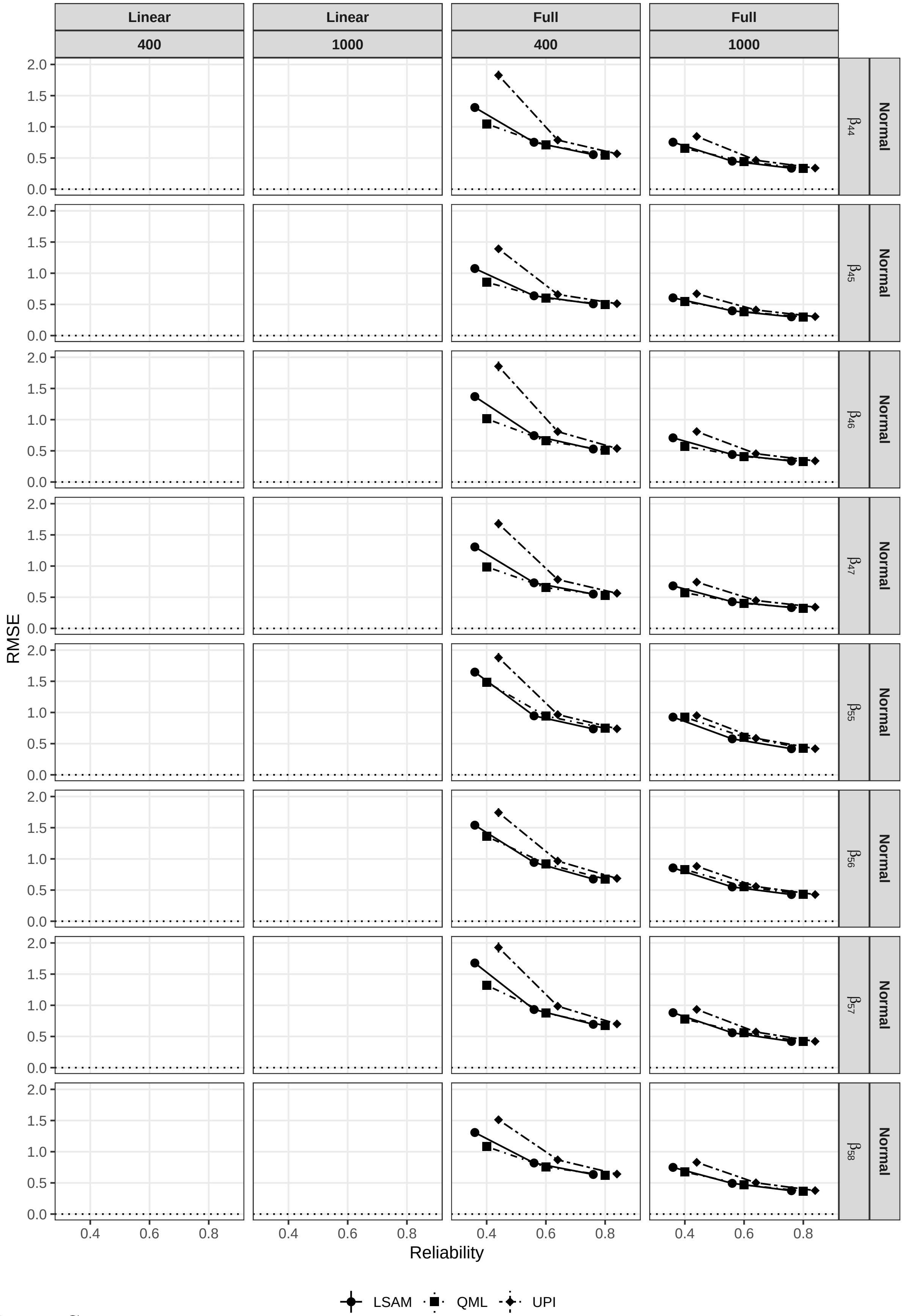

Figure S13

Relative RMSE for Simulation 2 (Normal)

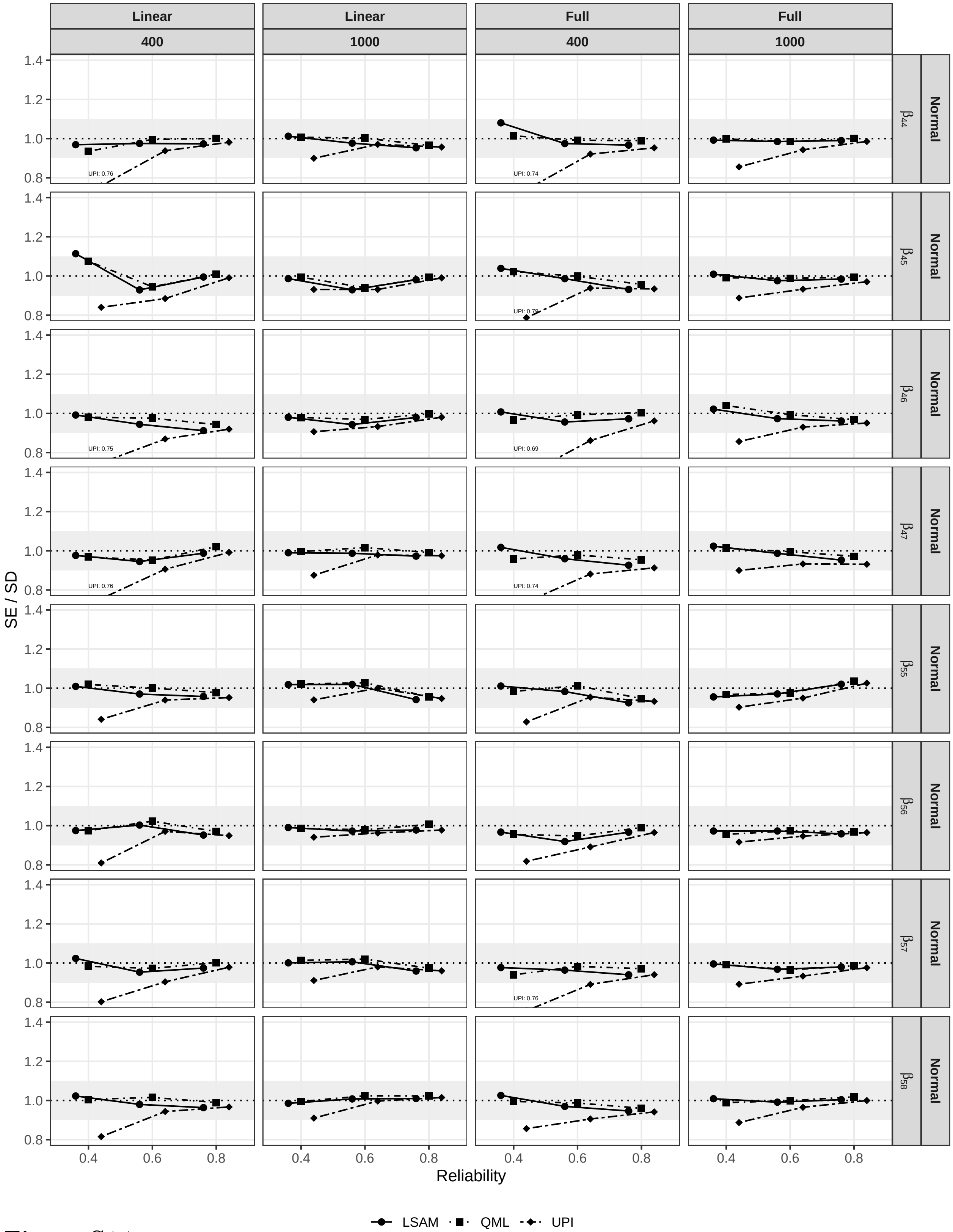

Figure S14

SE/SD ratio for Simulation 2 (Normal)

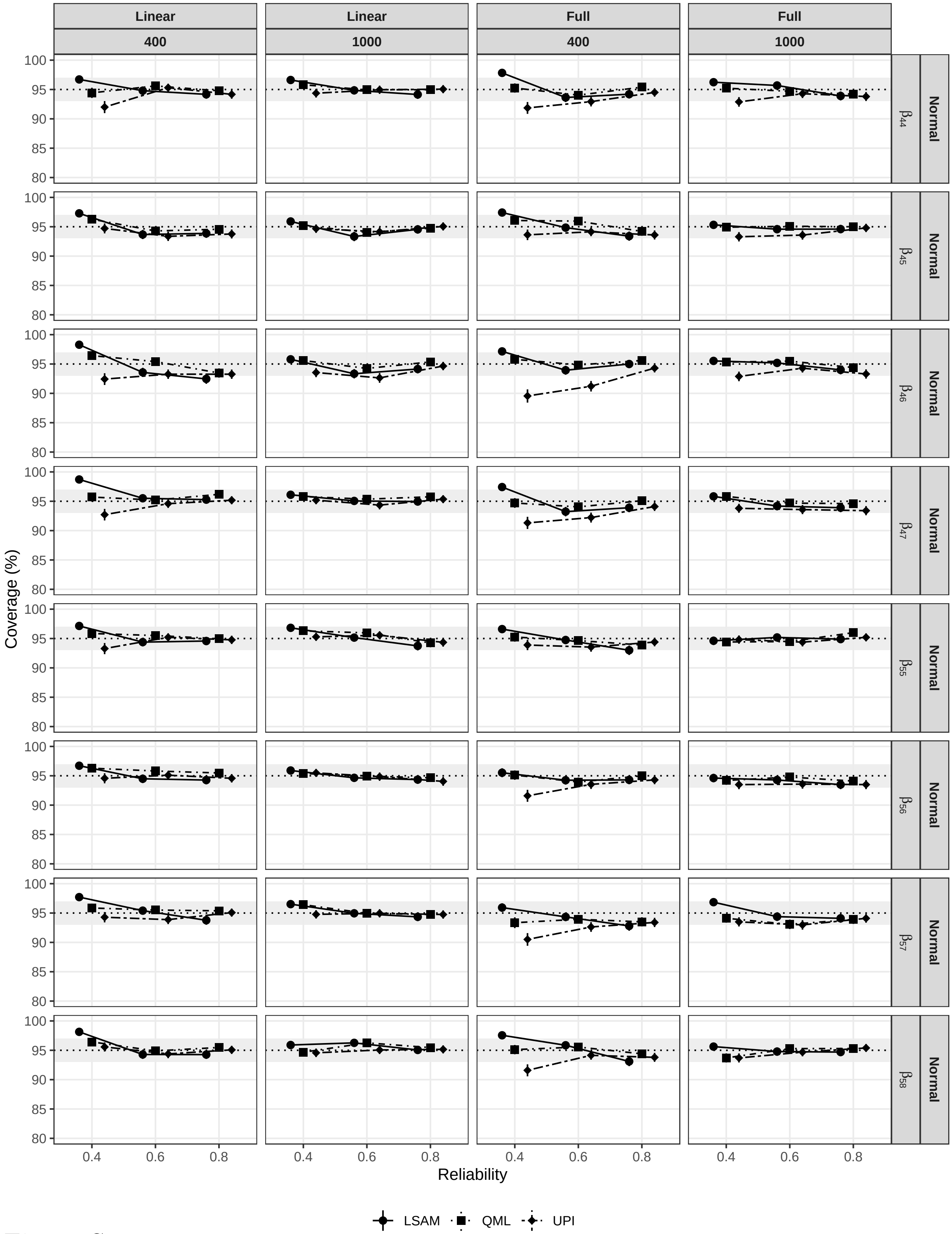

Figure S15

Coverage for Simulation 2 (Normal)

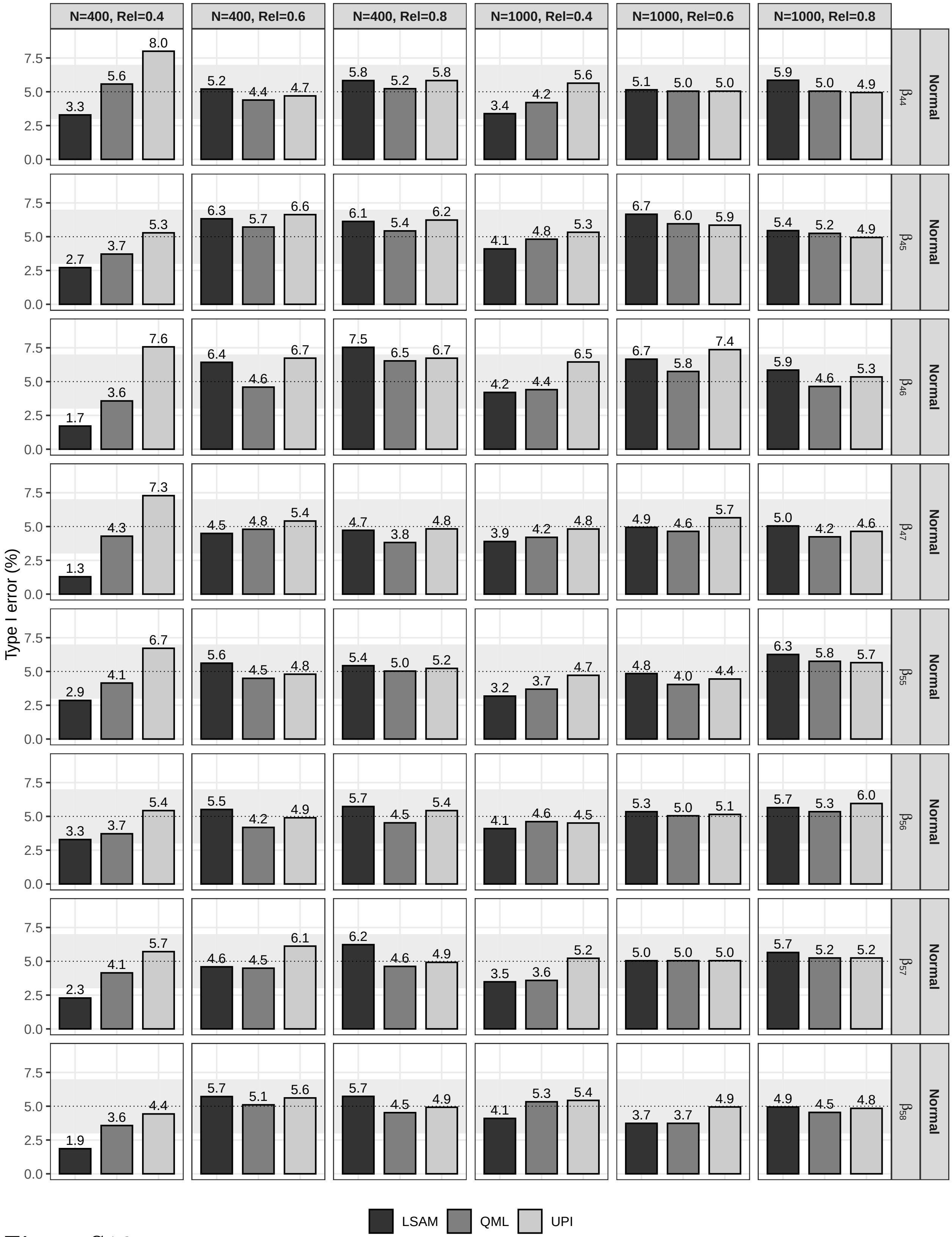

Figure S16

Type I error rate for Simulation 2 (Normal)

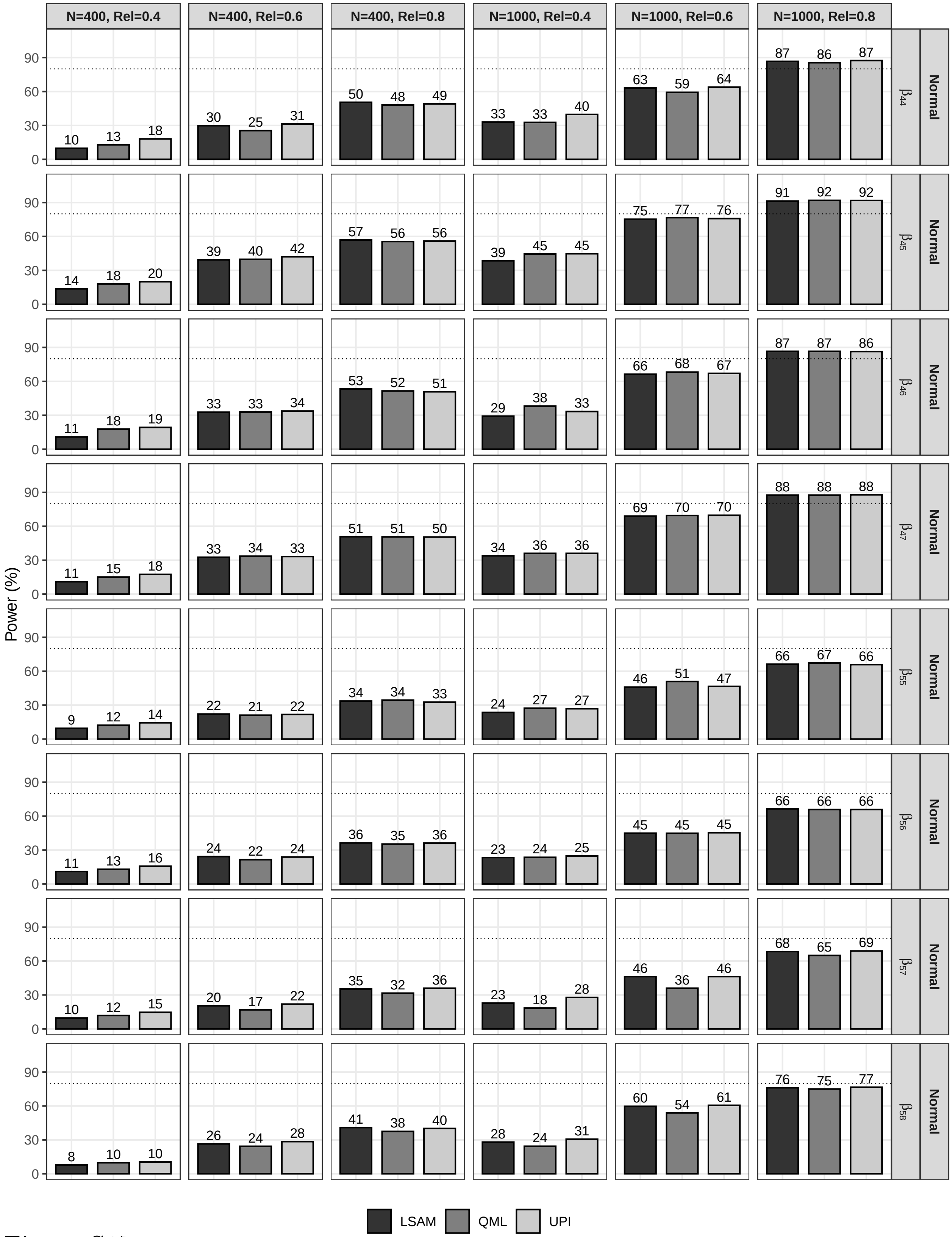

Figure S17

Empirical power for Simulation 2 (Normal)

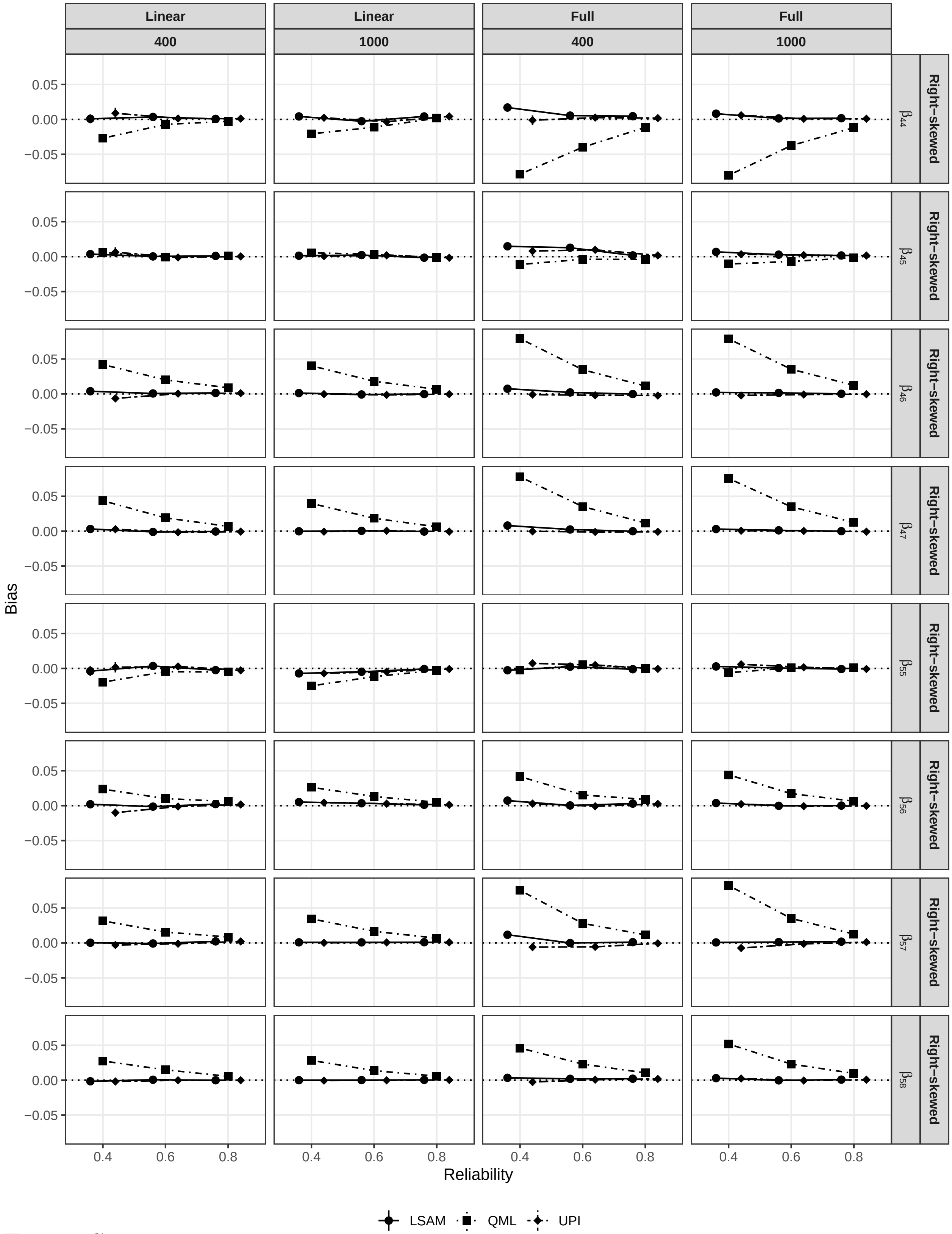

Figure S18

Absolute bias for Simulation 2 (Right-skewed)

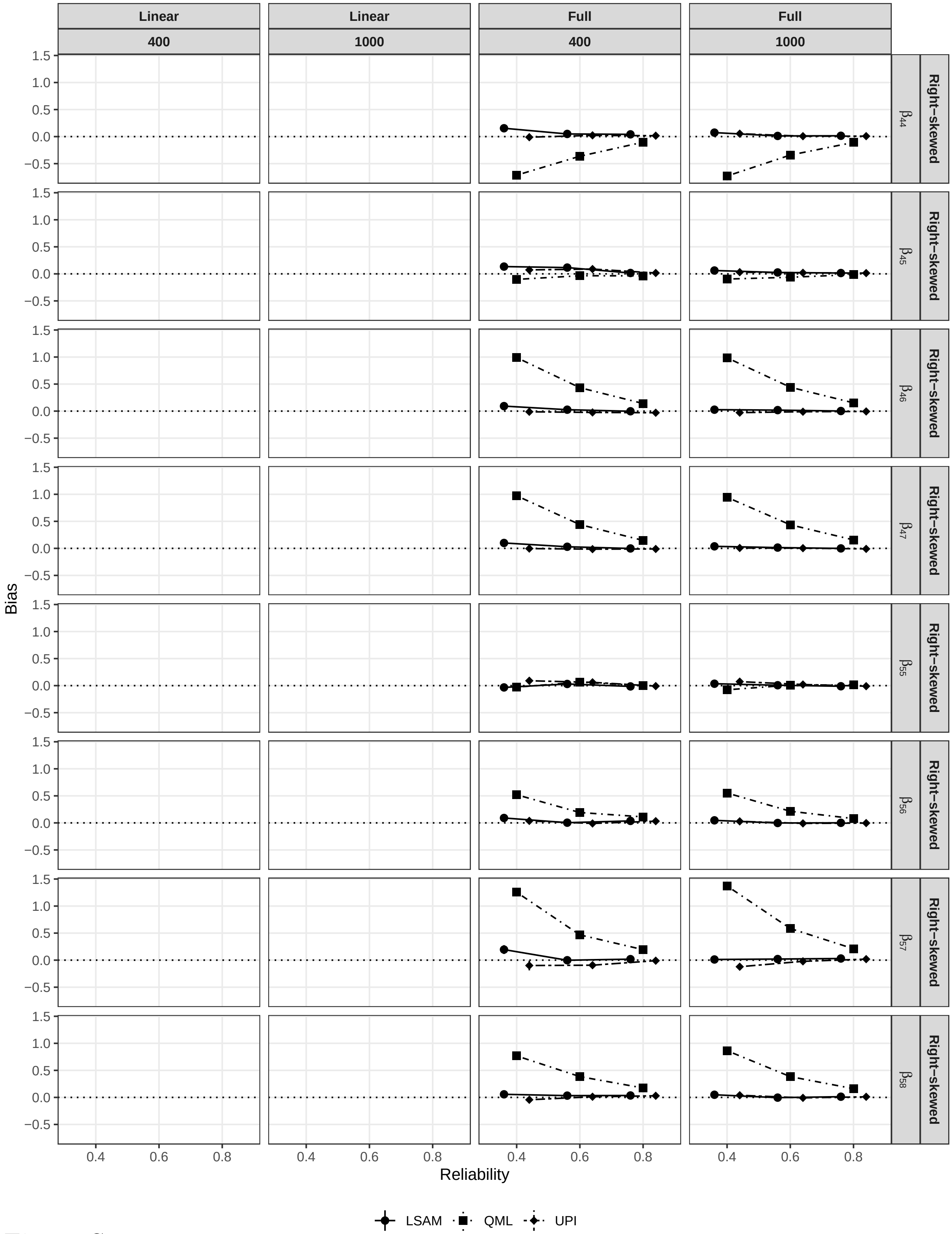

Figure S19

Relative bias for Simulation 2 (Right-skewed)

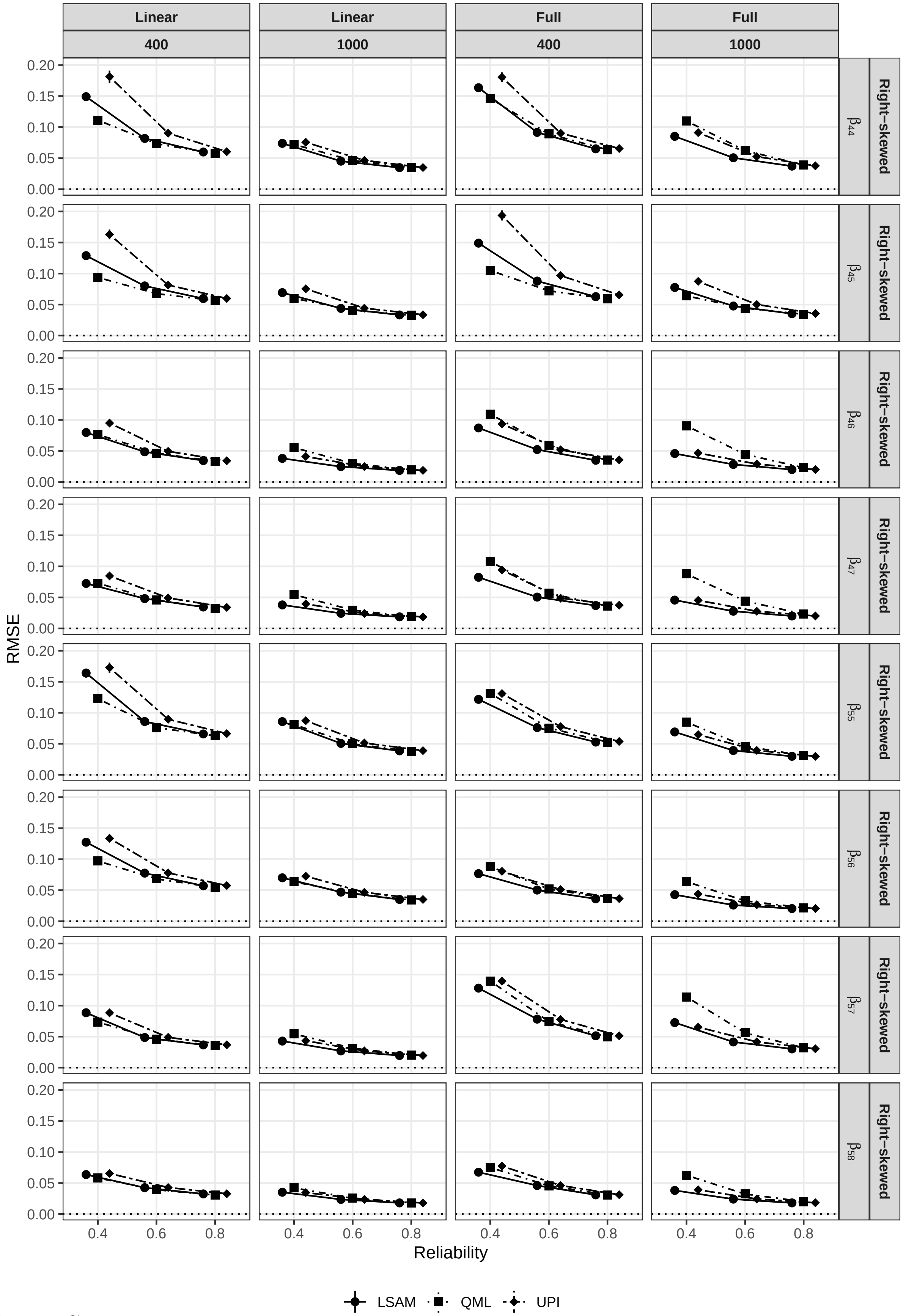

Figure S20

RMSE for Simulation 2 (Right-skewed)

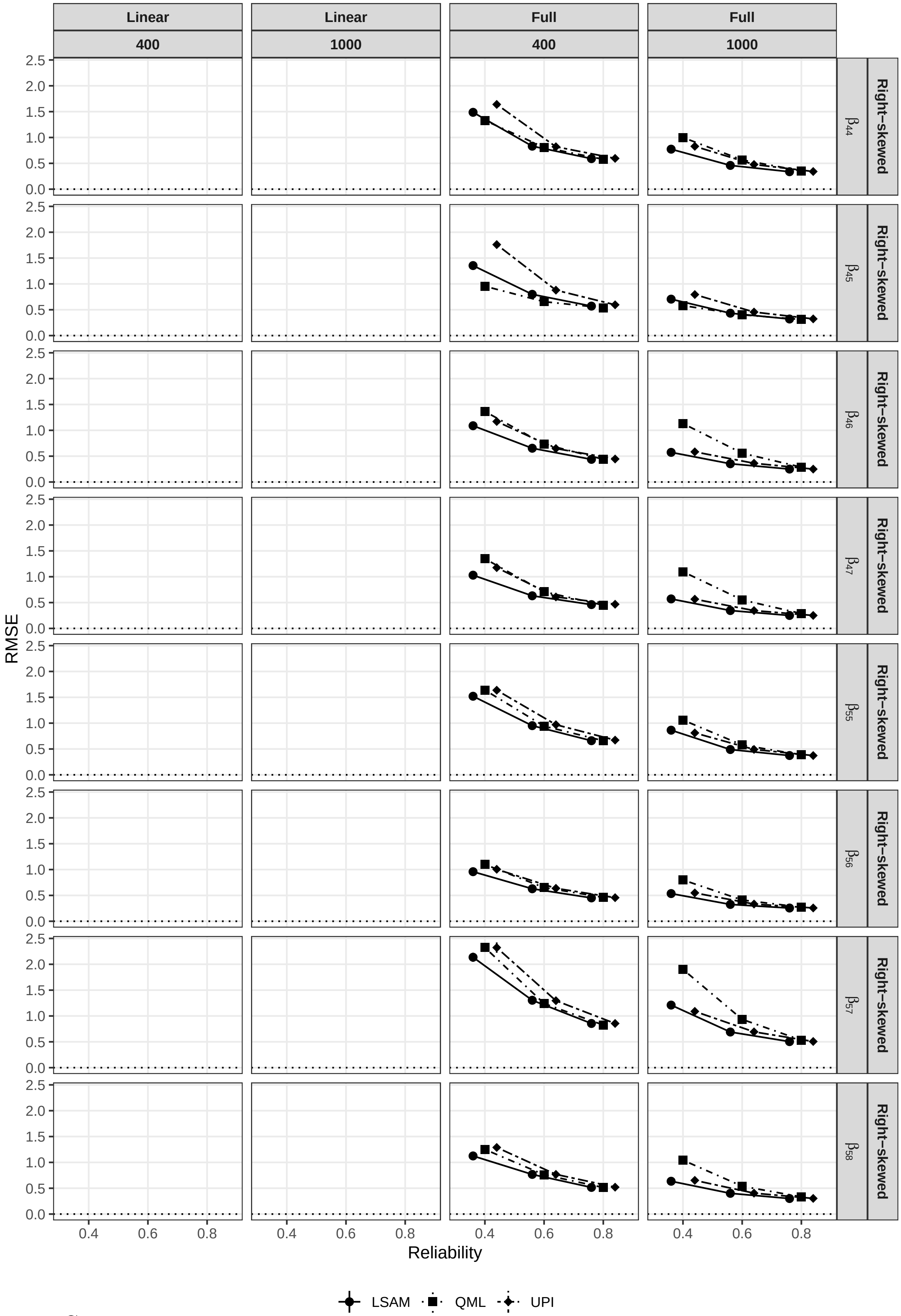

Figure S21

Relative RMSE for Simulation 2 (Right-skewed)

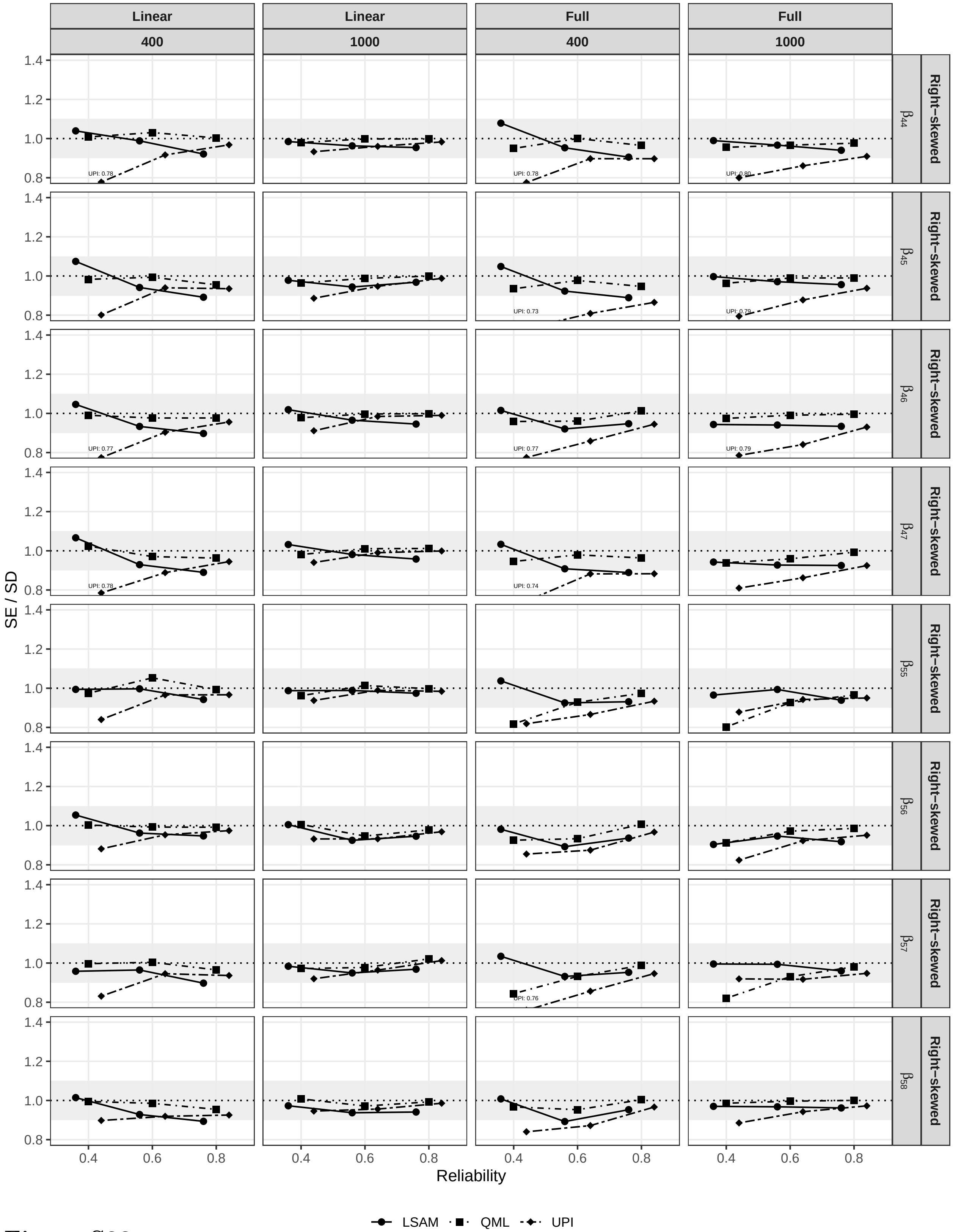

Figure S22

SE/SD ratio for Simulation 2 (Right-skewed)

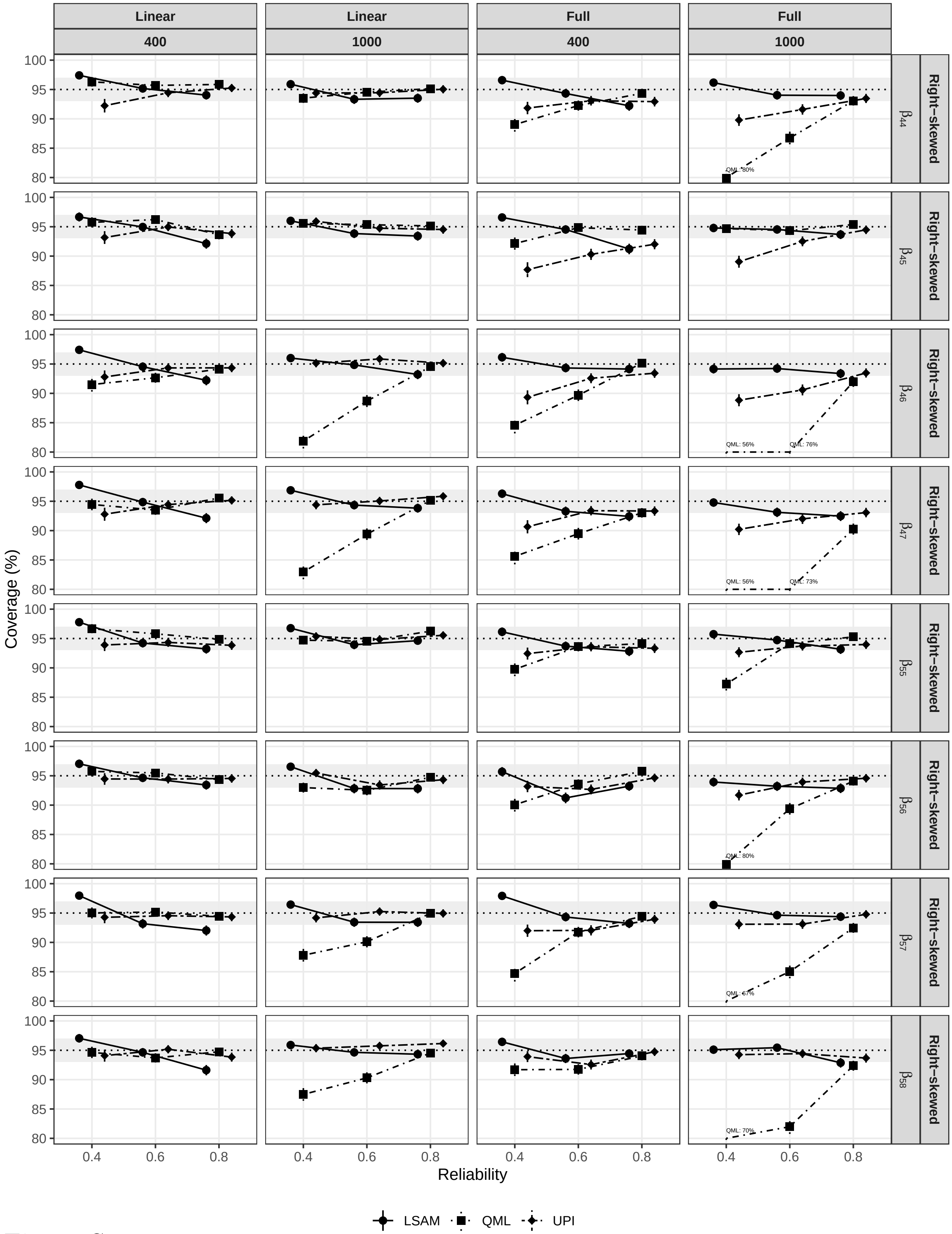

Figure S23

Coverage for Simulation 2 (Right-skewed)

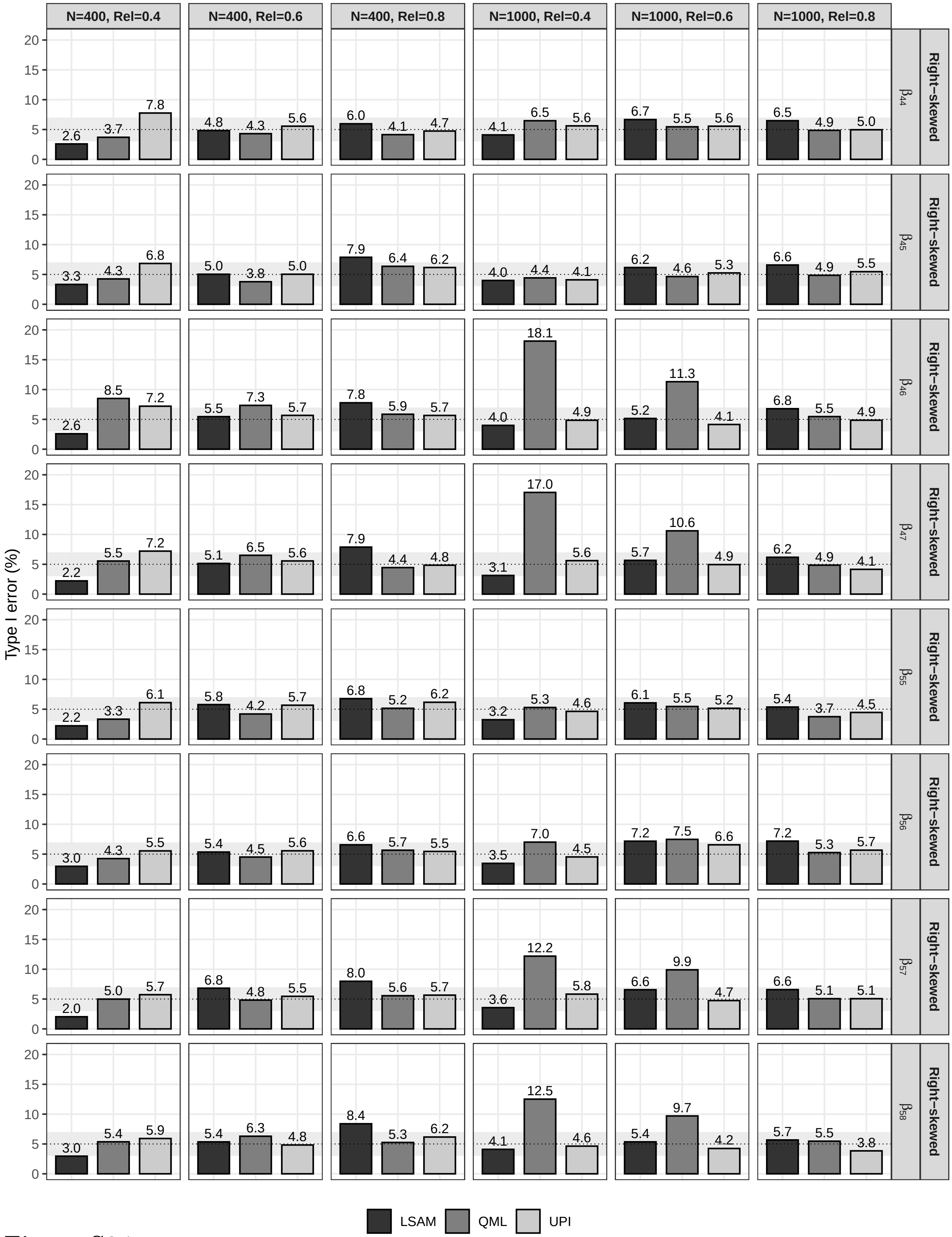

Figure S24

Type I error rate for Simulation 2 (Right-skewed)

Figure S25

Empirical power for Simulation 2 (Right-skewed)

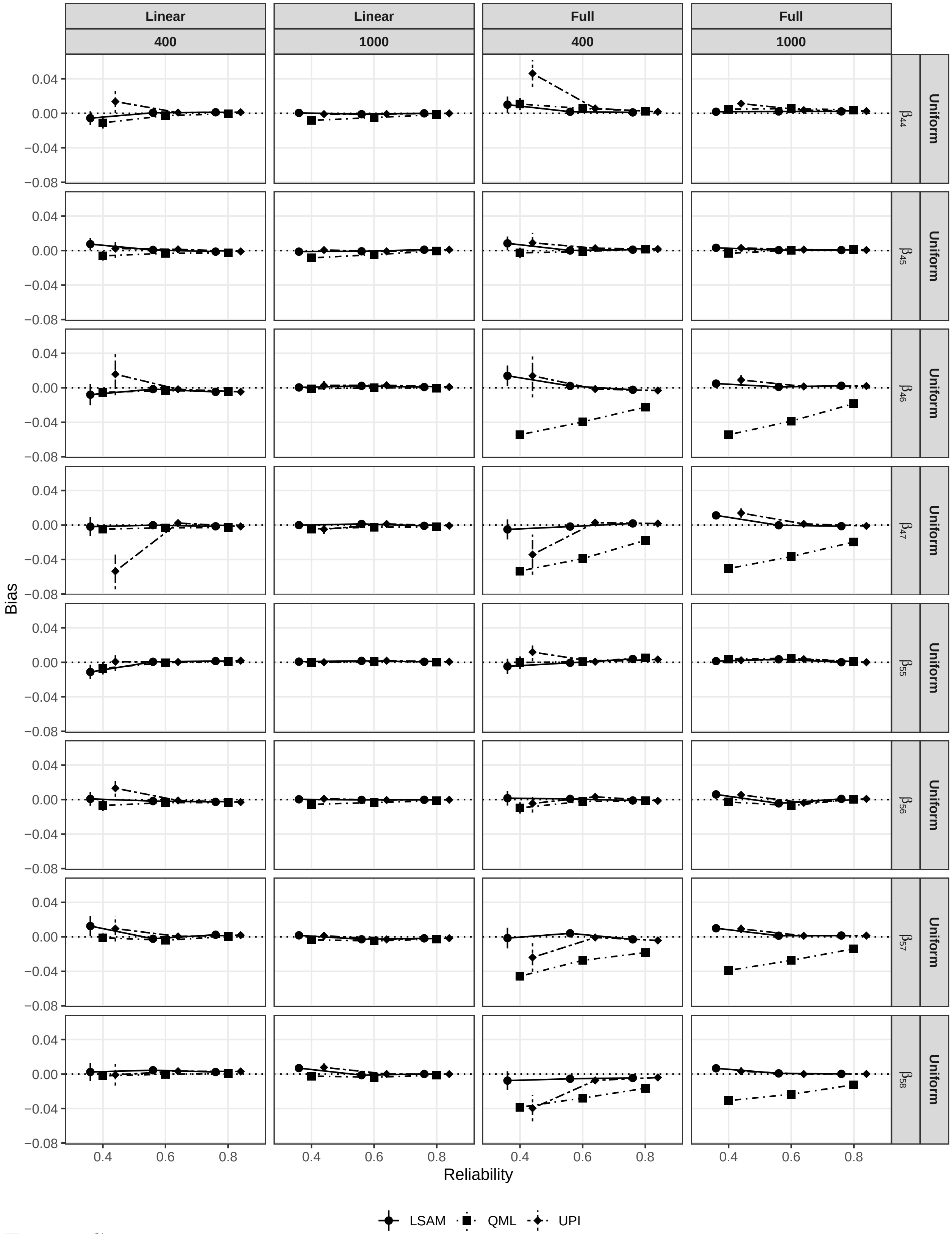

Figure S26

Absolute bias for Simulation 2 (Uniform)

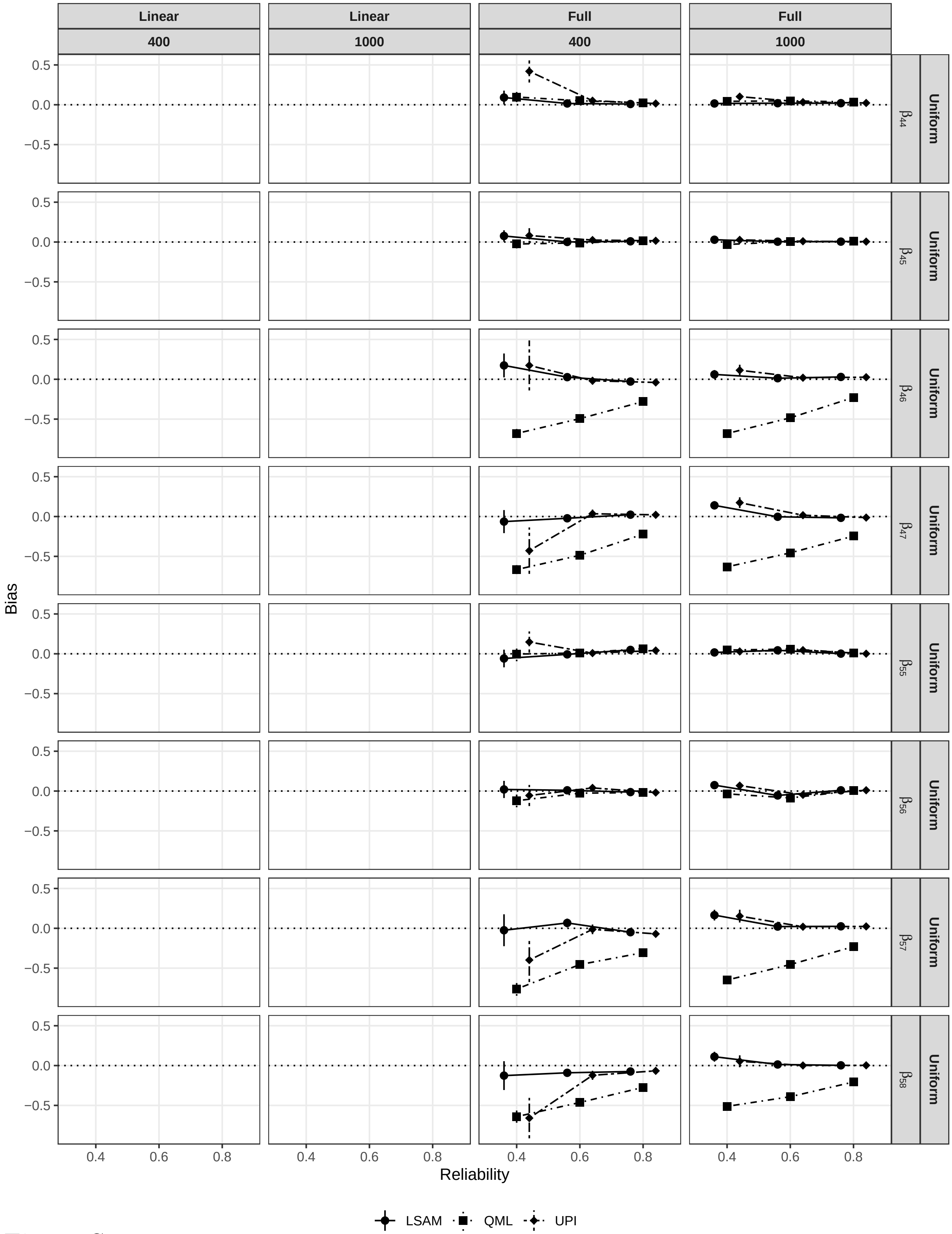

Figure S27

Relative bias for Simulation 2 (Uniform)

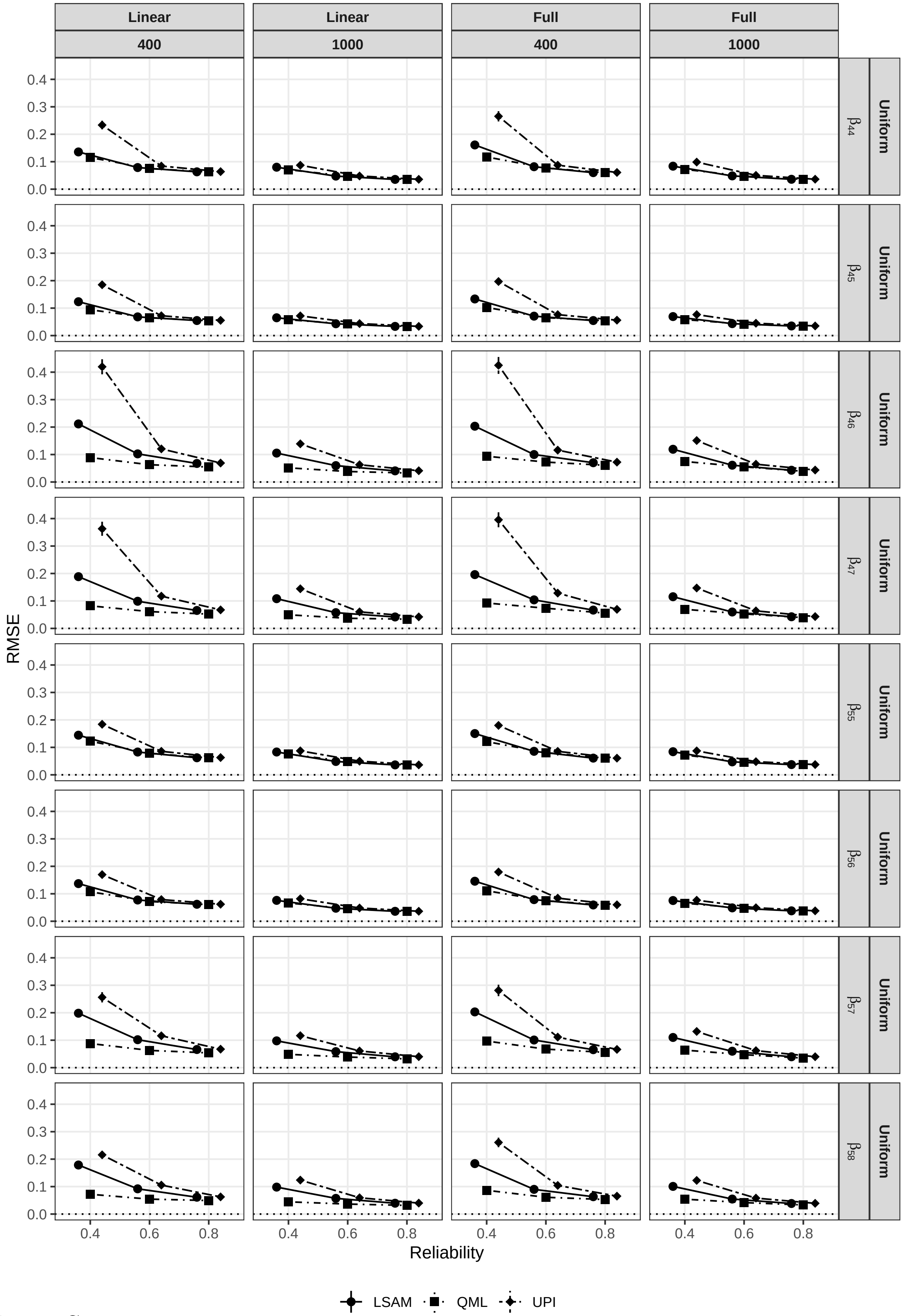

Figure S28

RMSE for Simulation 2 (Uniform)

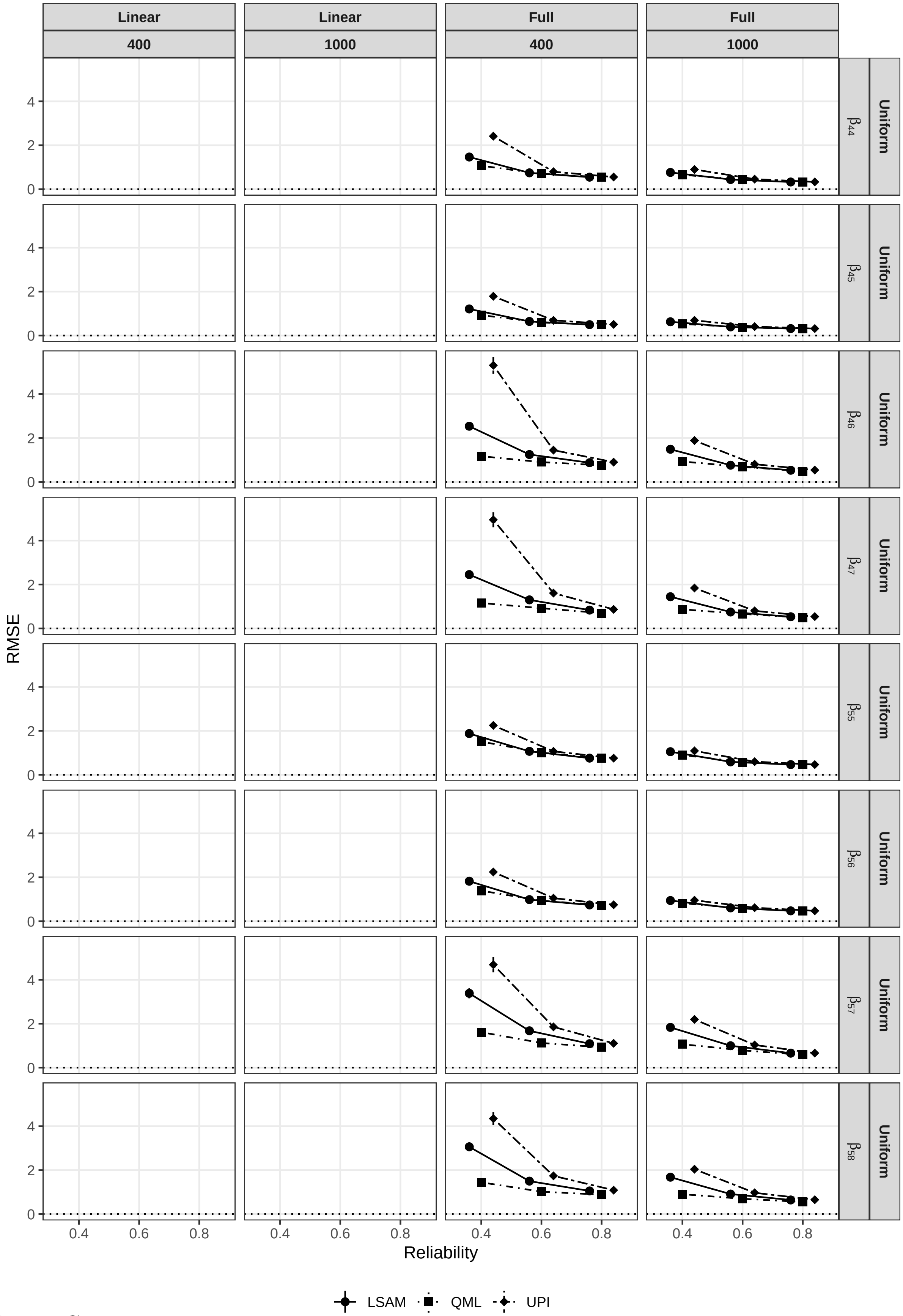

Figure S29

Relative RMSE for Simulation 2 (Uniform)

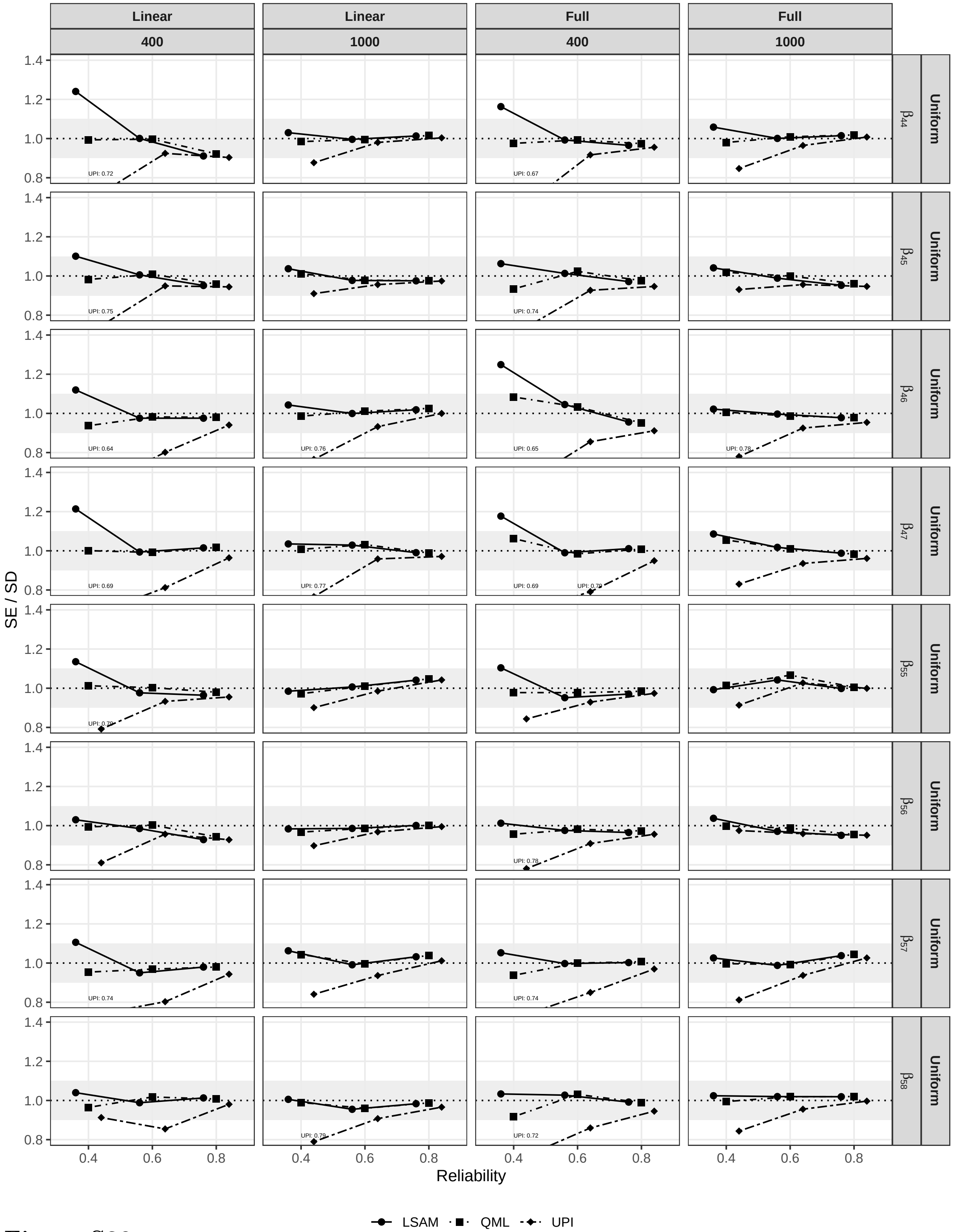

Figure S30

SE/SD ratio for Simulation 2 (Uniform)

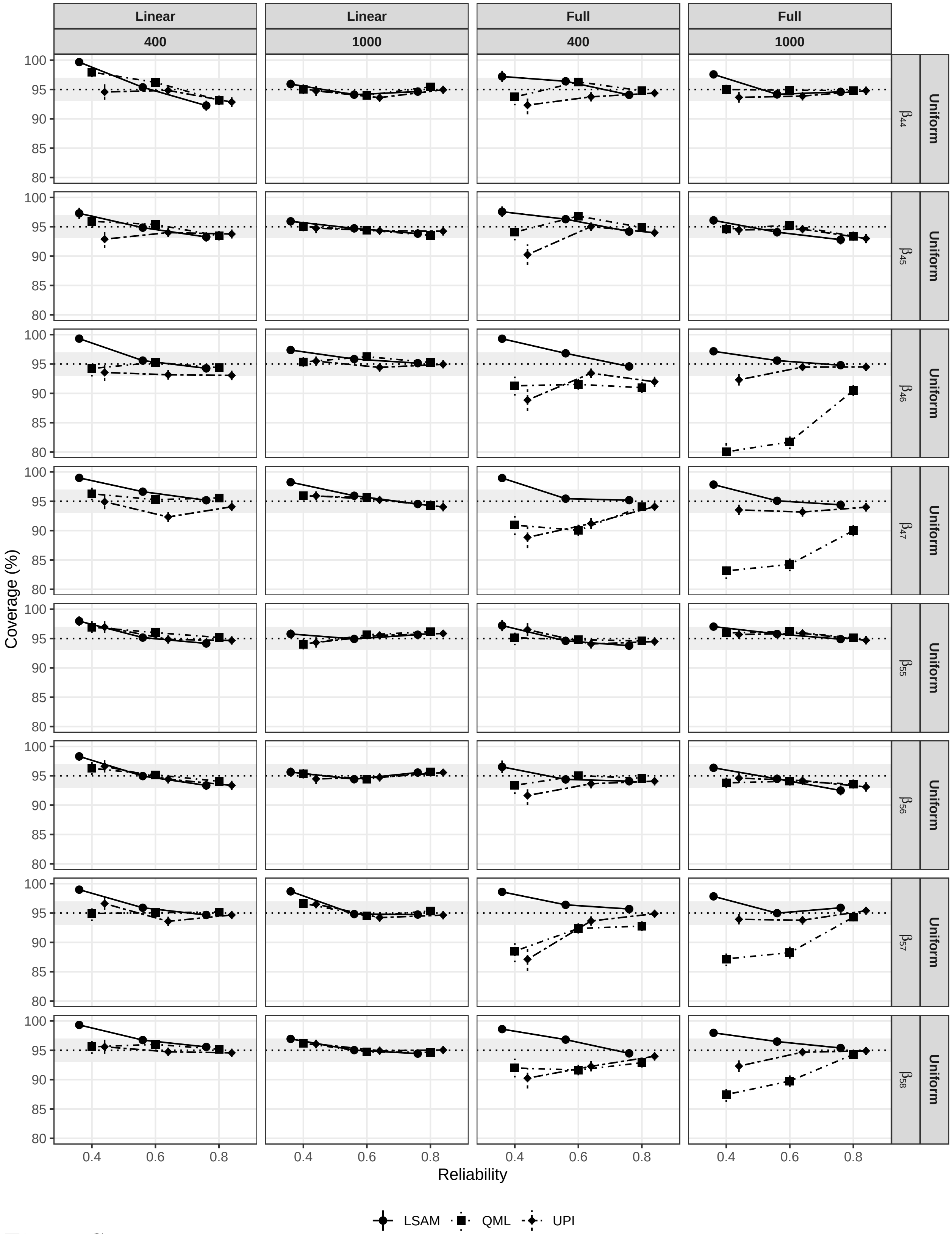

Figure S31

Coverage for Simulation 2 (Uniform)

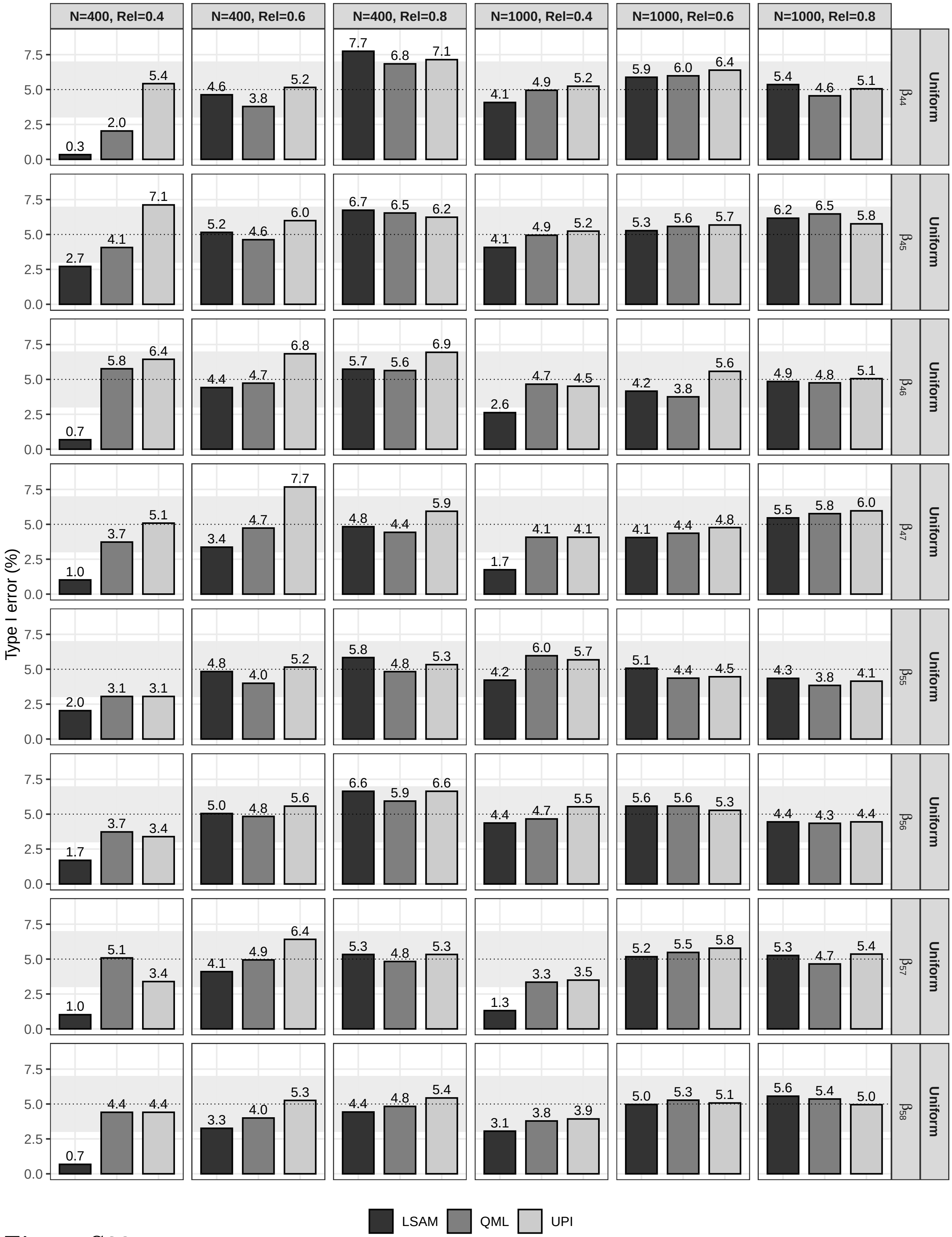

Figure S32

Type I error rate for Simulation 2 (Uniform)

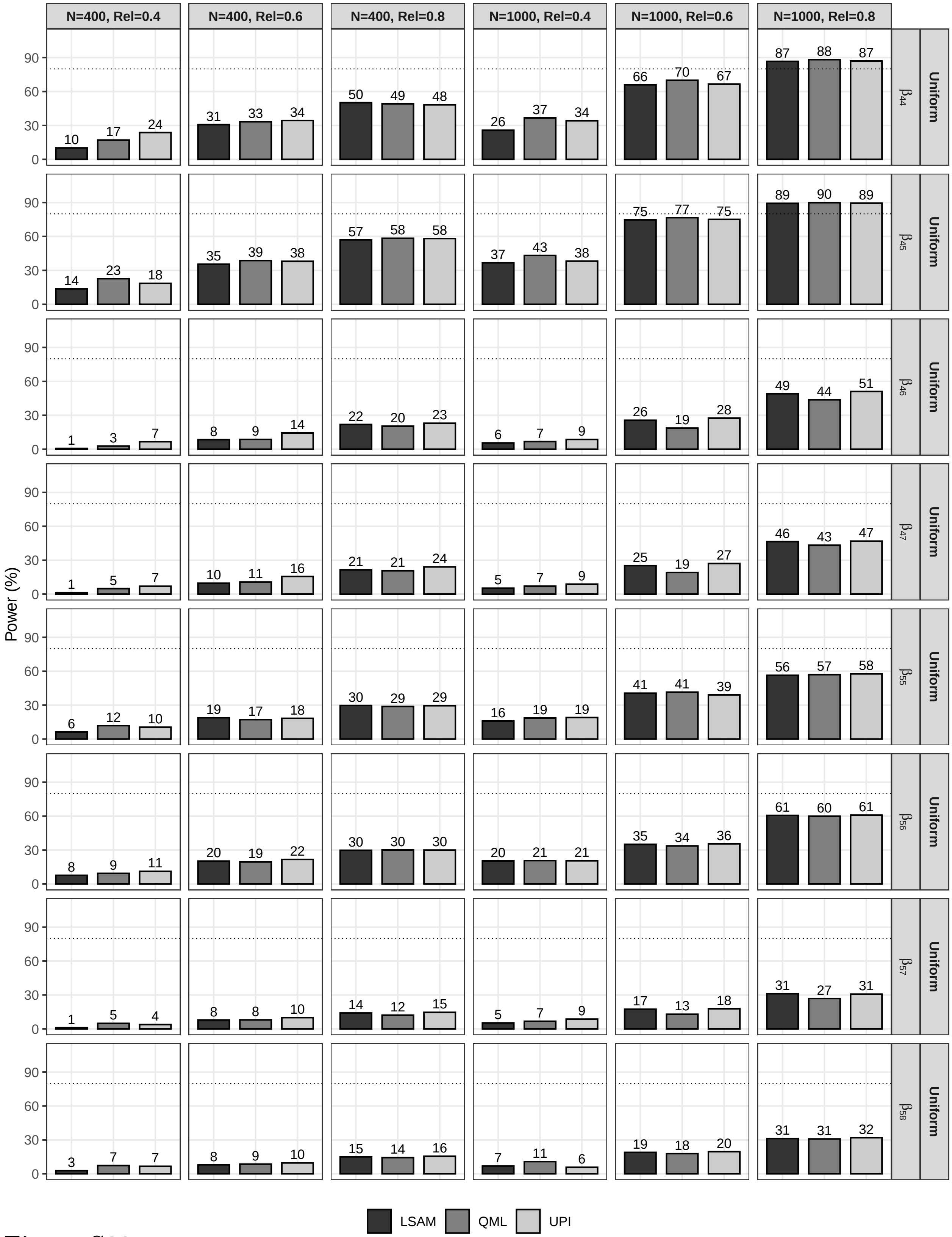

Figure S33

Empirical power for Simulation 2 (Uniform)